

PDQUEST™

User Guide for Version 6.2 *Windows and Macintosh*

PDQUEST User Guide

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Preface

1. About This Document

This user guide is designed to be used as a reference in your everyday use of PDQUEST™. It provides detailed information about the tools and commands of PDQUEST for the Windows and Macintosh platforms. Any platform differences in procedures and commands are noted in the text.

This guide assumes you have a working knowledge of your computer operating system and its conventions, including how to use a mouse and standard menus and commands, and how to open, save, and close files. For help with any of these techniques, see the documentation that came with your computer.

This guide uses certain text conventions to describe specific commands and functions.

Example	Indicates
File > Open	Choosing the Open command under the File menu.
Dragging	Positioning the cursor on an object and holding down the left mouse button while you move the mouse.
CTRL+S	Holding down the Control key while typing the letter s.
Right-click/ Left-click/ Double-click	Clicking the right mouse button/ Clicking the left mouse button/ Clicking the left mouse button twice.

Some of the illustrations of menus and dialog boxes found in this manual are taken from the Windows version of the software, and some are taken from the

Macintosh version. Both versions of a menu or dialog box will be shown only when there is a significant difference between the two.

2. Overview of 2-D Gel Electrophoresis

2-D gel electrophoresis combines the techniques of isoelectric focusing with SDS polyacrylamide gel electrophoresis. Since these two separation methods rely on independent properties of proteins, chemical (IEF) and physical (SDS), this procedure can resolve complex biological samples and polypeptide mixes. The extraordinary resolution capabilities of SDS-PAGE make it useful for studying complex systems in great detail.

2-D gels generate high resolution separation of proteins. The isoelectric focusing (IEF) dimension is most often run in cylindrical tube gels. After focusing, the cylindrical gel is placed on top of an SDS-PAGE slab gel which is then run as usual. Proteins can be visualized by metabolic radiolabeling or a variety of staining methods.

Gene Expression Studies

2-D gel electrophoresis is an excellent technique for the study of differential gene expression under various growth conditions. Since the expression and regulation of individual proteins can be detected, 2-D gels are an indirect way to monitor gene activity. They allow for the investigation of quantitative as well as qualitative changes in cellular protein expression.

The environmental conditions of a cell can be changed in order to determine optimal growth conditions as well as monitor the cell's response to different stresses. Environmental conditions that can stress the cell include changes in temperature, pH, and nutrient availability. Some examples of the chemical stresses that can be placed on a cell include drug and hormone treatments. Since protein structure and function are the direct result of gene expression, the loss or change of a protein as detected by a 2-D gel can be extrapolated back to events occurring at the DNA level.

Many questions encountered when genes are inserted and expressed in bacteria, yeast, and other cell types can be answered with 2-D analysis: Is the

cell making the protein? Is the cell's progeny making the protein? Is the protein being made but not secreted? Have mutations occurred?

Medical Applications

2-D gels can also have important applications in medical research. For example, this technique can be used to verify the presence of specific protein markers that are linked to genetic diseases and disease states. Used in conjunction with other tests, 2-D gels can be part of medical screening procedures associated with mutations and teratology linked to genetic damage.

Growth factors are being studied for their role in the regulation of cell growth. 2-D gels can be used to evaluate quantitative and qualitative changes in cellular proteins in response to growth factor stimulation.

2-D gels allow the visualization of proteins whose expression is altered as the result of cell transformation, introducing oncogenes into the host genome. Assessment of phosphorylation, sulfation, or other secondary modifications could reveal functional protein pathways affected by oncogene expression. This information could contribute to a better understanding of cell growth and regulation.

Sample Experiments

Valuable information can be learned by exposing cells to a set of specific experimental conditions and subsequently examining their biological response.

A preliminary in vitro experiment that is useful when beginning 2-D gel work is to radiolabel a cell's proteins to steady-state with ^{35}S -methionine. Running gels of these proteins will help to determine the commonly expressed proteins in the cell under normal conditions.

Subsequently, other amino acids such as ^3H -leucine, ^3H -proline, ^3H -lysine, etc., can be used to radiolabel the proteins to steady-state. This will label any proteins that do not contain methionine and were, therefore, not detected in the first experiment, and will provide preliminary information on the amino acid composition.

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Other experiments complementary to 2-D gels include: cell fractionation procedures, and post-translational modifications (phosphorylation, methylation, etc.). Data from these experiments can be added to a database, accumulating information about these proteins.

3. Bio-Rad Listens

The staff at Bio-Rad are receptive to your suggestions. Many of the new features and enhancements in this version of PDQUEST are a direct result of conversations with our customers. Please let us know what you would like to see in the next version of PDQUEST by faxing, calling, or e-mailing our Technical Services staff. You can also use Solobug (installed with PDQUEST) to make software feature requests.

1. Introduction

1.1 Overview of PDQUEST

PDQUEST is a software package for imaging, analyzing, and databasing 2-D electrophoresis gels.

The software runs in a Windows or Macintosh environment and has a graphical interface with standard pull-down menus, toolbars, and keyboard commands.

PDQUEST can acquire images of gels using any of several Bio-Rad imaging systems. An image of a gel is captured using the controls in the imaging device window and displayed on your computer screen. The scanned gel can then be cropped, rotated, etc. using the image editing controls.

Spot Detection, Analysis and Databasing

The Spot Detection Wizard guides you through the process of identifying and quantifying the spots in the gel image.

After detection, gels in the same experimental series are placed in a matchset for comparison, statistical analysis, and databasing. Histograms allow you to quickly compare the quantities of the same spot in all the gels in a matchset. Spots can also be compared qualitatively, organized into user-defined sets for further analysis, and annotated and databased for easy identification. Spots from different experimental series can be organized and compared in high-level matchsets.

PDQUEST can be used to simultaneously analyze thousands of spots on hundreds of gels. Data can be exported to other applications such as spreadsheets for further analysis.

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Mass Spec Analysis

PDQUEST is part of Bio-Rad's **ProteomeWorks** protein analysis package, and controls Bio-Rad's ProteomeWorks Spot Cutter. You can cut spots from gels or membranes using PDQUEST, digest them, and perform advanced protein analysis using MicroMass's mass spectrometry instruments and software. Data from MicroMass can then be imported back into PDQUEST to be included in spot annotations.

And More...

Scan files acquired in PDQUEST can be analyzed using other Bio-Rad Discovery Series software applications, such as Quantity One or Diversity Database. Scans can be converted into TIFF format for easy compatibility with other applications.

1.2 Digital Data and Signal Intensity

The Bio-Rad imaging devices supported by PDQUEST are light and/or radiation detectors that convert signals from biological samples into digital data. PDQUEST then displays the digital data on your computer screen, in the form of gray scale or color images.

A data object as displayed on the computer is composed of tiny individual screen pixels. Each pixel has an X and Y coordinate, and a value Z. The X and Y coordinates are the pixel's horizontal and vertical positions on the image, and the Z value is the signal intensity of the pixel.

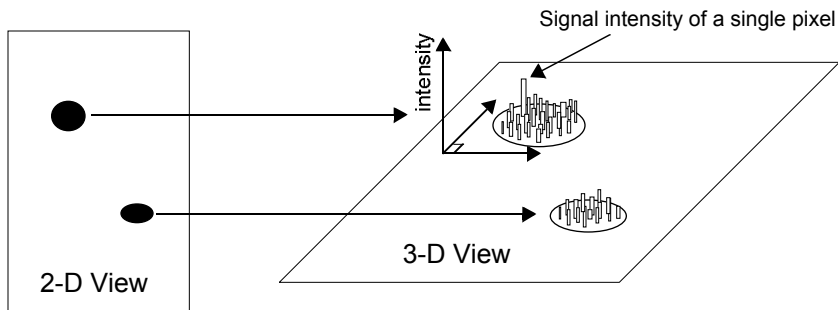


Fig. 1-1. Representation of the pixels in two digitally imaged spots in a gel.

For a data object to be visible and quantifiable, the intensity of its clustered pixels must be higher than the intensity of the pixels that make up the background of the image. The total intensity of a data object is the sum of the intensities of all the pixels that make up the object. The mean intensity of a data object is the total intensity divided by the number of pixels in the object.

The units of signal intensity are Optical Density (O.D.) in the case of the GS-700 and GS-710 densitometers, the Gel Doc and Chemi Doc with a white light source, or the Fluor-S and Fluor-S MAX MultiImagers with white light illumination. Signal intensity is expressed in counts when using the Personal FX or FX, or in the case of the Gel Doc, Chemi Doc, Fluor-S, or Fluor-S MAX when using the UV light source.

1.3 Steps Involved in PDQUEST

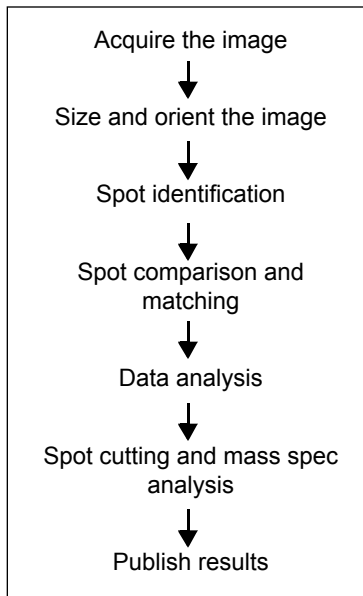


Fig. 1-2. Steps involved in using PDQUEST.

Image Acquisition (Appendices A–I)

PDQUEST can acquire images of gels using Bio-Rad's densitometers, storage phosphor imagers, and camera-based imaging systems.

First you open the acquisition window for your imaging device, and capture the image using the controls. Gel images are saved on your hard disk, network file server, or removable storage media. The displayed image in PDQUEST is ready for analysis.

Image Sizing and Orientation (Chapter 3)

In this step, you adjust the size and orientation of the image by using the cropping and rotating tools on the Image menu.

Spot Identification (Chapter 4)

Now you are ready to identify the spots in the gel. The Spot Detection Wizard automates the process of selecting the proper spot detection parameters for your gels. Using the Wizard, you select the parameters, study the results of spot detection, then adjust the parameters until you have identified all spots of interest in your gels.

When spots are detected in PDQUEST, the original gel image is filtered and smoothed to clarify the spots, then three-dimensional Gaussian spots are created from the clarified spots. The end result is three separate images: the original unaltered scan (2-D Scan), the filtered and processed scan (Filtered image), and a synthetic image containing the Gaussian spots (Gaussian image).

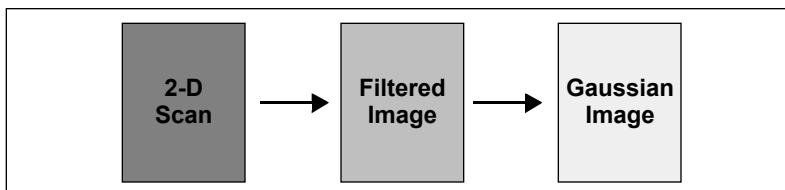


Fig. 1-3. Images created during spot detection.

All spot matching and analysis in PDQUEST are performed on Gaussian spots.

What Are Gaussian Spots?

Fuzzy, streaked, or overlapping spots in a 2-D gel can be difficult to accurately distinguish and quantify. Because the image profile of an ideal spot confirms to a Gaussian curve, PDQUEST uses Gaussian modeling to generate spots that can be precisely identified and quantitated.

A Gaussian spot is a three-dimensional representation of an original scanned spot. Gaussian curves are fitted to the scanned spot in the X and Y dimensions, and then additional modeling is performed to create the final Gaussian spot.

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Using Gaussian modeling, you can accurately quantitate overlapping spots, spots in gel streaks, and multiple spots in dense clusters.

Matching and Editing (Chapters 5-6)

After you have detected the spots in a gel or set of gels, you are ready to create a matchset. A matchset is composed of the Gaussian and Filtered images of the gel(s) in an experiment.

In a matchset, the protein spots from the different gels are matched to each other and are included in a synthetic image called a matchset standard. The standard includes all the information about the spots in the matchset.

As you match the spots in a matchset, you will correct the results of automatic spot detection by comparing the Gaussian spots with the original spots in the Filtered image.

Data Analysis (Chapters 7-8)

PDQUEST provides a variety of analytical tools to help you determine which spots are statistically and scientifically meaningful.

You can normalize the spot quantities in different gels for more accurate comparison. You can define replicate groups of duplicate gels. You can create groups of spots that are quantitatively, qualitatively, and statistically significant using analysis sets. You can compare the similarity of gels using scatter plots, and review the quantitation of individual spots using the Spot Review Tool.

Many of these tools are interactive with the matchset, so you can study the actual spots in the images as you review their quantities and other data.

You can create high-level matchsets to compare the results of different experiments. And the powerful annotation tool allows you to annotate your spots, link to Internet protein databases or other files, and create HTML pages of spot data.

Spot Cutting and Mass Spec Analysis

PDQUEST is part of Bio-Rad's ProteomeWorks protein analysis package, and controls Bio-Rad's ProteomeWorks Spot Cutter. You can cut spots from gels or membranes, digest them, and perform advanced protein analysis using MicroMass's mass spectrometry instruments and software. Data from MicroMass can then be imported back into PDQUEST to be included in spot annotations.

Publish Results

When your analysis is complete, you can print your experimental data or export it to another system for further analysis.

1.4 Computer Requirements

This software will run on a PC as a Windows 98, NT 4.0, or 2000 application, or on a Macintosh PowerPC.

The amount of computer memory required for using the program is mainly determined by the file size of the images you will scan and analyze. Images scanned at high resolution can be quite large. For this reason, we recommend that you archive images on a network file server or removable storage media.

PC

The following are the **minimum** system requirements for installing and running on a PC:

Operating system:	Windows 98, NT 4.0, or 2000.
Processor:	Pentium 166.
RAM:	64 MB for Gel Doc, Chemi Doc, and Fluor-S imaging systems. 128 MB for FX, Personal FX, and densitometers.

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Hard disk space:	3 GB. Recommended: Removable storage media (such as an Iomega Jaz drive) or a network file server.
Monitor:	17" monitor, 1024 x 768 resolution, 256 colors.
SCSI:	Required for all Bio-Rad imaging devices except the Gel Doc and Chemi Doc. Adaptec SCSI card with EZ-SCSI software.
Printer:	Optional.

Macintosh

Note: The default amount of memory assigned to this program on the Macintosh is 64 MB. If the total RAM in your Macintosh is 64 MB or less (the minimum recommended amount is 128 MB), you should reduce the amount of memory assigned to the program to 10 MB less than your total RAM. With the application icon selected, go to File > Get Info in your Finder to reduce the memory requirements for the application. See your Macintosh computer documentation for details.

The following are the **minimum** system requirements for installing and running on a Macintosh:

Operating system:	System 8.0 or higher.
Processor/Model:	PowerPC Macintosh 9500.
RAM:	64 MB for Gel Doc, Chemi Doc, and Fluor-S imaging systems. 128 MB for FX, Personal FX, and densitometers.
Hard disk space:	3 GB. Recommended: Removable storage media (such as an Iomega Jaz drive) or network file server.
Monitor:	17" monitor, 1024 x 768 resolution, 256 colors.
SCSI:	Required to run all Bio-Rad imaging devices except the Gel Doc and Chemi Doc. Macintosh has built-in SCSI.
Printer:	Optional.

1.5 Installing on a PC

The software can be installed on a PC from a CD-ROM or you can download the installation program from the Internet. Make sure that Windows is up and running on your computer before attempting to install.

Note: Windows NT and 2000: You must be a member of the Administrators group to install Discovery Series software on a computer. After installation, members of the Users group must have "write" access to the Discovery Series folder to use the software.

Installing from a CD-ROM

Insert the Discovery Series™ CD-ROM into the CD-ROM drive on your computer. The installer will start automatically.

Downloading from the Internet

You can download the software from the Internet using Bio-Rad's Web site. Go to www.discover.bio-rad.com, navigate to the Discovery Series download page, and select from the list of Windows applications to download. Follow the instructions to download the installer onto your computer. Then you can double-click on the Setup.exe icon on your desktop to begin running the installer.



Fig. 1-4. Setup.exe icon (Windows).

Installed Files and Directories

The installation program will guide you through a series of screens. The installer will create a default directory tree under Program Files on your hard disk called Bio-Rad/The Discovery Series/Bin (you can select your own directory if you wish). The main program will be placed in the Bin directory.

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Additional directories for storing user profiles and sample images will also be created

The installer will place an application icon on your desktop and create a Discovery Series folder in the Programs folder on your Windows Start menu.

Finally, after installation is complete, the installer will ask you if you want to start running the program.

1.6 Installing on a Macintosh

The software can be installed on a Macintosh from a CD-ROM or you can download the installation program from the Internet.

Installing from a CD-ROM

Insert the Discovery Series CD-ROM into the CD-ROM drive on your Macintosh. Double-click on the CD icon on your desktop, then double-click on the Install icon in the CD window.



Fig. 1-5. Installation program icon (Macintosh).

Downloading the Installation Program from the Internet

You can download the software from the Internet using Bio-Rad's Web site. Go to www.discover.bio-rad.com, navigate to the Discovery Series download page, and select from the list of Macintosh applications to download. Follow the instructions to download the installer and place it on your desktop. Then you can double-click on the installer icon to begin running the installer.

Installed Files and Folders

The installer will guide you through a series of screens. The installer will create a folder on your hard drive that contains the main application and associated sample images (you can select a different folder if you wish). The installation will also create a folder called The Discovery Series in the Preferences folder in your System Folder; this contains the on-line Help and various system files.

Once installation is complete, the folder containing the application icon will appear open on your desktop.

1.7 Starting the Program

PC

The installation program creates an application icon on your desktop. To start the program, double-click on this icon.

The installer also creates a Discovery Series folder in the Programs folder on your Windows Start menu. You can start the program by selecting the application in this folder.



Fig. 1-6. Application icon.

Macintosh

After installation, the main application folder will be open on your desktop. To start the program, double-click on the application icon inside the folder.

1.7.a Software License

When the program first opens, you should see a “Welcome” dialog box that shows the current status of your software license.

This program is protected by a software licensing system. You can have full use of the program for 30 days free of charge, after which you must purchase the software and obtain a password for continued use.

Most users will be able to start the program and begin the free trial period immediately after installation. If you open the software and see a “Free Trial” message, you can skip to section 1.7.c, Trial Period.

Unable to Obtain Authorization

If you attempt to start the program and receive an “Unable to obtain authorization” message, you will need a hardware security key (HSK) to run the program. HSKs for both PC and Macintosh are included with the full CD-ROM package, or are available from Bio-Rad for downloaded software. If you receive this message, turn off your computer, attach the appropriate HSK as described in the following section, and restart the computer and program.

Note: Network License holders: Your network administrator is responsible for the software license. If you have difficulty starting and running the program, contact your administrator.

1.7.b Attaching the Hardware Security Key (If Necessary)

Before attaching the HSK, you should first attempt to start the software without the HSK. If the program opens successfully, you can skip this section.

Note: Some parallel port devices such as zip drives may be incompatible with HSKs. Please check with your peripherals vendor.

PC



Fig. 1-7. PC Hardware Security Key

Before proceeding, please turn off your computer. If you have a printer attached to your computer's parallel port, please turn that off as well.

The HSK attaches to the parallel port on the back of your PC. If a printer cable is attached to this port, disconnect it. After you have attached the HSK, you can attach the printer cable to the key itself and restart your computer and printer.

The code for the PC hardware security key is EYYCY. This is printed on the key itself.

You will also need to install the system driver that allows the computer to "read" the HSK.

Note: Windows NT and 2000 users must be in the local administrator group to install the HSK driver.

To install the driver, click on your Windows Start menu and select Programs > The Discovery Series. Select Install HASP Hardware Security Key driver to begin installation.

Note: Windows 98 users must reboot their computer after installing the HSK. Windows NT and 2000 users do not have to reboot.

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Macintosh

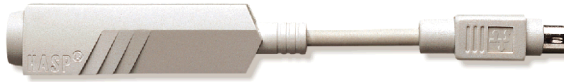


Fig. 1-8. Macintosh Hardware Security Key

Before proceeding, please turn off your Macintosh.

The Macintosh HSK must be inserted in the Apple Desktop Bus (ADB) path. The ADB port is located on the back of your Macintosh.



Fig. 1-9. Apple Desktop Bus icon on back of Macintosh.

The HSK can be inserted at any point in the ADB path—between the computer and the keyboard, between the keyboard and the mouse, between the keyboard and the monitor, etc. After you have attached the HSK, you can restart your computer.

The code for the Macintosh HSK is QCDIY. This code is printed on the key itself.

1.7.c Trial Period

When you start the program, the Software License screen will open. The type of screen you see will depend on whether you have an HSK, Network License, or neither.

Free Trial (No HSK or Network License)

If you do not have an HSK attached and do not have a Network License, the “Free Trial” screen will be displayed.

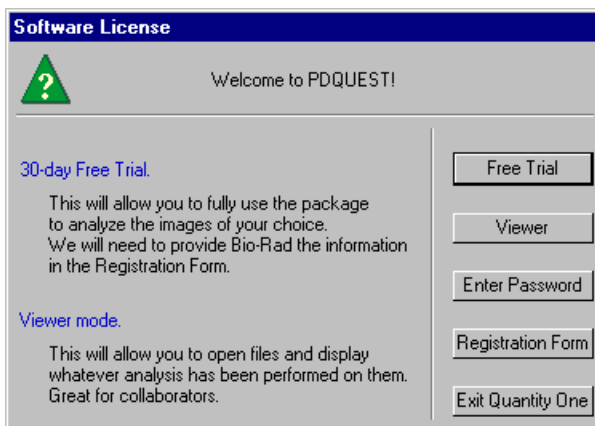


Fig. 1-10. Main Registration Screen without HPK

Click on the Free Trial button. This will open the Software License Registration Form (see next section).

Temporary License (With HSK or Network License)

If you have an HSK attached or are using a Network License, the Software License screen will reflect the fact that you receive a 30-day temporary license ("Your license will expire on _____").

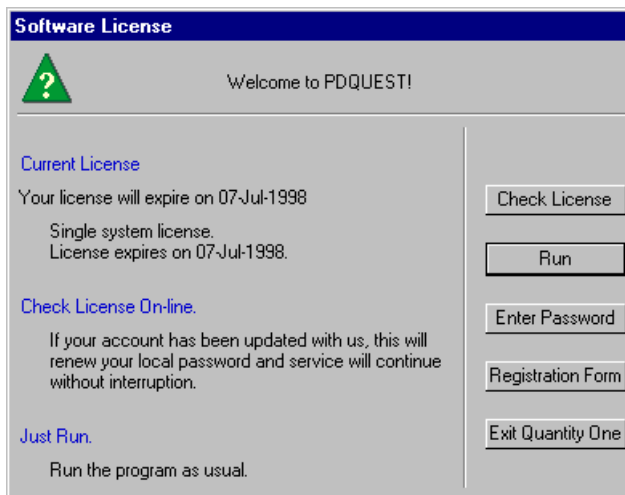


Fig. 1-11. Main Registration Screen with HPK.

Click on the Run button to begin using the software.

If you are using an HSK, some time during the 30-day license period, click on the Registration Form button to register your software. If your 30-day period has expired, a Free Trial button will appear when you open the software. Click on this button to open the Software License Registration Form and register your software.

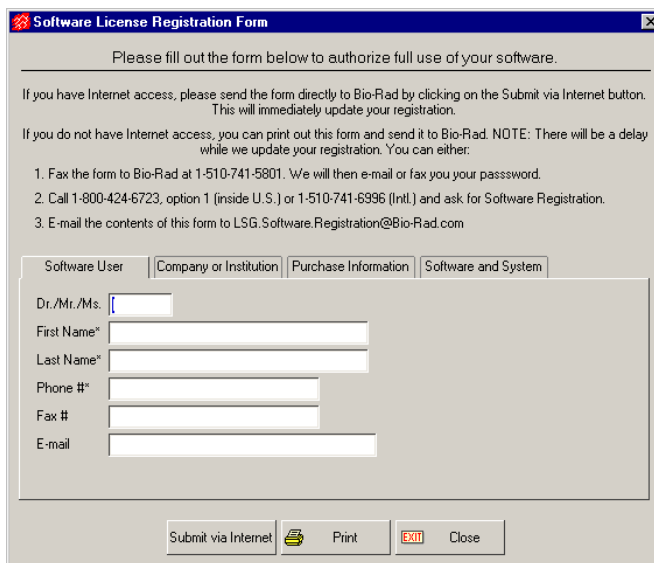
If you are using a Network License, any time during the initial 30 days click on the Check License button. Your full Network License will be activated.

Note: Network License holders: If your Network License is not activated when you click on Check License, notify your network administrator.

1.7.d Software Registration

Note: Network License holders do not need to register their software, and can skip the following section.

To obtain a full individual software license, you will need to fill out the information in the Software License Registration Form.



The image shows a screenshot of a software window titled "Software License Registration Form". The window has a blue title bar with a red 'X' icon on the left and a standard window control button on the right. The main content area is light gray and contains the following text:

Please fill out the form below to authorize full use of your software.

If you have Internet access, please send the form directly to Bio-Rad by clicking on the Submit via Internet button. This will immediately update your registration.

If you do not have Internet access, you can print out this form and send it to Bio-Rad. NOTE: There will be a delay while we update your registration. You can either:

1. Fax the form to Bio-Rad at 1-510-741-5801. We will then e-mail or fax you your password.
2. Call 1-800-424-6723, option 1 (inside U.S.) or 1-510-741-6996 (Intl.) and ask for Software Registration.
3. E-mail the contents of this form to LSG.Software.Registration@Bio-Rad.com

Below the text is a tabbed interface with four tabs: "Software User", "Company or Institution", "Purchase Information", and "Software and System". The "Software User" tab is selected. It contains the following labels and input fields:

- Dr./Mr./Ms. [dropdown menu]
- First Name* [text box]
- Last Name* [text box]
- Phone #* [text box]
- Fax # [text box]
- E-mail [text box]

At the bottom of the window is a row of buttons: "Submit via Internet" (with a small icon of a computer and a globe), "Print" (with a printer icon), "EXIT" (in red text), and "Close".

Fig. 1-12. Software License Registration Form.

Note: If you do not yet have a Purchase Order Number or Software Serial Number, you may leave these fields blank to receive a trial license.

Registering by Internet

If you have Internet access on your computer (the same computer on which you loaded the software), you can register quickly and easily.

In the Software License Registration Form, click on the Submit via Internet button.

Your information will be submitted automatically over the Internet, and a password will be generated automatically and sent back to your computer. Simply continue to run the application as before.

This password will be good for 30 days. During this period, if you have already submitted your Software Serial Number, click on Check License in the

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Software License screen to update your license. (To access the Software License screen from within the application, select Help > Register.)

If you have not yet submitted your Software Serial Number, open the form again, enter the serial number, and resubmit it over the Internet. In 1–2 days, click on the Check License button to update your license.

Registering by Fax or E-mail

If you do not have Internet access, click on the Print button in the Software License Registration Form and fax the form to Bio-Rad at the number listed on the form. Alternatively, you can enter the contents of the form into an e-mail and send it to Bio-Rad at the address listed in the Registration Form.

If you include your Software Serial Number in the form, Bio-Rad will contact you by fax or e-mail in 2–3 days with a full license.

If you do not include your Software Serial Number in the form, Bio-Rad will contact you in 2–3 days with a temporary 30-day license. Then, some time during the 30-day period, enter the serial number in the form and fax the revised form to Bio-Rad. Bio-Rad will contact you by fax or e-mail in 2–3 days with a full license.

Entering a Password

If you fax or e-mail your registration information, you will receive a password from Bio-Rad. You must enter this password manually.

To enter your password, click on Enter Password in the Software License screen. If you are not currently in the Software License screen, select Register from the Help menu.

Fig. 1-13. Enter Password screen.

In the Enter Password screen, type in your password in the field.

Once you have typed in the correct password, the OK light next to the password field will change to green and the Enter button will activate. Click on Enter to run the program.

1.8 Contacting Bio-Rad

Bio-Rad technical service hours are from 8:00 a.m. to 4:00 p.m., Pacific Standard Time in the U.S.

Phone: (800) 424-6723, option 2, option 3

(510) 741-6576

Fax: (510) 741-5802

E-mail: LSG.TechServ.US@Bio-Rad.com (in the U.S.)

LSG.TechServ.Intl@Bio-Rad.com (International)

For software registration, phone:

(800) 424-6723, option 1, or (510) 741-6996

2. General Operation

2.1 Graphical Interface

2.1.a Menu Bar

PDQUEST has a standard menu bar with pulldown menus that contain all the major features and functions available in the software.

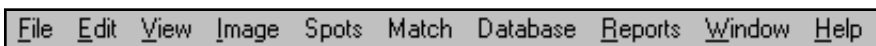


Fig. 2-1. Menu bar.

The pulldown menus contain groups of related functions:

- File—General file commands (Open, Save, Print, Export), imaging device acquisition windows.
- Edit—Calibration, normalization, preferences, file paths.
- View—General viewing tools (Zoom Box, Grab), quick analysis tools (Density in Box, Plot Density), image subwindow tools.
- Image—Transform function, image processing tools (Crop, Flip, Subtract Background).
- Spots—Spot detection, editing, and display tools.
- Match—Spot matching and matchset functions.
- Analysis—Analysis set functions, annotation tools, Mr pI standards, replicate groups functions, other spot analysis tools.
- Identification—Spot cutting and mass spectrometry analysis tools.
- Reports—Graphs and reports.
- Window—Tile windows, Imitate Zoom, Configure Subwindows.
- Help—On-line Help, keyboard layout, software registration.

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Below the menu bar is the main toolbar, containing some of the most commonly used commands. Next to the main toolbar are the status boxes, which provide information about cursor selection and toolbar buttons.

2.1.b Main Toolbar

The main toolbar in PDQUEST appears below the menu bar. It includes buttons for the main file commands (Open, Save, Print) and essential viewing tools (Zoom Box, Grab, etc.), and buttons to open the secondary toolbars.

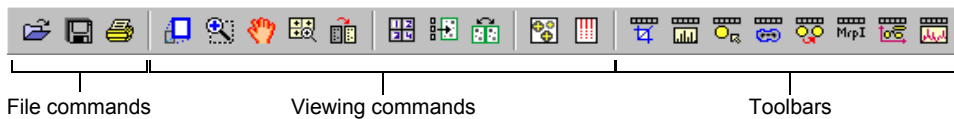


Fig. 2-2. Main toolbar.

Tool Help

If you hold the cursor over a toolbar icon, the name of the command will pop up below the icon. This utility is called Tool Help. Tool Help appears on a time delay basis that can be specified in the Display Preferences dialog box under Edit > Preferences. You can also specify how long the Tool Help will remain displayed.

2.1.c Status Boxes

There are two status boxes in PDQUEST. These appear to the right of the main toolbar.

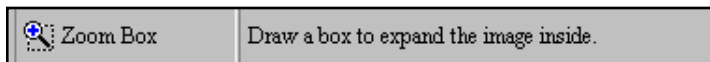


Fig. 2-3. Status boxes.

The first box displays any function that is assigned to the mouse (see section 2.2, Mouse-assignable Tools). If you select a command such as Zoom Box, the

name and icon of that command will appear in this status box and remain there until another mouse function is selected or the mouse is deassigned.

The second status box is designed to supplement Tool Help (see above). It provides additional information about the toolbar buttons. If you hold your cursor over a button, a short explanation about that command will be displayed in this second status box.

2.1.d Secondary Toolbars

PDQUEST has secondary toolbars that contain icons for groups of associated functions. You can open these toolbars from the main toolbar or from the View > Toolbars pulldown menu.

The secondary toolbars can be toggled between vertical, horizontal, and expanded formats by clicking on the resize button on the toolbar itself.

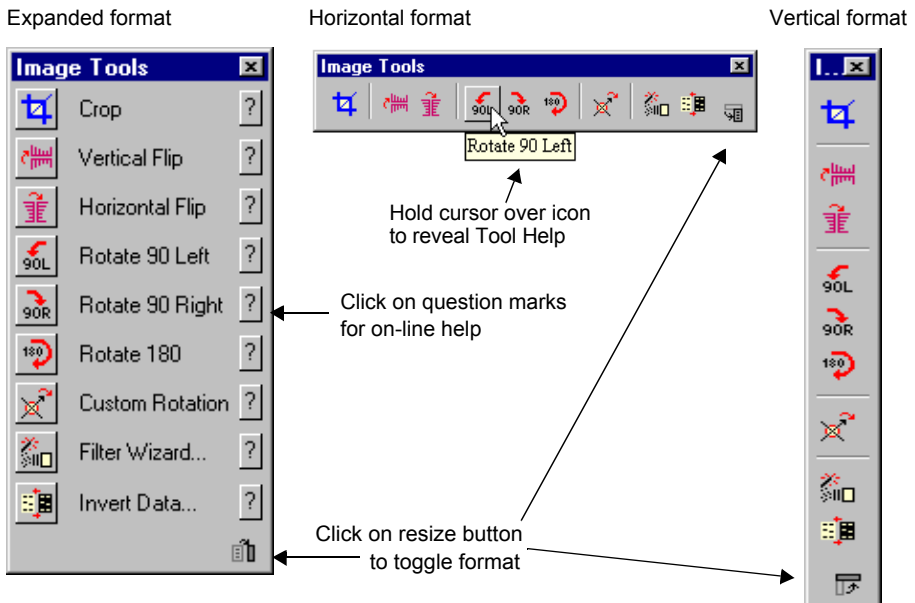


Fig. 2-4. Secondary toolbar formats and features.

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The expanded toolbar format shows the name of each of the commands. Click on the “?” button next to the name to display on-line Help for that command.

2.1.e Quick Guide

The PDQUEST Quick Guide (accessed from the Help menu or main toolbar) is a tool palette that is organized to guide you through the major functions of the software. Like the secondary toolbars, it can be toggled between vertical, horizontal, and expanded formats by clicking on the resize button.

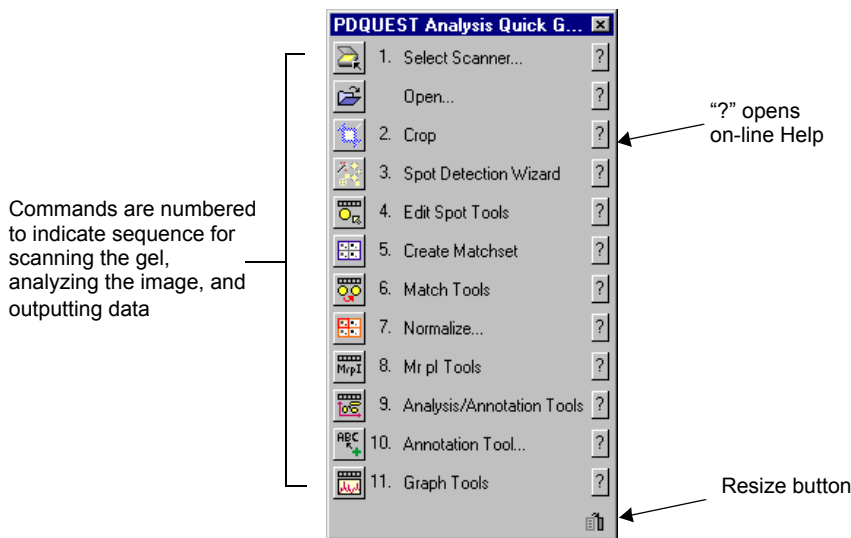


Fig. 2-5. PDQUEST Quick Guide

In the expanded format, the Quick Guide commands are numbered as well as named, so that the order of operation is clear. Simply follow the steps and the Quick Guide will lead you through the analysis. Note that many of the buttons open toolbars of related commands (e.g., Match Tools).

As with the secondary toolbars, you can click on the “?” next to the name of a function to display the Help text.

2.2 Mouse-assignable Tools

PDQUEST has a number of commands that don't perform actions right away, but instead "assign" a function to your mouse (e.g., Zoom Box, Density at Cursor, Add Spot). You first select one of these tools from a menu or toolbar, then click or drag on the image to execute the command.

Note: Mouse assignable tools selected using the keyboard have slightly different behavior.

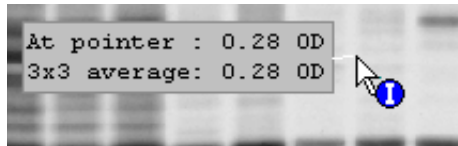


Fig. 2-6. Example of a mouse-assignable tool: Density at Cursor

When a "mouse-assignable" function is selected, the cursor appearance changes. The name and icon of the function appear in the status box next to the main toolbar (see section 2.1.c).

To *deassign* a function from the mouse, click on the toolbar button of the assigned function or click in the status box displaying the assigned function. You can also double-click on the Hide Overlays button.

Note: Some commands in PDQUEST *become* mouse-assignable if you are displaying multiple subwindows. For example, if you select Spots > Spot Crosshairs with a single image displayed in a window, crosshairs will simply be displayed. However, if you select Spots > Spot Crosshairs with multiple subwindows displayed, the command will be assigned to your mouse and you must click on a subwindow to display the crosshairs in that subwindow.

2.3 Keyboard Shortcuts

Many of the functions may be executed using keyboard commands (e.g., pressing the F1 key will activate the View Entire Image command). A list of

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key combinations and their associated commands will be displayed if you select Keyboard Layout from the Help menu.

The pulldown menus also list the shortcut keys for the menu commands.

Note: Mouse-assignable commands (see above) behave differently if you assign them using the keyboard versus selecting them from the menus or toolbars. For example, to use the Zoom Box command as a keyboard command, position your cursor on the image where you want to begin to create the magnifying box, then press the F2 key. The command is assigned to your mouse and immediately activated, so that when you move your cursor over the image, the zoom box is created. When you click the mouse button once, the defined region is magnified and the tool is automatically deassigned from your mouse.

2.4 File Commands and Functions

This section describes the basic file commands and functions of PDQUEST. These are selected from the File menu.

Note: Print and Export commands are described in Chapter 13.

2.4.a File Types in PDQUEST

The types of files supported by PDQUEST are:

2-D scans (.gsc)

Filtered images (.gim)

Gaussian images (.gsp)

Calstrip scans (.csc)

Matchsets (.ms)

Backup scans (.sbk) (Backups cannot be opened in Windows; you must change the extension to .gsc using Windows Explorer to open the file).

Scansets

A scanset consists of different exposures and/or versions of the same gel image. In PDQUEST, you can have multiple exposures of the same gel; calibration strips from the different exposures; and Filtered and Gaussian images generated from the original scan(s) when you detect spots. All of these are saved as separate images, but they are part of the same scanset. They have the same root file name, and when you open them they are loaded into subwindows of the same image window.

For example, the following image files are all in the same scanset:

Proteins v1 x1.gsc (2-D scan, exposure 1)

Proteins v1 x2.gsc (2-D scan, exposure 2)

Proteins v1 x1.csc (Calstrip scan, exposure 1)

Proteins v1 x2.csc (Calstrip scan, exposure 2)

Proteins v1.gim (Filtered image)

Proteins v1.gsp (Gaussian image).

2.4.b Opening Files

The Open command on the File menu or main toolbar opens previously saved images and matchsets.

Note: This version of PDQUEST will load any images and matchsets created with earlier versions of PDQUEST.

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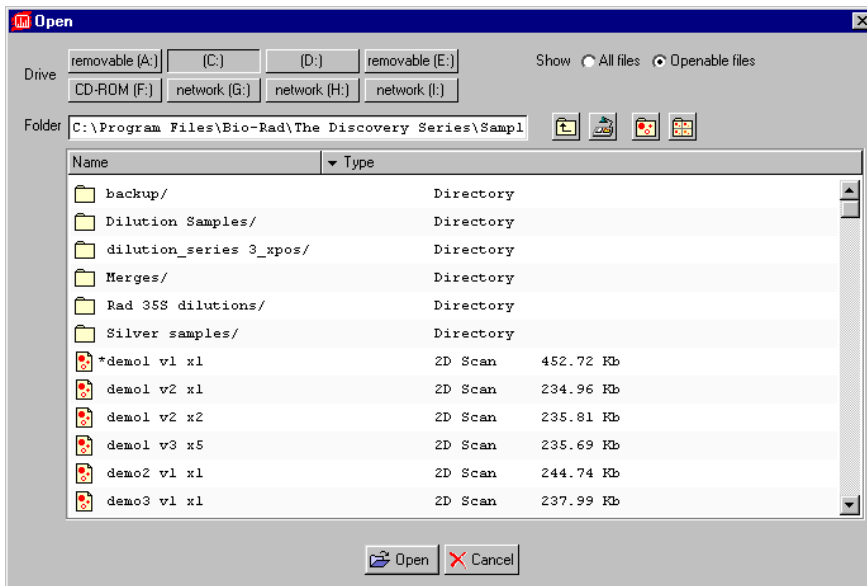


Fig. 2-7. Open dialog box.

In the PDQUEST Open dialog box:

- Already open images are marked with an asterisk.
- File types are listed next to each image name.
- Matchsets are listed as single files, even though they are really collections of images.

You can show All Files or only Openable files by selecting the appropriate option button. (See section 2.4.a, File Types in PDQUEST.)

In the dialog box, open an image by double-clicking on its name or selecting the file name and clicking on the Open button.

To select multiple files to open, use the standard ctrl-click and shift-click key combinations in Windows, or shift-click on the Macintosh. To open all the files in a directory, click on the Open button with no file names selected. You will be prompted to open all files.

Note: Images in a scanset will be loaded into the same window.

Click on the Parent Folder button to go up one level in the directory tree. Click on the Desktop button to open the desktop folder. You can also type the full file path into the Folder field.

To look for files on other drives, click on the appropriate drive letter button at the top of the dialog.

Note: The drive buttons only include mapped drives. If you need to access an unmapped (network) drive, you can type the drive name directly into the Folder field using UNC notation (e.g., \\remote server name\shared directory name\...\file name). Or you can create a shortcut to the drive on your desktop.

To go directly to the folder where your images are stored, click on the Image Folder button. To go to the folder containing your matchsets, click on Matchset Folder button. The locations of these folders are specified in Path Preferences under Edit > Preferences.

An image created in the Windows version of PDQUEST can be opened in the Macintosh version, and visa versa. However, you must add the appropriate extension to your Macintosh files to open them in Windows (e.g., .gsc for 2-D scans, .gim for Filtered images, .csc for calstrip scans, etc.).

PDQUEST comes with a selection of sample images. In Windows, these files may be found in the Discovery Series/Sample Images/2D directory. On the Macintosh, they are stored in the Sample Images folder in the PDQUEST folder. These sample images are related, and can be used to create a matchset.

Opening TIFF Images

The Open command can also be used to import TIFF images from other software applications.

TIFF images do not contain all of the tagged information that would normally be included in a Discovery Series image file (e.g., scanner type, date, color, etc.). For this reason, the Image Info and Print > Image Report commands will not display all such tagged information from TIFF files.

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There are many types of TIFF formats that exist on the market. Not all are supported by the Discovery Series. There are two broad categories of TIFF files that are supported:

1. 8-bit Grayscale. Most scanners have an option between line art, full color, and grayscale formats. Select grayscale for use with the Discovery Series software. In a grayscale format, each pixel is assigned a value from 0 to 255, with each value corresponding to a particular shade of gray. Hewlett Packard[™], Microtek[™], and Sharp[™] each make scanners that produce compatible 8-bit grayscale images.
2. 16-bit Grayscale. Bio-Rad's Molecular Imager (storage phosphor) systems use 16-bit pixel values to describe intensity of scale. Molecular Dynamics[™] imagers also use 16-bit pixel values. The Discovery Series understands these formats and can interpret images from both Bio-Rad and Molecular Dynamics storage phosphor systems.

Note: PDQUEST can import 8- and 16-bit TIFF images from both Macintosh and PC platforms.

TIFF files that are *not* supported include:

1. 1-bit Line Art. This format is generally used for scanning text for optical character recognition or line drawings. Each pixel in an image is read as either black or white. Because the software needs to read continuous gradations to perform gel analysis, this on-off pixel format is not used.
2. 24-bit Full Color or 256 Indexed Color. These formats are frequently used for retouching photographs and are currently unsupported in the Discovery Series, although most scanners that are capable of producing 24-bit and indexed color images will be able to produce grayscale scans as well.
3. Compressed Files. The software does not read compressed TIFF images. Since most programs offer compression as a selectable option, files intended for compatibility with the Discovery Series should be formatted with the compression option turned off.

2.4.c Closing files

To unload a single image from a window or subwindow, select the window or subwindow, then select File > Unload. If the image is in a subwindow, the image will close but the larger window will remain open; if the image is in its own window, the entire window will close. If the file has changed since it was last saved, you will be prompted to save the changes before closing.

File > Close will close the entire active window, including any subwindows. If any of the images in the window has changed, you will be prompted to save the changes before closing.

File > Close All closes all open images in all windows. If you have made changes to any of the images, a message box will open for each unsaved image giving you the option of saving those changes.

2.4.d Saving Files

File > Save will save a new or old scan, scanset, or matchset to your hard disk, network drive, or other storage media.

Note: In Windows, new PDQUEST images will be given a .gsc extension when they are first saved.

File > Save As can be used to save scans, scansets, and matchsets under different names and different directories.

Note that in the case of scansets (see section 2.4.a), all the different files in the set must be loaded when you select Save As if you want the root name to be changed for all of them.

Save All on the main toolbar or File menu saves all images, scansets, and/or matchsets that are currently open.

2.4.e Change Version/Exposure

Change Version/Exposure allows you to change the type, version and/or exposure number of a scan file.

Note: The root name of the file will remain unchanged.

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Note that different Discovery Series software applications read different file types. Quantity One and Diversity Database can read 1-D scan files, while DNACode reads DNA scans. Only 2-D and Calstrip scans can be read by PDQUEST.

With the scan image open, select Change Version/Exposure from the File menu.

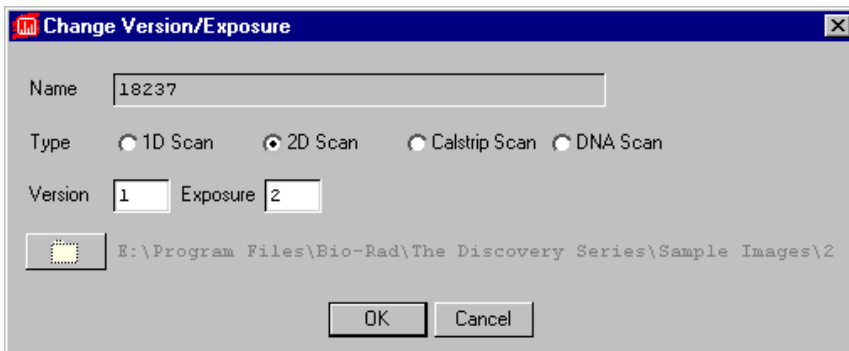


Fig. 2-8. Change Version/Exposure dialog box.

In the dialog box, select the type of scan file and enter a new version and/or exposure number in the appropriate field(s), and click on OK. You must save the image to save your changes.

Saving creates a new file with the same root name; any previously saved type/ version/ exposure of the image will remain unchanged.

2.4.f Revert to Saved

File > Revert to Saved will recover the last saved version of the image or matchset you are working on. This is a quick way to undo any alterations you have made since you last saved it.

Any changes you have made since last saving the file will be lost. (Any open dialog box will also be closed.) A message box will warn you before completing the command.

2.4.g Image Info

File > Image Info displays general information about your image, including the scan date, scan area, number of pixels in the image, data range, and the size of the file. Gaussian image info includes the number of spots, total data in the spots, and other information. There is also a field where you can type in a file description or comments.

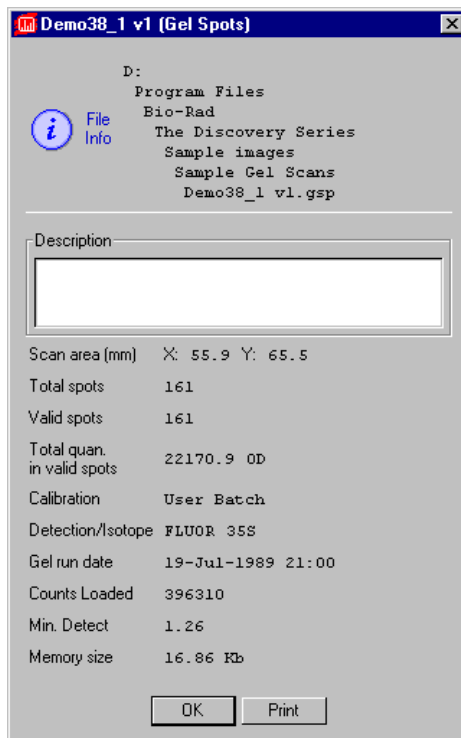


Fig. 2-9. Image Info box.

To print the file information, click on the Print button.

Changing the Image Dimensions

You can change the dimensions of certain scans using the Image Info dialog box.

Note: This feature is only available for scans captured by a camera (such as the Gel Doc or Fluor-S) or imported TIF images in which the dimensions are not already specified.

When you select File > Image Info for these types of images, the Image Info dialog box will include fields for changing the image dimensions.

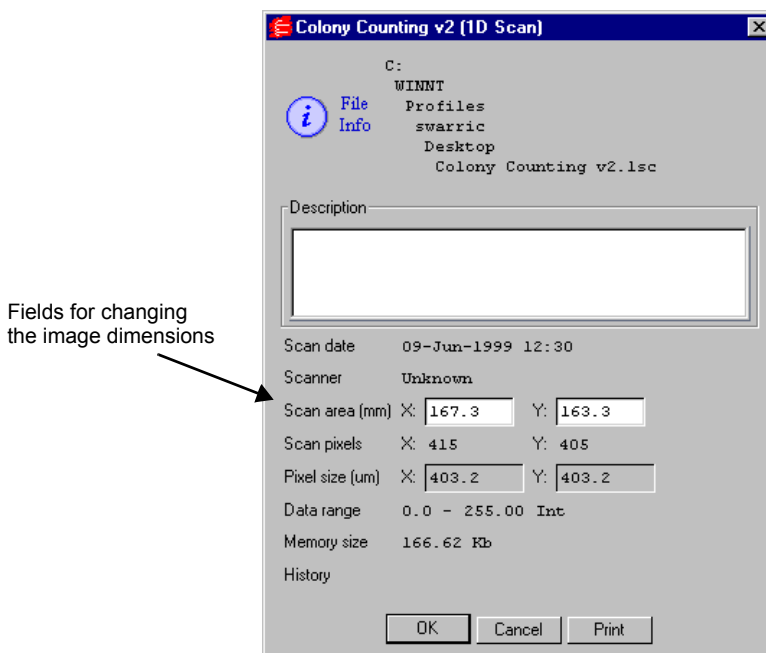


Fig. 2-10. Image Info dialog box with fields for changing the image dimensions.

Enter the new image dimensions (in millimeters) in the appropriate fields. Note that the pixel size in the image (in micrometers) will change to retain the same number of pixels in the image.

2.4.h Reduce File Size

Scan files can be quite large, and computer systems do not have unlimited memory or storage space. If you are having difficulty loading or storing a particular scan, you may want to reduce the size of the file by reducing the number of pixels in the image. (You can also trim unneeded parts of an image to reduce its memory size. See section 3.12, Cropping Images.)

Note: This command only works on 2-D scans.

This function is comparable to scanning at a lower resolution, in that you are increasing the size of the pixels in the image, thereby reducing the total number of pixels and thus memory size.

Note: Reducing the file size of an image will result in some loss of resolution. In most cases this will not affect quantitation. In general, as long as the pixel size remains less than 10 percent of the size of the objects in your image, changing the pixel size will not affect quantitation.

Select File > Reduce File Size. The dialog box shows you the size of the pixels in the image (Pixel Size: X by Y microns), the number of pixels in the image (Pixel Count: X by Y pixels), and the Memory Size of the image.

As you increase the size of the pixels, the pixel count will decrease, as will the memory size. You can increase the pixel size in either dimension (see the following figure for an example).

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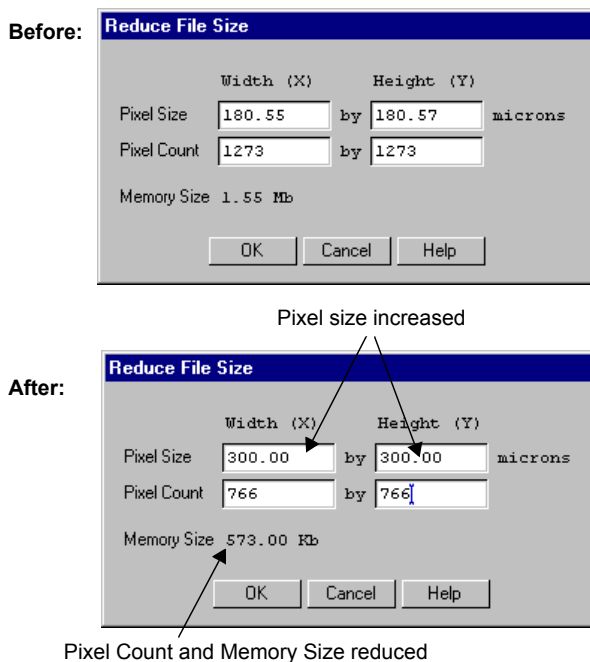


Fig. 2-11. Reduce File Size dialog box, before and after pixel size increase.

Note: Asymmetric pixel reduction (i.e., making pixels smaller in one dimension than the other) is not recommended for 2-D gels, since you want to resolve spots in both dimensions.

When you are finished, click on the OK button.

A pop-up box will give you the option of reducing the file size of the displayed image or making a copy of the image and then reducing the copy's size.

Reducing the file size is an irreversible process. For that reason, we suggest you make a copy of the image and reduce its file size. That way, if you lose too much resolution, you can simply delete the copy and try again. Once you are happy with the reduced image, don't forget to delete the original. The goal is to save space!

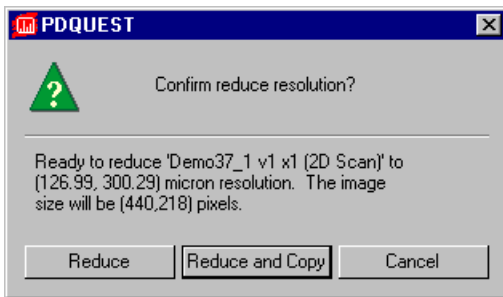


Fig. 2-12. Confirm Reduce File Size pop-up box.

If you choose to make a copy of the image, you will be asked to enter a name for the new copy before the operation is performed.

2.4.i Imaging Device Acquisition Windows

The File menu contains a list of Bio-Rad imaging devices supported by the Discovery Series software. These are:

1. Gel Doc 1000/2000
2. Chemi Doc
3. GS-700 Imaging Densitometer
4. GS-710 Imaging Densitometer
5. GS-800 Imaging Densitometer
6. Fluor-S MultiImager
7. Fluor-S MAX MultiImager
8. Personal Molecular Imager FX
9. Molecular Imager FX

Selecting a name in the File menu will open the acquisition window that allows you to scan using that instrument.

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See the individual appendices on the imaging devices for more details.

2.4.j Exiting PDQUEST

File > Exit quits the application. You will be prompted to save any unsaved changes to images or matchsets.

2.5 Preferences

The Preferences dialog box allows you to customize basic features of your system. Select Edit > Preferences to open the dialog.

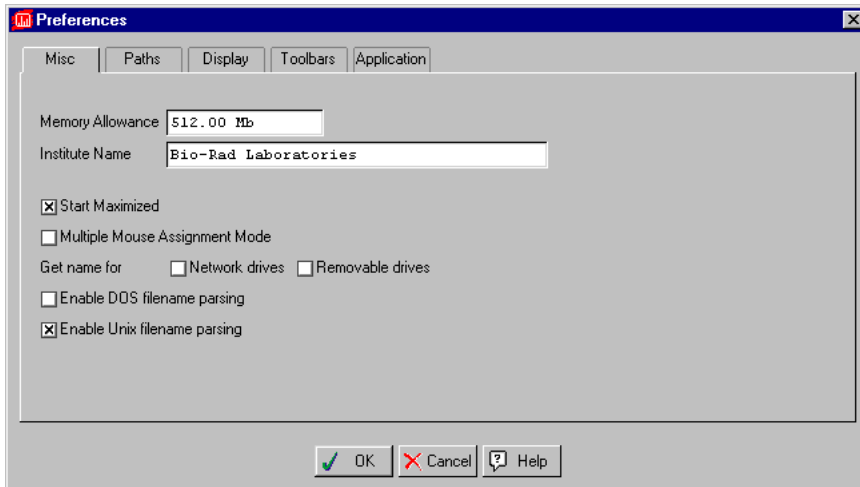


Fig. 2-13. Preferences dialog box.

Click on the appropriate tab to access groups of related preferences. After you have selected your preferences, click on OK to implement them. Some of your preferences will only be implemented by exiting and restarting the program. A message will notify you if this is necessary.

2.5.a Misc Preferences

Click on the Misc tab to access the following preferences.

The Memory Allowance field allows you to specify the amount of virtual memory allocated for the application at start up. The default value of 512 megabytes is recommended. If you receive a warning message when opening the program that the amount of virtual memory is set too high, you can enter a smaller value in this field. However, this should be considered a temporary fix.

Your institution name can be included in reports if you enter it in the Institute Name field.

Under Windows, the Start Maximized checkbox determines whether the main window occupies the entire computer screen when first opened. If this is unchecked, the menu and status bars will appear across the top of the screen and any toolbars will appear “floating” on the screen.

Multiple Mouse Assignment Mode is designed for backward compatibility with previous versions of PDQUEST. It is not documented in this manual.

If Network Drives and Removable Drives are checked, the names of these drives will appear in the Open dialog along with the drive letters.

If Enable DOS File Name Parsing is checked, for 8-character file names ending in two digits, the final two digits are interpreted as version and exposure numbers. For example, the file name IMAGE-11.gsc would be parsed as IMAGE ver 1 xpo 1.gsc. This is designed to enable backwards compatibility for users with DOS image files. You should only check this box if you are using these image files.

Enable UNIX File Name Parsing is similar to DOS file name parsing. Windows and Macintosh users are unlikely to run into difficulties with UNIX parsing, therefore this setting is checked by default.

2.5.b Path Preferences

Click on the Paths tab to set the file paths for various shortcuts and links throughout PDQUEST.

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To set a particular file path, click on the Browse button next to the path field and navigate to the appropriate directory, then click on Select to set the path. Click on Default to return the path to the default setting.

The Images path connects to the main directory where you keep your image files. This path is accessed when you click on the Image folder shortcut button in the Open, Save As, and related dialogs.

The Matchsets path connects to the main directory where you keep your matchset files. This path is accessed when you click on the Matchset folder shortcut button in the Open, Save As, and related dialogs.

The Log path sets the directory where spot detection log files are saved. Select Spot Detection Report from the Spots menu to access the log files in this directory.

The Micromass Projects path sets the directory where your Micromass project files are stored. This path is used to link to Micromass project files using the Generate MassLynx Batch File and Import MassLynx Results tools.

The Micromass Install path sets the directory where your Micromass processing parameters are stored. This path is used to link to Micromass parameter files using the Generate MassLynx Batch File function.

The Temporary path sets the directory where temporary image files and other files generated by PDQUEST are stored.

2.5.c Display Preferences

Click on the Display tab to access the following preferences.

Zoom % allows you to specify the percentage by which an image moves closer or farther away when you use the Zoom functions. This percentage is based on the size of the image.

Pan % determines the percentage by which the image moves side to side or up and down when you use the arrow keys. This percentage is based on the size of the image.

If you want to apply a uniform background to Gaussian images so they more closely resemble the original gels, enter a background optical density (O.D.) value in the text box next to the Gel Spot Bkgd prompt.

In Windows, Jump Cursor on Alert will automatically jump the cursor to the OK button when an alert box pops up. Deselect the checkbox to turn off this feature.

By default, a subwindow is selected when you move your cursor over it. If you would rather click on a subwindow to select it, select Click to Focus Subwindow.

Note: Click to Focus Subwindow will change the behavior of mouse-assignable tools. Some commands that were previously mouse-assignable will be performed immediately on the active subwindow; other commands will require you to click once in a subwindow to select it, then click or drag again to perform the action.

Auto “All Same” duplicates the zoom and positioning commands made in one subwindow in all displayed subwindows. For example, with Auto “All Same” selected, Zoom In on one subwindow; the other open subwindows will also zoom in.

Show Filtered Images by Default will automatically display the Filtered images when you first open a matchset (provided the matchset includes Filtered images).

If Bold Match Symbols is selected, landmarks, manual matches, added spots, etc. will appear larger and bolder on the screen.

If Filled Spot Ellipses is checked, marked spots (e.g., spots marked using the Saturated Spots command) will be displayed as filled ellipses. If unchecked, they will be displayed as unfilled ellipses.

Solid Label Background will automatically display spot labels in a filled-background box.

If Small Window Labels is checked, any text in the windows/subwindows will be displayed in a smaller font.

In Gaussian images, specify whether you want to see only valid spots or valid spots and combined spots by clicking on the appropriate button next to the Default Display Set prompt. Note that valid spots are spots that have not been cancelled. If you elect to display combined spots, they will appear in the Gaussian image as they do in the Filtered image, although the quantitation will be combined.

2.5.d Toolbar Preferences

These settings determine the behavior and positioning of the secondary toolbars and Quick Guide. Click on the Toolbar tab to access these preferences.

If the Show Quick Guide checkbox is selected, the quick guide will open automatically when you open PDQUEST.

If Align Quick Guide with Document is selected, the quick guide will pop up flush with the edge of your documents. Otherwise, it will appear flush with the edge of the screen.

The Guides Always on Top checkbox determines whether the quick guide and toolbars will always be displayed on top of images, or whether they can be hidden behind images.

Quick Guide Placement and Toolbar Placement determine which side of the screen the quick guide and toolbars will first open.

The Placement Behavior setting determines whether a quick guide or toolbar will always pop up in the same place and format (Always Auto), or whether it will pop up in the last location it was moved to and the last format selected (Save Prior).

The Toolbar Orientation option buttons specify whether toolbars will first appear in a vertical, horizontal, or expanded format when you open PDQUEST.

Tool Help Delay allows you to specify the amount of time (in seconds) the cursor must remain over a toolbar icon before the Tool Help appears. First-time users may want to specify a short delay to learn the names of the toolbar functions, while experienced users can specify a longer delay once they are familiar with the icons.

Persistence determines how long the Tool Help will linger on the screen after you move the cursor off a toolbar icon.

2.5.e Application Preferences

Click on the Application tab to access these preferences.

Enter up to three letters for the Matchset Name Prefix and a number for the Matchset Name Sequence #.

Each matchset that you create will be identified by a prefix followed by a unique sequence number. The prefix remains the same for all the matchsets while the sequence number increases by 1 each time a new matchset is created.

Once the prefix and the sequence number have been initially set, you probably will not need to change them.

If Auto-match after Landmark is selected, PDQUEST will automatically attempt to match the spots in your images when you create a landmark.

Finally, indicate the orientation in which your gels are run by clicking on either the Acid-Base or Base-Acid button next to the pH Gradient prompt.

When you create a matchset, you can give it a default name. The Matchset Name Prefix field determines the prefix of the default name, while the Matchset Name Sequence # determines the number added to the prefix to complete the name. The sequence number increases by 1 each time a new matchset is created using the default name.

If Auto-match after Landmark is selected, PDQUEST will auto-match all gels in the matchset each time you place a landmark in all the gels.

Indicate the orientation in which your gels are run by clicking on either the Acid-Base or Base-Acid button next to the pH Gradient prompt.

2.6 Mouse and Mouse Macros

These functions are designed for backwards compatibility with previous versions of PDQUEST. They are not recommended for new users and are not documented in this manual.

2.7 User Settings

If PDQUEST is on a workstation that has multiple users, each user can have his or her own set of PDQUEST preferences and settings.

In multiple-user situations, PDQUEST's preferences and settings are associated with individual user names. On a PC with Windows 95, 98, or NT 4.0, your user name is the name you use to log onto the computer. On a Macintosh, your user name is the Owner Name under the File Sharing control panel.

Note: If you do not log onto your PC under Windows 95/98 or do not have a Owner Name on your Macintosh, then you do not have a user name and your preferences and settings will be saved in a generic file.

3. Viewing and Editing Images

This chapter describes the tools for displaying images in windows and subwindows, and magnifying and optimizing images. This chapter also describes the tools for cropping, flipping, and rotating images and filtering image noise.

These tools may be found on the View, Edit, Image, and Window menus.

3.1 Windows and Subwindows

PDQUEST displays the multiple images in a matchset or a scanset in subwindows of the same image window. Subwindows appear as separate, smaller windows within the main image window. Subwindows are designed to allow you to easily compare and edit matchset and scanset images at the same time.

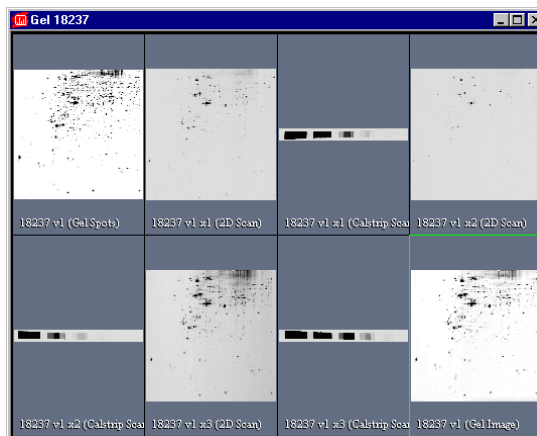


Fig. 3-1. A single image window displaying the eight members of a scanset.

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Note: A matchset is a single file composed of separate images, whereas the members of a scanset are separate files. You can open and close scanset images (i.e., different exposures or versions of the same image) individually, even though they are loaded into the same window.

	*18237 v1	Gel Image
	*18237 v1	Gel Spots
	*18237 v1 x1	2D Scan
	18237 v1 x1	Calstrip Scan
	18237 v1 x2	2D Scan
	*18237 v1 x2	Calstrip Scan
	18237 v1 x3	2D Scan
	18237 v1 x3	Calstrip Scan

Fig. 3-2. Gel 18237 has eight scanset images, all of which are loaded into the same window.

To perform an operation on a particular image in a matchset or scanset, simply move your cursor over its subwindow. A green border appears around the selected subwindow. The border shifts as you move the cursor over different subwindows.

Alternatively, you can select Click to Focus Subwindow under Edit > Preferences, in which case you must click on a subwindow to select it.

Note that some commands in PDQUEST only affect the selected subwindow image, whereas others affect some or all of the images in the matchset or scanset simultaneously. Many commands require you to first select the tool from the menu, keyboard, toolbar, or right-click menu, then click or drag in a subwindow to perform the action (see section 2.2, Mouse-assignable Tools). If Click to Focus Subwindow is selected, some commands that were previously mouse-assignable are performed immediately on the active subwindow; you do not have to click or drag.

The descriptions of the individual commands describe their behavior in subwindows.

PDQUEST includes tools for configuring subwindows and assigning images to particular subwindows.

3.2 Configuring Subwindows

To control how matchsets and scansets are partitioned into subwindows, click on the Configure Subwindows button on the main toolbar (also Ctrl+F1 and Windows menu). The Configure Subwindows dialog box will open.

The Loaded Images buttons show the types of files that are open (2D scans, Gaussian images, Matchsets, etc.) and list how many images are in each type. Click on a button to display only images of that type in the subwindows.

Click on All to display all available image types in subwindows.

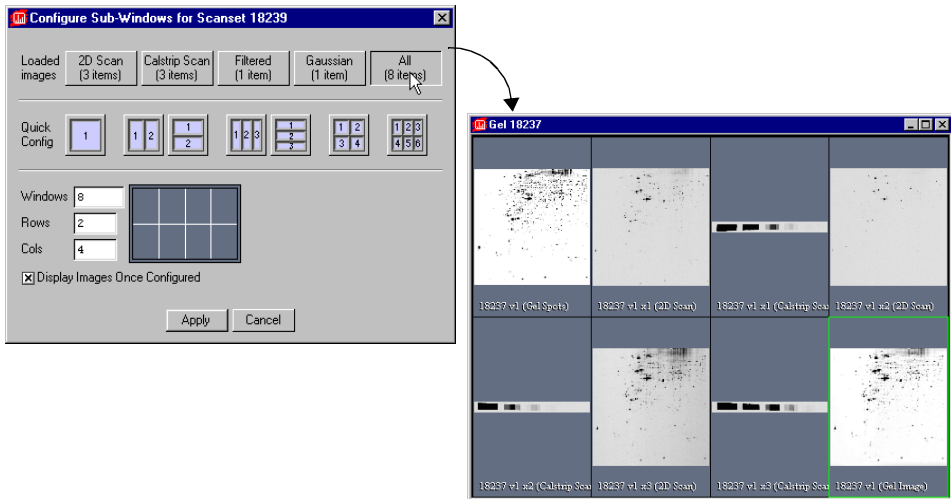


Fig. 3-3. Configuring the subwindows of a scanset.

The Quick Config buttons immediately partition the subwindows into the configuration depicted on the button. (Note that if you have selected a Loaded Images option, only images of that type will be displayed.)

Otherwise, select a configuration by typing values in the Windows, Rows, and Cols fields. The Windows field specifies how many subwindows to create. Rows and Cols specify how to layer the subwindows.

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For large matchsets with many members, you can deselect the Display Images Once Configured checkbox. Then, when you apply a new configuration, the images will not be immediately refreshed in the windows. Click in a subwindow to make the image in that subwindow appear.

Note: Deselect this checkbox only if you have a slow computer and are displaying many images.

Click on the Apply button to implement your configuration selections.

3.3 Assigning and Interchanging Images

You can use the Assign and Interchange commands to select the image you want to display in a particular window or subwindow.

Assigning Images

To assign an image to a particular subwindow, select the Assign tool from the main toolbar or Window menu and click in the subwindow, or use the Ctrl+A key command in desired subwindow. This will open a pop-up list of all the images currently loaded in the matchset or scanset.

Select an image name from the list; the image will appear in the selected window.

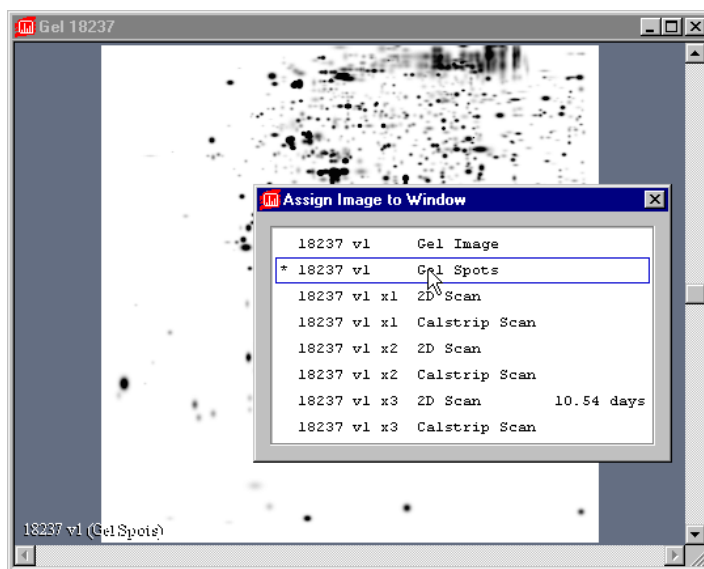


Fig. 3-4. Assign Image lists all the loaded images; select one to display it.

Interchanging Images

Interchange Images switches between the 2-D scan, Filtered image, and Gaussian image of the same gel within a selected subwindow. With the images loaded, select the tool from the main toolbar or View menu and click in the subwindow, or use the F11 key command in the subwindow. The type of image currently displayed is listed at the bottom of each subwindow.

Interchange All Images (Shift+F11, View menu) switches between the Filtered images and Gaussian images in all the displayed subwindows of a matchset.

3.4 Tiling Windows

The tile commands under the Window menu are used to arrange image windows on the screen.

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Note: The tile commands only affect image windows, not subwindows.

Select Tile to resize all open windows so that all the images are maximized. The windows will be arranged on the screen from left to right and top to bottom.

Tile Vertical will resize all open windows and arrange them in columns across the screen.

Tile Horizontal will resize all open windows and arrange them in rows across the screen.

3.5 Magnifying Images

The magnifying tools in PDQUEST can be accessed from the main toolbar or View menu, or by key command.

Note that, by default, if you magnify a region in one image of a matchset or scanset, the same region in all the other subwindows will be magnified. To turn off this feature, deselect the Auto “All Same” checkbox under Display Preferences (Edit > Preferences).

Ctrl+F3 will undo your last magnification or other viewing command.

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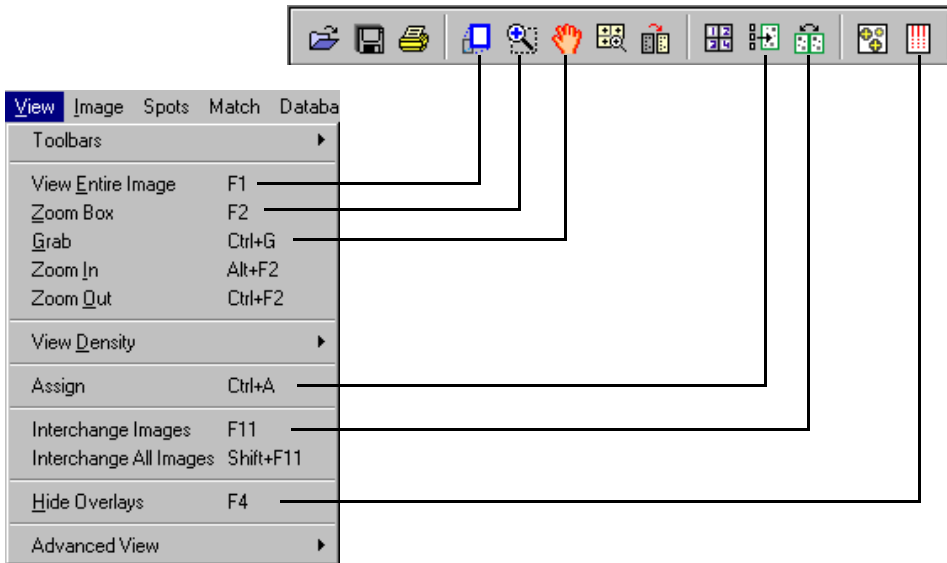


Fig. 3-5. Viewing functions on View menu and main toolbar.

Zoom Box

Use the Zoom Box tool to magnify a specific region of an image, matchset, or scanset.

Select Zoom Box from the main toolbar or View menu, or position your cursor on the image and press F2. Drag a box on the image, then release the mouse button. The area you selected will be magnified.

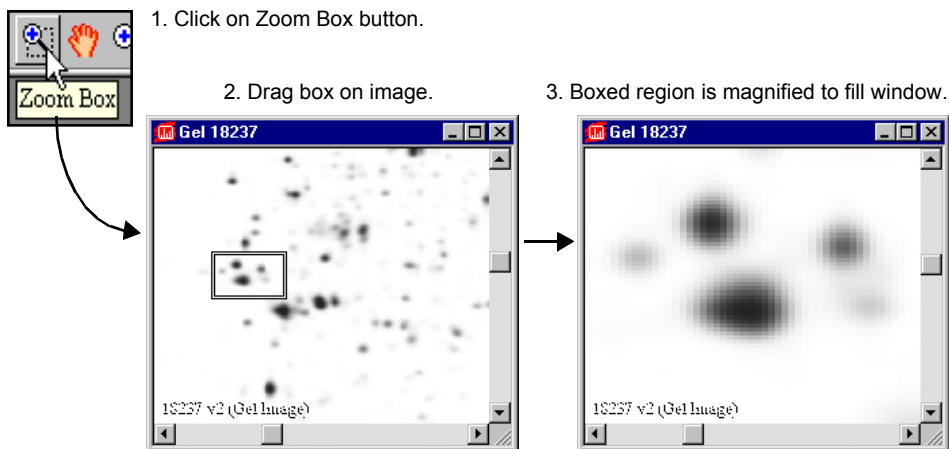


Fig. 3-6. Zoom Box tool.

Zoom In/Zoom Out

Zoom In (Alt+F2, View menu and toolbar) will magnify an image, or all the images in a matchset or scanset, by a defined percentage. The percent magnification is specified under Display Preferences (Edit > Preferences).

Zoom Out (Ctrl+F2, View menu and toolbar) will reduce the magnification by the same percentage.

View Entire Image

If you have magnified part of an image or moved part of an image out of view, this command will restore the original, full view. Select View Entire Image from the main toolbar or View menu and click on the image, or position your cursor on the image and press F1.

Imitate Zoom

Imitate Zoom applies the magnification and positioning of a selected window or subwindow to all open windows and subwindows. This command only works on comparable images with similar dimensions.

Note: Imitate Zoom is different than the Auto “All Same” setting under Display Preferences, in that it affects all open windows, not just the subwindows in a single window.

Select a window or subwindow you want to imitate, then select Imitate Zoom from the Window menu. The other windows will change their display to match that of the active window.

Magnifying Sections of an Image

To magnify one of nine sections of an image or matchset/scanset, go to the 3x3 Sections submenu on the View > Advanced View submenu, and select a region (Upper Left, Middle, Lower Right, etc.) That region will be magnified in the image or matchset/scanset.

Location in Gel

Location in Gel on the View > Advanced View submenu displays a graphical overlay on a subwindow or all subwindows showing the region that is magnified (green box) relative to the entire image (blue box). Select the command, then click on the image.

3.6 Positioning Images

The image positioning tools in PDQUEST can be accessed from the main toolbar or View menu, or by key command.

Note that, by default, if you reposition one image of a matchset or scanset in a subwindow, all the other subwindows will be repositioned as well. To turn off this feature, deselect the Auto “All Same” checkbox under Display Preferences (Edit > Preferences).

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Ctrl+F3 will “undo” your last positioning or other viewing command.

Grab

Use this tool to drag the image in a window.

Select Grab from the main toolbar or View menu, or position your cursor on the image and press Ctrl+G. Drag the image, then release the mouse button. The image will be repositioned.

Arrow Keys

You can shift the displayed region in a subwindow or all subwindows by using the Arrow keys on your keyboard. With a window or subwindow selected, click on an arrow button. The image will shift in the direction of the arrow click. The amount the image shifts is determined by the Pan % setting in Display Preferences under Edit > Preferences.

Centering an Image

You can center an image or images in a matchset/scanset quickly and easily using the At Cursor command (F3 on the keyboard; also on the View > Advanced View submenu). This is useful if you are comparing the same region of spots on multiple images and want to center the image windows on the same spot.

Position your cursor on a point on the image, then press F3. The image will shift so that that point is at the center of the image window.

Displaying Sections of an Image

To display one of nine sections of an image or matchset/scanset, go the 3x3 Sections submenu on the View > Advanced View submenu, and select a region (Upper Left, Middle, Lower Right, etc.) That region will be magnified in the image or matchset/scanset.

Location in Gel

Location in Gel on the View > Advanced View submenu displays a graphical overlay on a subwindow or all subwindows showing the region that is displayed (green box) relative to the entire image (blue box). Select the command, then click on the image.

3.7 Hiding Overlays

To conceal spot markings, graphs, information boxes, etc. in a subwindow or multiple subwindows, select Hide Overlays from the main toolbar or View menu and click in the subwindow, or use the F4 key command in the subwindow.

Note that, by default, hiding the overlays in one image of a matchset or scanset will hide them in all the other subwindows. To turn off this feature, deselect the Auto “All Same” checkbox under Display Preferences (Edit > Preferences).

Note: Shift+F4 (Hide All Overlays) will conceal overlays in all subwindows regardless of whether Auto “All Same” is selected.

3.8 Density Tools

The tools under the View > View Density submenu and on the Density Tools toolbar are designed to provide a quick measure of the signal intensity of the data in 2-D scans or Filtered images. Note that these commands will not work on Gaussian images.

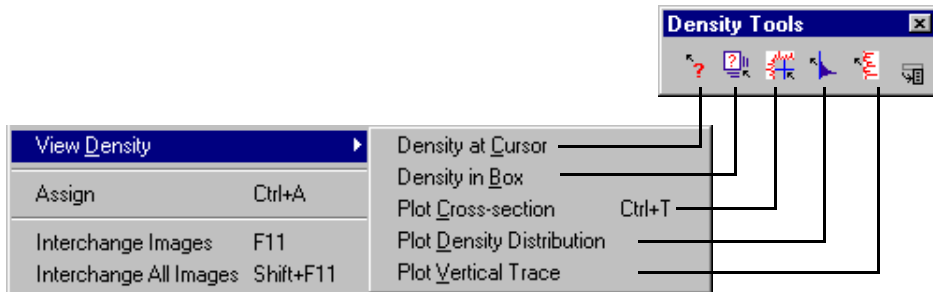


Fig. 3-7. Density submenu and toolbar.

Density at Cursor

To display the signal intensity at a point on the image, select Density at Cursor from the submenu or toolbar and click on the image. The pop-up box will list the intensity in image units (ODs or counts), and will also show the average intensity for a 3 x 3 image pixel box centered on that point.

Density in Box

To display the total signal intensity inside a specified region on the image, select Density in Box from the submenu or toolbar and drag a box around the area you want to measure. The area of the box (in square millimeters), the total image intensity (volume) inside the box (in image units—ODs or counts), and the average intensity will be displayed in a pop-up box.

Plot Density Distribution

Plot Density Distribution displays a histogram of the signal intensity distribution for the entire image. Select the command from the menu or toolbar, and click on the image.

Plot Cross-section

To display a cross-sectional intensity trace centered on a point on the image, select Plot Cross-section from the submenu or toolbar, or use the Ctrl+T key

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command, and click or drag on the image. The trace will be displayed, and will be continuously updated as you move your cursor over the image. The intensity (in ODs or counts) at the cursor point is listed, as is the maximum intensity along the lines of the cross-section.

Note: The sampling width for the density traces is one image pixel.

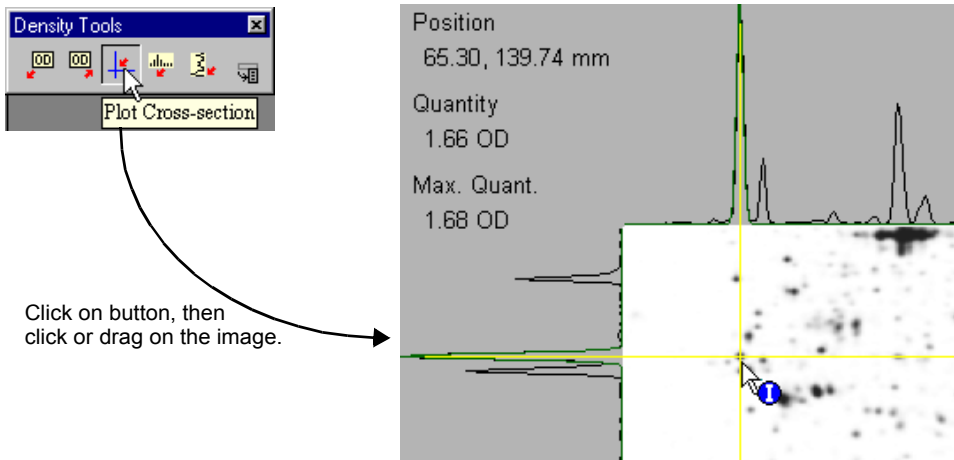


Fig. 3-8. Plot Cross-section tool.

Plot Vertical Trace

Plot Vertical Trace plots an intensity trace down a vertical line centered on the point where you click your mouse. Select the command from the menu or toolbar, and click or drag on the image.

Note: The sampling width for the density traces is one pixel.

3.9 Colors

Edit > Colors opens a dialog box in which you can adjust the colors of the image, windows, buttons, etc.



Fig. 3-9. Colors dialog box.

Selecting a Color Group

Within the Colors dialog box, click on the Color Group button to select the colors of the particular group of objects you want to change (e.g., pop-up boxes, image colors, etc.).

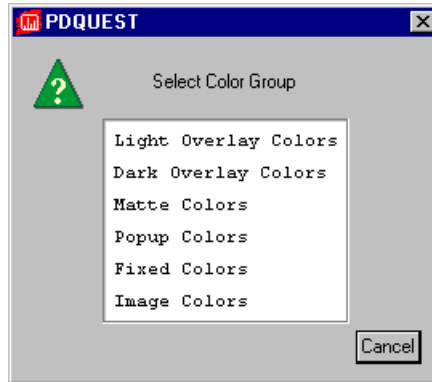


Fig. 3-10. List of Color Groups.

Click on a color group in the list to select it.

Changing a Color

After you have selected the color group to change, click on the specific color button to change. The Color Edit dialog box will open, allowing you to adjust the red-green-blue (RGB) values of the color you selected.

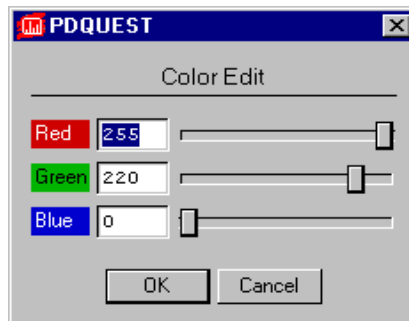


Fig. 3-11. Color Edit dialog box.

Drag the sliders or enter a value in the fields. The color of the button will change with your adjustments.

Saving/Selecting a Defined Set of Colors

After you have changed the colors within color groups, you can save these settings for future use on other images. The Colormap Name field displays the name of a defined set of colors and color groups. There are several standard colormaps, or you can create your own.

To select a predefined colormap, click on the Load button. From the list displayed, click on the set of colors you want to apply.

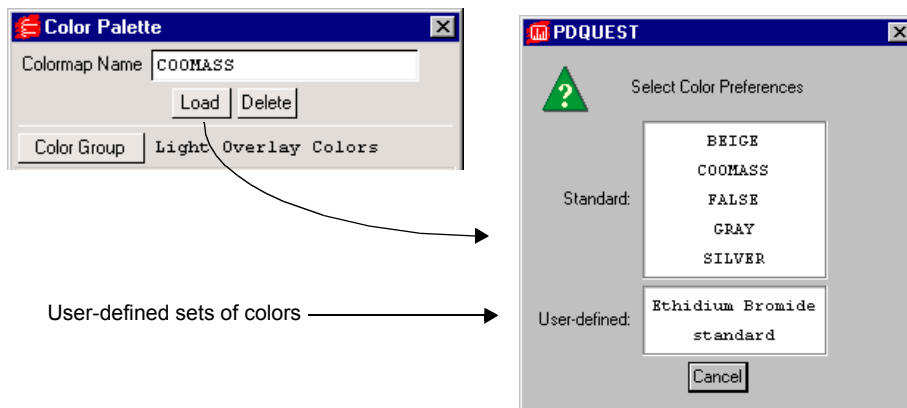


Fig. 3-12. Selecting a Colormap.

To create your own colormap, adjust the colors within the color groups as described above and type in a new colormap name. Click on OK to apply your changes.

To remove a colormap, click on the Delete button. Select the colormap to be deleted from the displayed list. A pop-up box will ask you to confirm the deletion.

If you want to return to the standard colormap for the selected group, click on the Reset button. All colors will be returned to their default values.

3.10 Multi-Channel Viewer

You can use the Multi-Channel Viewer to distinguish different types and levels of fluorescence in a gel that has been imaged at different wavelengths. The Multi-Channel Viewer can be used to merge the information from up to three different images of the same gel.

Note: The viewer requires that the images being compared are exactly the size and in exactly the same position. When capturing the images, you should be careful not to move the gel between exposures. If your images are not exactly the same size, you can use the Crop tool to resize them.

With at least one image open, select Multi-Channel Viewer from the View menu. The topmost open image will be displayed in the viewer window using the Red channel.

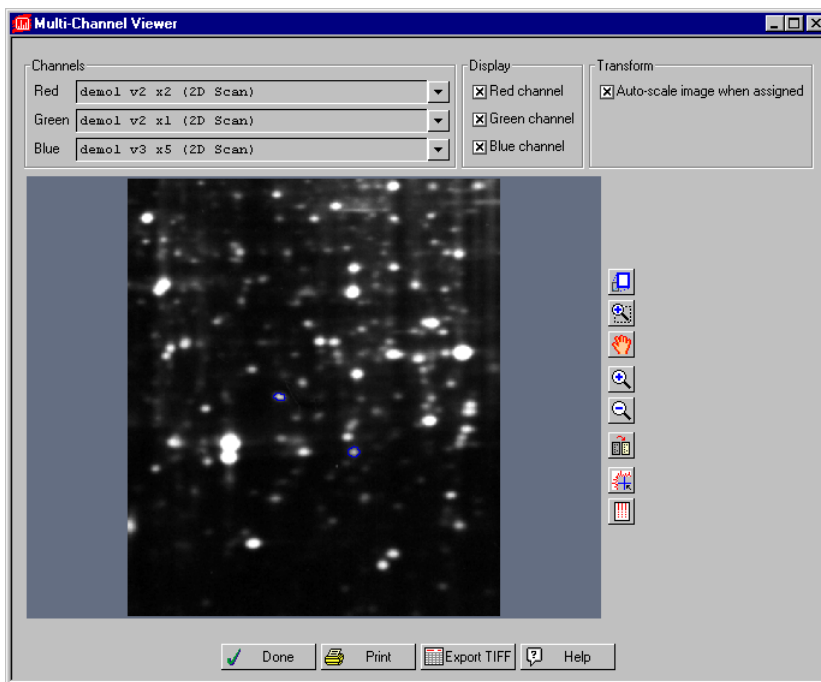


Fig. 3-13. Multi-Channel Viewer.

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Note: The color channel used to display an image in the viewer has no relation to the filter used when capturing the image. The red, green, and blue channels are simply designed to distinguish different images.

The image name is displayed in the Red name field at the top of the viewer, and the Red Channel checkbox will appear selected.

To add another image, make sure the image is open and click on the pulldown button next to the Green or Blue name field. Select the image name from the pulldown list. Add a third image using the same procedure.

You can reassign the different images to different channels using the pulldown buttons to the right of the name fields. Select <clear> from the pulldown list to remove an image from that channel of the viewer.

Viewing Options

To remove a particular color channel from the display, click in the checkbox associated with that channel to deselect it.

Selecting the Auto-Scale Image When Assigned checkbox will automatically adjust the brightness and contrast of each loaded image based on the data range in the image. It invokes the Auto-Scale command from the Transform window when an image is first opened in the viewer. Note that this setting affects only how the image is displayed in the viewer, not the actual data.

Note: If you deselect this checkbox, any images currently displayed will remain auto-scaled. Click on the Transform button in the viewer and click on the Reset button in the Transform window to undo auto-scaling.

Buttons for various viewing tools are included in the Multi-channel viewer. Commands such as Zoom Box and Grab will change the display of all the images in the viewer at once.

Note that if you open the Transform window, you can adjust the display of each channel independently, by selecting the appropriate channel option button in the Transform window. Similarly, the Plot Cross-Section command will report the intensity of each channel separately.

Exporting and Printing

You can export a 24-bit TIFF image of your merged view by clicking on the Export button. This will open a version of the Export to TIFF dialog. Note that you cannot export image data from the Multi-channel Viewer—only the current view of the image (designated as Publishing Mode in the Export dialog). The colors in the viewer will be preserved in the exported TIFF image.

You can also print a copy of your merged view to a color or grayscale printer by clicking on the Print button.

3.11 Transform

The Transform controls are used to adjust image brightness and contrast and optimize the image display. These controls affect the display only, not the underlying data.

Select Transform from the Image menu or main toolbar and click in the subwindow, or use the Ctrl+B key command in the subwindow.

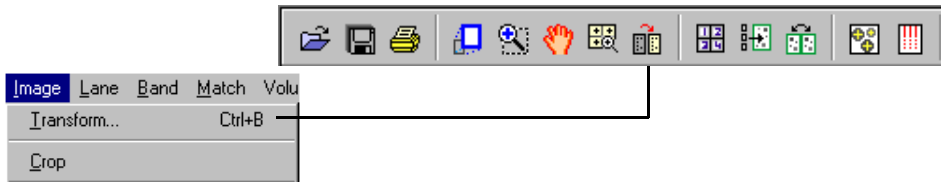


Fig. 3-14. Transform command.

The preview window in the Transform dialog shows a smaller view of the selected image. Changes are automatically reflected in the preview window, and are only applied to the main image(s) when you click on OK or (PDQUEST only) Apply All.

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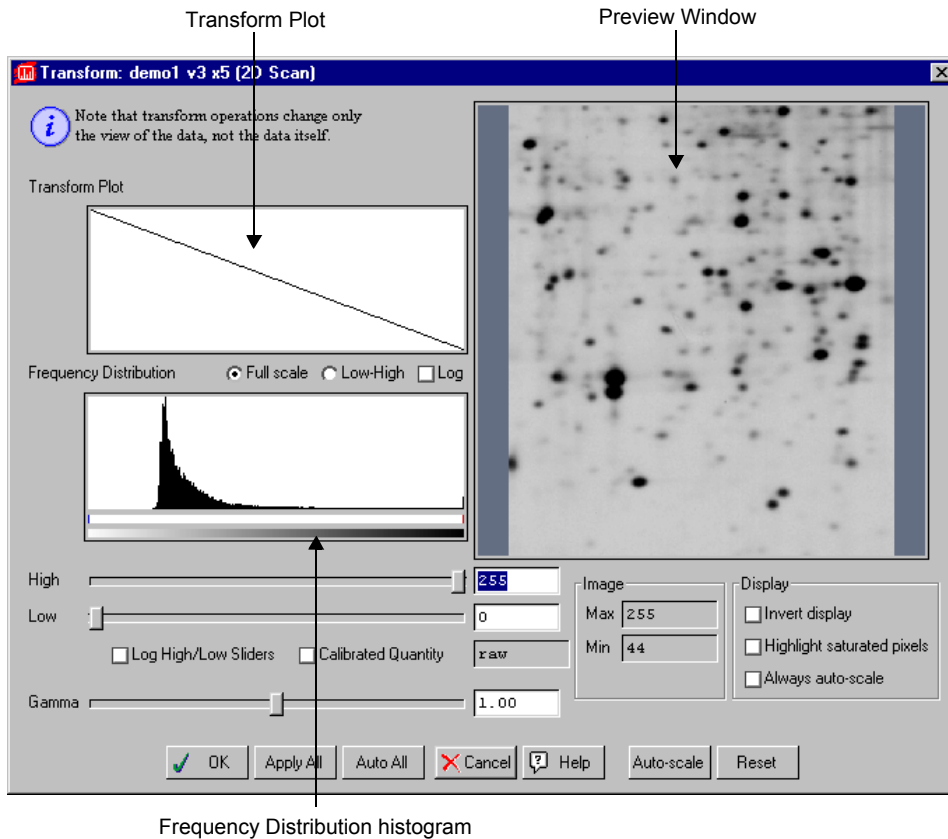


Fig. 3-15. Transform dialog box.

You can use regular viewing tools like Zoom Box and View Entire Image in the preview window to focus on particular regions of interest. You can also use Grab tool and arrow keys to move the image within the preview window.

The Frequency Distribution histogram shows the total data range in the image and the amount of data at each point in the range. In a typical scan, there is a signal spike at the left ("gray") end of the histogram due to background noise.

The Transform Plot is a logarithmic representation of how the raw pixel data are mapped to the pixels of your computer screen.

Note: If you are using the Transform window in conjunction with the Multi-channel Viewer, you can adjust the display of each channel independently by selecting the appropriate channel option button below the preview window.

3.11.a Transform Controls

Auto-scale

Click on the Auto-scale button to optimize the selected image automatically. The lightest part of the image is set to the minimum intensity (white), and the darkest is set to the maximum intensity (black). This maximizes the display range for the image. You can then “fine-tune” the display using the sliders.

Select the Always Auto-scale checkbox to automatically auto-scale every image that you open. This setting disables the remaining controls in the dialog box; the software will examine the data range in every image and optimize it accordingly.

Auto All will auto-scale all the images in the selected scanset or matchset and close the Transform dialog.

High/Low Sliders

Use the High and Low sliders to optimize the image manually. Drag the High slider handle to the left to make weak signals appear darker. Drag the Low slider handle to the right to reduce image background.

As you drag, examine the Frequency Distribution histogram. Everything to the left of the low slider will be remapped to minimum intensity, while everything to the right of the high slider will be remapped to maximum intensity. Position the markers at the ends of the data range in the histogram, and use the low slider to cut off the “spike” of background noise.

You can also type specific high and low values in the text boxes next to the sliders. Clicking anywhere on the slider bars will move the sliders incrementally.

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If your image data is in a narrow range (as shown in the histogram), select the Log High/Low Sliders checkbox. This allows for finer adjustments with the slider handles.

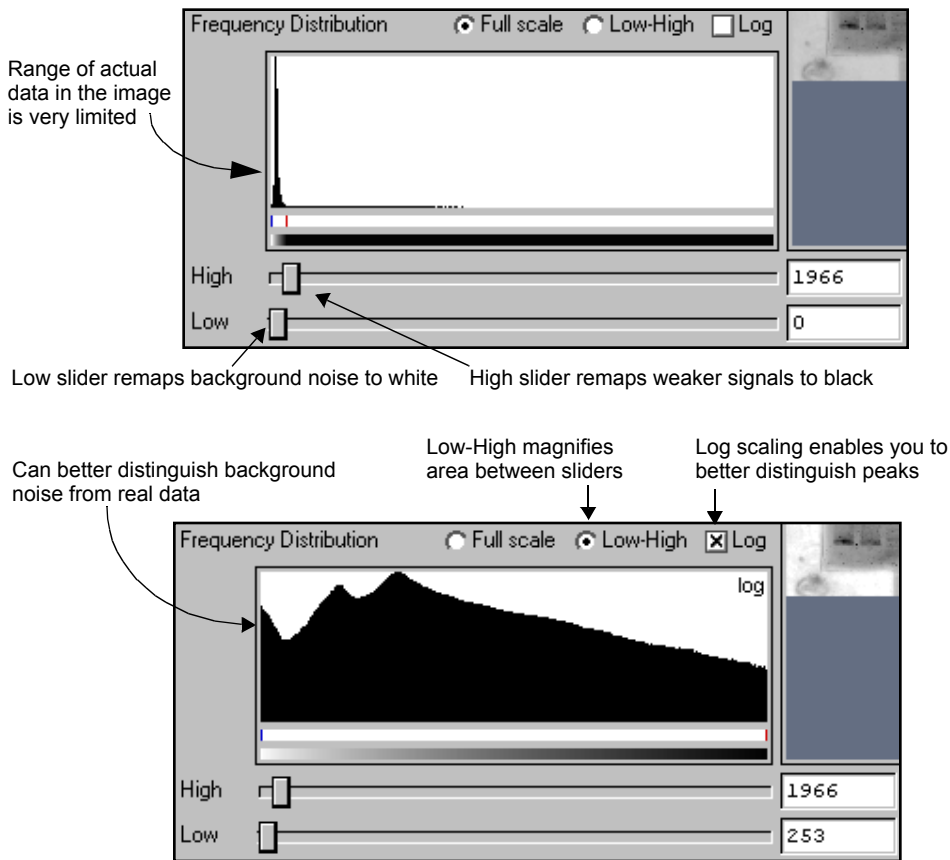


Fig. 3-16. Two views of the Frequency Distribution histogram.

Gamma Slider

Some images may be more effectively visualized if their data are mapped to the computer screen in a nonlinear fashion. Adjusting the Gamma slider

handle expands or compresses the contrast range at the dark or light end of the range, and this is reflected in the transform plot and preview window.

Reset

If at any time you want to return to an unmodified view of the scan data, click on Reset.

Applying the Settings

Click on OK to apply the selected settings to the image in the preview window. Click on Apply All to apply the settings to all the subwindows in the selected scanset or matchset, or click on Auto All to auto-scale all the subwindows. These buttons will close the Transform window and displayed the transformed image(s).

3.11.b Other Transform Features

Full Scale/Low-High

The Full Scale/Low-High option buttons adjust how the range of image data is displayed in the frequency distribution histogram and transform plot. They do not change how the image is displayed in the preview window.

Full Scale adjusts the frequency distribution and transform plot displays so they show the full intensity range of the image.

Low-High magnifies the range between the low and high sliders. This makes it easier to view your data if it does not occupy the full intensity range of the image.

Log

The Log checkbox changes the way the data is displayed in the histogram so you can better discern subtle changes in signal intensity.

Image Max/Min

Image Max and Min display the range of signal intensity in the image.

The image units are determined by the type of scanner used to create the image. In the case of densitometers, you can select Calibrated Quantity to display your image units in O.D.s.

Invert Display

The Invert Display checkbox flips light bands on a dark background to dark bands on a light background, and visa versa. Once, again, the actual data will not change—only the image.

Highlight Saturated Pixels

When the Highlight Saturated Pixels checkbox is selected, areas of the image with saturated signal intensity are highlighted in red. This setting does not apply to Gaussian images.

3.12 Cropping Images

When matching spots between the gels in a matchset, the matchset images should be all the same size and shape. If the original scans of these gels are different sizes, you will need to crop them before creating the matchset.

Note: You can only crop 2-D scans, not Filtered or Gaussian images.

The crop tools can also be used to reduce the file size of images. (To reduce the file size of images by lowering the resolution, see Reduce File Size.)

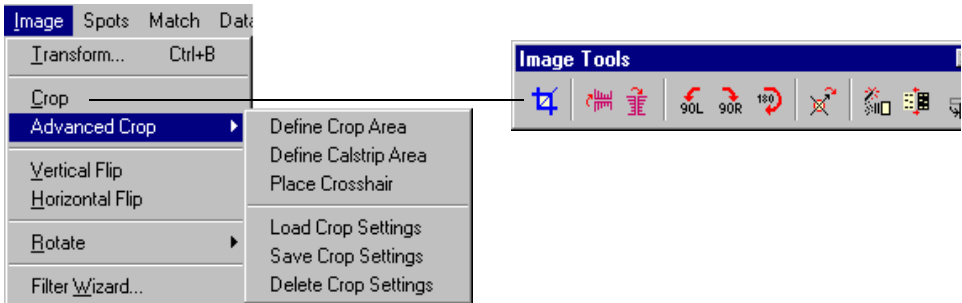


Fig. 3-17. Crop commands.

Select Crop from the Image menu or toolbar, and drag the cursor on the image, creating a box. Everything outside the crop box will be deleted.

The size of the crop area (in millimeters and number of pixels) is shown at the bottom of the crop box, as is the file size of the image inside the crop area.

- To *reposition* the crop box, position your cursor at the center of the box. The cursor appearance will change to a multidirectional arrow symbol. You can then drag the box to a new position.
- To *resize* the box, position your cursor on a box border or corner. The cursor appearance will change to a bidirectional arrow. You can then drag that border or corner in or out, resizing the box.
- To *redraw* the box, position your cursor outside the box. The cursor appearance will change back to the Crop tool, and you can draw another box, replacing the one you just drew.

To complete the crop, position your cursor inside the box slightly off-center. The cursor appearance will change to a scissors symbol. Click once to perform the crop.

A pop-up box will ask you whether you want to: (1) crop the original image, (2) save a copy of the area inside the crop box as a separate image, keeping the original image intact, or (3) cancel out of the cropping operation.

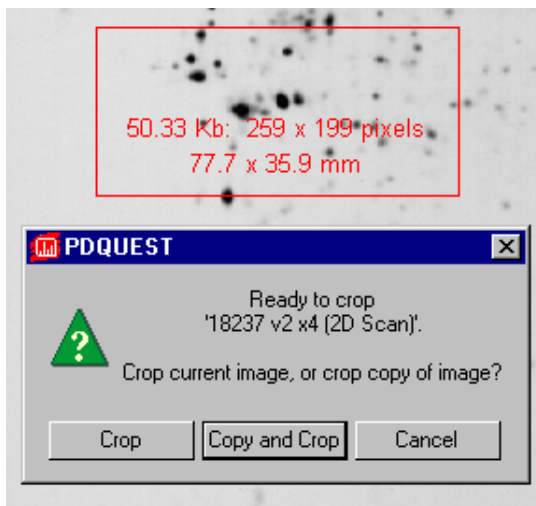


Fig. 3-18. Crop box and pop-up Crop dialog.

If you select Crop, your cropped image will be displayed in the image window. If you select the Copy and Crop button, a dialog box will open in which you can enter the name of the cropped image, its version number, exposure number, and/or file type. Click on OK, and the image window will display both the original image and the cropped image in subwindows.

3.12.a Advanced Crop Commands

To ensure that your scans are exactly the same size and shape, you can use the tools on the Image > Advanced Crop submenu to save the crop box from one scan and apply it to others. These tools also allow you to crop a gel and calstrip out of the same scan.

To define and save a crop box and apply it to another scan:

1. Select Define Crop Area from the submenu and drag a crop box on an image. Position the box as described in Cropping Images.
2. Select Place Crosshair from the submenu and click on a landmark inside the box that is present in all the gels you want to crop. This could be a

spot or some other image detail. The crosshair will make it easier to position the box in the other images so that it encloses the same area.

3. Select Save Crop Settings from the submenu, enter a name for the current crop settings in the pop-up box, and click on Apply.
4. Complete the crop action in the current image by positioning your cursor inside the box slightly off-center and clicking to perform the crop, as described in Cropping Images.
5. Open or select the next image you want to crop, select Load Crop Settings from the submenu, and select the name of the settings you saved. The crop box and crosshair will appear on the image.
6. Reposition the crop box so that the crosshair is correctly aligned with the appropriate image object, then complete the crop as described in step 4 above.

To delete any crop settings you have saved, select Delete Crop Settings from the submenu and choose the settings to be deleted from the list.

Cropping Calstrips

In an image containing both a gel and a calstrip, you can define two crop boxes at once—one for the gel, and one for the calstrip.

First draw your crop box on the image as described in step 1 above, then select Define Calstrip Area from the submenu and draw a second crop box around the calstrip portion of the image. When you complete the crop action in one of the boxes, both will be cropped, and the calstrip area will be labeled and saved as a calstrip scan.

If you save the crop settings, both boxes will be saved, and both will be applied to a new image when you load the settings.

Be careful to define the regular crop box before defining the calstrip area. Also, only use the Define Calstrip Area command on calstrips, because the resulting cropped area will be defined as a calstrip image (i.e., with a .csc extension in Windows).

3.13 Flipping and Rotating Images

The matching functions in PDQUEST will tolerate minor distortions in the shape and orientation of images. However, if your images are grossly misaligned, you may have to flip and/or rotate them. These commands are located on the Image menu and toolbar.

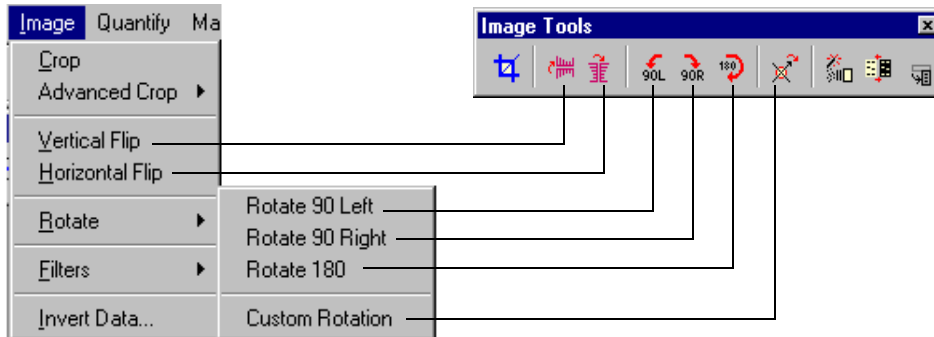


Fig. 3-19. Flipping and rotating commands.

Note: These actions will erase any analysis you have performed on an image. You will be asked to confirm the flip/rotation before the command is executed.

Flipping

To flip the image right-to-left, select Horizontal Flip from the Image menu or toolbar. To flip the image top-to-bottom, select Vertical Flip.

90° Rotations

Select Rotate 90 Left, Rotate 90 Right, or Rotate 180 from the Image > Rotate menu or Image toolbar to perform the specified rotation. You will be asked to confirm your choice before the command is executed.

Custom Rotation

If you need to rotate your image in increments other than 90° , you can use the Custom Rotation command.

Select Custom Rotation from the Image > Rotate menu or Image toolbar. A green “plus” sign will appear next to your cursor. Click on the image you want to rotate and a circular overlay with an orange arrow will appear. A small dialog box also will open, indicating the angle of rotation in degrees and radians.

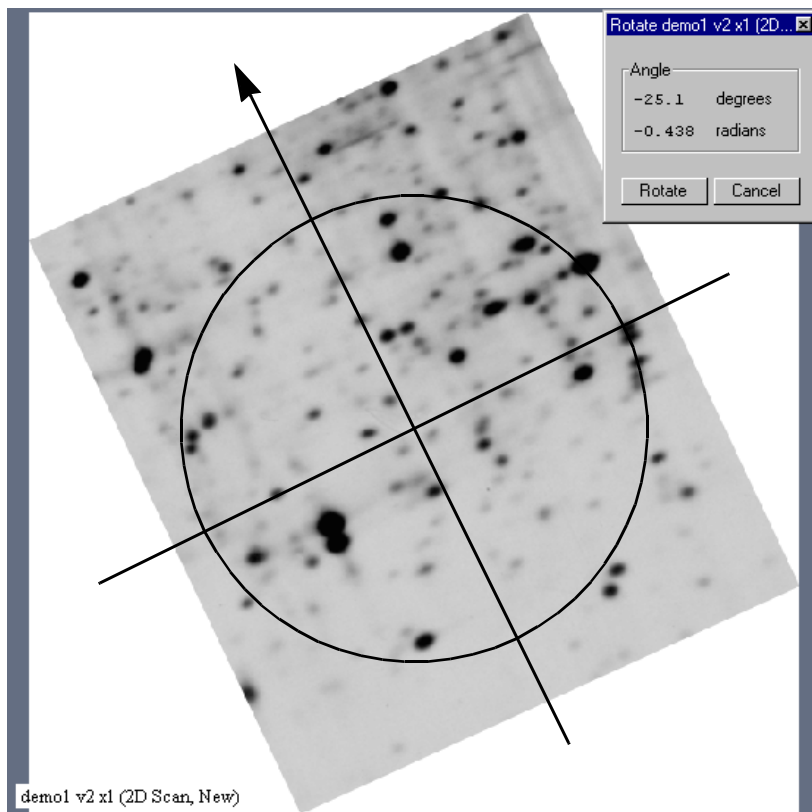


Fig. 3-20. Custom rotation; the arrow points in the direction of the new top of

the image.

To perform the rotation, position your cursor on the arrowhead and drag. As you drag, the arrow will rotate and the angle in the dialog box will change. Position the arrow so that it points in the direction of the new top of the image. You can “fine-tune” your rotation as much as you like.

Note: If you want to center your arrow on a particular point on the image (e.g., to align along a particular lane), you can use the F3 key command. Position your cursor on the point on the image you want to center on, and press the F3 key. The image will move so that the center of the arrow and your cursor point are aligned.

To complete the rotation, click on the Rotate button in the small dialog box. A subwindow will open containing the rotated image. You will then have the option of renaming your new image and changing the version number.

If you are not satisfied with your rotated image, simply delete it and start over.

Note: Because an image is composed of square or rectangular pixels, Custom Rotation has to perform some minor smoothing on the image to turn it at a non-90° angle. In addition, any analysis performed on the image cannot be rotated and will be lost.

3.14 Filtering Images

Filtering is a process that removes small noise features on an image while leaving larger features (like data) relatively unaffected. A wide range of filters are available for removing different types of noise from images. Depending on the nature of your data, you will probably need to use only one or two of the available filters. However, you should experiment with several different filters before selecting the ones that work best for your images.

The filtering commands are located on the Image menu and toolbar.

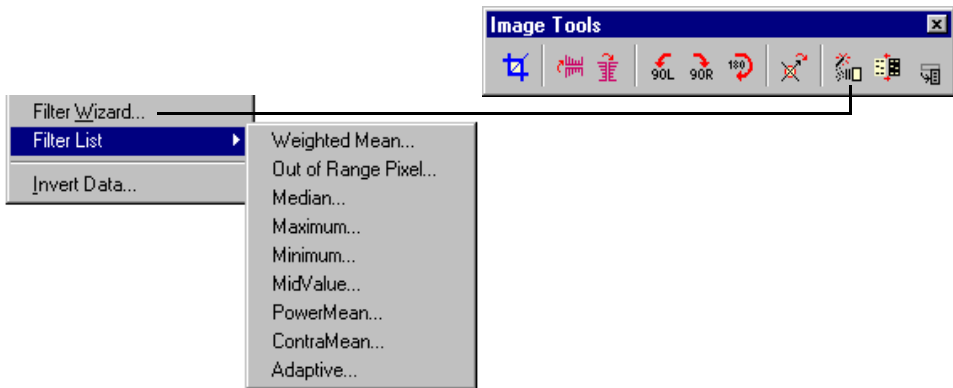


Fig. 3-21. Filtering commands.

Note: Since filtering is an irreversible process, you will be asked if you want to create a copy of the original image before you filter. If you are experimenting with various filters, you should create copies of your image and compare them side-by-side. If you filter the original image and save it, *you cannot return to the original, unfiltered state.*

3.14.a Filter Wizard

The Filter Wizard is designed to guide you through the filter selection process. First, you identify the type of noise in your image. Next, select the size of the filter to use on that noise. Finally, filter the image.

To open the Wizard, select Image > Filter Wizard or click on the Filter Wizard button on the Image Tools toolbar.

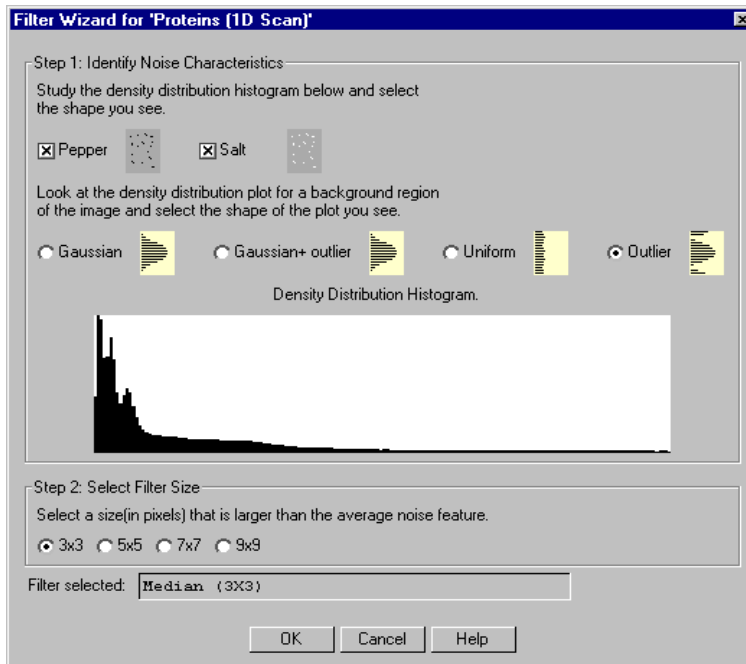


Fig. 3-22. Filter Wizard dialog box.

The Wizard contains settings for identifying the different types of noise in the image. It also includes a density distribution histogram of the noise in the image to aid in filter selection.

Step I: Identify Noise Characteristics

The first step in the Wizard is to identify the type of noise in your image. Examine both the image and the density distribution histogram, then select one, both, or neither of the following checkboxes:

- **Salt.** This type of noise appears as specks that are lighter than the surrounding background. The density distribution histogram of this type of noise displays noise peaks at the high end of the range (right end of the plot). This type of noise is common in electronic cameras with malfunctioning pixels. It can also be caused by dust or lint in the imaging

optics or scratches on photographic film. Salt is a type of outlier noise (see below).

- **Pepper.** This type of noise appears as specks that are darker than the surrounding background. The distribution histogram of this type of noise displays noise peaks at the low end of the range (left end of the plot). Its causes are similar to those of salt noise. Pepper is a type of outlier noise (see below).

Next, select one of the following option buttons to describe additional features of your noise.

- **Gaussian.** The distribution histogram of this type of noise has a Gaussian profile, usually at the bottom of the data range. This type of noise is usually an electronic artifact created by cameras and sensors, or by a combination of independent unknown noise sources.
- **Uniform noise.** This type of noise appears in the histogram as a uniform layer of noise across the data range of the image.
- **Outlier noise.** This category of noise includes salt and pepper noise (see above). The distribution histogram of this type of noise displays noise peaks at the high and low ends of the range.

After you have identified the type of noise, go to Step 2.

Step 2: Select Filter Size

Image noise is filtered by means of a filtering window (or kernel), which is measured in pixels. This filtering window slides across the image, processing the pixels within it.

The available filter dimensions range from 3 x 3 pixels to 9 x 9 pixels. To select an appropriate size, magnify a background region of your image so that you can see the individual pixels. The filter size you select should be larger than the average noise feature but smaller than your data features.

Note: A smaller filter will alter your image less than a larger filter. Large filters can result in better suppression of noise, but can also blur desirable features in the image.

Step 3: Begin Filtering

After you have completed your selections, the filter name and size will be displayed at the bottom of the Filter Wizard dialog box.

To begin filtering, click on the OK button. Because filtering is an irreversible process, a pop-up box will give you the option of filtering the original image, creating a copy of the image to filter, or cancelling out of this operation.

If you choose to Copy and Filter, you will be asked to enter a new name and/or version number for the new copy before the operation is performed. Once this information has been entered, the filtering operation will be performed.

3.14.b Selecting a Filter Directly

If you already know the type and size of filter you want, you can select it directly by selecting Image > Filters > Filter List. The pulldown menu includes all the filters available in the Discovery Series.

The types of filters in the Discovery Series are:

- **Weighted Mean.** This filter is useful for reducing Gaussian noise; it has little effect on salt and pepper. It calculates the weighted mean of the pixels within the filtering window and uses it to replace the value of the pixel being processed.
- **Out of Range Pixel.** This filter is useful for suppressing salt and pepper noise; its effect on Gaussian noise is minimal. This filter calculates the mean of the pixel values in the filtering window, including the pixel being processed. If the difference between the mean and the individual pixel value is above a certain threshold, then the individual value is replaced by the mean.
- **Median.** Also useful for suppressing salt and pepper noise, this filter has little effect on Gaussian noise. It calculates the median value of the pixels within the filtering window and uses it to replace the value of the pixel being processed. The median filter produces very little blurring if a small-sized window is selected.
- **Maximum.** This filter is useful for eliminating pepper noise in an image (it worsens the effect of salt noise). It replaces the value of the pixel being

processed with the maximum value of the pixels within the filtering window.

- **Minimum.** This filter replaces the value of the pixel being processed with the minimum pixel value within the filtering window. This filter is useful for eliminating salt noise in an image (it worsens the effect of pepper).
- **MidValue.** This filter is useful for suppressing uniform noise within an image; however, it worsens the effect of pepper and salt. This filter replaces the value of the pixel being processed with the mean of the maximum and minimum pixel values within the filtering window.
- **Power Mean.** This filter is useful for suppressing salt and Gaussian noise within an image (it worsens the effect of pepper noise). It replaces the value of the pixel being processed with the power mean of the pixel values within the filtering window.
- **ContraMean.** This filter is useful for suppressing pepper and Gaussian noise within an image (it worsens the effect of salt). It replaces the value of the pixel being processed with the contra-harmonic mean of the pixel values within the filtering window.
- **Adaptive.** This filter is useful for suppressing Gaussian noise and salt and/or pepper within an image. If your image contains a mix of salt and pepper, select this filter.

To begin filtering, select a filter type from the pull-down list. A pop-up box will ask you to select a filter size.

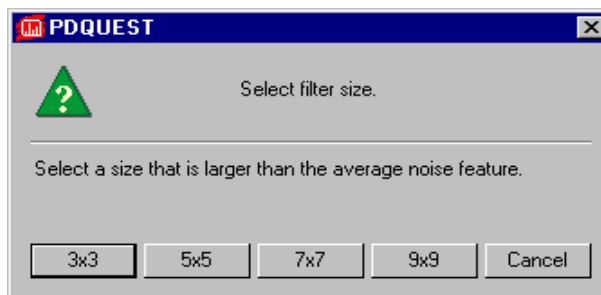


Fig. 3-23. Selecting a filter size.

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The available filter dimensions range from 3 x 3 pixels to 9 x 9 pixels. (See the previous section for guidance on selecting a size.)

Because filtering is an irreversible process, a pop-up box will give you the option of filtering the original image, creating a copy of the image to filter, or cancelling out of this operation.

If you choose to copy and filter, you will be asked to enter a new name and/or version number for the new copy before the operation is performed. Once this information has been entered, the filtering operation will be performed.

3.15 Invert Data

If your image has light spots on a dark background (i.e., the signal intensity of the background is greater than the signal intensity of the sample), you will need to use the Invert Data function before you can analyze the image.

Note: The Invert checkbox in the Transform dialog box (section 3.11.b) inverts only the appearance of the image.

To invert your image data, select Image > Invert Data. You may need to use the Transform function to adjust the appearance of your inverted image.

This command is reversible. Select Invert Data again to revert to the original data.

3.16 Text Overlays

If you want to create and display textual notes directly on your image, select Text Overlay Tools from the View > Toolbars submenu. This will open the Text Overlay Tools toolbar.

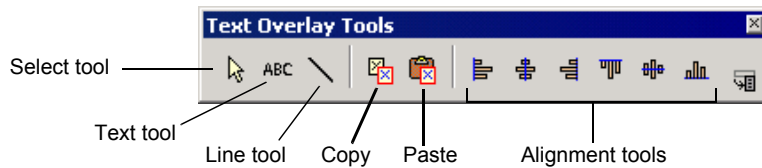


Fig. 3-24. Text Overlay Tools toolbar.

Creating a Text Overlay

To create a text overlay, click on the Text Tool, then click on the image at the spot where you want the text to appear. This opens the Text Overlay Properties dialog box.

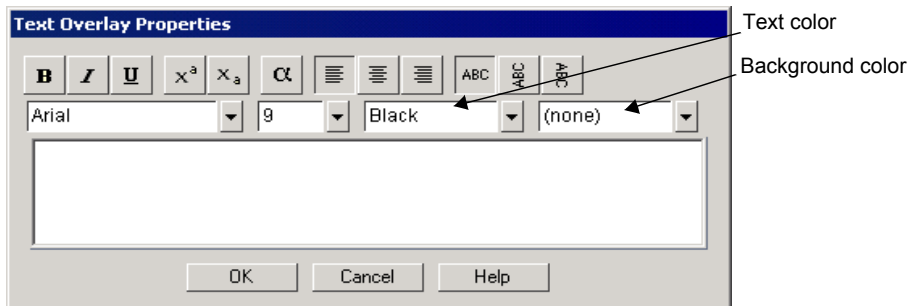


Fig. 3-25. Text Overlay Properties dialog box.

To enter your text, simply begin typing in the main field. The buttons in the dialog allow you to select the properties of your text, including format, alignment, and justification.

The pull-down boxes (from left to right) allow you to select the font style, font size, color of the text, and color of the background within the text box.

Once you have typed your text, click on OK.

When you exit the Text Overlay Properties dialog by clicking OK, the text you typed will appear on the image at the spot where you originally clicked.

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You can create as many textual overlays as you want.

Editing a Text Overlay

To edit a text overlay, make sure the Text Tool or Select Tool is assigned to your mouse, then double-click on the overlay to open the Properties dialog. The existing text will be displayed and can be edited.

Line Tool

You can use the Line Tool on the Text Overlay toolbar to draw a line between text and an image feature, or between any two points of interest on your image.

Click on the Line Tool button, then drag on your image to create the line. You can create as many lines as you want.

To resize or adjust a line, make sure the Line Tool or Select Tool is assigned to your mouse, then position your cursor on one end of the line (marked by a circle) and drag.

To add arrowheads to a line, make sure the Line Tool or Select Tool is assigned to your mouse, then double-click on the middle of the line. A dialog box will pop up with options to add arrowheads to one or both ends of the line.

Moving and Copying Text Overlays and Lines

You can move, copy, or delete a single overlay/line or a group of overlays/lines within an image. You can also copy and paste between images.

First, you must select the object(s). Click on the Select Tool button on the Text Overlay toolbar. To select a single overlay or line, click on it. To select multiple objects, either drag a box around them or hold down the SHIFT key while you click on them one at a time. When dragging to select a group of objects, make sure that you completely surround all the objects to be selected.

Each selected overlay/line will have a green border.

- To move the selected object(s), position your cursor over the selection and drag.

Viewing and Editing Images

- To copy within an image, hold down the CTRL key while dragging the selected object(s). The copy will be created and dragged to the new position.
- To delete the selected object(s), press the DELETE key.
- To copy between images, click on the Copy to Clipboard button on the Text Overlay toolbar, then open or select the image you want to copy to and click on the Paste from Clipboard button. The copied object(s) will be pasted into the new image in the same relative position they were copied from.

Note: If you are pasting into an image with a different pixel size (i.e., resolution), you will receive a message that the placement of the copy may not be exact. Click on OK to complete the paste, then position the pasted objects manually.

Viewing Previously Created Text Overlays/Lines

To display previously created text overlays and/or lines after opening an image, click on the Text Overlay Tools button on the main toolbar.

If you have concealed all your overlays using Hide Overlays, clicking on any of the buttons on the Text Overlay Tools toolbar will display the hidden text.

4. Detecting and Editing Spots

This chapter describes how to detect and edit spots using PDQUEST.

4.1 Selecting Spot Detection Parameters

The Spot Detection Wizard is used to select the parameters for detecting spots in gel scans.

Because gels vary in stain type, spot intensity, clarity, streaking, etc., different parameters are needed to detect spots on different gels. In general, you can use the same parameters to detect spots on all gels that use the same stain and have been run and scanned in the same manner.

To select the detection parameters for a particular type of gel, make sure that a representative gel scan is open and active in PDQUEST, then select Spot Detection Wizard from the Spots menu or quick guide.

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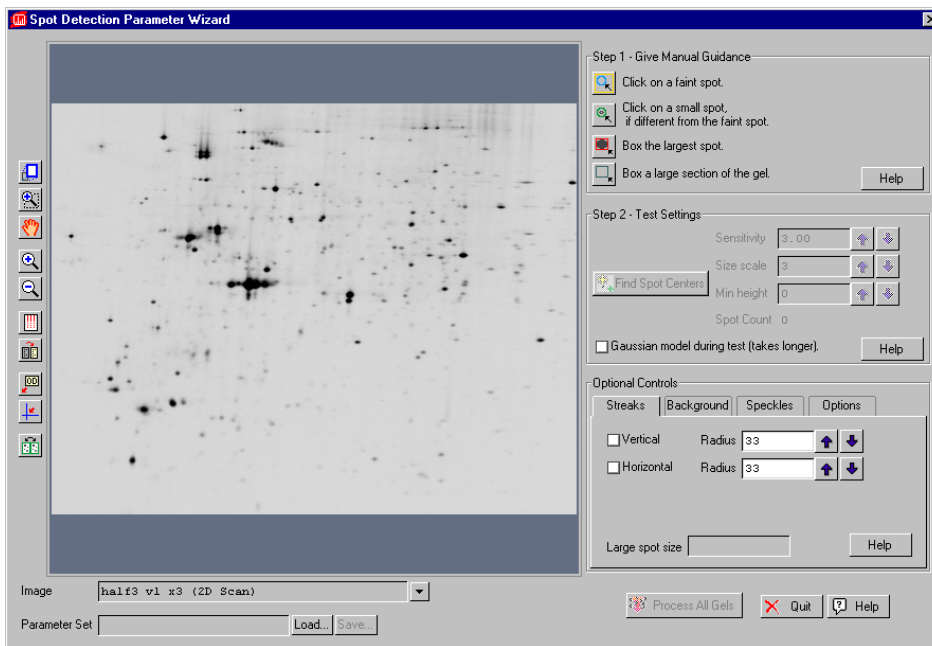


Fig. 4-1. Spot Detection Wizard.

The Wizard will open, and the name of the scan will be displayed below the preview window. Click on the pulldown button next to the image name to load a different open image into the Wizard. Any unsaved parameters will be lost.

Detecting and Editing Spots

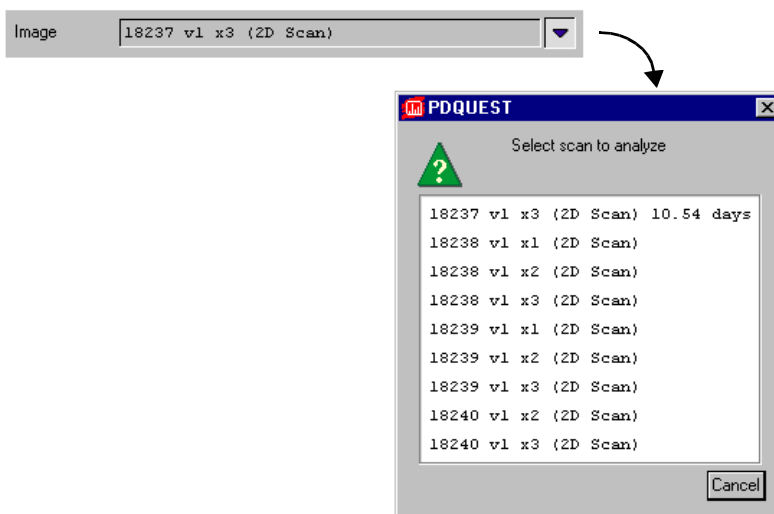


Fig. 4-2. Selecting a new image in the Spot Detection Wizard.

The control panel in the Wizard is designed to guide you through the parameter selection process:

4.1.a Step 1. Give Manual Guidance

In this step, you mark features on the gel scan that will be used to automatically select the correct spot detection parameters.

1. Click on the Faint Spot button, then click on the faintest spot in the scan that you want to detect. This will set the sensitivity and minimum peak value parameters.
2. If the smallest spot you want to detect is smaller than the faintest spot, click on the Small Spot button, then click on the smallest spot. This will set the size scale parameter. In most cases, the smallest and faintest spots are the same, and you can skip this step.
3. Click on the Large Spot button, then draw a box around the largest spot on the image that you want to detect. This sets the radius of the background subtraction rolling ball and the streak removal rolling disk.

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4. Finally, click on the Box Large Section button, then draw a box around a large representative region of the image, including spots, streaks, background gradations, etc. This selects the correct noise filter.

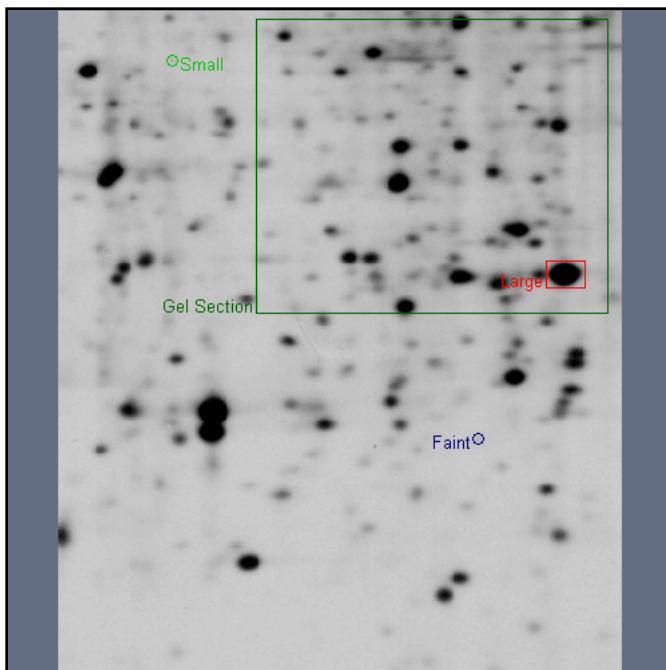


Fig. 4-3. Image with large, small, and faint spots and large region defined.

4.1.b Step 2. Test Settings

In this step, you first see how many spots are detected based on the settings in Step 1, then you adjust those settings until all significant spots are identified.

Note: Before proceeding, if there are streaks in your image, select the Vertical and/or Horizontal streak removal checkboxes under Optional Controls.

Click on the Find Spot Centers button. Noise smoothing, background subtraction, and (if selected) streak removal will be performed, and the spot centers will appear marked with crosshairs.

Detecting and Editing Spots

Review the results of spot detection by studying the crosshairs on the image. The number of spots detected is displayed in the Spot Count field.

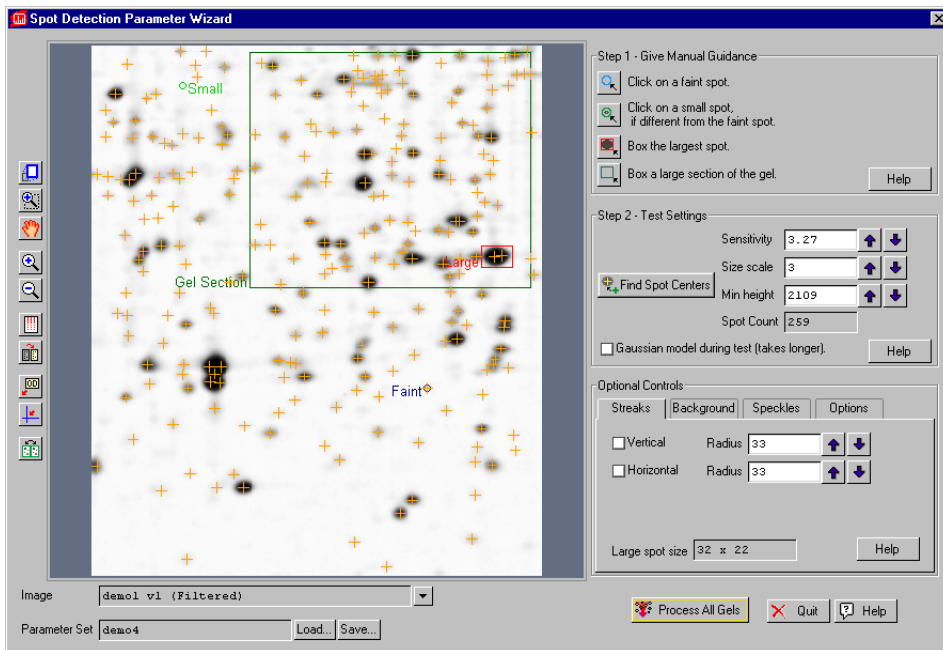


Fig. 4-4. Spot crosshairs marked in the Spot Detection Wizard.

If all the spots you want to identify have been detected, you can proceed to save this parameter set. If significant spots have been missed, you can change the faint, small, and large spot selections in Step 1, or change the Sensitivity, Min Peak, and Size Scale directly.

Note: Adjustments to these settings will automatically redetect spots.

Sensitivity

If significant faint spots in the scan have been missed, you can use the Faint Spot tool in Step 1 to change your faint spot selection, or you can increase the Sensitivity setting by typing a number directly in the field (next to the Find

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Spot Centers button) or clicking on the adjustment arrows. Higher sensitivity will detect more spots. If image noise has been detected as spots, decrease the sensitivity or perform additional speckle removal.

Min Peak

If increasing the Sensitivity setting or selecting a new Faint Spot still fails to find some faint spots, you can lower the Min Peak value. This is the minimum peak intensity that a spot must have to be identified. Again, image noise may be detected as spots if the Min Peak is too low.

Size Scale

The Size Scale field helps distinguish trends in signal intensity (spots) from random intensity fluctuations (noise). It uses the size of objects in the image (in pixels) to determine the nature of those objects. The initial size scale setting is based on the Small Spot you selected in Step 1.

If a gel image has noticeable noise features, a larger size scale may be preferable. If smoothing has adequately removed noise, small values are preferred so that small spots will be detected. You can enter odd values from 3 to 9 in the field or adjust the value using the arrow buttons.

Gaussian Model During Test

If the Gaussian Model During Test checkbox is selected, two-dimensional Gaussian fitting will be performed on all spots when you find spot centers. It will take longer to detect spots; however, spot detection will be more accurate for overlapping spots.

Note: Fewer spots may be found if Gaussian Model During Test is selected, because large spots that would be identified as overlapping spot clusters can be correctly resolved with Gaussian modeling.

4.1.c Optional Controls

Most of the background removal and smoothing settings under Optional Controls are set automatically based on your selections in Step 1. You should try to Find Spot Centers first before adjusting these controls.

Note: If your image has streaks, you should select streak removal under the Streaks tab when finding the spot centers.

Background

Click on the Background tab to access the settings for subtracting background intensity from the image.

Background subtraction is required for spot detection. The default method, Floating Ball, and default ball radius size are highly recommended. Radius size is calculated automatically based on the largest spot you want to detect (selected in Step 1).

If Floating Ball results in too much background subtraction or undesirable image artifacts, you can select the Roller Ball method. Typically, the roller ball method results in more image artifacts than the floating ball.

The Bi-Disk method is slightly faster than the floating or roller balls, and will perform adequately for images with very smooth, even backgrounds. However, bi-disk will perform less well for images with uneven backgrounds.

Speckles

Click on the Speckles tab to access the settings for removing random image noise.

Image smoothing in your scans is optional but highly recommended. The default state of the Perform Smoothing checkbox is checked. To deactivate this function, deselect the checkbox by clicking in it.

The Filter Type is initially set based on the section of the image you boxed in Step 1. The Kernel Size is set based on the faintest/smallest spot you want to detect.

Note: You should first attempt to find spot centers using these default settings before attempting to adjust them.

If you know the type of noise in your images, you can select a different filter type and kernel size. Section 3.14, Filtering Images, provides additional guidance on filter selection.

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- **Median.** This filter is useful for suppressing salt and pepper noise. It has little effect on Gaussian noise. This filter produces very little blurring if a small-sized window is selected.
- **Weighted Mean.** This filter is useful for reducing Gaussian noise. It has little effect on salt and pepper noise.
- **Power Mean.** This filter is useful for suppressing salt and Gaussian noise within an image (it worsens the effect of pepper noise).
- **ContraMean.** This filter is useful for suppressing pepper and Gaussian noise within an image (it worsens the effect of salt).

To adjust the Kernel Size, magnify a background region of your image using the Zoom Box tool so that you can see the individual pixels. The filter size you select should be larger than the average noise feature but smaller than the smallest spots.

Streaks

Protein gels frequently display streaking in areas of high spot concentration. Click on the Streaks tab to access the streak removal settings.

To remove vertical and/or horizontal streaks from your image, select the appropriate checkbox(es). The disk radius size is determined automatically based on the size of the largest spot you want to detect (Step 1).

The default disk radius size is conservative. If, after you find spot centers, you would like to remove more streaking from the image, reduce the size of each disk radius by about half and find spot centers again.

Fix Faint Spots

Fix Faint Spots is recommended for more accurate quantitation of faint spots. This corrects spots below a certain intensity over background (e.g., 0.2 OD) by making their X and Y sigma values the same as those of the spots in the same Y-coordinate region of the film. This overcomes the difficulty of obtaining an accurate fit when very few pixels are available.

4.1.d Saving/Loading Parameters

After you have selected the best detection parameters for the image, save them by clicking on the Save button (next to Parameter Set below the image window). Enter an appropriate name for the set (e.g., the type of stain used in the gel) in the pop-up box. Click on OK. The parameter set will now be available for processing all similarly detected gel scans.

To load previously saved parameters, click on the Load button and select from the list of parameter sets. When the parameters are loaded, you can proceed to Find Spot Centers on the loaded image. (Each parameter set is saved as an “.adp” file in the Discovery Series <user name>.prm directory. To delete a parameter set, delete its .adp file.)

4.1.e Processing Gels

Click on the Process All Gels button to close the Spot Detection Wizard and open the Detect Spots dialog box.

If you have not saved the current detection parameters, you will be prompted to do so before you exit the Wizard.

The Quit button will close the Wizard without opening the Auto-Detect Spots form. If you have not found spot centers, any unsaved parameters will be lost.

4.2 Detecting Spots on All Gels

After you have created and saved a set of detection parameters using the Spot Detection Wizard, you are ready to detect spots on your images. In general, you can use the same parameters to detect spots on all gels that use the same stain and have been run and scanned in the same manner.

Click on Process All Gels in the Wizard to open the Detect Spots dialog box. You can also open the dialog directly by selecting Detect Spots on All Gels from the Spots menu.

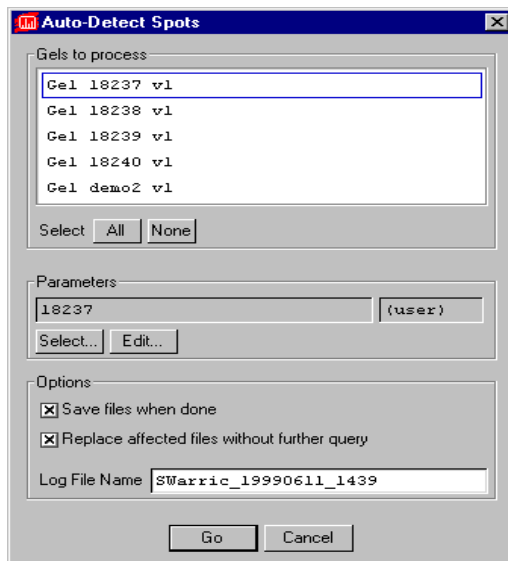


Fig. 4-5. Auto-Detect Spots control panel.

All currently loaded gel scans will be displayed in the panel.

Note: Only the root name and version number of each gel scan will be displayed. Calstrip scans and multiple exposures will not be listed separately.

Select the scan or scans you want to process by clicking on them individually, or clicking on the All or None buttons.

Select the appropriate set of detection parameters by clicking on the Select button and selecting from the pop-up list of saved parameters. The parameters you select will be displayed in the Parameters field.

Note: If you clicked on Process All Gels in the Spot Detection Wizard, the detection parameters selected in Wizard will be displayed in the Parameters field.

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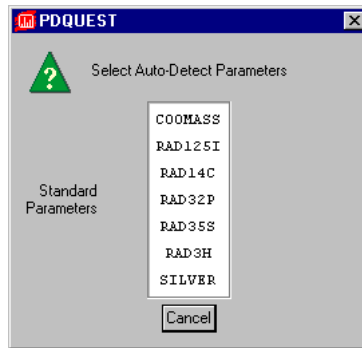


Fig. 4-6. List of parameters for spot detection.

Click on Edit to return to the Spot Detection Wizard and change the parameters.

To automatically save the results of image processing, select the "Save files when done" checkbox.

If these gel scans have been processed before, specify whether you want to replace the existing files by selecting or deselecting the "Replace affected files without further query" checkbox.

You can review the results of spot detection in a log file that is created when you process your gel(s). The Log File Name field displays a default name for the log file. Accept this name or enter a new name.

Click on the Go button to proceed with spot detection. A status screen will display the progress of spot detection.

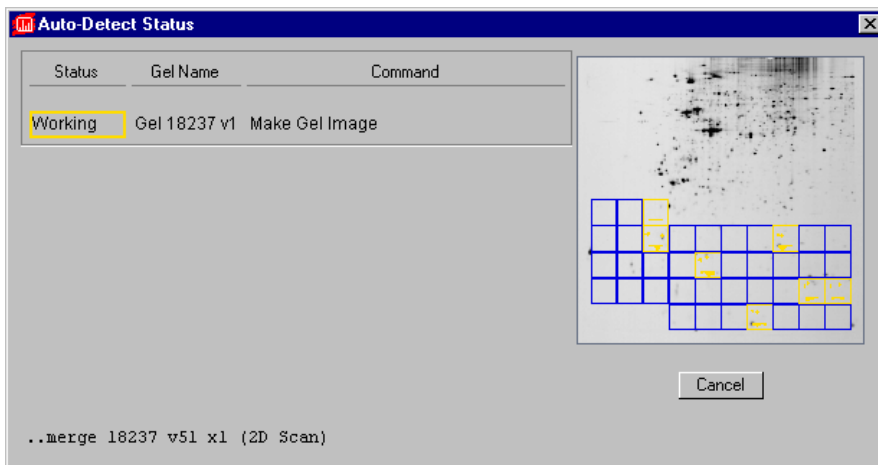


Fig. 4-7. Displaying the progress of spot detection.

Continuously updated messages display information about the individual steps as they are performed. A preview image shows a graphical depiction of each step as it occurs.

Click on Cancel to interrupt spot detection before it is finished. Any steps already performed will be lost.

The result of spot detection is a Filtered image and a Gaussian image for each gel scan you have processed (described in the following section). These images are used in subsequent spot analysis.

When detection is complete, a pop-up box will ask if you want to proceed with creating a matchset (see Chapter 5). You can create a matchset immediately, or review the results of spot detection and create a matchset later.

If you did not select "Save files when done" in the Detect Spots dialog, save the spot detection results by clicking on the Save button on the main toolbar.

Spot Detection Report

To review the log file created during spot detection, select Spot Detection Report from the Spots menu. This report documents when spot detection began, which gels were detected (or failed to detect), which steps were performed, and when spot detection was completed.

4.3 Filtered and Gaussian Images

When spots are detected in PDQUEST, the original gel scan is filtered and smoothed to clarify the spots, then three-dimensional Gaussian spots are created from the clarified spots. The end result is three separate images: the original 2-D scan, which remains unchanged; the Filtered image, which is a copy of the original scan that has been filtered and processed; and the Gaussian image, which is a synthetic image containing the Gaussian spots.

All quantitation and other analysis are performed on the Gaussian image. PDQUEST preserves all three images so that at any point you can go back and reexamine the original scan and the Filtered image to compare the original spots with the synthetic spots in the Gaussian image.

Note: Any spots not detected in the Filtered image will not be included in the Gaussian image. You can add spots to the Gaussian image during spot editing, as described in following sections.

A 2-D Scan, Filtered image, and Gaussian image of the same gel are part of the same scanset; they all share the same root file name with different extensions. Therefore, performing spot detection on an original 2-D scan named Proteins.gsc would result in a Filtered image named Proteins.gim and a Gaussian image named Proteins.gsp.

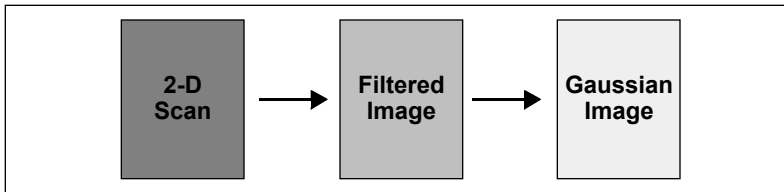


Fig. 4-8. Images created during spot detection.

After spot detection, the 2-D scan, Filtered image, and Gaussian image will be displayed in subwindows of the same window. The file name and type of each image are displayed at the bottom of each subwindow.

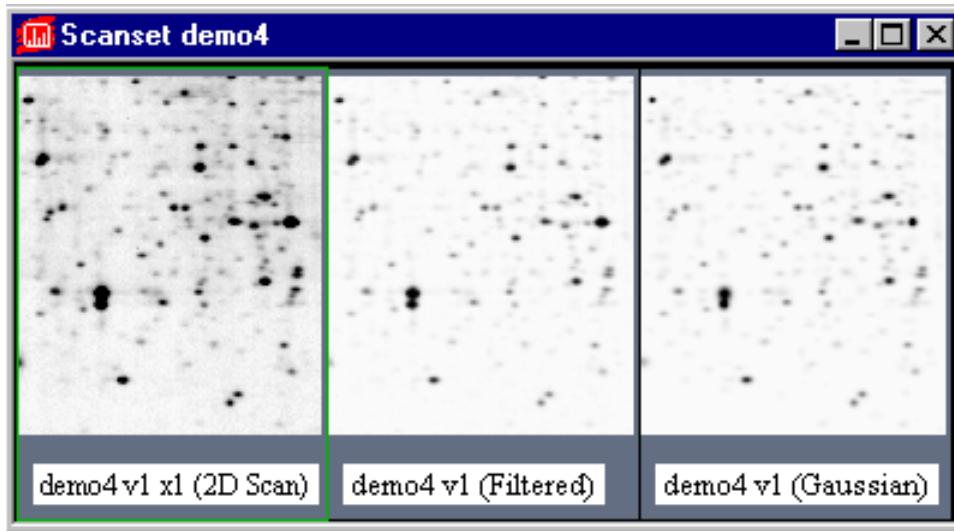


Fig. 4-9. Example of a scanset.

Note: Use the Assign tool and Configure Subwindows command on the main toolbar to arrange images in subwindows. See Chapter 3 for more information about viewing commands.

What Are Gaussian Spots?

Fuzzy, streaked, or overlapping spots in a 2-D gel can be difficult to accurately quantify. Because the image profile of an ideal spot conforms to a Gaussian curve, PDQUEST uses Gaussian modeling to create “ideal” spots that can be easily identified and quantitated.

A Gaussian spot is a precise three-dimensional representation of an original scanned spot. Gaussian curves are fitted to the scanned spot in the X and Y dimensions, and then additional modeling is performed to create the final Gaussian spot.

Using Gaussian modeling, you can accurately quantitate overlapping spots, spots in gel streaks, and multiple spots in clusters.

Note that spots created using the spot boundary tools (section 4.6) are not Gaussian modeled, even though they appear in the Gaussian image.

4.4 Spot Crosshairs and Ellipses

Spot crosshairs mark the centers of spots in the Gaussian image, while spot ellipses show their shapes. The tools for displaying spot crosshairs and ellipses are located on the Spots menu.

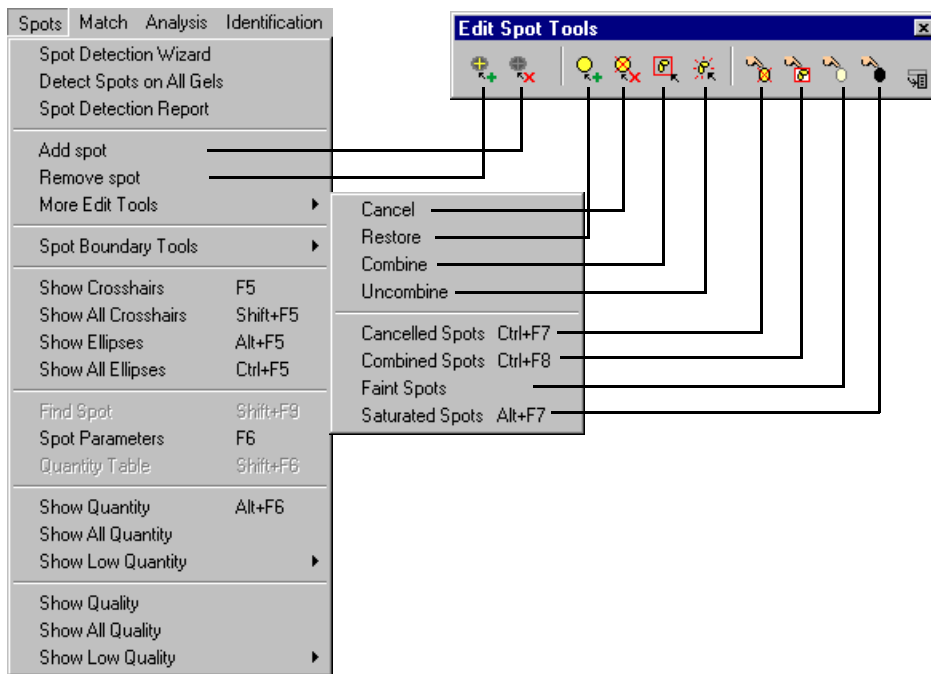


Fig. 4-10. Spots menu and Edit Spot toolbar.

Note that, in a scanset, the Gaussian image must be loaded to display crosshairs and ellipses. With the Gaussian image loaded, you can also display the crosshairs and ellipses in the Filtered image and 2-D Scan in the same scanset.

Show Crosshairs (F5, Spots menu) is used to display the crosshairs in a subwindow. Use the tool or the key command in the subwindow to toggle the display. (In a scanset, the Gaussian image must be loaded.) To display crosshairs in all subwindows, select Show All Crosshairs (Shift+F5, Spots menu).

Note: Spots without crosshairs in the Filtered and 2-D Scan images have not been detected and are not included in the Gaussian image.

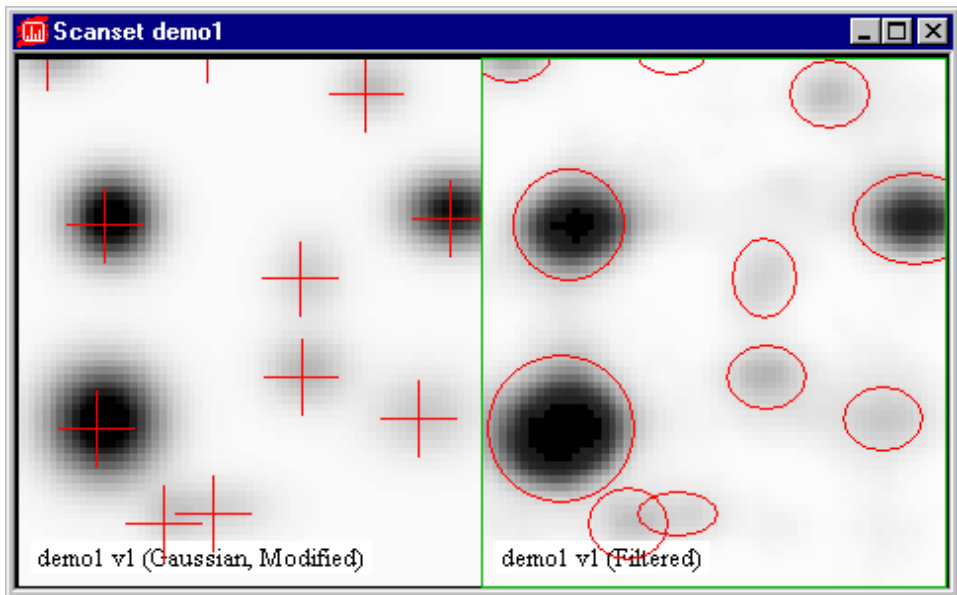


Fig. 4-11. Examples of crosshairs and ellipses in a scanset.

Show Ellipses (Alt+F5, Spots menu) shows the shape of each spot generated by Gaussian fitting. Use the tool or the key command in the subwindow to toggle the display. Show All Ellipses (Ctrl+F5, Spots menu) shows the spot ellipses in all the subwindows.

Hide Overlays (F4, main toolbar) will hide the crosshairs and ellipses on the image. Use the tool or the key command in the subwindow to hide the overlays.

4.5 Adding and Removing Spots

If spot detection has missed some spots or identified too many spots in a cluster, you can add and remove spots in the Gaussian image.

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Note: You can add and remove spots in the matchset and/or the scanset. Note that, after a matchset has been created, subsequent changes in the scanset are not automatically transferred to the matchset, and visa versa.

Make sure that both the Filtered image and Gaussian image are loaded before adding and removing spots. In a matchset, use Interchange Images (F11, main toolbar) to toggle between the two images.

The tools for adding and removing spots are located on the Spots menu and Edit Spots toolbar.

Adding Spots

If a spot in the Filtered image has not been added to the Gaussian image, select Add Spot from the Spots menu or Edit Spots toolbar, then click on the spot in the Filtered image. The spot will be Gaussian modeled and added to the Gaussian image.

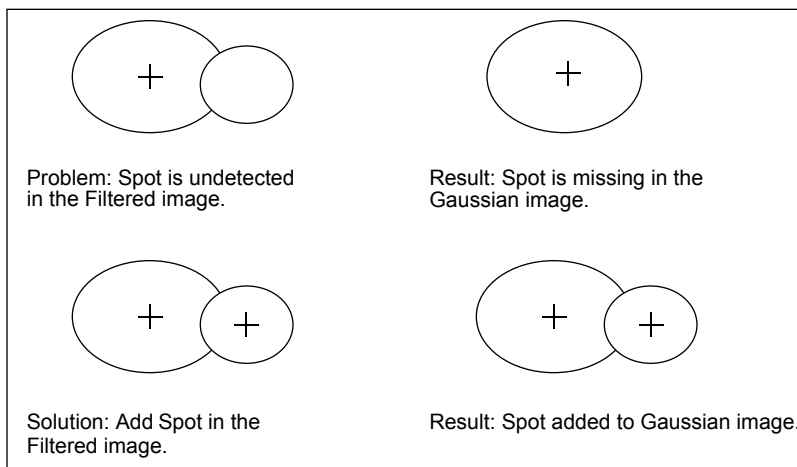


Fig. 4-12. Adding a spot.

Note: Use Plot Cross-section (Ctrl+t, View > View Density submenu) to aid in accurately positioning new spots.

Note that the spot boundary tools (section 4.6) can be used to add non-Gaussian modeled spots to the Gaussian image.

Removing Spots

To remove incorrectly positioned or misidentified spots from the Gaussian image, select Remove Spot from the Spots menu or toolbar, then click on individual spots or drag a box around a group of spots in the Filtered image. The spot(s) will disappear from the Gaussian image and the spot crosshair(s) will disappear from the Filtered image.

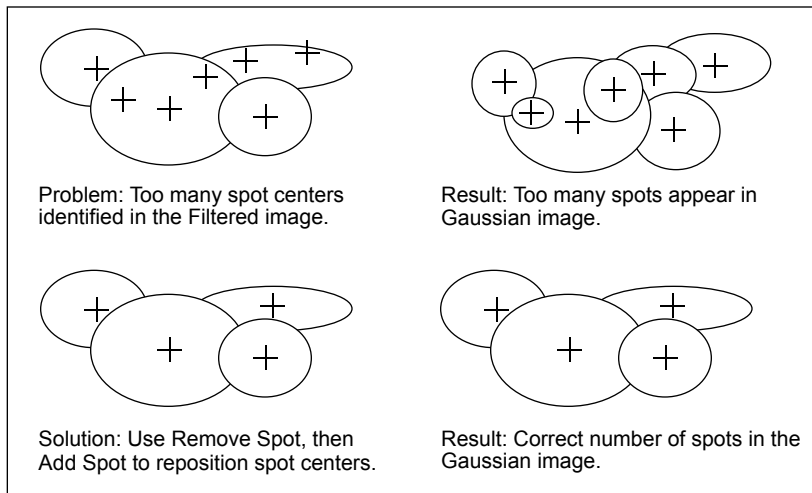


Fig. 4-13. Removing incorrectly identified spots in a cluster.

4.6 Spot Boundary Tools

If a spot is saturated, irregularly shaped, or otherwise of poor quality, then Gaussian modeling will be unable to accurately determine quantity. In these cases, you can manually define the spot in the Filtered image using the spot boundary tools.

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Note: For most spots, Gaussian modeling results in more accurate quantitation than defining the spot boundary.

When you define a spot boundary, the region inside the boundary is identified as the spot and is added to the Gaussian image. Spot quantity is calculated by summing the intensities of the pixels inside the defined boundary. You can define the boundary of any spot in a Filtered image.

Spots defined using the boundary tools are not Gaussian modeled, even though they appear in the Gaussian image. However, their quantities can be compared with regular, Gaussian-modeled spots.

Note: Creating a spot boundary will overwrite the results of spot detection and Gaussian modeling. However, regular spot detection will not overwrite any spots defined using the boundary tools.

You can open the Spot Boundary Tools toolbar from the main menu, or select commands from the Spots > Spot Boundary Tools submenu.

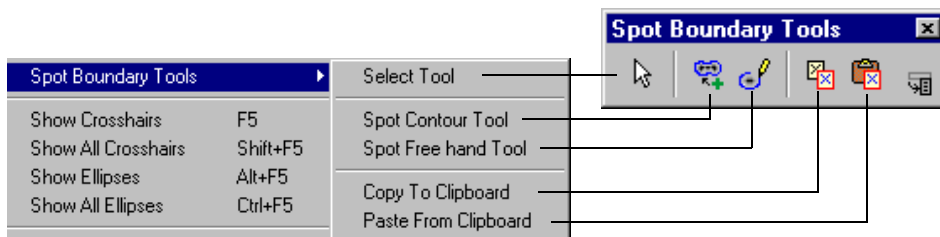


Fig. 4-14. Spot boundary tools.

Spot Contour Tool

To use this tool, first magnify the spot you want to quantify in the Filtered image, then select Spot Contour Tool from the submenu or toolbar.

Click and drag the cursor from the inside of the spot outward to change the shape of the contour as you move over pixels of different intensity; note that the contour is created by enclosing all pixels whose intensity is greater than or equal to the intensity of the pixel at the cursor.

Detecting and Editing Spots

When you release the mouse button, the boundary is created. The contour should completely surround the spot you want to quantify.

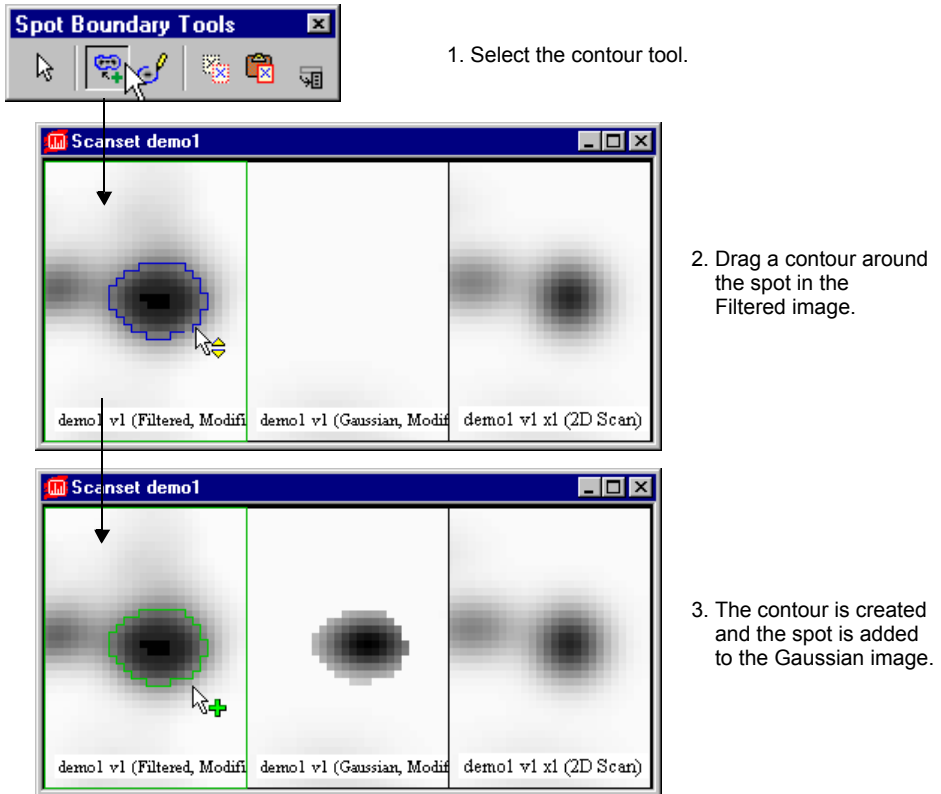


Fig. 4-15. Creating a spot contour.

To edit the boundary, position your cursor on the border with the contour tool. Your cursor will change to a pencil. Drag across the line; a new white line will appear. When you recross the old line, a new contour will be created.

Spot Freehand Tool

To use this tool, first magnify the spot you want to quantify (you must be able to see the individual pixels in the image). Then select the Spot Freehand Tool from the toolbar or submenu, and click and drag around the edge of the spot to enclose it in a boundary. When your line crosses itself, a freehand boundary is created.

If you make a mistake while drawing, simply backtrack with the mouse. The line you draw should completely surround the spot you want to quantify.

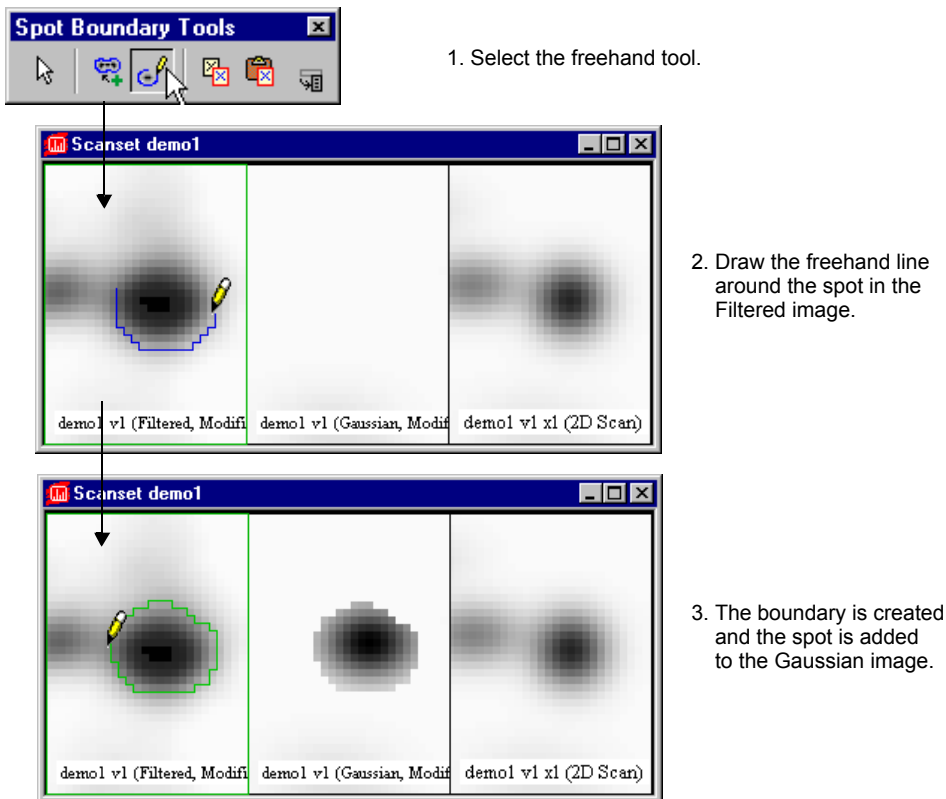


Fig. 4-16. Creating a freehand boundary.

To edit the boundary, position your cursor on the border and drag across the line with the freehand tool; a new white line will appear. When you recross the old line, a new boundary will be created.

Deleting and Moving Spot Boundaries

To remove a spot boundary, you can use the Remove Spot tool, or you can click on the spot with the Contour, Freehand, or Select tool to select it (the boundary will change to green) and press the Delete key on your keyboard. The spot will also be deleted in the Gaussian image.

To reposition a spot boundary, make sure the Contour, Freehand, or Select tool is active, then position your cursor in the middle of the spot boundary. The cursor will change to a multidirectional arrow; simply drag the boundary to a new location. The spot will also appear relocated in the Gaussian image.

Copying Spot Boundaries Between Images

To copy a spot boundary between Filtered images, first select it (the boundary will change to green), then select Copy to Clipboard from the submenu or toolbar. Open or select the image you want to copy to, and select Paste from Clipboard from the submenu or toolbar. The spot boundary will appear in the same relative position in the new image. The boundary will also appear in the corresponding Gaussian image.

Note that the two images should have the same pixel size and approximate dimensions. You may have to reposition the pasted boundary as described above.

4.7 Cancelling Spots

Cancelling a spot in a matchset or scanset preserves its location but does not include it in matching or any other analysis. This is different than removing a spot from the Gaussian image entirely.

Select Cancel from the Spots > More Edit Tools submenu or toolbar and click on the spot that you want to cancel. It will disappear from the Gaussian image (it will remain in the Filtered image, but will not be included in any analysis).

To highlight all cancelled spots in the Gaussian image, select Cancelled Spots (Ctrl+F7) from the Spots > More Edit Tools submenu or Edit Spot toolbar.

To restore a cancelled spot, select Restore from the Spots > More Edit Tools submenu or toolbar and click on the position of the cancelled spot. The spot will reappear.

To cancel all the spots that have not been matched in a matchset, select Cancel Unmatched Spots from the Match > Edit Matches submenu and click in the matchset gel.

To restore all unmatched cancelled spots in a matchset, select Restore Unmatched Spots from the Match > Edit Matches submenu and click in the matchset gel.

4.8 Combining Spots

Combining spots combines the quantitation of a cluster of spots into one spot. This can be performed on Filtered images in a matchset or scanset.

Select Combine from the Spots > More Edit Tools submenu or Edit Spots toolbar, and then draw a box around the cluster of spots that you want to combine in the Filtered image. After you draw the box, click on the point in the box where you want the combined spot to be placed. The cluster will appear as one spot in the Gaussian image; it will still appear as individual spots in the Filtered image, but the quantitation will be combined.

To resolve a combined spot into its original cluster, select Un-combine from the submenu or toolbar and draw a box around the combined spot in the Filtered or Gaussian image. The spot will be uncombined into its original multiple spots.

Combined Spots (Ctrl+F8, Spots > More Edit Tools submenu, Edit Spots toolbar) highlights all combined spots in the matchset.

4.9 Saturated and Faint Spots

Saturated Spots

Spot detection may not correctly fit spots in saturated regions of the image. Therefore, although saturated spots will appear in the Gaussian image, they are not quantified using Gaussian modeling (their quantity will be reported as zero). You should use the spot boundary tools (section 4.6) to quantify these spots.

To highlight spots that are saturated in the image, use the Saturated Spots command (Alt+F7, Spots > More Edit Tools submenu, Edit Spots toolbar) in any image window. The saturated spots will appear highlighted and the number of saturated spots will be listed at the bottom of the window.

You can also use the Highlight Saturated Pixels in the Transform window to identify regions of saturation in the image.

Faint Spots

Faint Spots on the Spots > More Edit Tools submenu and Edit Spot toolbar highlights those spots whose signal falls within the bottom 5 percent of the intensity range of the image. Select the command from the submenu or toolbar, then click in the image. The faint spots will be highlighted in the image and the number of faint spots will be listed at the bottom of the image.

4.10 Spot Quantity

Spot Quantity is the total intensity of a defined spot. This corresponds to the amount of protein in the actual spot in the gel (see section 1.2, Digital Data and Signal Intensity).

For spots defined using the boundary tools (section 4.6), this is simply the sum of the intensities of the pixels inside the boundary. For Gaussian spots, it is calculated during spot detection and Gaussian fitting.

The formula for calculating the quantity of Gaussian spots in PDQUEST is:

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Spot height * $\pi * \sigma_x * \sigma_y$ where:

Spot height (also known as peak value) is the peak of the Gaussian representation of the spot. It is measured in ODs or counts/image units², depending on the type of imaging device used.

Note: One image unit (IU) = 100 micrometers or 0.1 millimeter. This scaling factor is used in to make the numbers in the formula more manageable.

σ_x is the standard deviation of the Gaussian distribution of the spot in the direction of the x axis, and σ_y is the standard deviation in the direction of the y axis. These are measured in IUs.

Note that this formula results in units of OD * IU² for images measured in ODs, and units of counts for images measured in counts.

Note: This formula results in more accurate quantitation of imaged spots than summing the intensities of the pixels in the spot. Measuring the area under the Gaussian model compensates for peak intensity values that approach image saturation, as well as background intensity that submerges the intensity of a spot at its edges.

To display the quantities of the spots in a single image, use Show Quantity (Alt+F6, Spots menu) in the image window. To display the quantities of all the spots in all the subwindows of a selected window, select Show All Quantity from the Spots menu.

Low Quantity Spots

You can define a set of spots that do not meet a threshold quantity level.

From the Spots > Low Quantity Spots submenu, select Define Low Quan. Set, then click in the matchset or scanset Gaussian image. In the pop-up box, enter a threshold quantity value. All spots below this threshold will be flagged as low quantity.

When you click on OK, a pop-up box will list the number of low quantity spots in the image and their total quantity. The total number of spots in the image and their total quantity will also be listed.

Detecting and Editing Spots

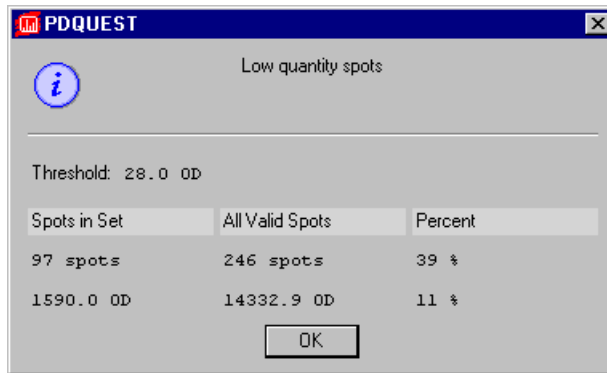


Fig. 4-17. Pop-up box displaying low quantity spot information.

The low quantity spots will appear highlighted in the image.

To re-highlight the low quantity spots in the set, select Spots > Low Quantity Spots > Mark Low Quan. Spots and click in the image.

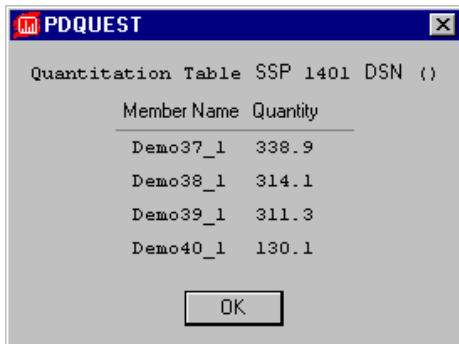
To cancel all low quantity spots in the set, select Spots > Low Quantity Spots > Cancel Low Quan. Spots and click in the matchset or scanset Gaussian image.

To restore the cancelled low quantity spots, select Spots > Low Quantity Spots > Restore Low Quan. Spots and click in the matchset or scanset Gaussian image.

Quantity Table

You can display the quantity of a spot in each matchset gel (or replicate group) using the Quantity Table function.

Select Quantity Table from the Spots menu, then click on the spot whose quantities in different gels you want to compare. Alternatively, position the cursor on the spot and click Shift+F6. A pop-up box will display the quantities of that spot in table format.

A screenshot of a PDQUEST software window titled "PDQUEST". Inside the window, there is a table titled "Quantitation Table SSP 1401 DSN {}". The table has two columns: "Member Name" and "Quantity". It lists four rows of data: Demo37_1 with quantity 338.9, Demo38_1 with 314.1, Demo39_1 with 311.3, and Demo40_1 with 130.1. At the bottom of the window is an "OK" button.

Member Name	Quantity
Demo37_1	338.9
Demo38_1	314.1
Demo39_1	311.3
Demo40_1	130.1

Fig. 4-18. Quantity table.

If Replicate Group Quantitation is turned on (section 7.5), the average quantity of that spot in each replicate group will be displayed, as will the coefficient of variance (CV) and the number of gels in the replicate group (Count).

4.11 Spot Quality

Spot quality is a number ranging from 0 to 100 that is calculated based on five attributes of each spot:

- Gaussian fit. PDQUEST determines how well the spot fits the Gaussian model and assigns a value based on this fit.
- X streaking. The software determines how much of the spot is affected by gel streaking in the x direction.
- Y streaking. The software determines how much of the spot is affected by gel streaking in the y direction.
- Overlap. The software determines the percentage of the spot that overlaps with another spot or spots.
- Linear range of scanner. PDQUEST determines whether the peak intensity value of the spot is within the linear range of the scanner.

Detecting and Editing Spots

Each spot attribute is evaluated and weighted to produce a single numerical value for Spot Quality. If a spot fits the Gaussian model perfectly, has no streaking in the x or y direction, does not overlap with any other spot, and has a peak intensity within the linear range of the scanner, PDQUEST will assign a value of 100 to the spot.

To display the quality of all the spots in a single image, select Show Quality from the Spots menu and click on the image. To display the quality of all the spots in all the subwindows of a selected window, select Spots > Show All Quality.

Note: Spots of low quality may be more accurately quantitated using the spot boundary tools (section 4.6). Spots created using the boundary tools are not assigned a quality value.

Low Quality Spots

You can identify a set of spots that do not meet a threshold quality level. You can then flag those spots as low quality and/or redefine them using the spot boundary tools (section 4.6).

From the Spots > Low Quality Spots submenu, select Define Low Qual. Set, then click in the matchset or scanset Gaussian image. In the pop-up box, enter a threshold quality value. All spots below this threshold will be flagged as low quality.

When you click on OK, the low quality spots will appear highlighted in the image, and the number of low quality spots will be displayed at the bottom of the image window.

To re-highlight the low quality spots in the set, select Spots > Low Quality Spots > Mark Low Qual. Spots and click in the image.

To cancel all low quality spots in the set, select Spots > Low Quality Spots > Cancel Low Qual. Spots and click in the matchset or scanset Gaussian image.

To restore the cancelled low quality spots, select Spots > Low Quality Spots > Restore Low Qual. Spots and click in the matchset or scanset Gaussian image.

4.12 Spot Parameters

The Spot Parameters information box provides a variety of information about an individual spot, including its quantity, quality, peak value, location in the gel, Mr pI data (if any), and the sigmas of its Gaussian curve.

Select Spot Parameters from the Spots menu, then click on a spot in a Gaussian image to display the data for that spot. Alternatively, position your cursor on the spot and click on F6.

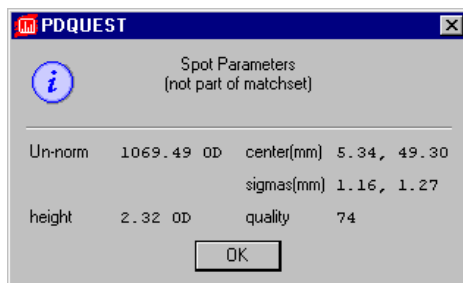


Fig. 4-19. Spot Parameters information box.

4.13 Finding Spots

You can find spots in a matchset by SSP number or DSN (see sections 7.7 and 7.8).

Select Find Spot from the Spots menu (Shift+F9), click in the matchset member or standard you want to search, and enter the SSP or DSN number in the pop-up box.

Click on OK to complete the search. The spot will appear highlighted and numbered on the image.

5. Matchsets

After the protein spots in your gels have been detected, you are ready to compare spots across gels. To do so, you must create a matchset.

A matchset is PDQUEST's mechanism for comparing and analyzing the spots in an experiment. It contains copies of the Gaussian and Filtered images of the gel(s) in an experiment.

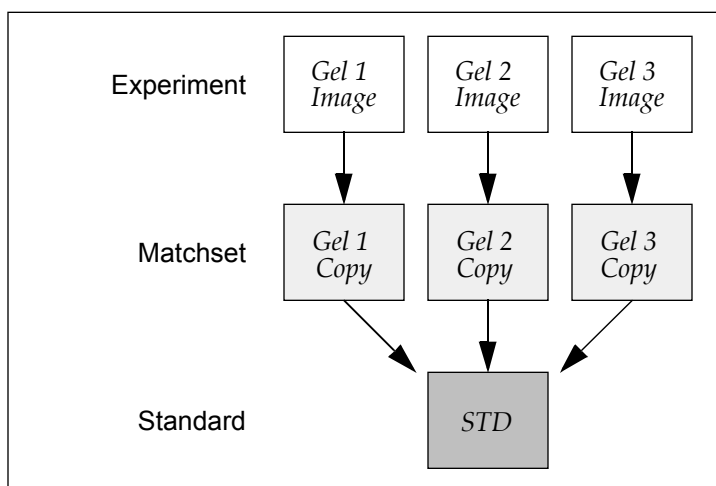


Fig. 5-1. How experiments are organized into matchsets.

A matchset can consist of one gel or many gels, depending on the type and size of the experiment. The matchset is displayed in a single window, with subwindows for the images of the member gels, and a subwindow for the matchset standard.

The **standard** is a synthetic image that contains the spot data from all the gels in the matchset. All spot comparisons and analysis are coordinated through the standard. Standards from different matchsets can be grouped into high-level matchsets for further comparison (see section 5.4).

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Using a matchset, you can make quantitative and qualitative spot comparisons across gels, calculate molecular weight/isoelectric point values, annotate spots, group them into analysis sets, combine the data from multiple gels into replicate groups, and select spots for excision and mass spec analysis.

You can also use a matchset to combine gels with different pH gradients into a larger cybergel.

Note: While matchsets can contain gels with different quantitation units, quantitative spot comparisons can only be made between gels that have the same units (i.e., spot quantities measured in O.D.s cannot be compared to spot quantities measured in counts). Qualitative comparisons are still valid.

The commands for creating and editing matchsets are located on the Match menu.

5.1 Creating a Matchset

A matchset is created from the Gaussian and Filtered images of the gel(s) in an experiment. You will be prompted to create a matchset automatically after spot detection is complete. If you have detected spots in all the gels in your experiment, select Yes to create the matchset immediately. If not, select No to create it later.

(To create a matchset later, select Create Matchset from the Match > Create/Edit Matchset submenu.)

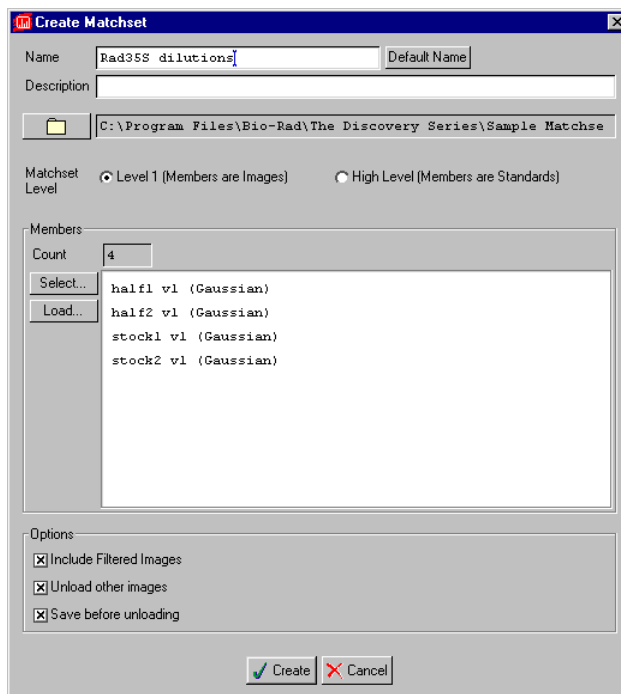


Fig. 5-2. Create Matchset dialog.

At the top of the Create Matchset dialog, enter a name for the matchset in the Name field. To assign your matchset a default name (the default prefix can be set under Edit > Preferences), click on the Default Name button.

You can enter a description of the matchset in the text box next to the Description prompt.

To select the directory in which the matchset will be saved, click on the Directory button, navigate to the appropriate directory, and click on the Select button.

Next, specify whether you are creating a Level 1 matchset or a High Level matchset. Level 1 matchsets are created from the Gaussian images of scanned gels. High-level matchsets are created from other matchsets (see section 5.4).

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If you are creating a level 1 matchset, select the Gaussian images to be included in the matchset

If the images are already open in PDQUEST, click on the button labeled Select. A list of currently open Gaussian images will be displayed. Select the images that you want include and click on Done. The images will be included in the Members list.

To include Gaussian images that are not currently open, click on the Load button to open the Add New Member dialog, navigate to the directory containing the images you want, and select the images.

Note: In the Add New Member dialog, other matchsets will be displayed as folders, from which you can also select Gaussian images; the images will be copied from those matchsets into the matchset you are creating.

Click on Add Selected to make your selection(s). The names of the files you selected will appear in the Members list.

Filtered images are included in the matchset by default. Including the Filtered images allows you to compare the Gaussian spots with the original spots while matching, and edit the results of spot detection. (Invariably, during matching you will notice missed or misdetected spots.) To not include Filtered images, deselect the Include Filtered Images checkbox.

Note: You should include Filtered images in the matchset, unless file size is a severe problem. If you do not include Filtered images, you should carefully review each Gaussian image before creating the matchset for missed or misdetected spots.

Any other open images will be closed before the matchset is created unless you deselect the Unload Other Images checkbox.

To save any open images before they are unloaded, select the Save Before Unloading checkbox.

After you have selected the matchset members and options, click on the Create button. You will be prompted to select the template for the matchset standard.

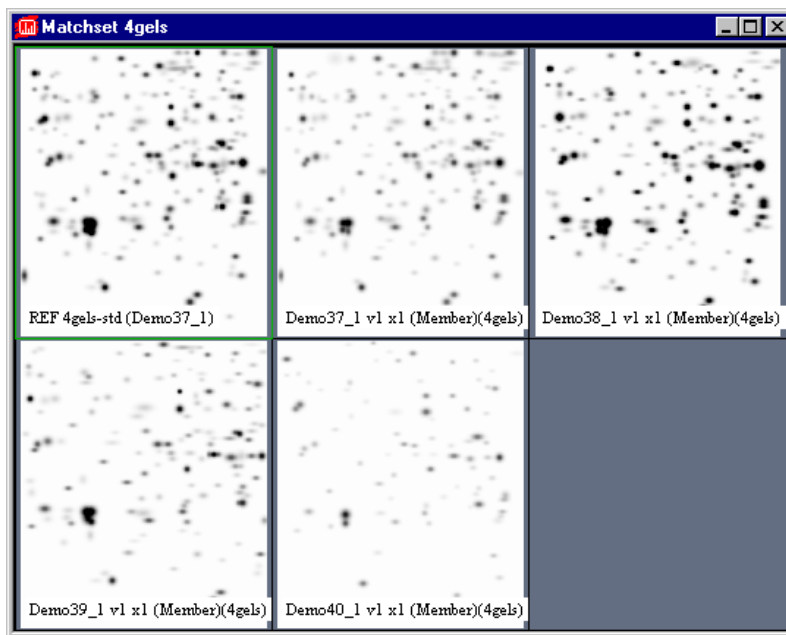


Fig. 5-3. Example of a matchset. The Standard is in the upper left subwindow.

5.2 Selecting the Standard Template

When you first create a matchset, you are prompted to select one of the images to serve as the template for the matchset standard.

The matchset standard is a synthetic image that contains the spot data from all the gels in the matchset. When you select a gel to serve as the standard template, a copy of that Gaussian image is created and designated the standard. You then match the spots in the remaining gels with those in the standard, and add the unique spots in those gels to the standard.

There are several possible criteria for selecting the template gel:

Number of Spots. To review the number of spots in each gel, display each Gaussian image and select Image Info (Ctrl+I). The info box will list the

number of spots in the image and the total data in those spots. Choosing the gel with the most spots as the template minimizes the number of spots that you will need to add from the remaining gels.

- **Representative Gel.** The matchset standard is a representation of all the data in your matchset and is often used in reports. Therefore, you may not want to use an atypical gel as a template, even if it has the most spots. In this case, we suggest that you choose the most representative gel in your biological system.
- **Control Gel.** Choose a control gel as the matchset standard so that it can be included in every experiment you plan to analyze and used as the standard in every matchset. This makes it easier to compare results between different experiments.
- **Spot Quality.** Editing the matchset standard and matching other gels to it are easier when the standard consists of high quality spots.
- **Cybergel.** If you are creating a cybergel standard from gels with different pH gradients, it may be easiest to add spots from each gel in sequence (i.e., use the gel with the lowest pH as the template).

When you first create a matchset, the Select Standard Member pop-up box will prompt you to select a template for the standard. Choose from the list of gels, or click on Cancel to create the matchset standard later.

If you elect to create a standard later, make sure the matchset is open, then select Make/Replace Std Image from the Match > Create/Edit Matchset submenu and select the template gel from the pop-up list.

With your standard created, you are ready to begin matching and editing spots in the matchset (see the following chapter).

5.3 Displaying and Editing Matchsets

Displaying Matchsets

A matchset is displayed in a single window, with the matchset members in the different subwindows. The matchset standard initially appears in the upper left subwindow and has the name *REF <matchset name>std.*

As with scansets, you can configure the subwindows in a matchset using the Configure Subwindows command (section 3.2), and assign images to different subwindows using the Assign command (section 3.3).

The Gaussian image and Filtered image (if included) of each member gel are both loaded in the same subwindow. To toggle between the two images, use Interchange Images (F11) or Interchange All Images (Shift+F11). The member name and type of image are listed at the bottom of each subwindow.

Editing Matchsets

The commands for editing the structure of a matchset are located on the Match > Create/ Edit Matchset submenu.

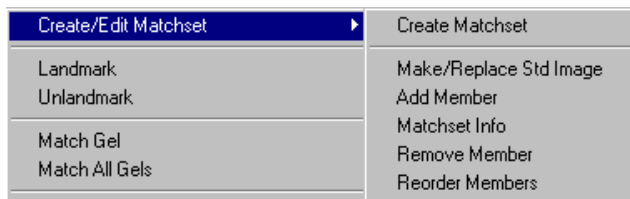


Fig. 5-4. Create/Edit Matchset submenu.

The commands for editing the structure of a matchset are located on the Match > Create/ Edit Matchset submenu.

Select Make/ Replace Std Image to add a standard to a matchset or replace the standard in a matchset. You will be prompted to select the standard template from a list of matchset gels.

Note: Any annotations, analysis sets, etc. in the matchset will be lost if you replace the standard template gel.

To add a member to a matchset, select Add Member. A version of the Create Matchset dialog box will open. Add a new gel to the list of gels in the matchset using the steps described in Creating a Matchset.

To remove a matchset member, select Remove Member and select the gel to be removed from the pop-up list. You will receive a warning that this command cannot be undone.

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Note: Any spots you have added to the matchset standard from the removed gel will remain in the standard.

Matchset Info opens a dialog box in which you can enter a description and comments about your matchset.

Reorder Matchset opens a dialog box in which you can select a member name and move it up or down in the matchset hierarchy. This hierarchy affects the order in which the matchset members are displayed on the screen (top to bottom, left to right). Most importantly, the order of the members determines the order of the bars in a histogram showing a spot's quantity in each member of a matchset.

5.4 High-Level Matchsets

PDQUEST allows you to create matchsets from matchsets, so you can build a database by comparing and summarizing the results of a series of experiments.

High-level matchsets are matchsets created from the standards of lower level matchsets. A lower level matchset might be a matchset created from your original gels (a "level 1" matchset), or it could be another high-level matchset. Many experimental protocols can be designed to take advantage of high-level matching.

For example, one experiment may find all the proteins that change during cell growth in culture, while another may identify nuclear proteins by cell fractionation. These two experiments can be analyzed individually in level 1 matchsets, each with their own standards. These two standards could then be combined into a high-level matchset with its own standard. Data from the two experiments can now be compared, and the high-level matchset might shed light on growth-related proteins found in the nucleus.

High-level matchsets are actively linked to the matchsets used to create them. If you edit spots in the member images of a level 1 matchset, those changes will take immediate effect in the high-level matchset.

For example, when working with a high-level matchset, you may notice that a spot is present in one member and not the other. You could open the level 1

matchset with the missing spot and see if that spot is present in any of the gels. If it is, you could add it to the standard of the level 1 matchset and it would appear in the corresponding member in the high-level matchset, ready to be matched.

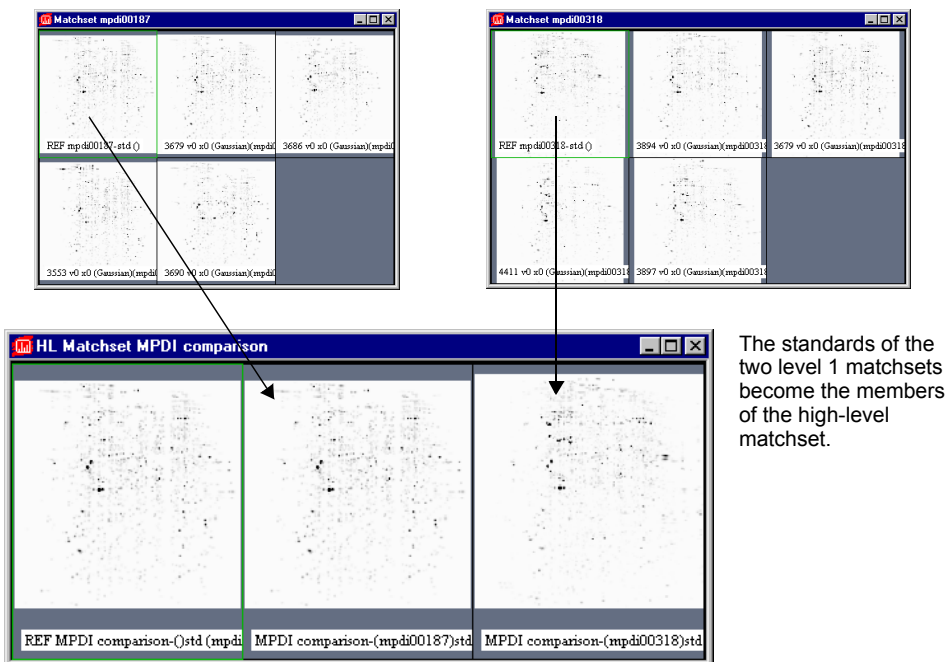


Fig. 5-5. Example of a high-level matchset.

Creating a High-Level Matchset

High-level matchsets are created in the same way as level 1 matchsets (see section 5.1), except:

1. You select the High Level option button instead of Level 1.
2. You select matchsets as members, instead of images.

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Note that you cannot include Filtered images in a high-level matchset, because high-level matchsets are made up of standards, which do not have Filtered images.

Because high-level matchsets are linked to the lower-level matchsets they are created from, they should not be moved from the file directory in which they are created. If you want to create a high-level matchset to be shared among a group of users, you should move the lower-level matchsets to a shared file server and create it there.

Note: If a high-level matchset and/or its constituent matchsets have been moved from their original location, you can edit the high-level matchset file to make it openable. First open the matchset folder using Windows Explorer or Macintosh Finder, then open the **descript** file (a text file) inside the matchset. In the descript file, the path links to the lower-level matchsets are shown. You can change these paths to the new location to make the high-level matchset openable.

To create a high-level matchset, go to the Create/Edit Matchset submenu of the Match menu and select Create Matchset.

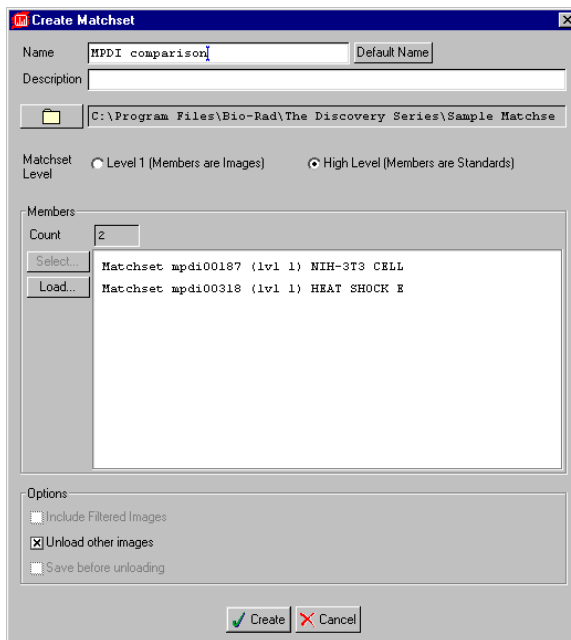


Fig. 5-6. Creating a high-level matchset.

To select the matchsets for inclusion, click on the Load button. This will open the Select New Member dialog.

Select the matchsets you want to include by clicking on them, then click on the Add Selected button. The names of the matchsets will appear in the Members list.

Note: The matchsets you select must be writable (i.e., not locked or read-only). See your Macintosh or Windows documentation for instructions on making files writable.

When you have made your selections, click on the Create button.

You will prompted to select a template to serve as the standard for the high-level matchset. This is similar to selecting the standard template in a level 1 matchset (see section 5.2). As with level 1 matchsets, you can create the

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standard later using Make/Replace Std Image on the Match > Create/Edit Matchset submenu.

Note that when you select the template for the standard, you will be prompted to copy the SSP numbers from that template into the high-level standard. Click on Yes to copy the numbers, or No to assign new numbers. You can also transfer SSP numbers and/or DSNs from the matchset member of your choice later.

The high-level matchset is displayed in a single window with subwindows, just like a level 1 matchset.

You can add members to, remove members from, and reorder members in high-level matchsets just like level 1 matchsets. These commands are described in Displaying and Editing Matchsets.

Matching and Editing High-Level Matchsets

You landmark and match spots in a high-level matchset just as you do in a level 1 matchset. See the following chapter for details.

6. Matching Spots

After you have created a matchset and selected the standard template, you are ready to begin matching spots and adding unique spots in the gels to the standard.

6.1 Landmarking Spots

Before you can begin matching spots in the matchset, you must landmark key spots in all the gels. Landmarks are reference spots used to align and position the gels for matching.

When choosing spots to landmark, you should look for spots that are well-resolved and present in all members of the matchset (except when creating a standard out of gels with different pH gradients; see below). Avoid spots in clusters, in case the spot detection mechanism has misidentified the peaks in the cluster. Unless you are creating a standard out of gels with different pH gradients, you should landmark spots in all corners of the gels, as well as the middle of the gels.

To begin landmarking, first magnify a region of the matchset gels and identify a well-resolved spot that appears in all the gels. Study the region under different levels of magnification to make sure you have identified the same spot in all the gels.

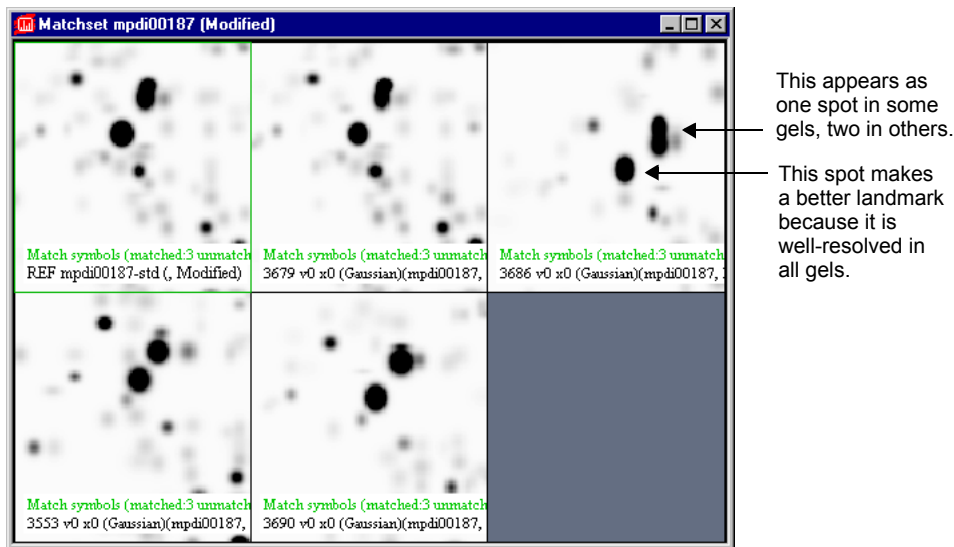


Fig. 6-1. Selecting a spot to landmark.

Select Landmark from the Match menu, Match toolbar, or right-click menu, and click on the spot in the matchset standard.

The cursor will jump to the same area in the first member gel. If the landmarking program can identify the same spot in that member, it will appear marked with a box. Click on it to place the landmark. Landmarked spots are marked with green triangles.

Matching Spots

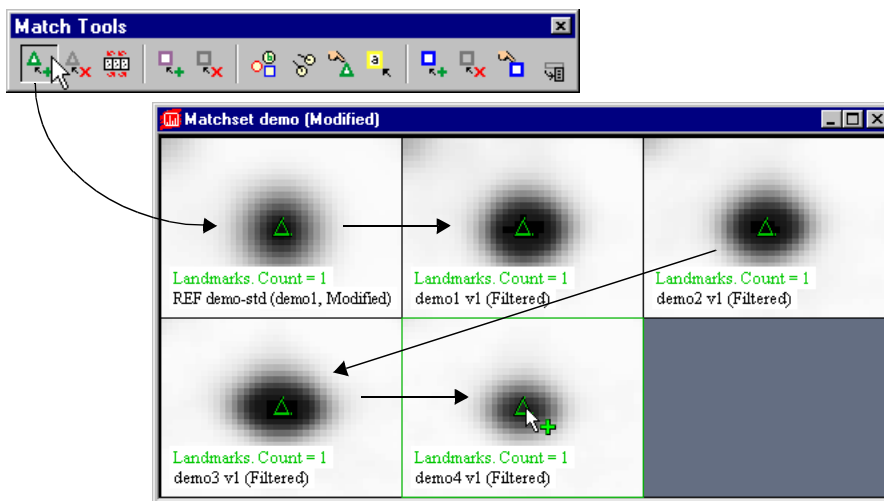


Fig. 6-2. Landmarking a spot.

If the program is unsure about positioning the landmark, the area in which the spot was expected will be circled. Locate the appropriate spot and click on it.

As each landmark is placed, the cursor will jump to the next gel. Landmark the spot in all members of the matchset (except when creating a standard out of gels with different pH gradients; see below).

If you accidentally landmark the wrong spot in a gel, simply go back and click on the correct spot. The landmark will move.

To remove a landmark from all the matchset gels, select Unlandmark from the Match menu or toolbar and click on the incorrect landmark in the standard. To remove a landmark from a single gel, click in that gel.

To highlight all the landmarks in a matchset, use the Mark Landmarks command (F8, Match menu, Match toolbar) in the standard image.

As you add landmarks, study the auto-matching in all regions of the gels. Add more landmarks to regions where auto-matching has missed spots.

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The number of landmarks required will vary depending on the number and quality of the gels in the matchset. Gels with well-resolved spots and little distortion will require fewer landmarks.

Landmarking Gels with Different pH Gradients

If you are combining gels with different pH gradients into the same standard, you will not be able to landmark the same spot in all the gels. For example, to create a cybergel from three gels with different pH gradients, you would place landmarks in the overlapping region of the first two gels, then add the spots from the unique region of the second gel to the standard. Then you would landmark the overlapping region of the last two gels.

6.2 Matching

In a matchset, spot data from the member gels are included in the standard via matching. All the significant spots in your matchset should be matched with spots in the standard.

Using landmarks, PDQUEST will automatically match most of the spots in the matchset. You then review the remaining unmatched spots in the member gels and either manually match them with existing spots in the standard, cancel them, or add them as new spots to the standard.

Automatic Matching

By default, auto-matching occurs immediately after each landmark is placed in all the matchset members. To turn off this feature, deselect Auto-match After Landmark in Application Preferences under Edit > Preferences.

Note: Most matchset gels require at least two landmarks before auto-matching will occur. (The gel that was used as the template for the standard may be auto-matched after only one landmark is placed).

To auto-match all the gels in the matchset, select Match All Gels from the Match menu or toolbar. To auto-match a single gel with the standard, select Match Gel from the Match menu and click in the gel subwindow.

Matching Spots

Each matched spot is marked with a green letter in the standard and the matched gels. Unmatched spots in the member gels are marked with red circles. Auto-matching should successfully match most of the spots in the matchset.

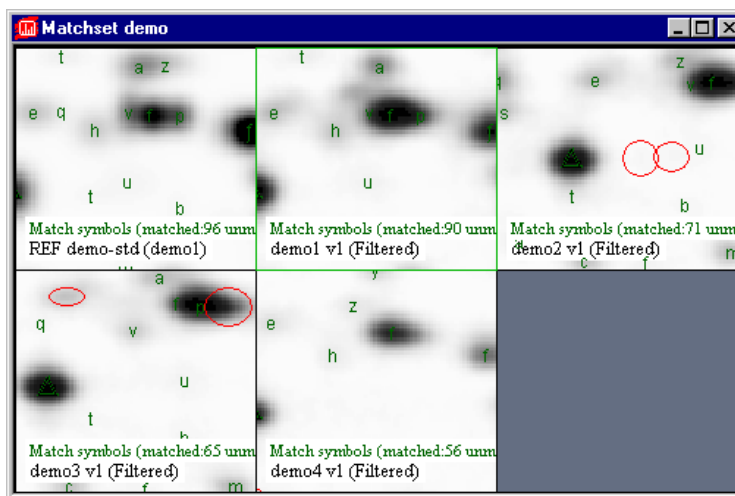


Fig. 6-3. Matched spots are marked with letters; unmatched spots are circled.

If multiple unmatched spots appear in certain areas of the gels, review these areas using the Show Offsets command, and add more landmarks to facilitate auto-matching.

Manual Matching

After you have landmarked numerous spots and reviewed the results of auto-matching, you are ready to identify the significant spots that could not be auto-matched. To highlight these spots in a member gel, use the Mark Unmatched Spots command (Shift+F8, Match > Edit Matches submenu) in the gel's subwindow. Unmatched spots are marked with red circles. Use the Zoom Box tool to study each unmatched spot in detail in each gel. (See also Reviewing Matches.)

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If a spot was missed during spot detection, you must add it to the Gaussian image, then match. If a spot was not included in the standard, you must add it to the standard. If an unmatched spot is not significant, you can cancel it.

Otherwise, if auto-matching has missed a spot that should have been matched, you can landmark it to improve the alignment and modeling of the standard, or you can manually match it.

Note: Manual matching will not change the modeling of the standard, and therefore will not facilitate additional matching.

Select Manual Match from the Match toolbar or Match > Edit Matches submenu and click on the spot in the standard, then click on the same spot in all the member images where that spots appears. The spot will be marked with a purple box to indicate that it has been manually matched.

To unmatch a manual match, select Manual Unmatch from the toolbar or submenu and click on the manually matched spot in a member image.

Removing Matches

To remove all automatic matches from a member image, select Remove Auto Matches from the Match > Edit Matches submenu and click on the member image. Manual matches and landmarks will not be removed. Because this command is irreversible, you will be prompted to complete the operation.

To remove all matches (automatic and manual), as well as landmarks, from a member image, select Remove All Matches from the Match > Edit Matches submenu and click on the member image. Because this command is irreversible, you will be prompted to complete the operation.

6.3 Adding Spots to the Standard

Recall that all spots initially included in the matchset standard come from the template gel used to create the standard. After you have matched these spots in all the gels, you are ready to add the unique spots from the other gels to the standard. These may be spots induced by your experimental protocols, or they may be spots that fall outside the pH gradient of the current standard.

Auto-Adding Spots

To automatically add the unmatched spots in a matchset member to the standard, select Auto-Add Spots to Standard from the Match menu.

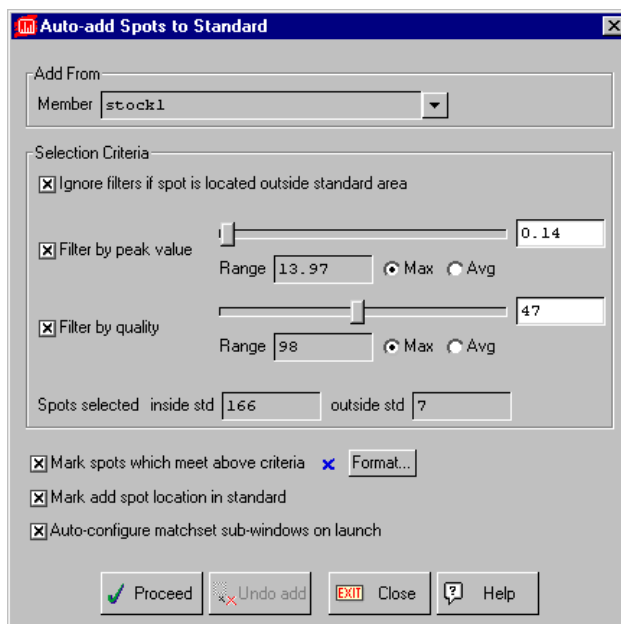


Fig. 6-4. Auto-add Spots to Standard dialog.

In the dialog box, select the matchset member from the Member pulldown list. The matchset will reconfigure to display the member and standard in adjacent subwindows, with the unmatched spots selected and their corresponding locations in the standard highlighted.

If you are adding spots from a gel with a different PH gradient than the current standard, you can select the Ignore Filters If Spot Is Located Outside Standard Area checkbox. Then, the standard will be expanded and all unmatched spots that fall outside the area of the standard will be added regardless of quantity or quality.

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You can add all the unmatched spots or only those spots that meet a threshold peak value and/or quality value. To set a peak value threshold, select Filter by Peak Value and drag the slider to the left, or enter a threshold value directly in the box to the left of the slider. As you adjust the threshold, spots that no longer meet that threshold will be unselected.

The filter range can be set by selecting the Max option button (up to the maximum peak value in the gel) or the Avg button (up to the average peak value).

The Filter By Quality setting is similar to Filter by Peak Value, except that it filters by spot quality.

The number of Selected Spots appears below the filter settings. Inside std shows the number of spots that fall within the area of the current standard image, while Outside Std shows the number of spots that fall outside that area.

When you are ready to add spots from the gel, click on the Proceed button. The selected spots will be added and the gel will be rematched with the standard. If some spots could not be added (e.g., because they were too close to existing spots), this will be noted in a pop-up box. You can add these manually.

To remove all spots added from the selected gel, click on Undo Add. You will be warned before the action is completed.

Manually Adding Spots

To manually add spots from a member gel to the standard, first display the unmatched spots in the gel using Show Symbols (F7) or Mark Unmatched Spots (Shift+F8). Use the Zoom Box tool to magnify a spot.

Select Add Spot to Standard from the Match menu or toolbar, and click on the center of the unmatched spot in the member. The cursor will jump to the same area in the standard (calculated from the surrounding matches). When you click in the standard, the spot will appear and a match will automatically be made.

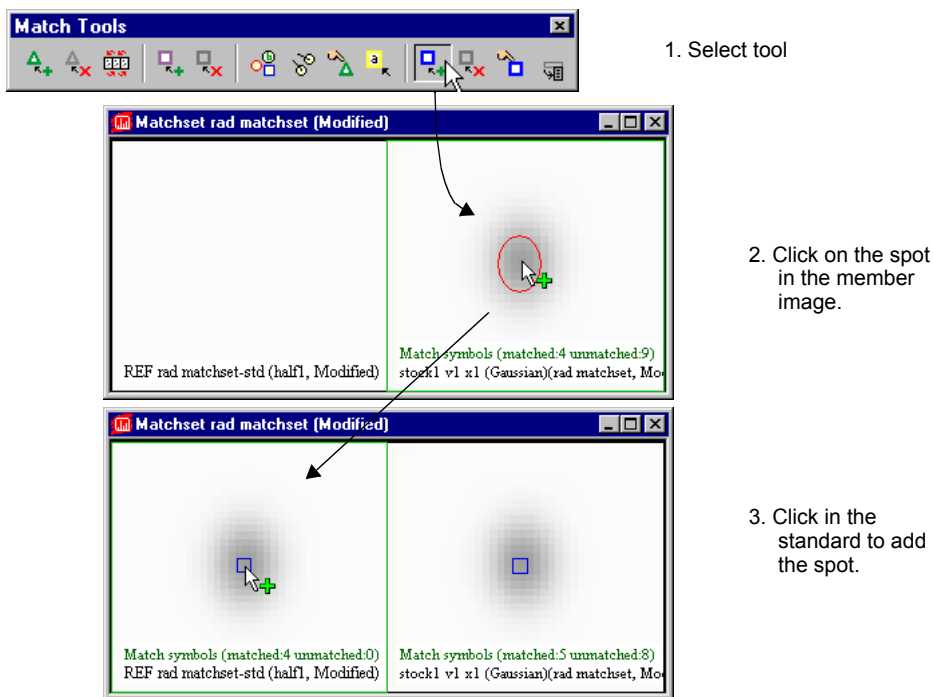


Fig. 6-5. Adding a spot to the standard.

To remove an added spot, select Remove Added Spot from the menu or toolbar and click on the spot in the standard. The spot will be removed and unmatched.

Mark Added Spots (Alt+F8) on the menu or toolbar highlights the spots that you have added to the standard. Select the command and click in the subwindow. Added spots are marked by blue squares.

6.4 Reviewing Matches

PDQUEST include several tools for reviewing matched and unmatched spots.

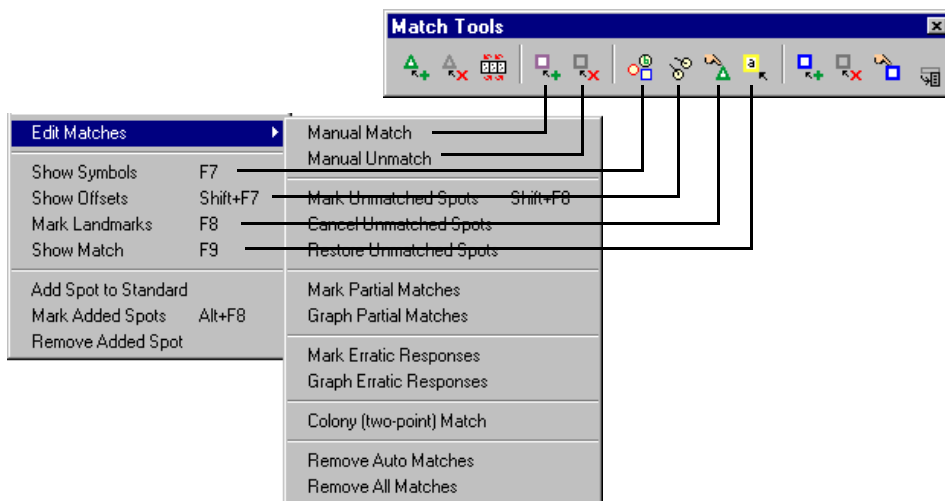


Fig. 6-6. Edit Matches commands.

Displaying Matched and Unmatched Spots

Show Symbols highlights all the different types of matches and unmatched spots using different labels. Select the tool from the Match menu or toolbar and click in the standard or member gel subwindow, or use the F7 key command in the appropriate subwindow.

- Auto-matched spots are labeled with green letters.
- Manually matched spots are labeled with purple boxes.
- Unmatched spots are circled in red. These are also displayed using the Mark Unmatched Spots command (Shift+F8) on the Match >Edit Matches submenu.
- Landmarks are labeled with green triangles. These are also displayed using the Mark Landmarks command (F8) on the Match menu and toolbar.
- Spots added to the standard are labeled with blue boxes. These are also displayed using the Mark Added Spots command (Alt+F8) on the Match menu and toolbar.

To highlight individual matched spots, select Show Match from the Match menu or toolbar and click on a spot in the standard, or use the F9 key command on the spot. The spot will appear highlighted with a yellow box in all the gels in which it has been matched. In gels where it is unmatched, a circle will appear where the spot would be located.

Vector Offsets

Vector offsets are lines showing the distance and direction between the spot centers in the member gels and the corresponding spot centers in the standard. They provide a visual tool for judging gel alignment and distortion and the accuracy of matching.

Select Show Offsets from the Match menu or toolbar and click in a subwindow to display the offset lines, or use the Shift+F7 key command in the subwindow.

Look for the following conditions that indicate possible matching problems:

- Offset lines that cross each other (this is a key problem).
- Grossly nonparallel lines in local regions.
- Lines of unequal length in local regions.
- Areas of the gel with few offsets; this indicates unmatched regions.

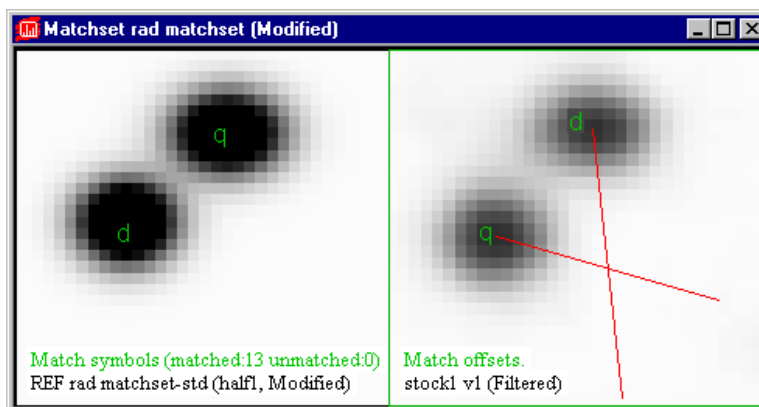


Fig. 6-7. Vector offsets indicate that these two spots have been mismatched.

Partial Matches

Partial matches are spots in the matchset standard that are not matched in one or more member images. You should review these spots to see if they were undetected, mismatched, cancelled, or simply not present.

Select Mark Partial Matches from the Match > Edit Matches submenu and click in the standard. The partial matches will appear circled in the standard.

Graph Partial Matches displays histograms of each partially matched spot in the standard. Select the command from the submenu and click in the standard. A missing bar in the histogram indicates that the spot is unmatched in a matchset gel.

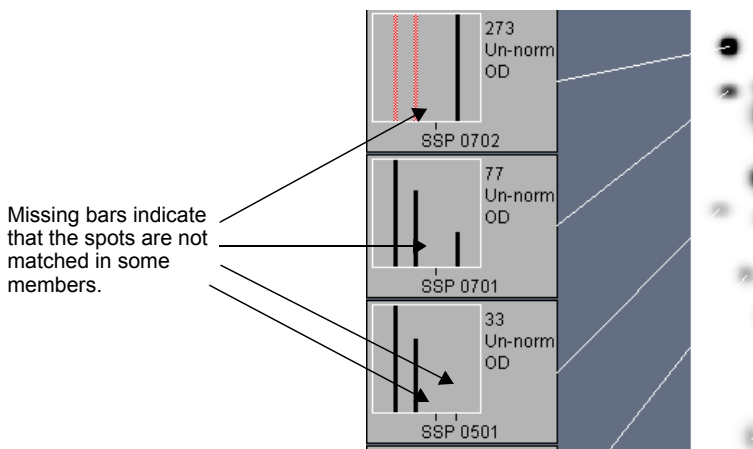


Fig. 6-8. Graph Partial Matches; each bar represents the spot's quantity in a gel.

Note: Histograms can only be displayed for one region of an image at a time. Use the Graph Advance (Alt+.) and Graph Retreat (Alt+,) commands to shift between regions.

Erratic Spots

Erratic spots are matched spots whose quantities vary significantly between member gels. To mark these, select Mark Erratic Responses from the Match > Edit Matches submenu and click in the standard. Enter a fold change factor in

the pop-up box. (For example, a change factor of 3 will highlight spots whose quantity varies by a factor of three or more among member images.) When you click on Apply, the selected spots will appear circled in the standard

Graph Erratic Responses displays histograms of each erratic spot in the standard. Select the command from the submenu, click in the standard, enter a fold change factor in the pop-up box, and click on Apply to display the graphs.

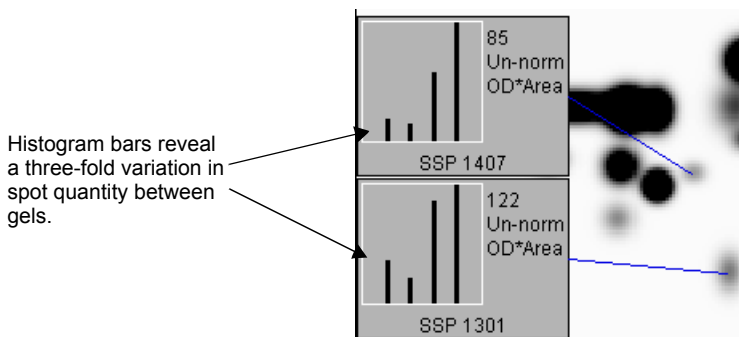


Fig. 6-9. Graph Erratic Responses; each bar represents the spot's quantity in a gel.

Note: Histograms can only be displayed for one region of an image at a time. Use the Graph Advance (Alt+.) and Graph Retreat (Alt+,) commands to shift between regions.

6.5 Colony Matching

If you have a matchset containing colony plate images, you can match colonies quickly and easily using the Colony (two-point) Match command on the Match > Edit Matches submenu.

Unlike the regular matching function, you only need to define two landmarks on your member images to use this command.

Note: Also unlike the regular matching function, this command will automatically correct for any rotational errors in the imaged plates

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(common in colony plates). You do not need to perform additional image rotation.

After you have created a matchset and defined two landmarks in the member images (see above), select Colony (two-point) Match from the submenu and click on a member image to automatically match the colonies. The other matching commands also work in colony plates.

7. Analysis Tools

The tools on the Analysis menu are designed to help you identify and compare spots of interest in a matchset. Note that analysis sets and annotations are covered in the following chapter.

7.1 Spot Review Tool

The Spot Review Tool displays histograms of all the spots in a matchset. The bars in each histogram represent the spot's quantity in each member of the matchset. If replicate group quantitation is selected, the bars represent the spot's quantity in each group. (See section 12.1 for a more complete description of histograms.)

This tool is interactive, so you can use the graphs to identify spots in the matchset for matching and editing or create an analysis set out of selected spots.

With a matchset open, select Spot Review Tool from the Analysis menu or Graph toolbar. The review tool will open and the matchset will be reconfigured alongside it. This automatic window configuration can be set under the report options (see below).

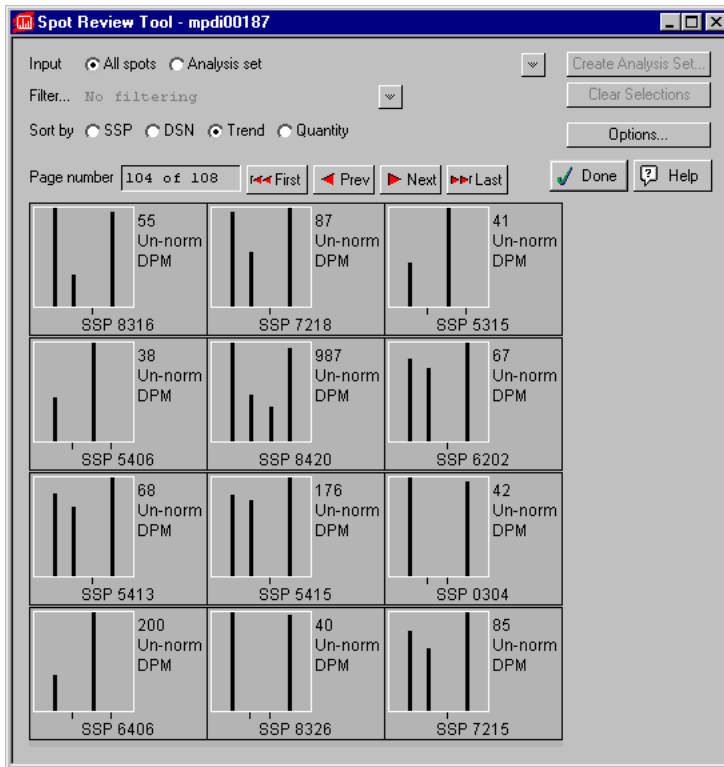


Fig. 7-1. Spot Review Tool.

At the top of the review tool window, select All Spots to display histograms of all the spots in the matchset or Analysis Set to display histograms of only the spots in a specified set. If you select the analysis set option, a pop-up list of available analysis sets will be displayed. Select the set to display from the list.

You can reorder the graphs by clicking on the Sort By buttons. You can sort by SSP number (default) or DSN (if defined). If you select Trend, the graphs will be arranged based on a linear regression model of spot presence and quantity in each member. If you select Quantity, the graphs will be arranged starting with spots of greatest quantity to spots of least quantity.

To scroll through the pages of histograms, click on the arrow buttons above the graphs.

If you click on a histogram of a spot, that spot will be magnified and highlighted in all the matchset gels (see options below). You can then review the spot detection and matching using the other tools in PDQUEST.

If you click on a histogram, the histogram itself will be highlighted by a yellow border, indicating that it is selected. If you click on other histograms, they will be selected as well.

Note: To highlight and magnify spots in the matchset without selecting the histograms, shift-click on each histogram.

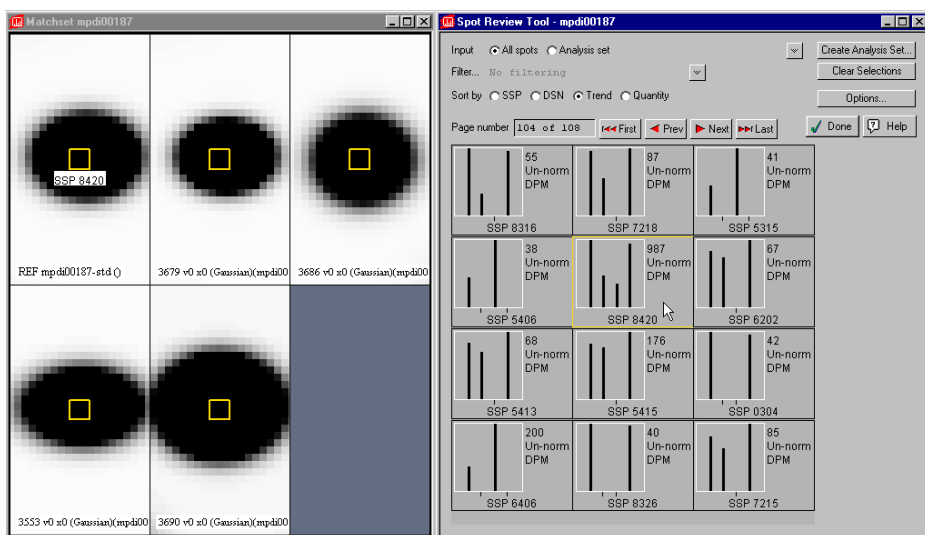


Fig. 7-2. Reviewing spots using the Review Tool.

You can create an arbitrary analysis set (section 8.1) out of spots whose histograms are selected. First, select the spots by clicking on the histograms. If you make a mistake, click on a selected graph to deselect it or click on the Clear Selections button to deselect all the graphs.

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After selecting all the graphs of interest, click on the Create Analysis Set button and enter a name for the set in the pop-up box. Click on Create to create the set. You can then select the set in the review tool as described above.

Click on the Options button to change the display options of the report. When you click on a spot graph, that spot will be highlighted (Highlight spot on images), centered in the subwindow (Center images on selected spot), and magnified (Auto-zoom images) by default. Also by default, the review tool window and matchset window will be displayed side-by-side when you open the tool (Configure windows on launch). Deselect these options to turn off the particular functionality.

To close the review tool window, click on Done.

7.2 Image Stack Tool

The Image Stack Tool allows you to scroll through the images in a scanset or matchset layered on top of one another. Using this tool, you can easily compare spots that appear, disappear, or change size in gels run under different experimental conditions.

Note: Your images should be close to the same size with spots in the same relative positions to use this tool.

With all your matchset or scanset images open, select Image Stack Tool from the Analysis menu. The stack tool will open.

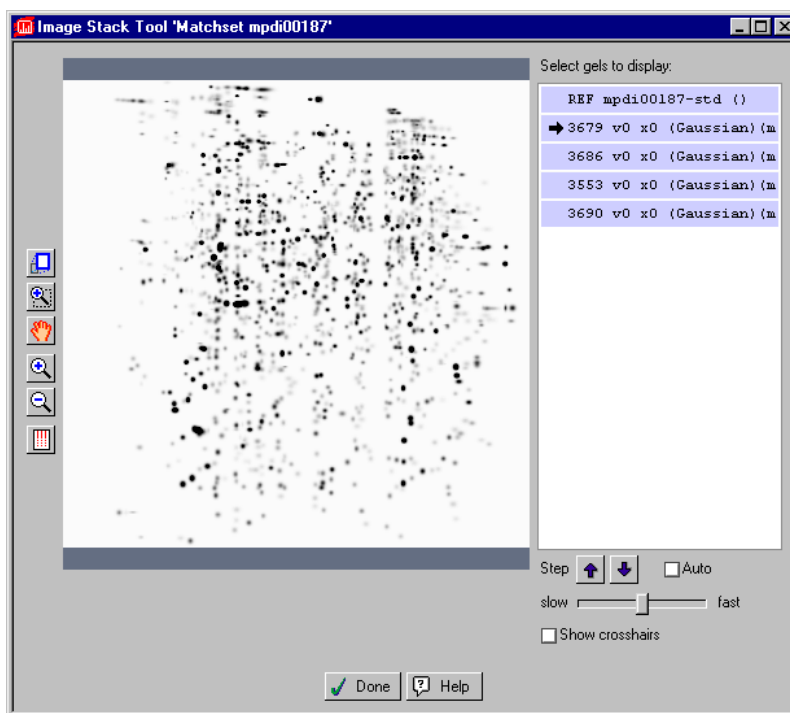


Fig. 7-3. Image Stack Tool.

In the stack tool window, all available gels will be listed in the field to the right of the display window. To select an image to display, click on a gel name. The name will appear highlighted with an arrow and the image will appear in the window.

Click on another gel name to display that image.

Using the controls below the list of names, you can "step" through the images in the stacker. First, select some or all of the gel names using standard shift-click or control-click key commands. When multiple names are selected, the Step arrow buttons will become active. Click on the arrow buttons to scroll through the list of selected gels.

Alternatively, click on the Auto checkbox next to the arrow buttons to begin automatically scrolling through the list. You can adjust the auto-scroll speed using the Slow-Fast slider. To stop auto-scrolling, deselect the checkbox.

To display the spot crosshairs on the images in a matchset, select the Show Crosshairs checkbox.

Buttons for various viewing tools are included in the stack tool. These commands will change the display of all the images in the stacker at once (e.g., zooming in on one image will magnify the same relative area in all the images).

7.3 Scatter Plot Tool

The Scatter Plot Tool shows the relatedness of two gels in a matchset. It can also be used to print scatter plots for all the gels in a matchset, or a selected analysis set in a matchset.

With a matchset open, select Scatter Plot Tool from the Analysis menu. When the scatter plot window opens, the first two gels in the matchset are selected by default. Select different gels to compare from the pull-down list buttons next to X-Axis Gel and Y-Axis Gel.

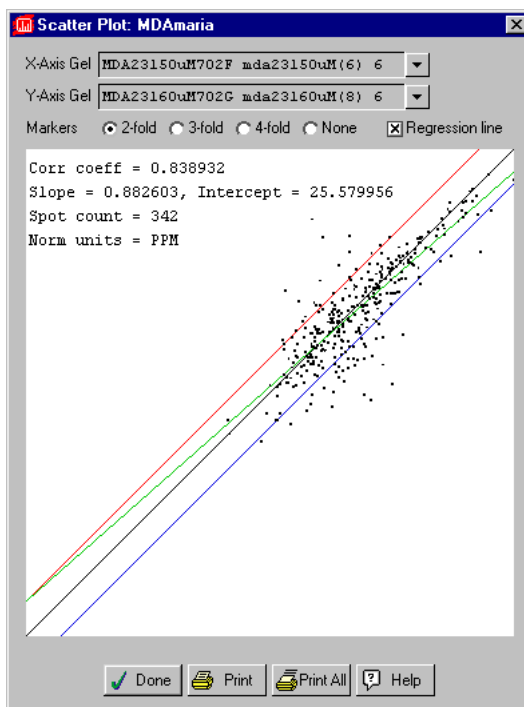


Fig. 7-4. Scatter Plot Tool.

The quantity of each spot in the first gel (X-axis) is plotted on a log scale against its quantity in the second gel (Y-axis). If the spot's quantity is the same in both gels, its point will fall on the black center line in the graph (slope = 1.00). If the quantities are not the same, the point will fall above or below the center line.

The regression line generated from the plot is shown in green. This is displayed when the Regression Line checkbox is selected.

The correlation coefficient for the regression line is displayed on the graph. A coefficient of 1.00 indicates that the two gels are perfectly similar, while a low coefficient value (e.g., 0.40) indicates that the two gels are not very similar. The number of spots that appear in both gels is also listed on the graph.

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If you select the 2-fold, 3-fold, or 4-fold option button, parallel red and blue lines will appear on the graph. All spots between these lines have quantities in the two gels that fall within the selected fold-factor range.

If you click on a point in the graph, that spot will be highlighted and magnified in the actual matchset.

To print the currently displayed scatter plot, click on the Print button in the window.

To print scatter plots comparing every gel in the matchset with every other gel, click on the Print All button. In the pop-up dialog, you can enter a report title, researcher name, and the name of the research institution. Next, specify whether you want scatter plots of All Matched Spots or only the spots in a Specified Analysis Set. If you select the analysis set option, a pop-up box will list the available analysis sets in the matchset. Select a set and click on OK.

Click on the Go button to print. The plots will be printed in a report format.

7.4 Mr pI Data

In a matchset, you can enter the molecular weight and isoelectric point values for your known protein spots. With this data, PDQUEST can calculate Mr pI values for all the spots in the matchset.

The functions needed to enter and display Mr pI data are found in the Analysis > MrpI Standards submenu and the MrpI Tools toolbar.

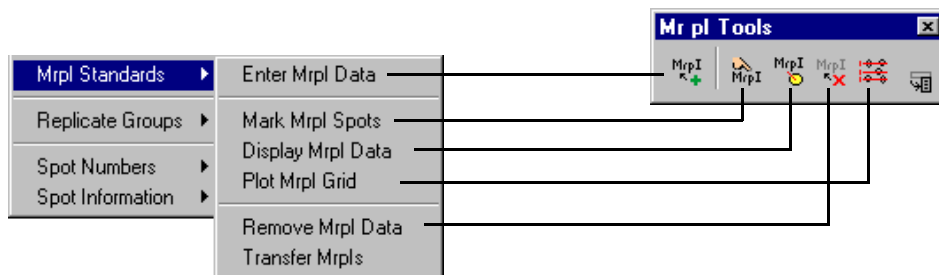


Fig. 7-5. MrpI commands.

Note: PDQUEST needs to know the pH gradient of the gel in order to calculate isoelectric point values. To enter the correct orientation, go to the Preferences dialog box (Edit > Preferences) and click on either Acid-Base or Base-Acid next to the pH Gradient prompt.

Entering Mr pI Values for Known Spots

To determine Mr pI values for all the spots in the matchset, you first need to enter the values for a few known spots.

Select Enter MrpI Data from the Analysis > MrpI Standards submenu or toolbar, then click on a spot in the matchset standard whose molecular weight and/or isoelectric point values are known. A dialog box will ask you to enter the values for that spot.

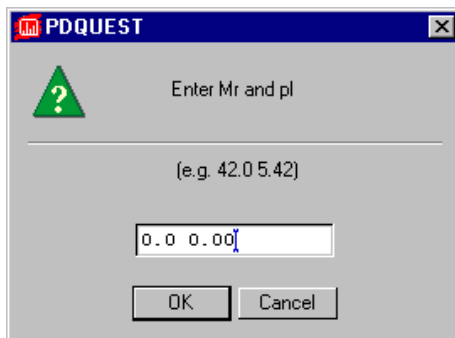


Fig. 7-6. Pop-up box in which Mr pI data are entered.

Type the molecular weight first, followed by a space, followed by the isoelectric point. Click OK when you are done.

If you make a mistake entering the data, you can select Enter MrpI Data again and click on the spot. The values you entered will appear in the dialog box and you can make the necessary corrections.

If you want to remove the MrpI data for a spot, select Remove MrpI Data and click on the spot.

Displaying Mr pI Data

To see the proteins for which you have entered Mr pI data, you can use either Mark MrpI Spots or Display MrpI Data. The first will highlight the spots for which you entered data. The second will display Mr pI values next to the spots.

For a graphical display of the overall trend of Mr pI values across the entire gel, first make sure the standard subwindow is active, then select Plot MrpI Grid from the Database > MrpI Standards submenu or toolbar.

A grid of Mr pI axes will appear over the standard. You can use the grid to verify that the values that you entered make sense; you can also use it to estimate the Mr pI values of proteins of interest.

To see the exact Mr pI values of a spot, select Spot Parameters from the Spots menu or toolbar and click on the spot, or use the F6 key command on the spot. In the pop-up box, the Mr pI values will be displayed.

Note: If the isoelectric point value of the spot falls outside the highest and lowest known values, the pI value will not be calculated. Empirical results have shown that extrapolated pI values are unreliable, due to gel distortions and other factors.

Transferring MrpI Data in a High-Level Matchset

Transfer MrpIs allows you to apply the molecular weight and isoelectric point values from one high-level matchset member to the other members. This makes it easier to correlate data from the member matchsets. (See section 5.4 for more information on high-level matchsets.)

With your high-level matchset open, select Transfer MrpIs from the Analysis > MrpI Standards submenu. In the first pop-up box, select the matchset member you want to copy from; in the second box, select the member you want to copy to. You will be asked to confirm the transfer.

Repeat the procedure to apply the values to all the high-level matchset members.

7.5 Replicate Groups

Since most experiments are carried out in duplicate, data analysis can be performed on each gel individually or on sets of duplicate gels grouped together. The Replicate Groups function allows you to group sets of duplicate gels and determine the average quantities of their protein spots.

Note: All duplicate gels must be in the same matchset before you create a replicate group.

To make a replicate group, make sure the matchset with the duplicates is open and active, then go to the Analysis menu, open the Replicate Groups submenu, and select Create Group.

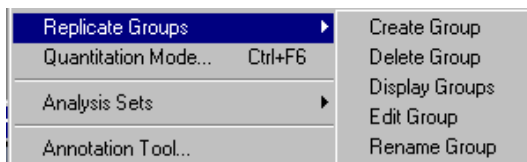


Fig. 7-7. Replicate Groups submenu.

In the dialog box, enter a name for the replicate group, then click on each duplicate gel in the list to be included in the group. The gel names will appear highlighted. Click on Go to create the group.

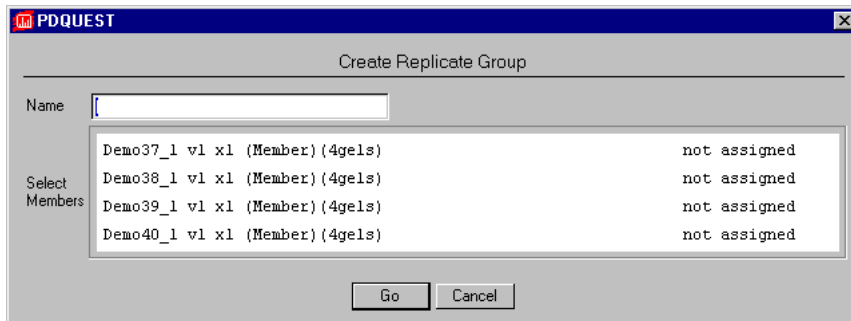


Fig. 7-8. Create Replicate Group dialog box.

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In the upper left status bar in PDQUEST, you will see the total number of gels in this group. Repeat this procedure to designate other groups in the matchset.

To delete a replicate group, select Delete Group from the Analysis > Replicate Groups submenu, then select the group to delete from the list.

To edit a replicate group, select Edit Group from the submenu. In the dialog, add or remove gels from the list.

To rename a replicate group, select Rename Group from the submenu, select the group to rename in the dialog, then enter a new name for the group in the pop-up box.

Quantitating a Replicate Group

After you have created a replicate group, you can calculate the average spot quantities in the members of the group by selecting Quantitation Mode under the Edit menu (CTRL+F6) and clicking on the Replicate Group Quantitation button in the pop-up form.

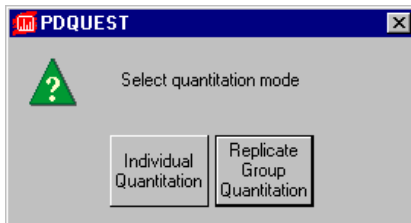
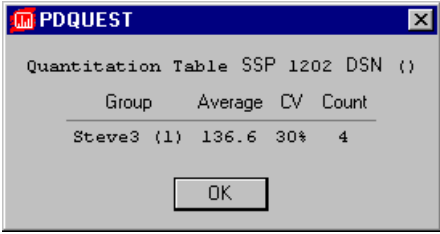


Fig. 7-9. Quantitation Mode pop-up.

Replicate group quantitation can be displayed using the Quantity Table function (Shift+F6, Spots menu).

Click on the spot whose quantitation you want to see. If you selected Replicate Group Quantitation, the average value of that spot's quantitation will be displayed, as will the coefficient of variance (CV) and the number of gels in the replicate group (Count).



Group	Average	CV	Count
Steve3 (1)	136.6	30%	4

Fig. 7-10. Quantity table displaying replicate group quantitation.

7.6 Normalization

When comparing gels in a matchset, there is often some variation in spot size and intensity between gels that is not due to differential protein expression. This variation can be caused by a number of factors, including:

- Inconsistent cell numbers (cell densities) in the sample prep,
- Inconsistent sample prep efficiency due to variations in reagents, protocols, etc.,
- Handling errors resulting in some sample loss during prep,
- Pipetting errors during sample prep,
- Sample loss during gel loading,
- Inconsistent staining times between gels,
- Inconsistent labeling efficiency with radioisotopes,
- Inconsistent excitation/emission efficiencies between different fluorescent tags,
- Inconsistent detection energy sources between gels during image acquisition, and
- Inconsistent exposure times between gels during image acquisition.

To accurately compare spot quantities between gels, you must compensate for these nonexpression-related variations in spot intensity. This is called normalization.

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With a matchset open, select Normalize from the Analysis menu.

Matchset: Demo2

☒ Enable Normalization

Basis (denominator)

☐ Total of spots in analysis set 1

☐ Total of all valid spots

☐ Total density in gel image

☐ Specified value

☒ Counts loaded from Gel Record

☐ Pipetting error compensation

Scaling

☒ PPM (x 1000000)

☐ Percent (x 100)

☐ User specified

Scale factor: Units:

Member information

Gel Name	Analysis Set	Valid Spots	Total Density	Specified Value	Counts Loaded	Comp.
demo1	724.6	25357.5		396310.0	396310.0	1.000
demo2	232.0	21644.7		396310.0	396310.0	1.000
demo3	685.5	19929.7		396310.0	396310.0	1.000
demo4	448.1	11020.8		198160.0	198160.0	1.000

OK Cancel Help

Fig. 7-11. Normalization dialog box for member images.

In the dialog box, click on the Enable Normalization checkbox to activate normalization for the matchset.

After you have selected the settings described below, click on Done.

Normalization Formula

Normalization in PDQUEST is performed for each gel in a matchset using the following formula:

Normalized spot quantity = Raw spot quantity * Scaling factor * Pipetting error compensation factor (if any) / Normalization factor

Raw spot quantity is the unnormalized quantity (intensity) of each spot,

Scaling factor is a factor you select to give a meaningful value (see below), and

Normalization factor is calculated for each gel based on the normalization method you select (see below).

(Pipetting error compensation is only available if you have calibrated the gels using Calstrips and entered the number of counts loaded in the gel into the Gel Record. See Appendix J.)

7.6.a Scaling Factor

The scaling factor is a constant by which the normalized quantity is multiplied to give a more meaningful value. A common scaling factor is 10^6 , giving units of parts per million (PPM).

Under Scaling, specify the scaling factor that you would like to use by clicking on PPM ($\times 1000000$), Percent ($\times 100$), or User Specified. If you select User Specified, enter your own scaling factor and units in the appropriate fields.

7.6.b Normalization Methods

The normalization method you select will depend in part on how much you know about your gels and how they were prepared. For example, if you can readily identify housekeeping proteins on your gels, the Total of Spots in Analysis Set method is preferable. If you know that each gel was loaded with a different amount of sample, you can select Specified Value and enter the known variation factor.

Total Quantity in Analysis Set

This normalization method involves identifying a group of protein spots that are present in all matchset members and are conserved (i.e., their quantities should not vary between samples). These are proteins that you know are expressed in a very consistent manner at all times within a cell, with little or no regulation. These are commonly referred to as "housekeeping" proteins.

First, you create an arbitrary analysis set containing several of these housekeeping proteins from different regions of the matchset standard.

Note: To create an arbitrary analysis set, make sure your matchset standard is selected, then go to the Analysis menu and select Analysis Sets > Create

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Analysis Set. In the dialog box, click on the Type button and select Arbitrary from the list. In the Arbitrary Analysis Set dialog, enter a name for the set and select the No Spots option. Then click on Go. This creates an empty analysis set. To add spots to this set, choose Select Analysis Set from the Analysis > Analysis Sets submenu and select the name of the set you just created from the list. With the set selected, select Add Spot To Set from the Analysis > Analysis Sets submenu and click on an appropriate (i.e., housekeeping protein) spot. The spot will be highlighted, indicating that it has been added to the set. If you want to remove a spot from the analysis set, select the set, then select Remove Spot From Set from the Analysis > Analysis Sets submenu and click on a highlighted spot.

See section 8.1 for more information on creating analysis sets.

In the Normalization dialog box, select the Total Quantity in Analysis Set option, and select your set of housekeeping proteins from the pulldown list.

The total quantity of the spots in this analysis set will be displayed for each matchset gel in the table at the bottom of the dialog box.

The raw quantity of each spot in a gel will be divided by the total quantity of the spots in the analysis set in that gel, according to the normalization formula.

Note: The advantage of selecting multiple housekeeping proteins is that occasional minor variations in the size and/or intensity of a particular spot will be averaged out when the group of spots is considered is a whole.

Total Quantity in Valid Spots

In this normalization method, the raw quantity of each spot in a member gel is divided by the total quantity of all the spots in that gel that have been included in the Standard (see the normalization formula).

Select the Total Quantity in Valid Spots option button to activate this normalization method. This method assumes that few protein spots change within the experiment, and that the changes average out across the whole gel. It can be useful if very little information is known about the possible source of sample variation.

Total Density in Gel Image

In this method, the raw quantity of each spot in a member gel is divided by the total intensity value of all the pixels in the image (see the normalization formula).

Select the Total Density in Gel Image option button to activate this normalization method. This model assumes that the total density of an image (i.e., background density plus spot density) will be relatively consistent from gel to gel. It can be useful if very little information is known about the possible source of sample variation.

Specified Value

In this method, the raw quantity of each spot in a member gel is divided by a user-definable normalization factor for that gel.

Select the Specified Value option and enter a normalization value for each gel in the table at the bottom of the dialog box. This value will be used as the denominator in the normalization formula.

This model is useful if you want to correct for known variations in samples across gels (e.g., the amount of cells that went into an experiment, the amount of extract loaded, the length of time the images were detected, etc.).

Counts Loaded from Gel Record

In this method, the raw quantity of each spot in a member gel is divided by the total counts loaded into that gel, as specified in the Gel Record when you performed calibration.

Note: This method is only available if you have performed calibration on your matchset gels. (See Appendix J.)

The counts loaded in each gel are listed in the table at the bottom of the Normalization dialog; this value will be used as the denominator in the normalization formula. Select the Counts Loaded from Gel Record option button to activate this normalization method.

This method assumes that few protein spots change within the experiment, and that the changes average out across the whole gel. This model can be

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useful if you want to correct for known variations in the amount of sample loaded across gels, as measured prior to loading with a scintillation counter.

If you select this method, you can compensate for pipetting error in all matchset members by selecting the Pipetting Error Compensation checkbox.

Pipetting Error Compensation

When this checkbox is selected, the software will attempt to correct for minor systematic errors inherent in scintillation counting, pipetting, and exposing films. The normalization process is based on the assumption that these minor errors affect all proteins equally. It also assumes that few protein spots change within the experiment or that when changes do occur, they do not alter the random distribution from one sample to the next.

When you select this checkbox, a pop-up box will display the calculated compensation value for each matchset member. These values will also appear in the in the table at the bottom of the Normalize dialog.

7.7 Standard Spot Numbers

Standard spot (SSP) numbers are unique numbers assigned to each spot in the matchset standard. In general, spots in the lower left region of the image have low SSP numbers, while spots in the upper right region have high SSP numbers.

Note: SSP numbers are increasingly important when you begin comparing spots between different matchsets using high-level matchsets.

SSP numbers are automatically assigned when you create the matchset standard. To create SSP numbers, PDQUEST first divides the standard into rows and columns, placed so that there are approximately the same number of spots in each section and that no section has over 100 spots. The first two digits of a spot's SSP number correspond to its section's X and Y coordinates. Spots within a section are numbered sequentially.

Displaying SSP Numbers

The spot number tools are located on the Analysis > Spot Numbers submenu.

With the matchset open and active, select Plot SSP Grid from the submenu to display the SSP grid on the standard. Numbered vertical and horizontal grid lines will appear on the standard.

The grid is helpful for locating spots by SSP number. However, it is an approximation and should only be used as a guide. If enough spots are added to a single grid block to use up the 99 available numbers, numbers will be assigned from other blocks.

Display SSPs displays SSP numbers next to every spot in the active window. You may need to zoom in on the image to be able to read them.

Display All SSPs displays SSP numbers next to every spot in every open window.

Transferring SSP Numbers in a High-Level Matchset

Transfer SSPs is used to copy the SSP numbers from one high-level matchset member to the others. This makes it easier to correlate data from the member matchsets.

With the high-level matchset open, select Transfer SSPs from the submenu. In the first pop-up box, select the matchset member you want to copy from; in the second box, select the member you want to copy to. You will be asked to confirm the transfer.

When the transfer is complete, the number of SSP numbers transferred will be listed in an alert box. Repeat the procedure to apply the numbers to all the members.

7.8 Database Spot Numbers

Database spot numbers (DSNs) are an alternative to the SSP numbers. The DSN system provides a more flexible way to number protein spots in a standard. A coordinate grid can be established for each axis, based on

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molecular weight and isoelectric point. The grid coordinates are identified with letters (A–R, excluding I and O because they look like numbers). The letters are intended to represent specific Mr pI ranges so that the numbering between different gels can be comparable. These ranges are as follows:

Mr values:	0-10=A	87-117=J
	11-14=B	118-158=K
	15-19=C	159-215=L
	20-25=D	216-293=M
	26-34=E	294-398=N
	35-46=F	399-541=P
Mr values (cont):	47-63=G	542-736=Q
	64-86=H	737-1000=R
pI values:	0-2=A	7.0-7.5=J
	2-3=B	7.5-8.0=K
	3-4=C	8.0-8.5=L
	4-5=D	8.5-9.0=M
	5.0-5.5=E	9-10=N
	5.5-6.0=F	10-11=P
	6.0-6.5=G	11-12=Q
	6.5-7.0=H	12-14=R

DSNs have the format ABnn. “A” is the molecular weight grid letter, “B” is the isoelectric point grid letter, and “nn” is the number of the spot within that section of the grid.

Assigning DSNs

After you have entered Mr pI values for known spots, you can create a DSN grid using this data by selecting Assign DSN Grid from the Analysis > Spot Numbers submenu. This automatically assigns a DSN number to each spot in the standard.

To display the grid (this does not occur automatically when the grid is created), click in the standard and select Plot DSN Grid. A grid of vertical and horizontal axes will appear, with the letter assignments of each axis displayed in the window margins.

To see the spot assignments, select Display DSNs. To read the numbers, you will probably need to zoom in on a smaller region of image.

Transfer DSNs is used to copy the DSN numbers from one higher level matchset to another (see).

Transferring DSNs in a High-Level Matchset

Transfer DSNs is used to copy the DSNs from one high-level matchset member to the others. This makes it easier to correlate data from the member matchsets.

With the high-level matchset open, select Transfer DSNs from the submenu. In the first pop-up box, select the matchset member you want to copy from; in the second box, select the member you want to copy to. You will be asked to confirm the transfer.

When the transfer is complete, the number of DSNs transferred will be listed in an alert box. Repeat the procedure to apply the numbers to all the members.

8. Analysis Sets and Annotations

Analysis sets and annotations are tools to help you organize and analyze your spot data.

Analysis sets allow you to create groups of spots that are statistically and biologically significant. You can also create sets of spots to cut using Bio-Rad's ProteomeWorks Spot Cutter.

Annotations allow you to add notes, comments, and other pertinent information to spots. Using annotations, you can create links to protein databases or files, import mass spec data, and store spot information in HTML format for easy portability and uploading to Web sites.

8.1 Analysis Sets

The analysis set tools allow you to create and study sets of proteins that are statistically and scientifically significant. You also use analysis sets to identify spots to cut using Bio-Rad's ProteomeWorks Spot Cutter.

There are six different kinds of analysis sets:

Quantitative Analysis Sets: Are composed of spots whose quantitation has increased or decreased "x" fold (where x is a user-determined value) or whose quantitation has changed above or within the fold change factor that you specify.

Qualitative Analysis Sets: Are composed of spots that are present in one gel but not in another (i.e., spots that have been turned "on" or "off").

Arbitrary Analysis Sets: Are composed of spots that you manually select.

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Boolean Analysis Sets: Are created by comparing two other analysis sets. For example, set C could be made up of those spots that are present in both set A and set B.

Statistical Analysis Sets: Are composed of spots that are significant based on the statistical test that you select.

The tools for creating and editing analysis sets are located on the Analysis > Analysis Sets submenu and the Analysis/ Annotation Tools toolbar.

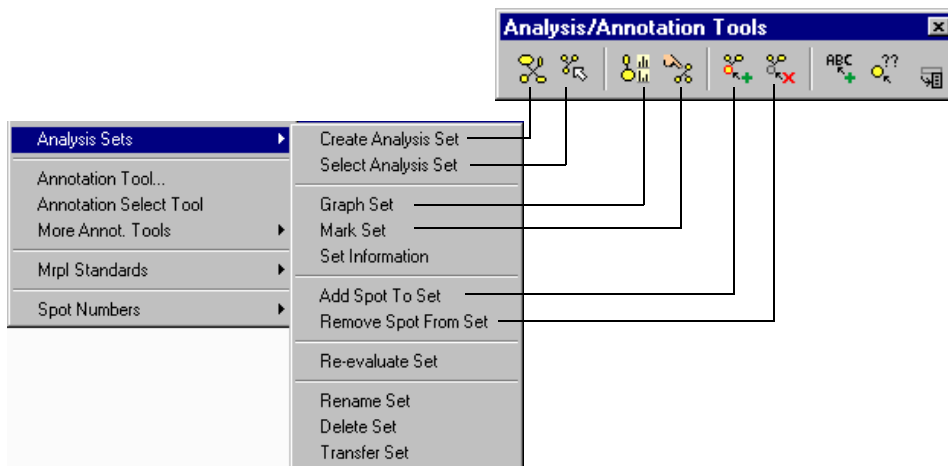


Fig. 8-1. Analysis Set tools.

8.1.a Creating an Analysis Set

With a matchset open, select Create Analysis Set from the Analysis > Analysis Sets submenu.

The analysis set dialog box will open.

Fig. 8-2. Example of a Create Analysis Set dialog.

If you want to update an analysis set that already exists, click on the Name button at the top of the Analysis Set dialog box. A list of existing sets will be displayed. Select the name of the set you want to update by clicking on it. The dialog box will be reconfigured and the information about that analysis set will be displayed.

The first step in creating the analysis set is selecting the type you want to create. In the Analysis Set dialog box, click on the Type button.

A list of analysis set types will appear. Click on a type to select it. (See the following sections for descriptions of the different types of Analysis Sets.)

The list will disappear and the analysis set dialog box will reconfigure to display the type you selected. You will see the name of your selection displayed next to the Type button.

Selecting the Operands to Compare

All analysis sets except arbitrary involve comparisons: How does a protein or group of proteins (in a gel or replicate group) behave under one set of

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conditions versus another? To answer this question, you need to specify which sample should be compared with which other sample.

First, indicate whether you want to look at the quantitation of individual gels or the average quantitation of groups of gels. Click on the Compare button and select either Gels or Groups from the list displayed. Your choice will appear to the right of the Compare button.

Next, click on the A button and choose a sample from the list displayed. (If you selected Gels to compare, a list of all the gels will be displayed. If you selected Groups, a list of replicate groups will be displayed.)

After clicking on your choice, you will see it displayed next to the A button.

Repeat this procedure with the B button to enter the second sample.

After you have entered the settings for an analysis set, click on the Go button to create the set. The dialog box will disappear. On the matchset standard, the spots that are included in the analysis set will be highlighted.

Displaying the Spots that Are Included in a Set

After you have entered the settings for an Analysis Set, click on the Go button to create the set. The dialog box will disappear. On the Standard, the spots that are included in the Analysis Set will be highlighted.

Once an analysis set is active, select Mark Set from the Analysis Sets submenu or toolbar to highlight all the spots contained in that analysis set.

Another way to display the proteins in an analysis set is with the Graph Set function (in the Analysis > Analysis Sets submenu). This function will display histograms of the spot quantitation along the left and right margins of the screen. (For more information on histograms, see section 12.1)

To clear the display, click on Hide Overlays on the main toolbar.

Editing an Analysis Set

To open an existing analysis set, choose Select Analysis Set from the Analysis > Analysis Sets submenu and select the set from the list.

You can also open an existing set from within the Analysis Set dialog box. Click on the Name button at the top of the dialog and select the name of the set you want to load from the list. The settings and information for that set will be displayed in the dialog box.

To delete a saved set, select Delete Set from the Analysis > Analysis Sets submenu and select the set to delete from the list of names.

To rename an analysis set, select Rename Set from the submenu and select the set to rename from the list. Enter a new name for the set in the pop-up box.

To recalculate the members of a set (e.g., if you have added or matched more spots), select Re-evaluate Set from the submenu. The set will be recalculated and the members will be displayed on the matchset standard.

8.1.b Quantitative Analysis Sets

Quantitative analysis sets allow you to perform fold change analyses. To create a Quantitative Analysis Set, select Analysis > Analysis Sets > Create Analysis Set. Quantitative Analysis Set is the default configuration of the analysis set dialog box.

Create Analysis Set

Name:

Description:

Type: Quantitative

Compare: Gels

A:

B:

Select spots which are:

- Increased B > A times 2.00
- Decreased B < A over 2.00
- Increased or Decreased B > A times 2.00
- Conserved B <= A times 2.00

Units: Quan. units OD Norm. units (comp) PPM Basis Counts Loaded

Go Cancel

Fig. 8-3. Quantitative Analysis Set dialog.

After entering your selections for comparison (e.g., control gels versus experimental gels), you need to decide what kind of quantitative changes you want to examine. You can select Increased, Decreased, Increased or Decreased, or Conserved.

Next to each choice is a diagrammatic representation of the type of change the test will find and an explanation of the type of quantitative criteria a spot must meet to be included in the analysis set.

Do you want to look for a protein whose quantitation went up under the experimental conditions as compared to the control? If so, click on the Increased button next to the Select spots which are prompt.

A represents all matched spots in sample A and B represents the corresponding matched spots in sample B. Comparisons are made using spot quantity only. Positional comparisons, etc., are not possible.

The comparison is performed as follows: A threshold value is established by multiplying and/or dividing A by the factor you specify. B is then compared to the threshold.

For example:

- If you ask to create a set by clicking on the Increased button and enter a factor of 2.0, the set will include spots whose quantity in B is at least twice that of the corresponding spot in A.
- If you click on the Increased button and enter a factor of 0.5, the set will include spots whose quantity in B is greater than half the quantity of the corresponding spot in A.
- Clicking on the Decreased button and entering a factor of 2.0 will find spots whose quantity in B is less than half the quantity of the corresponding spot in A.
- Creating a set by clicking on Decreased and entering a factor of 3.0 finds all the spots whose quantity in B is less than one-third the quantity of the same spot in A.
- If you are interested in finding the spots whose quantitation has significantly changed either way (either increased or decreased), select the Increased or Decreased option. The proteins found in this test have increased in quantitation above the upper bound or decreased below the lower bound. If you enter an upper bound of 2.0 and a lower bound of 3.0 you will find spots whose quantity in B is more than twice the quantity in A or less than one-third the quantity in A.
- To find spots whose quantitation in B has not changed significantly from A, click on the Conserved button. Enter factors to define the upper and the lower bounds. If, for example, you enter an upper bound factor of 3.0 and a lower bound factor of 2.0, the analysis set will include spots whose quantity in B is between three times and one-half times the quantity of those spots in A.

After entering the fold-change factor(s), the corresponding diagram will automatically update to show you the quantitative relationship these criteria define.

Click on the Go button at the bottom of the dialog box to create the set you have specified.

Be sure to enter a name for the set. If you forget, a pop-up box will remind you to do so.

8.1.c Qualitative Analysis Sets

Qualitative analysis sets contain spots that were not detected in one gel but were detected in the other. This test would detect a protein that was expressed under experimental conditions but was not expressed in the control sample, or visa versa.

Note: We strongly recommend that all the spots in such an analysis set be double-checked for accuracy. It is possible that a protein might actually be present in both gels but was missed during matching.

To create a qualitative analysis set, select Analysis > Analysis Sets > Create Analysis Set, click on the Type button in the dialog box, and select Qualitative from the list displayed. The dialog box will change to a qualitative analysis set configuration.

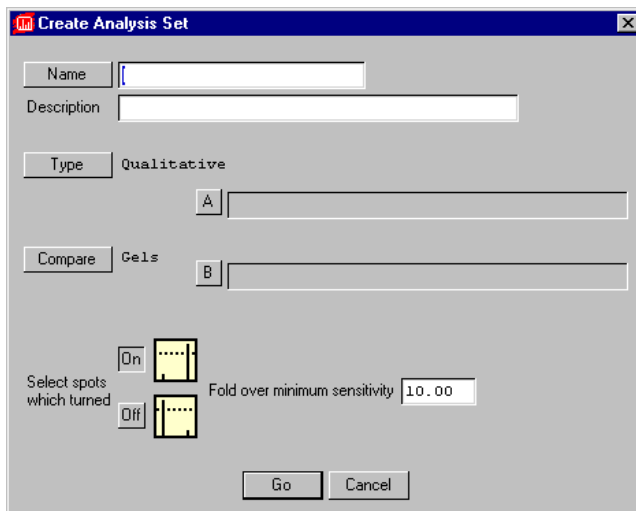


Fig. 8-4. Qualitative Analysis Set dialog.

Select the gels you want to compare, then select whether you want to highlight spots that were on in gel A/off in B (On in A) or on in gel B/off in A (On in B).

Finally, you need to tell PDQUEST what “turned on” or “turned off” means. How many times greater than the minimum detectable spot should the quantitation be for a spot to be included in this analysis set? Should it be five-fold greater than the minimum? If so, enter 5.00 next to the Fold over minimum sensitivity prompt.

Click on the Go button to find all the spots that fit the criteria that you specified. They will be highlighted on the matchset standard.

8.1.d Arbitrary Analysis Sets

Arbitrary analysis sets are composed of any group of spots that you choose. To create an arbitrary analysis set, select Analysis > Analysis Sets > Create Analysis Set, click on the Type button in the dialog box, and select Arbitrary from the list displayed.

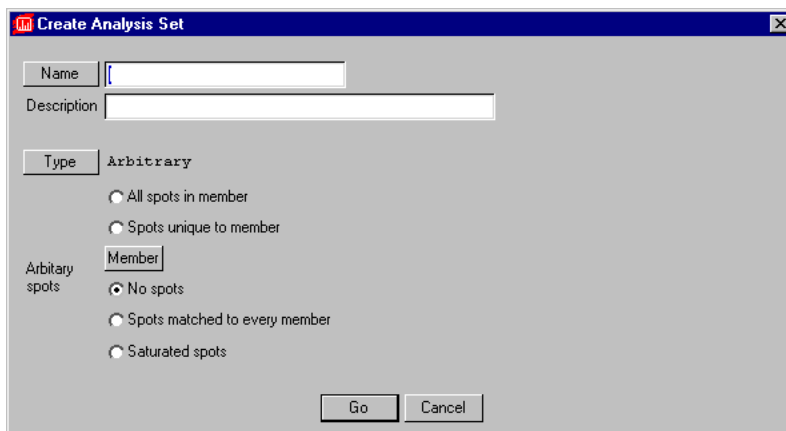


Fig. 8-5. Arbitrary Analysis Set dialog.

The first option (All spots in member) will create an analysis set containing all the spots in a selected member that are matched to the standard. Click on the Member button to display a list of the gels from which to choose. Click on your choice and it will be displayed next to the Member button. The analysis set will include all the spots in the member you selected that are matched to the standard.

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Spots unique to member will create an analysis set of those spots found only in the member that you specify and not in any of the other members of the matchset. Click on the Member button to display a list of gels from which to choose. Click on your selection and it will be displayed on the form next to the Member button.

Spots matched to every member will create an analysis set of spots that are matched in all the members.

Saturated spots will create an analysis set of all spots that are saturated in any member.

The No spots option creates an empty analysis set to which you can add any spots you choose. See the following section for detailed instructions.

Adding Spots to/Removing Spots from an Arbitrary Analysis Set

To add spots to an arbitrary analysis set, you first select the set you want to add to, then you click on the spots you want to add to it.

Note: Spots may only be manually added to arbitrary analysis sets.

Choose Select Analysis Set from the Analysis > Analysis Sets submenu or toolbar to display a list of analysis sets. Select the arbitrary analysis set you want to add spots to.

With the set selected, select Add Spot To Set from the Analysis > Analysis Sets submenu and click on a spot. The spot will be highlighted, indicating that it has been added to the analysis set. The name of the analysis set to which it has been added is displayed in the upper left corner of the window.

If you want to remove a spot from the analysis set, select the set, then select Remove Spot From Set from the Analysis > Analysis Sets submenu and click on an added spot. The spot will be unhighlighted, indicating that it has been removed.

8.1.e Boolean Analysis Sets

Boolean analysis sets are formed by combining two previously defined analysis sets using Boolean operators. You can create Boolean sets that include:

1. The intersection of two sets of spots (proteins present in both set *A* *and* set B).
2. The union of two analysis sets (spots in set A *or* in set B).
3. Spots that are unique to set A plus spots that are unique to set B.
4. Spots that are found only in set A.
5. Spots that are found only in set B.

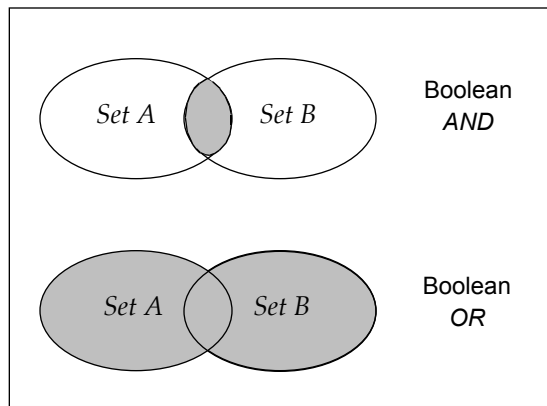


Fig. 8-6. Venn diagrams depicting the AND and OR boolean operations.

To create a Boolean analysis set, select Create Analysis Set, click on the Type button in the dialog box, and select Boolean from the list displayed.

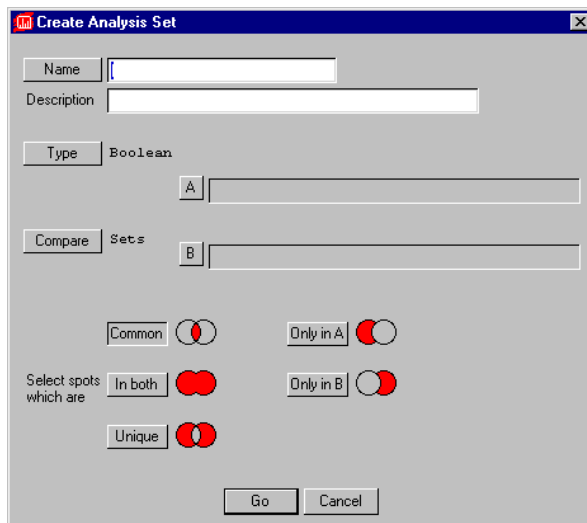


Fig. 8-7. Boolean Analysis Set dialog.

Select the two analysis sets you want to compare to one another from the lists displayed when you click the A and B buttons.

Enter the name and description to be given to the set.

Select the Boolean operator to be used to create the set by clicking on one of the buttons next to the “Select spots which are” prompt.

When you click on the Go button, the dialog box will disappear and the spots that meet the criteria that you specified will be highlighted in the matchset standard.

8.1.f Statistical Analysis Sets

Statistical analysis sets are composed of spots from replicate groups found to be significant according to the statistical test that you specify.

To create a statistical analysis set, select Create Analysis Set, click on the Type button in the dialog box, and select Statistic from the list displayed. The

dialog box will be reconfigured to let you select the replicate groups to be compared and the statistical test to be performed.

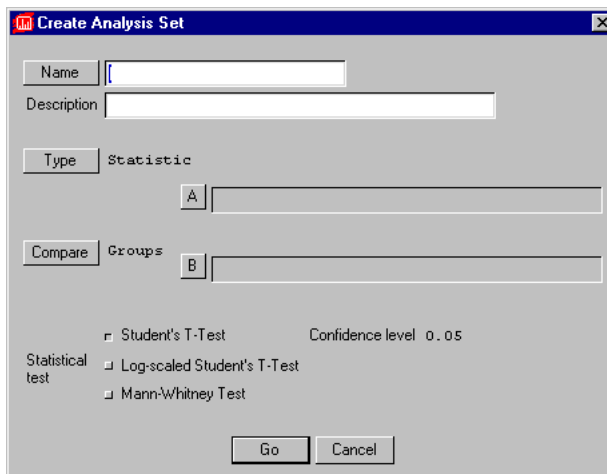


Fig. 8-8. Statistic Analysis Set dialog.

Statistical analysis sets only work with groups, because the tests require multiple gels. The Student's T Test and Log Student's T Test require at least two gels per group. The Mann-Whitney Test requires at least four gels per group.

Choose the groups by clicking on the A and B buttons as described in previous sections.

Indicate the statistical test you want to perform by clicking on the button next to the name of the test.

Enter a name for the analysis set at the top of the dialog box and click the Go button.

Significant protein spots will be highlighted on the matchset standard.

8.2 Annotation Tool

You can add notes, comments, and other pertinent information to each protein spot in a matchset using annotations. Annotations can be used to describe spots, group them into categories, and reference additional information in other files or Internet databases. You can import MassLynx mass spec data into annotations. Spot annotation data is automatically stored in HTML format for easy portability and uploading to Web sites.

8.2.a Opening the Annotation Tool

Spots are annotated using the Annotation Tool. The tool can be opened by selecting a spot or spots in the matchset, or you can open the tool with no spots selected.

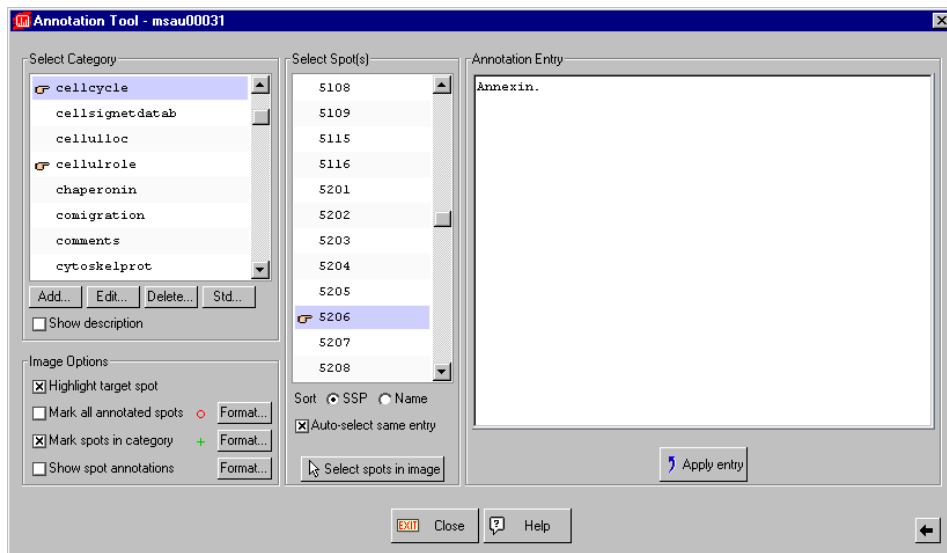


Fig. 8-9. Annotation Tool.

To open the tool with spot(s) selected, select Annotation Select Tool from the Analysis menu, and click on a spot in the matchset standard or drag a box around a cluster of spots. You can also right-click on a spot in the standard

and select Annotation Tool from the dropdown menu. The Annotation Tool will open with the SSP number(s) of the selected spot(s) highlighted.

Alternatively, select Annotation Tool from the Analysis menu or toolbar. The tool will open with no spots selected.

Note: If you are annotating a matchset for the first time, you will be prompted to add a list of standard categories when you first open the tool.

Using the tool, you select a category for each annotation, select the spot(s) that you want to annotate in that category, and then type the annotation entry. Note that the workflow is flexible; e.g., you can select the spot(s) first, then select a category. However, each annotation entry must be placed in a category and have at least one associated spot.

8.2.b Annotation Categories

Annotations are organized in categories. Some examples of annotation categories are protein name, amino acid composition, binding properties, translational regulation, etc.

You first select the category, then enter the annotation. A single spot may be annotated in multiple categories, depending on the amount and type of information you know about it.

Most categories contain simple text annotations. Specialized categories can be used to link spots to Internet protein databases or open files in other applications.

PDQUEST includes a number of standard categories. To add these to the Select Category list, click on the Std button below the list. The standard categories will appear in the list.

Creating and Editing Categories

To create a new category, click on the Add button below the Select Category list.

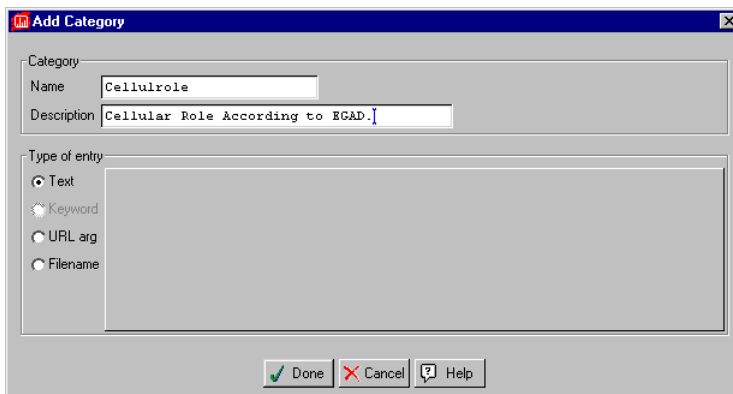


Fig. 8-10. Creating a text category.

In the dialog box, enter a Name for the category and any Description you want to include. If this category will contain textual annotations, the Text option button should be selected. You can also create specialized categories using the URL arg and Filename buttons (category types are described below).

When you click on Done, the category is created and will appear selected in the Select Category list.

To edit an existing category, select the category in the list and click on the Edit button. You can then change the category type and description. Note that the annotation entry may be specific to a particular category type; changing the type may render the entry meaningless.

To delete a category, select it, then click on Delete. You will be warned before the category and all its entries are deleted.

URL Categories

This type of category provides a link to an Internet protein database or other Web site; then the annotation entry references the specific file or Web page for a particular protein. You can create different URL categories to link to different databases, including Swissprot, Unigene, and GeneCards.

To create a URL category, open the Add Category dialog box, enter a name for the category, then select the URL arg option button and type a name for the database or Web page in the Link Name field. This is the name that will appear in the Browse Annotations page for a spot in this category.

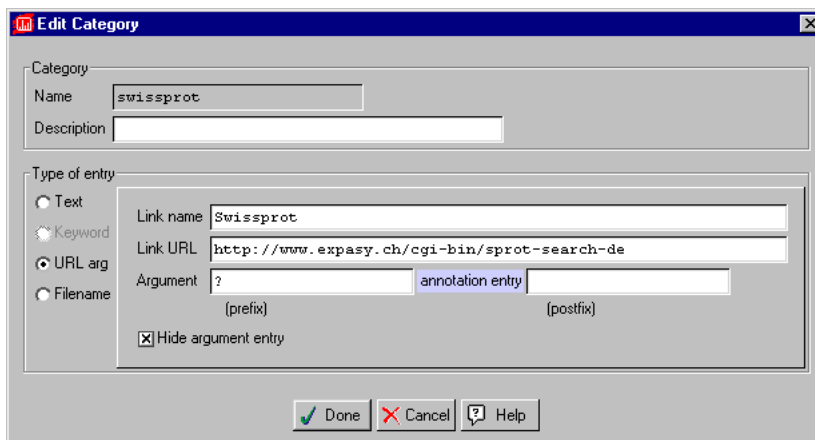


Fig. 8-11. Example of a URL category.

Type the entire link address in the Link URL field including "http://". For example, Swissprot is <http://www.expasy.ch/cgi-bin/sprot-search-de>.

Most Internet databases like Swissprot require a specific argument to access specific information for particular proteins. In the Argument fields, enter the appropriate argument prefix and/or postfix (including any spaces) for the site you are accessing. Note that <annot. entry> between the fields represents the annotation entry you will type in the main Annotation Tool dialog.

Typical arguments are:

	<u>Argument (prefix)</u>	<u>Argument (postfix)</u>
Swissprot	?	
Unigene	/query.cgi?TEXT=	&ORG=Hs
GeneCards	?search=	

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When you are finished creating the category, click on Done. When you create an annotation entry for a spot in this category, it should be the exact format necessary to complete the URL link to the page in the database for the specific protein information you are referencing.

To open a URL link, right-click on the spot and select Browse Annotations. The HTML page for the spot will open, and the URL category name will be displayed as a link. Click on this link to access the protein information in the database or Web site.

File Name Categories

This type of category provides a link between spots and files on the user's computer, network, or removeable disk drive. For example, you can open a Protein Probe file linked to a spot from within PDQUEST.

To create a file name category, open the Add Category dialog box, enter a name for the category (e.g., Protein Probe), then select the Filename option button. In the File Extension field, type the extension of the file type you want to link (e.g., Protein Probe files have the extension "prp"). Use the Browse button next to the Files in Folder field to select the directory where the files in this category are located.

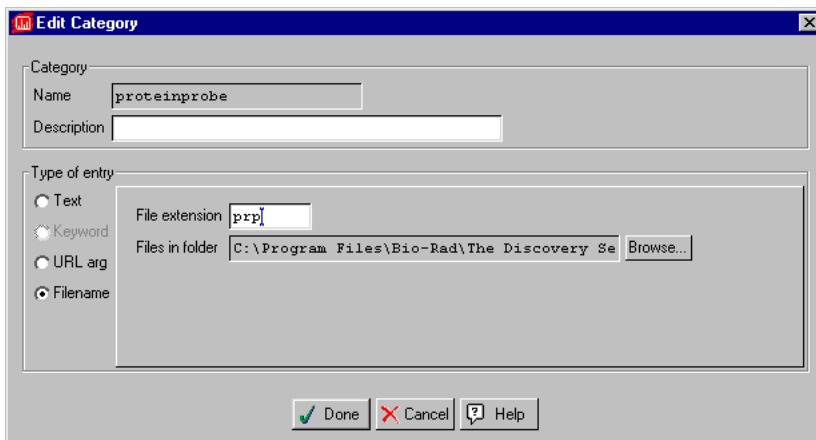


Fig. 8-12. Example of a file name category.

When you are finished, click on Done. In the annotation entry for a spot in this category, enter the exact file name you want to associate with the spot. This file must be located in the directory you specified in the Add Category dialog.

To open the file linked to a spot, right-click on the spot in the matchset and select Find File For. In the pop-up box, select the appropriate category name. This will launch the application associated with that file and open the file.

You can associate different files with the same spot, but they must be linked using different file name categories.

Protein Name Category

You can create a special category for identifying spots by their protein names instead of by SSP numbers in the Annotation Tool.

Open the Add Category dialog box and type "protname" in the Name field. Click on Done to create the category. In the annotation entry for a spot in this category, type the name of the protein for that spot.

When you select the Sort: Name option under the Select Spot(s) list in the Annotation Tool, the spots will be listed alphabetically by their protein names.

8.2.c Selecting a Category

After you have created a category, you can select it. Click on the category name in the Select Category list. Click on the Show Description checkbox to display the description of each category in the list.

All the spots that have been annotated in a selected category will be marked in the Select Spot(s) list.

8.2.d Selecting Spot(s)

In the Select Spot(s) list, click on a spot SSP number to select it, or shift-click or ctrl-click to select multiple spots that will have the same annotation entry. Alternatively, click on the Select Spots in Image button and click on a spot in

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the matchset standard to select it. To select multiple spots with the Select Spots tool, ctrl-click or drag on the spots in the standard.

Selected spots will appear highlighted in the list. You can sort the spot list by SSP number or Name. Names of spots are specified using a special category called protname, which you either create or add as a standard category. You type the protein name of a spot as an annotation entry in this category, and you can sort by it using the Name sort option.

Auto-select Same Entry will automatically select spots in the list that have the same entry in a specified category. With the checkbox and category selected, click on a spot in the list or the matchset; the other spots with the same entry will be selected too.

All categories containing annotations for the selected spot will be marked in the Select Category list. If multiple spots are selected, no categories will be marked.

8.2.e Entering an Annotation

After you have selected an annotation category and a spot or multiple spots, type the annotation in the Annotation Entry field.

If you are entering an annotation in a URL category, it should be the exact format necessary to complete the URL link to the page in the database for the specific protein information you are referencing.

If you are entering an annotation in a file name category, the entry should be the exact file name you want to link to the spot.

If you are entering an annotation in the protname category, the entry should be the protein name you want to associate with the spot.

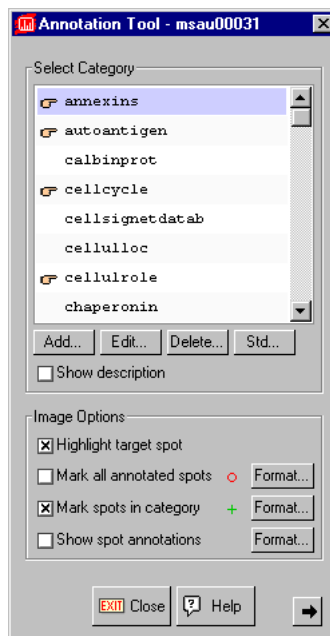
To apply an annotation entry, click on the Apply Entry button. The entry will be automatically applied if you select another spot or category to annotate.

To delete an annotation entry, select the entry text and press the Backspace or Delete key. Then click on Apply Entry or select another spot to apply the deletion.

8.2.f Reviewing Annotations in the Matchset

The Image Options in the Annotation Tool allow you to highlighted selected spots in the matchset standard. You can also display actual annotations on the standard image.

Note: When reviewing annotations in the matchset, it is useful to shrink the annotation tool window. Click on the arrow button in the lower left-hand corner of the Annotation Tool to collapse the tool; only the reviewing functions will be displayed.



← Click here to expand tool.

Fig. 8-13. Annotation Tool in reviewing mode.

Highlight Target Spot will highlight the spot(s) that are selected in the Annotation Tool. The spots will be marked with

Mark All Annotated Spots will mark all annotated spots in the matchset. Click on the Format button to select the style and color of the markings.

Mark Spots in Category will mark the spots that are annotated in the selected category. Click on the Format button to select the style and color of the markings.

Show Spot Annotations will display all the annotations in the selected category next to the annotated spots in the matchset. Click on the Format button to select the style and size of the annotations.

Reviewing Annotations

8.3 Browsing Annotations

You can click on an annotated spot in PDQUEST and open an HTML page displaying all the annotation information for the spot. You must have a Web browser installed on your computer to use this command.

Right-click on an annotated spot in the matchset standard and select Browse Annotations, or select the command from the Analysis > More Annot Tools submenu or Analysis toolbar and click on a spot in the standard. The default browser will open and display a page containing all the annotation information for the spot, listed by category. The Mr pI (if calculated) and protein name (if entered) of the spot are listed at the top of the page.

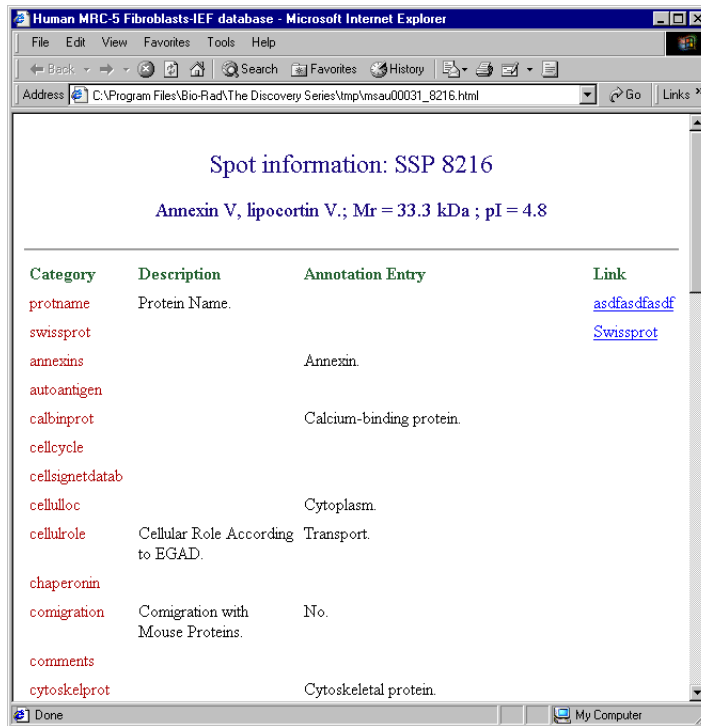


Fig. 8-14. Example of the annotation browser.

If you have created an annotation link for the spot in a URL category (see section 8.2.b), that link will appear in the page under the Link column. Click on this link to open the database or Web site specified by the category and link to the protein information.

Use the Print command in the browser window to print the annotation information. The displayed page is a temporary file that can be saved by using the browser's Save As command. You can then include saved pages on your Web site or e-mail them to colleagues.

8.4 Creating Annotations from Analysis Sets

Analysis sets are used to group spots in a matchset that are statistically and scientifically meaningful. After you have created an analysis set, you can create an annotation describing the spots in the set.

Note: You can create annotations for individual spots or other groups of spots using the Annotation Tool.

With the matchset open, select Create Annotation from Set from the Analysis > More Annot Tools submenu. The dialog box will open.

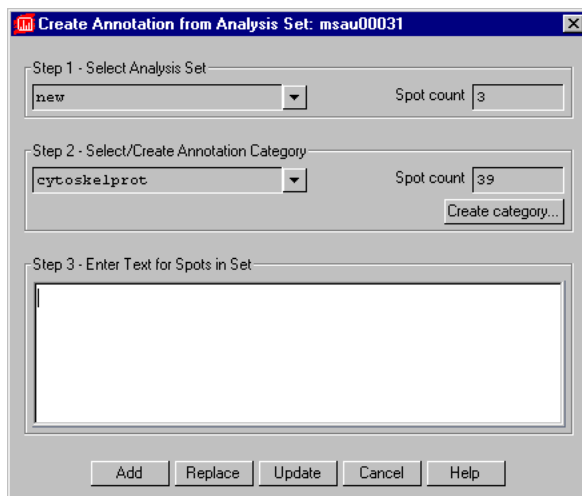


Fig. 8-15. Create Annotation from Analysis Set dialog box.

In Step 1, select the analysis set from the pulldown menu. The number of spots in the set will be displayed in the Spot Count field.

In Step 2, select an existing annotation category from the pulldown list, or create a new one by clicking on the Create Category button. See Annotation Categories for more information on creating categories. The initial spot count in the category will be displayed in the Spot Count field.

In Step 3, enter the annotation for the spots in the set. Note that this annotation will be applied to all the spots in the analysis set.

If you are creating an annotation in a specialized category (i.e., a URL or file name category), make sure the entry is in the appropriate format.

When you have entered the annotation, click on the Apply Entry button. An alert box will notify you how many spots were annotated in the category.

8.5 Printing Annotations

The functions for printing spot annotations are located on the Print Annotations submenu of the Analysis > More Annot Tools submenu. You can also print annotations from the Browse Annotations page.

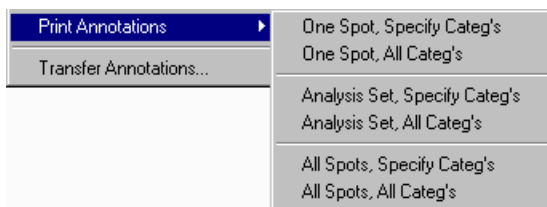


Fig. 8-16. Print Annotations submenu.

Using the commands in this submenu, you can print annotations for individual spots, the protein spots in an analysis set, or all the spots in the matchset standard. You can also print annotations for particular categories or all categories.

Printing Annotations for Individual Spots

To print the annotations for a particular spot, select One Spot, Specify Categ's or One Spot, All Categ's, then click on the spot in the standard. Note that Browse Annotations will also print out all the annotations for a particular spot.

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If you select One Spot, Specify Categ's, a pop-up list will prompt you to select annotation categories to print. Once you have made your selections, click on the Done button.

Printing Annotations for Analysis Sets

To print annotations for spots in a specific analysis set, select Analysis Set, Specify Categ's or Analysis Set, All Categ's from the Print Annotations submenu. A list of analysis sets will be displayed. Click on the name of the set whose annotations you want to print.

If you selected Analysis Set, Specify Categ's, a pop-up list of categories will appear. Make your selection(s), then click on Done.

Printing Annotations for All Spots

All Spots, Specify Categ's prints selected annotation categories for all the annotated spots.

All Spots, All Categ's prints the annotations in all available categories for all annotated spots.

8.6 Transferring Annotations

If a member of a high-level matchset includes annotations, you can transfer all or some of them to another high-level matchset member or to the high-level matchset standard.

Note: Only annotations for spots that are matched in the high-level matchset will be transferred. If no spots are matched between the members, only categories will be transferred.

With the high-level matchset open, select Transfer Annotations from the Analysis > More Annot Tools submenu.

In the dialog box, select the member matchset that you want to transfer from using the From Matchset pulldown list.

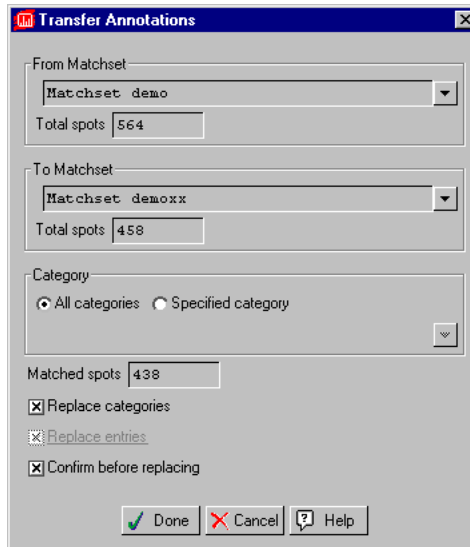


Fig. 8-17. Transferring annotations.

Select the member matchset that you want to transfer to using the To Matchset pulldown list. Note that when you select the second matchset, the total spots in each matchset will be listed, as well as the number of spots that have been matched between the matchsets (in the high-level matchset).

Next, select whether you want to transfer annotations in All Categories or just one category. If you select Specified Category, select a category from the pulldown list.

If the two matchsets share categories, selecting Replace Categories will replace each duplicate category and all its entries in the "to" matchset. Replace Entries will replace only the entries for shared spots in the "to" matchset. If neither checkbox is selected, the transfer will only add new entries; shared entries and categories will not be replaced.

Selecting Confirm Before Replacing will prompt you before each category / entry is replaced.

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Click on Done to complete the transfer. An alert box will list how many categories and entries were transferred. You can review the transferred annotations using the Annotation Tool.

9. Basic Excision Tool

The Basic Excision Tool is used to control Bio-Rad's ProteomeWorks Spot Cutter system for basic spot cutting operations. If you are cutting spots from matchsets for mass spec analysis, use the Integrated Excision Tool.

Before using this tool, make sure the spot cutter is connected with the computer, is switched on and has warmed up, and make sure the camera is connected and has been turned on.

Note: See the spot cutter hardware manual for instructions on installing the platform, camera, and associated hardware and software.

Select Basic Excision Tool from the Identification menu. The software will connect with the spot cutter platform and camera, and the Basic Excision Tool dialog box will open.

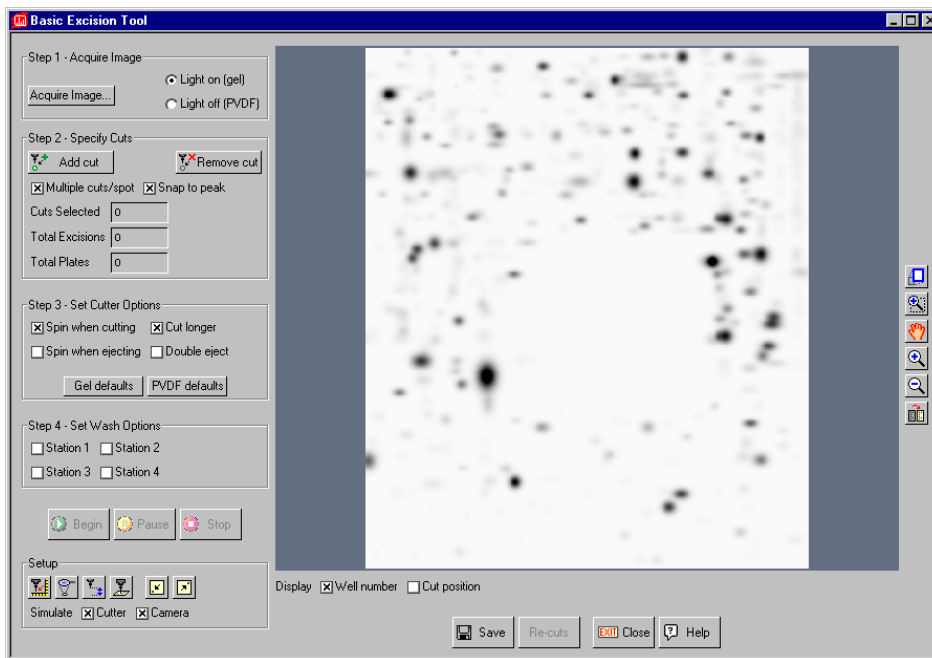


Fig. 9-1. Basic Excision Tool window.

The Basic Excision Tool includes a window for viewing the spot cutter camera image and a control panel for running the spot cutter.

9.1 Spot Cutter Setup

Before you can use the spot cutter for the first time, you must focus the camera and adjust the cutting tip height. You must also adjust the spot cutter settings and calibrate the instrument.

9.1.a Focus Cutter Camera

The Focus Cutter Camera dialog box controls the camera exposure setting. Focusing and aligning the camera are described in the spot cutter hardware manual.

Click on the Focus Cutter Camera button to open the dialog. The platform light on the spot cutter will turn on and the camera will focus on a small region at the center of the gel or membrane and begin taking continuous exposures. The exposures will appear in the Focus window at the rate of about 2 per second.

At the top of the dialog, click on Auto-Expose to reset the exposure time of the camera.

The Time field shows the time of each exposure in milliseconds. The Ceiling field shows the highest pixel value in the focused region. The Levels field shows the levels of gray scale in the image. All of these settings are automatically determined by the software, based on the position and aperture of the camera lens.

Manually adjust the camera lens and position as described in the hardware manual to optimize the image. When you are satisfied with the image and the levels of gray scale, click on Stop to close the Focus dialog.

9.1.b Set Cutting Tip Height

Click on the Set Cutting Tip Height button to adjust the distance the cutting tip drops down to cut a spot. The adjustment dialog box will open, and the cutting tip will move to the center of the gel.

Click on Lower Tip to drop the cutting tip to its lowest position. In this position, manually adjust the cutting tip height as described in the hardware manual until it presses lightly against the surface of the cutting mat.

Raise Tip raises the cutting tip. Test Cut performs a test cut in the center of the gel. Click on Done to accept the settings and return the cutting tip to home position.

The cutting tip height only needs to be set once, unless the spot cutter gets out of alignment.

9.1.c Spot Cutter Settings

Click on the Spot Cutter Settings button to open the settings dialog box for the cutter.

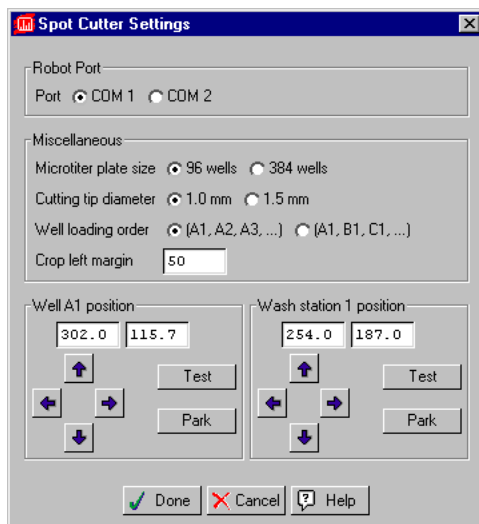


Fig. 9-2. Spot cutter settings.

In the dialog box, select the communications port to the spot cutter by clicking on COM 1 or COM 2.

Specify the microtiter plate size you are using by clicking on 96 wells or 384 wells.

Specify the diameter of the cutting tip by selecting 1.0 mm or 1.5 mm. Note that the cut request circles that you make on the image reflect the actual size of the cutting tip, and will change based on the tip size selection.

Specify the order in which you want to load the wells on the microplate by selecting A1, A2, A3, ... or A1, B1, C1, ...

Crop Left Margin repositions the image in the Basic Excision Tool window to the left. Enter a value (in screen pixels) in the field to reposition the image.

Note that you must click on Done to close the settings dialog and then acquire a new image for this change to take effect.

Well A1 position

The fields and buttons in this section are used to calibrate the position of the microplate for proper loading of samples. The number fields contain the X and Y coordinates (in millimeters) of well A1 on a plate in the spot cutter plate holder. Once you have set the position of this well, the cutter robot will automatically “know” the positions of the remaining wells.

To set the well A1 position, position your cursor over the number in the X or Y coordinate field and press Enter on your keyboard twice. The tip will move to that position without attempting to lower into the well. If the tip is not centered directly over the well A1 on the microplate, you will need to adjust the numbers in each field.

You can either type a new value directly in each field, or (for small adjustments) use the arrow buttons to adjust the tip by fractions of a millimeter. The cutting tip will move as you either enter new numbers or click on the buttons.

Note that the arrow buttons are oriented as if you were facing the front of the spot cutter; also, if you are facing the cutter, higher numbers in the first field will move the cutting tip to the right, while higher numbers in the second field will move the cutting tip toward the back of the instrument.

When you are satisfied with the position of the cutting tip, click on Test to lower the tip into the well. The tip should be centered inside the well.

Park returns to the tip to its home position.

Wash Station 1 position

Calibrate the position of Wash Station 1 on the cutting platform by following the steps described above for Well A1. Once you have set the position of station 1, the cutter robot will automatically “know” the positions of the remaining wash stations.

9.1.d Spot Cutter Calibration

Before you can begin excising spots, you must calibrate the spot cutter. This procedure requires a PVDF membrane.

Note: Before calibration, you should focus the cutter camera and align the wash station position. You should make sure the spot cutter camera image is properly centered using the Crop Left Margin setting.

Click on the Calibrate Cutter button. A series of dialog boxes will guide you through the calibration procedure. At any point in the procedure you can move back to repeat a step.

Note: Calibration creates several parameter and image files in the Install.prm directory in the Discovery Series directory on your hard drive. These files should not be moved, deleted or altered, or calibration will be affected.

When calibration is complete, you can proceed with normal use of the spot cutter. Once the cutter is calibrated, you do not have to repeat the procedure during normal day-to-day use. You may have to recalibrate if the instrument is moved.

9.2 Cutting Spots

The control panel is designed to guide you through the spot cutting procedure.

1. **Acquire Image.** In this step, you capture an image of your gel or membrane using the spot cutter camera.
2. **Specify Cuts.** In this step, you identify the spots in the image that you want to cut.
3. **Select the Cut Run Options.** These options can be adjusted for each cut run.
4. **Set Wash Options.** Select the wash stations to use after each cut by clicking on one or more checkboxes.

9.2.a Step 1. Acquiring the Image

Before acquiring an image using the spot cutter camera, select the Light On option if you are imaging a gel, or Light Off if you are imaging a PVDF membrane. (The platform light should be off for PVDFs, because the ambient light will produce a better image.) You must also focus the camera.

Click on Acquire Image. The camera will take a single image of the gel or membrane on the platform, and the image will appear in the window.

To adjust the brightness and contrast of the image, open the Transform window by clicking on the button to the right of the image window.

9.2.b Step 2. Specifying Cuts

In this step, you will click on spots in the image to identify which spots to cut.

When clicking on spots in the image, select the Snap to Peak checkbox to automatically center the cut circle on the spot peak (calculated from the spot intensity in the image). You can also use the Zoom Box tool included in the Basic Excision Tool to magnify each spot.

To make a single cut in a spot, make sure the Multiple Cuts/Spot checkbox is unchecked, then click on the Add Cut button and click on the spot in the image that you want to cut.

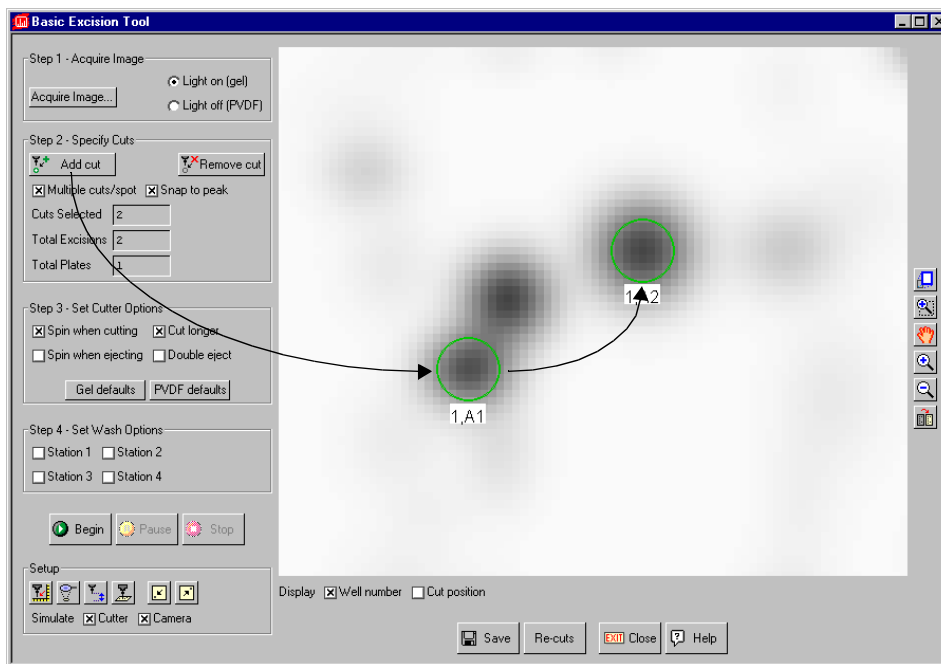


Fig. 9-3. Adding cuts.

If this is the first cut circle marked on the image, the Enter Plate Information dialog box will open, and you will be prompted to enter a name for the microplate and identify the first well you want to load. If you specify a starting well other than the default of A1, the skipped wells will be noted in the dialog box. Click on Done to make your selections.

A green circle will appear on the image, indicating the spot to be cut. Note that the circle is the size of the actual cutting tip (1.0 mm or 1.5 mm, as specified in the Spot Cutter Settings dialog box).

Continue clicking on spots in the image with Add Cut selected to make additional cuts. The number of cuts will appear in the Cuts Selected field.

If you specify more cuts than there are wells in the microplate, you will be prompted to name a new plate and starting well for subsequent cuts.

Multiple Cuts of the Same Spot

You can take multiple cuts of the same spot. The cuts can be placed in separate microtiter plate wells or in the same well (to increase the volume of sample in that well).

To place multiple cuts of the same spot in different wells, make sure the Multiple Cuts/Spot checkbox is unchecked, then click on several locations inside the spot to create individual cuts of the same spot. Drag these circles so that they touch but do not overlap.

Note: Overlapping circles will result in incomplete cuts that may not be fully extracted by the spot cutter.

To place multiple cuts of the same spot in the same microtiter plate well, make sure the Multiple Cuts/Spot checkbox is checked, click on the Add Cut button, then drag a box over the spot. Multiple cutting circles will appear grouped inside a red box. All these cuts will be placed in the same well.

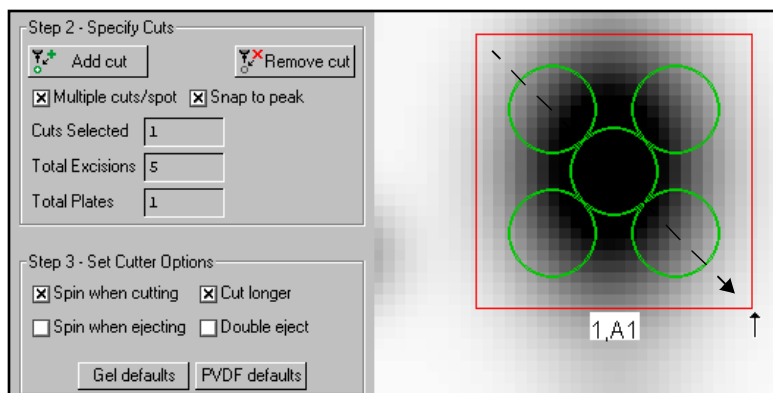


Fig. 9-4. Drag a box to make multiple cuts of the same spot.

Note: If the box you create contains too few cutting circles, simply drag over the area again to create a bigger box.

To move multiple circles inside a box, with Add Cut selected, position your cursor inside the center circle in the box. The appearance of the cursor will

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change to a multidirectional arrow. Drag the box to a new location and release the mouse button.

Moving a Cut Circle

To move a cut circle, make sure that Add Cut is selected, then position your cursor inside the cut circle. The appearance of the cursor will change to a multidirectional arrow. Drag the circle to a new location and release the mouse button.

Removing Cuts

To remove a cut circle from the image, position your cursor on the spot and right-click. You will be asked to confirm the deletion. To remove multiple cut circles, click on the Remove Cut button, then click on the cut circles, or drag a box around the circles.

Display Options

Click on the Well Number checkbox (below the image window) to display the well number associated with each cut request.

Click on the Cut Position checkbox to display the position of the cut (in millimeters) on the gel.

9.2.c Step 3. Set Cutter Options

Before beginning a cut run using the Basic Excision Tool, you must select the cutter options for that run.

If Spin When Cutting is selected, the cutting tip will spin as it cuts.

Spin When Ejecting will cause the cutting tip to spin as it ejects the cut into a microplate well.

Cut Longer will cause the cutting tip to hold in the “down” position longer when making a cut.

Double Eject will cause the cutting tip to emit two blasts of air when ejecting a cut into a microtiter plate well.

The Gel Defaults button automatically selects the most useful options when cutting gels. The PVDF Defaults button automatically selects the most useful options when cutting PVDF membranes.

9.2.d Step 4. Set Wash Options.

Select the wash stations to use after each cut by clicking on one or more checkboxes.

9.2.e Performing a Cut Run

After you have identified all the cuts you want to make and selected the appropriate options, click on the Begin button. You will be prompted to place a microplate in the plate holder.

As each cut is being made, it will be highlighted on the screen. After each cut has been made, the cut circle appears in white on the screen.

Click on Pause to pause the cut run. Note that the cutter will pause only after the most recent cut has been deposited in a well.

Click on Stop to stop the cut run. The cutter will stop only after the most recent cut has been deposited in a well. Any remaining cut requests will remain selected in the image.

When all cuts are complete, the cutting tip will return to home position and you will receive an alert message on the screen. At this point, you can review the cut holes in the gel/membrane. If all holes are complete, you can proceed with saving the image.

Recutting Spots

If the cutting tip has failed to cut some spots, click on the Recuts button below the image window. In the pop-up box, enter the numbers of your recut requests (microplate number, well number), separated by semicolons. Click on OK to close the recut request dialog, then click on Begin to perform the recuts.

9.3 Other Spot Cutter Controls

Click on the appropriate button to move the cutting tip to the front left corner of the spot cutter or the back right corner.

You can simulate the action of the cutter and its camera by clicking on the Simulate Cutter and Simulate Camera checkboxes. These checkboxes will cut off communication with the cutter and camera if selected. In simulation mode, you will be prompted to open a file image of a gel in place of the camera cutter image.

10. Integrated Excision Tool

The Integrated Excision Tool allows you to cut spots that have been matched in a matchset and placed in an analysis set (see section 8.1). First, you select the gels in the matchset that you want to cut from; then you use the Integrated tool to perform the cuts.

Finally, you can export the cut list to Micromass's MassLynx software for mass spectrometry analysis (see the next chapter).

10.1 Excision Gel Selection

The Excision Gel Selection tool is used to select the gel(s) in a matchset from which to cut spots using Bio-Rad's ProteomeWorks Spot Cutter System.

First, you must create an analysis set of spots (see section 8.1) in the matchset that you want to cut. Then you open the tool, select that analysis set, and identify which gels you want to cut from. This creates a cut list of spots and gels. Finally, you use the Integrated Excision Tool to perform the cuts.

Note: You must open the matchset containing the gels whose spots you want to cut and create an analysis set out of those spots before you can open the Excision Gel Selection tool.

With your matchset open, select Excision Gel Selection from the Identification menu. The Excision Gel Selection tool will open, and you will be prompted to select from the list of available analysis sets in the matchset.

Excision Gel Selection... (mpdi00318)

Step 1 - Select Analysis Set
 Set name: Spot count:

Step 2 - Set Selection Mode
☒ Auto-select Tolerance ratio:
☐ Single gel
☐ Manual ☒ Include spot quality Gels selected:

Table options: ☐ Show (quality) ☒ Display cuts
☐ Show ratio ☒ Auto-zoom on cut
☒ Show all gels ☐ Active gels only

Cut options: ☒ Multiple cuts/spot
 Max cuts/spot:

Step 3 - Specify cuts

SSP	3894	3679	4411	3897
0620	259.5	924.6	1003.8	790.3
0704	1401.8	2212.3	4019.8	3161.6
1306	275.4	574.6	619.2	553.8
1606	1442.8	746.1	--	5662.3
1612	1797.1	2720.2	--	5166.7
2315	296.4	456.8	651.9	564.4
2407	3094.5	--	7014.2	8062.6
2502	127.2	264.6	564.8	12711.1
2507	--	106.3	--	1709.8
2526	4077.0	4491.2	--	15728.9
Totals	0	2	4	12

Fig. 10-1. Excision Gel Selection dialog.

Select the analysis set you want to cut. The dialog box includes a table of the spots in the analysis set and their quantities in the different gels in the matchset. The first spot in the set is magnified and highlighted in each of the gels of the matchset.

10.1.a Step 1. Select Analysis Set.

To select a different analysis set, click on the pulldown button next to the analysis set name and select from the list. The number of spots in the set is displayed next to the name.

10.1.b Step 2. Set Selection Mode.

In this step, you can select parameters for automatically selecting which gels to cut.

Select the Auto-select option to automatically select the best gels to cut from based on spot quantity and quality. When you choose this option, the auto-selection parameters will become available, and the gels will be selected in the table based on these parameters.

Spot quantity usually varies among the gels in a matchset. In general, you want to cut from gels in which spot quantity is higher (to get more sample). The Tolerance ratio is used to determine the range of spot quantity that is acceptable when selecting which gel to cut a particular spot from. If the quantity of a spot in only one gel falls within the tolerance ratio, that gel will be auto-selected for the spot cut. However, if the quantity of a spot in two or more gels falls within the tolerance ratio, then the auto-select mechanism will consider which gel will best minimize the time and effort required to switch gels on the spot cutter platform.

For example, say that you have five gels in a matchset, and with a tolerance ratio of 1 (the lowest setting) auto-selection specifies different cuts in all five gels, requiring to you to change gels in the spot cutter five times. However, by increasing the tolerance ratio slightly, auto-selection finds that three gels contain spots whose quantities fall within the specified tolerance, and it adjusts the cuts so that only those three gels are selected. This will save you considerable time and effort switching gels in the spot cutter.

Select the Include Spot Quality checkbox to consider spot quality when auto-selecting spots. When this checkbox is selected, a well-resolved spot of sufficient quantity will be favored over a less well-resolved spot of higher quantity (within the specified tolerance ratio).

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To make all the cuts from the same gel, select the Single Gel option button. Then select the gel name from the pulldown box. All the cut requests listed in the table will be located in that gel.

To choose gels and spots manually, click on the Manual option button. The spots will appear selected in the table based solely on quantity, and you can then click on which spots to cut.

10.1.c Step 3. Specify Cuts.

In this step, you review the cut selection in the table and make changes as appropriate.

The table in the Excision Gel Selection tool lists the spots in the analysis set (by Standard Spot Number) and their quantities in the different gels. The columns are gels, the rows are spots. You can use this table to make cut requests or modify the auto-selection of cuts.

Note: Only those gels with cut requests will be listed in the table, unless the Show All Gels checkbox is selected.

The spots selected to be cut are highlighted in the table in blue. The table will initially show one cut request per spot—that is, one highlighted blue table cell per row.

If you click on the Standard Spot Number (SSP) at the beginning of each row, that spot will appear highlighted in the actual matchset. Study the size and quality of each spot in the gels in the matchset when selecting spots to cut.

You can manually add or remove cut requests by clicking on the table cells. Click on a blue cell to delete a cut request. Click on a white cell to make a cut request. To cut the same spot in different gels, click on multiple white cells in the same row; they will be highlighted in blue. You can add as many cut requests in as many gels in the matchset as you want.

At the bottom of the table, the total number of cut requests per gel is shown.

Table Options

Under Table Options, click on Show (Quality) to display the spot quality in the table cells in parentheses following the spot quantity.

Click on Show Ratio to display a column in the table showing the quantity tolerance ratio across all gels for each spot. This value is used in conjunction with the Tolerance Ratio setting, described above.

Show All Gels will show columns for all the gels in the matchset. If this is unselected, only those gels with cut requests will be listed in the table.

Image Options

Display Cuts marks each cut request with a red box in the matchset gel(s), and also shows the actual cut request circles. These circles indicate the region of the spot that will be cut. Note that if the Multiple Cuts/Spot checkbox is selected, multiple circles may appear on some or all spots.

Active Gels Only displays only those gels listed in the table. Note that if the Show All Gels checkbox is not selected, only those gels with cut requests are listed in the table, and consequently the matchset will reconfigure to show only those gels.

Auto-zoom on Cut will automatically magnify each spot you select in the table.

Multiple Cuts/Spot will automatically calculate the size of each spot that has a cut request and, if the spot is sufficiently large, specify multiple cuts of that spot. The multiple cut circles will be displayed in the matchset.

10.1.d Performing the Cuts

When you have made your selections, click on the Excise Spots button to close the Excision Gel Selection tool and open the Integrated Excision Tool.

10.2 Integrated Excision Tool

The Integrated Excision Tool is used to cut a set of spots from the gels in a matchset using Bio-Rad's ProteomeWorks Spot Cutter system. It provides all the features of the Basic Excision Tool, plus the ability to cut spots from PDQUEST matchsets.

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Before using this tool, you must have selected the spots and the gels to cut using the Excision Gel Selection tool, as described above. The matchset containing the gels must be open and active. Also, make sure the spot cutter is connected with the computer, is switched on and has warmed up, and make sure the camera is connected and has been turned on.

See the spot cutter hardware manual for instructions on installing the platform, camera, and associated hardware and software.

Select Integrated Excision Tool from the Identification menu. The software will connect with the spot cutter platform and camera, and the Integrated Excision Tool dialog box will open.

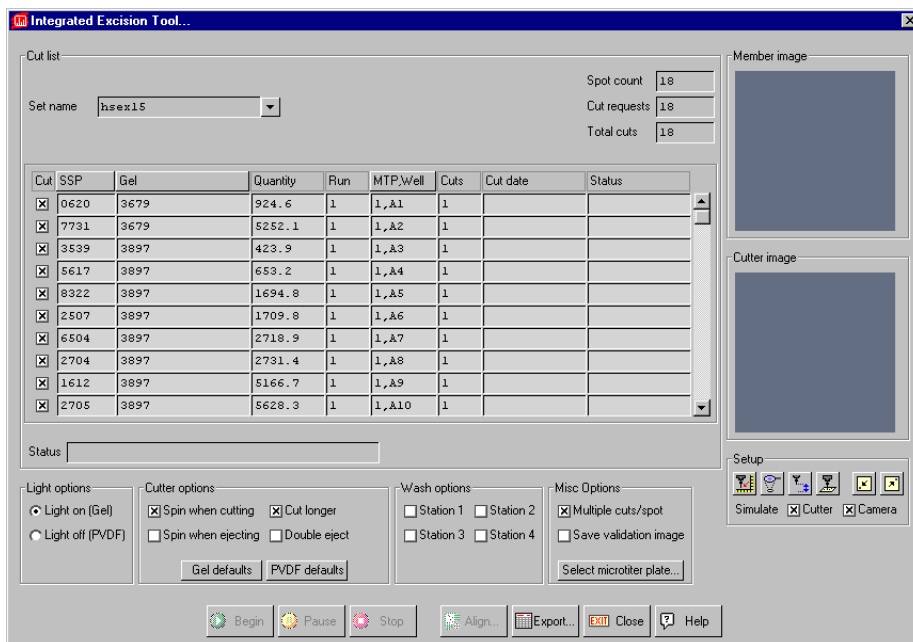


Fig. 10-2. Integrated Excision Tool.

10.2.a Spot Cutter Setup

Before you can use the spot cutter for the first time, you must focus the camera and adjust the cutting tip height. You must also adjust the spot cutter settings and calibrate the instrument.

Note: If you have already focused and calibrated the spot cutter using the Basic Excision Tool, you do not need to perform these steps again.

Focus Cutter Camera

The Focus Cutter Camera dialog box controls the camera exposure setting. Focusing and aligning the camera are described in the spot cutter hardware manual.

Click on the Focus Cutter Camera button to open the dialog. The platform light on the spot cutter will turn on and the camera will focus on a small region at the center of the gel or membrane and begin taking continuous exposures. The exposures will appear in the Focus window at the rate of about 2 per second.

At the top of the dialog, click on Auto-Expose to reset the exposure time of the camera.

The Time field shows the time of each exposure in milliseconds. The Ceiling field shows the highest pixel value in the focused region. The Levels field shows the levels of gray scale in the image. All of these settings are automatically determined by the software, based on the position and aperture of the camera lens.

Manually adjust the camera lens and position as described in the hardware manual to optimize the image. When you are satisfied with the image and the levels of gray scale, click on Stop to close the Focus dialog.

Set Cutting Tip Height

Click on the Set Cutting Tip Height button to adjust the distance the cutting tip drops down to cut a spot. The adjustment dialog box will open, and the cutting tip will move to the center of the gel.

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Click on Lower Tip to drop the cutting tip to its lowest position. In this position, manually adjust the cutting tip height as described in the hardware manual until it presses lightly against the surface of the cutting mat.

Raise Tip raises the cutting tip. Test Cut performs a test cut in the center of the gel. Click on Done to accept the settings and return the cutting tip to home position.

The cutting tip height only needs to be set once, unless the spot cutter gets out of alignment.

Spot Cutter Settings

Click on the Spot Cutter Settings button to open the settings dialog box for the cutter.

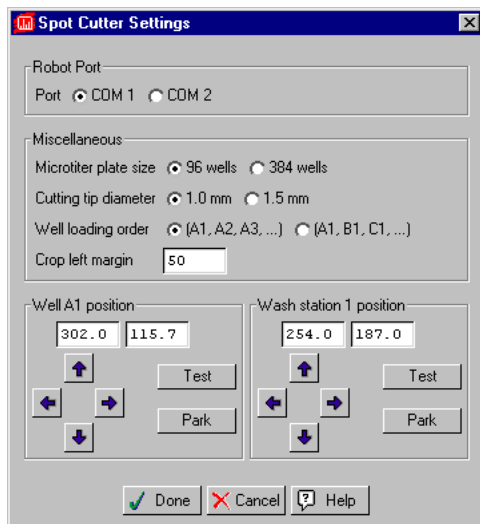


Fig. 10-3. Spot cutter settings.

In the dialog box, select the communications port to the spot cutter by clicking on COM 1 or COM 2.

Specify the microtiter plate size you are using by clicking on 96 wells or 384 wells.

Specify the diameter of the cutting tip by selecting 1.0 mm or 1.5 mm. Note that the cut request circles that you make on the image reflect the actual size of the cutting tip, and will change based on the tip size selection.

Specify the order in which you want to load the wells on the microplate by selecting A1, A2, A3, ... or A1, B1, C1, ...

Crop Left Margin repositions the image in the Basic Excision Tool window to the left. Enter a value (in screen pixels) in the field to reposition the image. Note that you must click on Done to close the settings dialog and then acquire a new image for this change to take effect.

Well A1 position

The fields and buttons in this section are used to calibrate the position of the microplate for proper loading of samples. The number fields contain the X and Y coordinates (in millimeters) of well A1 on a plate in the spot cutter plate holder. Once you have set the position of this well, the cutter robot will automatically “know” the positions of the remaining wells.

To set the well A1 position, position your cursor over the number in the X or Y coordinate field and press Enter on your keyboard twice. The tip will move to that position without attempting to lower into the well. If the tip is not centered directly over the well A1 on the microplate, you will need to adjust the numbers in each field.

You can either type a new value directly in each field, or (for small adjustments) use the arrow buttons to adjust the tip by fractions of a millimeter. The cutting tip will move as you either enter new numbers or click on the buttons.

Note that the arrow buttons are oriented as if you were facing the front of the spot cutter; also, if you are facing the cutter, higher numbers in the first field will move the cutting tip to the right, while higher numbers in the second field will move the cutting tip toward the back of the instrument.

When you are satisfied with the position of the cutting tip, click on Test to lower the tip into the well. The tip should be centered inside the well.

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Park returns to the tip to its home position.

Wash Station 1 position

Calibrate the position of Wash Station 1 on the cutting platform by following the steps described above for Well A1. Once you have set the position of station 1, the cutter robot will automatically “know” the positions of the remaining wash stations.

Spot Cutter Calibration

Before you can begin excising spots, you must calibrate the spot cutter. This procedure requires a PVDF membrane.

Note: Before calibration, you should focus the cutter camera and align the wash station position. You should make sure the spot cutter camera image is properly centered using the Crop Left Margin setting.

Click on the Calibrate Cutter button. A series of dialog boxes will guide you through the calibration procedure. At any point in the procedure you can move back to repeat a step.

Note: Calibration creates several parameter and image files in the Install.prm directory in the Discovery Series directory on your hard drive. These files should not be moved, deleted or altered, or calibration will be affected.

When calibration is complete, you can proceed with normal use of the spot cutter. Once the cutter is calibrated, you do not have to repeat the procedure during normal day-to-day use. You may have to recalibrate if the instrument is moved.

10.2.b Cutting Spots

The control panel is designed to guide you through the spot cutting procedure. Click on the link for more information about each step.

1. Select the Cut List. In this step, you select the cut list created using the Excision Gel Selection tool, and specify which cuts you want to perform for this run.

2. Select Cut Run Options. In this step, you select the options for the current cut run.
3. Select Microtiter Plate. In this step, you select a microtiter plate and specify the starting well.
4. Align the Images. Next, you click on the Begin button and align the matchset image and the spot cutter image.

Step 1. Select the Cut List

At the top of the Integrated Excision Tool dialog, click on the pulldown button to select from the available cut lists (created using the Excision Gel Selection tool).

The cut list will be displayed in table format in the middle of the dialog box. Each spot to be cut is listed in the first column, and the gel to be cut is listed in the second column. Additional columns list spot quantity, microplate well, number of cuts per spot, cut date, and cut status.

At the beginning of each row is a Cut checkbox, specifying the spots that will be cut in the current cut run. Deselect a checkbox to not perform that cut.

After each cut is performed, its status will be listed as "completed" and its cut checkbox will be deselected.

After a cut run is complete, you can reselect checkboxes in the list to recut any missed spots or cut spots that were skipped in the previous run.

The cuts are initially listed in the table in order of gel, so that all the cuts will be performed on one gel before you are prompted to load the next one. To sort the list by SSP number, click on the SSP button at the top of the table. To sort the list by microtiter plate well number, click on MTP Well button. (Note that these options may require you to switch gels on the cutting platform more often than if you sort the cuts by gel.) To resort the list by gel, click on Gel button in the table.

Step 2. Select the Cut Run Options

Before beginning a cut run using the Integrated Excision Tool, you must select the cutter options for that run.

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Select the Light On option if you are imaging a gel, or Light Off if you are imaging a PVDF membrane. (The platform light should be off for PVDFs, because the ambient light will produce a better image.)

Select Spin When Cutting to make the cutting tip spin as it cuts.

Spin When Ejecting will cause the cutting tip to spin as it ejects the cut into a microplate well.

Cut Longer will cause the cutting tip to hold in the 'down' position longer when making a cut.

Double Eject will cause the cutting tip to emit two blasts of air when ejecting a cut into a microtiter plate well.

The Gel Defaults button automatically selects the most useful options when cutting gels. The PVDF Defaults button automatically selects the most useful options when cutting PVDF membranes.

Select the wash stations to use after each cut by clicking on one or more checkboxes (Station 1, Station 2, etc.).

Under Misc. Options, select Multiple Cuts/Spot to specify multiple cuts of larger spots. Note that the software will automatically calculate the size of each spot based on its Gaussian profile in the matchset gel when determining whether to make multiple cuts.

Save Validation Image will create and save an image of the gel on the spot cutter platform when the cut run is complete. You will be prompted to name the validation image.

Step 3. Select the Microtiter Plate

Click on the Select Microtiter Plate button to select a plate name and starting well for the current cut run.

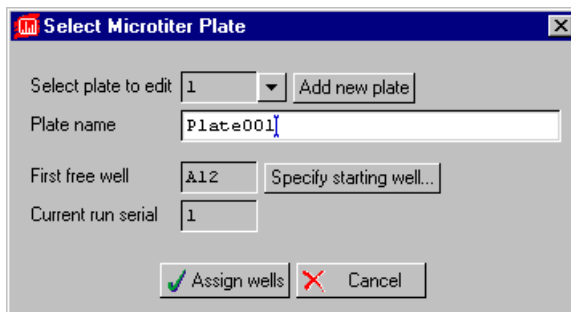


Fig. 10-4. Selecting a microtiter plate.

In the dialog box, a default plate name will appear in the Plate Name field. Note that by default plates are named sequentially (Plate001, Plate002, etc.). However, you can enter any name you want in the field.

Click on the Specify Starting Well button and, in the pop-up box, specify the first well you want to load in the current cut run. If you specify a starting well other than the default of A1, the skipped wells will be noted in the box. Click on Done to make your selection.

Note: Specifying a different starting well is essential if you want to load spots from different matchsets or different analysis sets on the same microtiter plate. You would load your spots from one matchset/analysis set on the plate, then select a different matchset/analysis set, skip the previously loaded wells.

To select a previous microtiter plate, click on the pulldown button next to Select Plate to Edit and select a plate name from the list. Note that if a previous cut run has been performed using this plate name, you cannot specify a new starting well. Any additional cuts added to this plate will be loaded into the remaining free wells; any recuts will be loaded into the appropriate previously loaded wells.

If a previous plate is selected and you want to define a new plate, click on the Add New Plate button. The new plate will be assigned a default name, displayed in the plate name field.

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The Current Run Serial field displays the serial number of the cut run. The cut run serial number increments whenever you perform a previous cut run on a new matchset gel or gels. For example, if you first cut spots from one gel in a matchset and do a cut run, then cut the same spots from a different gel in the matchset and do a run, the serial number for the second run would be 2.

When you have made your selections, click on the Assign Wells button to complete the microplate selection.

Step 4. Align the Images

After you have selected the cut list and the cut run settings, click on the Begin button. You will be prompted to place the first gel in the cut list on the spot cutter platform. Click on Gel Ready when you have positioned the gel.

You can also align the images by clicking on the Align button.

The spot cutter camera will acquire an image of the gel, and the Align Analysis and Cutter Images dialog box will open.

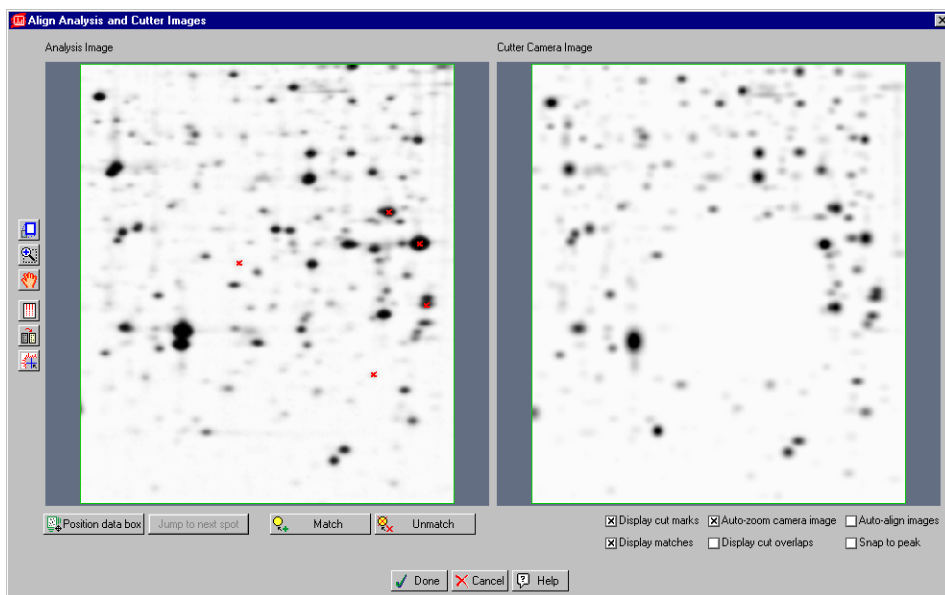


Fig. 10-5. Aligning the images.

In the dialog box, the left window displays the matchset image, and the right window displays the cutter camera image of the corresponding gel. Now you must match the actual gel spots in the camera cutter image with the cut list spots in the matchset image.

First, note that a green Data Area Box surrounds the matchset image, and a corresponding green box appears on the camera cutter image. Position your cursor over the right-hand (camera cutter) image box, and drag it so that it defines the same area as the box in the left image.

Now, click on the Zoom Box button included in the dialog box and magnify a spot to be cut in the left-hand image. It will be marked with a red X. Click on the Match button and click on this spot.

The cursor will jump to the right-hand window. Find the same spot in this image and click on it. The spot will be marked in both windows with a green circle, indicating that it has been matched and is ready to cut.

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Note: Note that the green circle represents the actual cut made by the cutting tip in the gel, and will vary in size depending on the cutting tip size selected under Spot Cutter Settings.

Click on the Advance button to jump to a magnified view of the next spot in the cut list. Repeat the above procedure for every spot to be cut in this gel. When all the cut spots have been matched, click on Done.

The following display options are available in the align images dialog box:

Display Cut Marks will mark the spots to be cut with red X's in the analysis image.

Display Matches will mark matched spots with green circles in both the analysis image and cutter camera image.

Auto-zoom Camera Image will automatically change the magnification and position of the (right-hand) camera cutter image to imitate the display in the analysis image.

Display Cut Overlaps will mark overlapping spots with ellipses in the Analysis image, and the percent overlap will be displayed next to the overlapped cluster.

Auto-align Images will attempt to automatically match each spot you click on in the Analysis image with the spot in the camera cutter image, based on previous matches.

Snap to Peak automatically centers the cut circle on the spot peak in the camera cutter image (calculated from the spot intensity).

Performing a Cut Run

When you click on Done in the Align Images window, the dialog box will close and you will be prompted to place a microplate in the plate holder. When you click on OK, the cut run will begin.

As each cut is being made, it will be highlighted in the table list. After each cut has been made, the cut will be marked as *completed* in the cut list, and the checkbox next to that spot will be automatically unselected.

When all the spots on one gel have been cut, you will be prompted to place the next gel on the spot cutter platform and align that gel image with the matchset image (see above). Repeat the steps described above until the cut run is complete for each gel.

Click on Pause to pause the cut run. Note that the cutter will pause only after the most recent cut has been deposited in a well.

Click on Stop to stop the cut run. The cutter will stop only after the most recent cut has been deposited in a well. Any remaining cuts will remain selected in the cut list.

When the cut run is finished, the cutting tip will return to home position. If you have selected the Save Validation Image checkbox, the validation image will be acquired and you will be prompted to save it. The validation image is a cutter camera image showing the cut spots.

Recutting Spots

After a cut run is complete, review the cut holes in the gel/membrane. You can then select Cut checkboxes in the cut list to recut any missed spots or cut spots that were skipped in the previous run. At any time, you can reopen an old cut list in the Integrated tool and perform recuts or cut previously skipped spots.

As before, click on Begin to begin the cut run.

10.2.c Exporting the Cut List

You can export the information in the cut list as a spreadsheet file. Click on the Export button, enter a name for the file, and select the directory in which to create the file. The export command will create a tab-delimited file that can be opened as a Microsoft Excel worksheet or as a simple text file.

11. Mass Spectrometry Analysis

After you have cut spots from a gel using the Integrated Excision Tool and digested them, you can select the parameters for processing them using Micromass's MALDI or MS/MS mass spec instruments. You can then import the mass spec data back into PDQUEST in the form of annotations.

11.1 Creating a MassLynx Batch File

After you have cut spots from a gel using the Integrated Excision Tool and digested them, you can select the parameters for processing them using Micromass's MALDI or MS/MS mass spec instruments.

First you set up a project file using MassLynx software, then you export the processing parameters from PDQUEST into the project file. After you have identified the proteins, you can import the results back into PDQUEST.

Note: See Micromass's MassLynx documentation for information on project files and how to use them.

Before you can generate a MassLynx batch file, you must map the directories where your Micromass projects and parameters are stored. Go the Edit menu, select Preferences, and click on the Paths tab. Map the directory where your MassLynx projects are stored using the Micromass Projects path. Map the directory where the mass spec parameters are stored using the Micromass Install path. See Path Preferences for more information.

Note: These paths must be mapped correctly, or you will not be able to access the proper batch file parameters.

With the matchset containing your cut spots open, select Generate MassLynx Batch File from the Identification menu to set up the batch file.

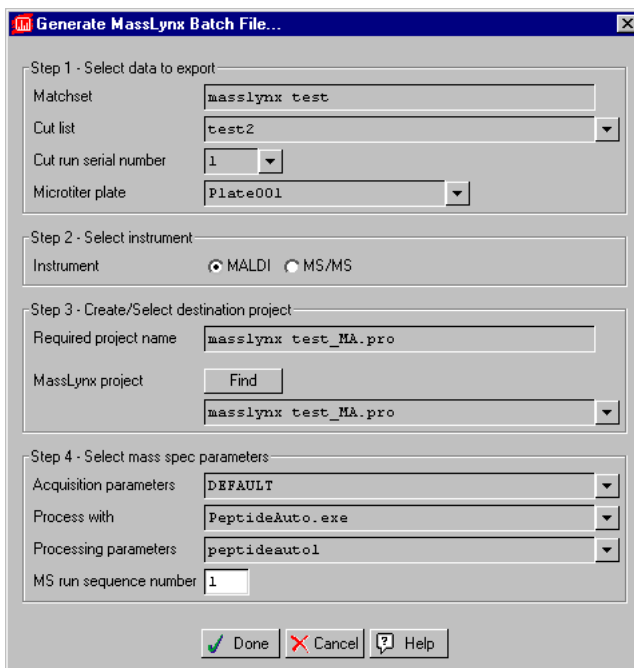


Fig. 11-1. Creating a MassLynx batch file.

Step 1

In the dialog box, the matchset name is listed next to Matchset. Select the cut list you want to analyze from the Cut List pulldown list. If you have performed more than one cut run on this cut list, select the number of the run from the Cut Run Serial Number pulldown list.

Next, select the plate used for the cut run using the Microtiter Plate pulldown list. If more than one plate was used for the run, select the first plate you want to process.

Step 2

Select the appropriate Micromass instrument using the MALDI or MS/MS option button.

Step 3

The Required Project Name is the name of the MassLynx project that must be created for this cut list. Open MassLynx and create a project with this name. See MassLynx documentation for information on how to create project files.

Note: We recommend that you create a MassLynx project file from an existing project or a default template, to ensure that the default parameter files are stored in the correct location inside the project.

When you have created a project, you can click on the Find button (if the path to the project directory is mapped correctly as described above) or navigate to it yourself by clicking on the MassLynx Project pulldown button.

Step 4

In this step, you specify the mass spec parameters to be used for processing the proteins. See Micromass documentation for descriptions of the parameters and their functions.

Note: You must map the Micromass paths as described above to specify the parameters.

Select the acquisition parameters using the Acquisition Parameters pulldown list.

Select the processing program you want to use using the Process With pulldown list.

Select the processing parameters using the Processing Parameters pulldown list.

You can specify a mass spec run number in the MS Run Sequence Number field.

When you have made all your selections, click on Done.

11.2 Importing MassLynx Results

After you have completed a mass spec analysis and processed the data using MassLynx, you can import the results back into PDQUEST and include them in annotations for the spots you have analyzed.

Note: Before importing MassLynx results, you must select the Bio-Rad output checkbox in the MassLynx project settings. To access this checkbox, go to the ProteinLynx menu in MassLynx and select Setup ProteinLynx. One of the setup steps is called Select Output. In this step, under Database Searching, select the Bio-Rad (full text output) checkbox.

With the appropriate matchset open, select Import MassLynx Results from the Identification menu.

Import MassLynx Results...

Step 1 - Select project

Matchset:

Instrument: ☒ MALDI ☐ MS/MS

Required project name:

MassLynx project:

Step 2 - Select run

Step 3 - Select sample

A,1	20-APR-1999
A,2	20-APR-1999
A,3	20-APR-1999

Step 4 - Select primary hit

6-PHOSPHOFRUCTOKINASE ISOZYME I - Escherichia coli.				
13 peptides	13.92% coverage	115.93kDa	5.41pH	score 20617.3
6-PHOSPHOFRUCTOKINASE ISOZYME I - ESCHERICHIA COLI.				
13 peptides	13.88% coverage	116.35kDa	5.41pH	score 20556.8
6-PHOSPHOFRUCTOKINASE ISOZYME I (EC 2.7.1.11) - ESCHERICHIA				
13 peptides	13.88% coverage	116.35kDa	5.41pH	score 20556.8
CVU89933 NID: g2190711 - Cloning vector pBI-G.				
13 peptides	13.79% coverage	117.03kDa	5.66pH	score 20417.1

Score: ☒ Joint probability ☐ MDWSE

Buttons: Apply, Protein Probe..., EXIT, Close, Help

Fig. 11-2. Importing MassLynx data.

In Step 1, the matchset name will appear in the Matchset field. Select the instrument used in the analysis by selecting the MALDI or MS/MS button.

The Required Project Name is based on the matchset name. Use the MassLynx Project pulldown list to select the project with this name from the appropriate directory.

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Note: If you have mapped the path to the Micromass project directory using the Preferences dialog (Edit > Preferences), the correct MassLynx project will be automatically selected. See Path Preferences.

In Step 2, the mass spec runs that have been performed in this project are listed. Click on a run name to select it.

In Step 3, the samples in the selected run are listed by microtiter plate well and sample date. Click on a sample to select it.

In Step 4, the primary hits for each sample are listed in alphabetical order. The number of peptides, percent coverage, molecular weight and pH are listed for each protein. You can display the Joint Probability or MOWSE score for each hit by selecting the appropriate option button at the bottom of the list.

Select the protein that you feel is the most likely identification, and click on Apply to import all the information into spot annotations in PDQUEST.

Note: The protein you select will be used as the name for the spot. (All the hits will be included in a special annotation category called Hit Table.) If you change your mind about the protein name, you can change the information directly in the Spot Annotations dialog, or reimport the results with a different selection.

Click on Protein Probe to launch the Protein Probe browser for each sample.

Annotation Categories Created

When you click on Apply, the following annotation categories are created, and the mass spec data for each spot is entered as an annotation in the appropriate category. (See section 8.2.b for more information on annotation categories.)

- Protein Name (The annotation entry in this category is the name of the protein selected by you from the hit list).
- Accession Number (The annotation entry in this category is the database number keyed to the selected protein name. Note that some databases (e.g., Swissprot) include accession numbers, while others do not. If the database used to identify the protein does not have accession numbers, this category is not created.)

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- Hit Table (The annotation entry in this category is the entire list of proteins identified for the spot.)
- Protein Probe (This file name category provides a link to the Protein Probe file for the spot.)
- Mass Spec Score (Both the Joint Probability and MOWSE scores are listed.)

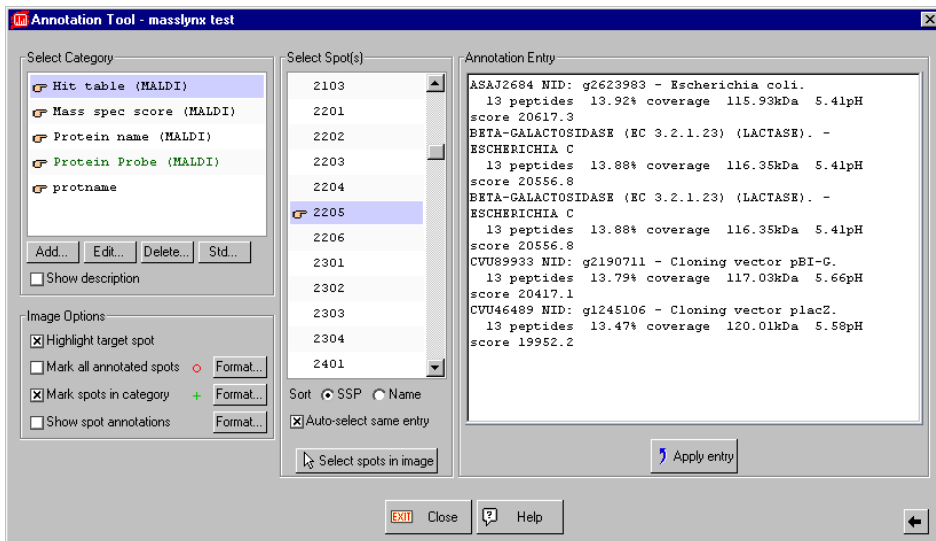


Fig. 11-3. Example of the Hit Table annotation category and entries.

In addition, the protein name is included in the standard protname category (which will be created if it doesn't already exist), so you can sort the spots by protein name in the Select Spot list.

Note: Each category is specific for the mass spec instrument used. For example, if you import information from both MALDI and MS/MS, you would create both a Hit Table (MALDI) category and a Hit Table (MS/MS) category.

11.3 Mass Spec Score Overlay

If you have imported mass spec data for spots using Import MassLynx Results, you can display overlays indicating the confidence level of the mass spec protein identification.

Select Mass Spec Score Overlay from the Identification menu.

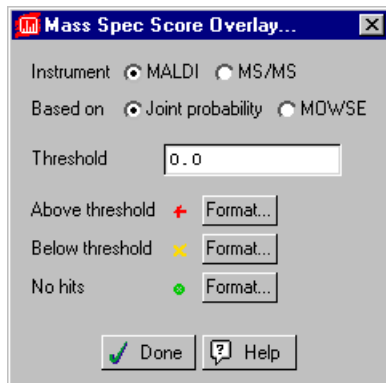


Fig. 11-4. Mass Spec Score Overlay dialog.

In the dialog box, indicate the instrument you want to report scores for by selecting MALDI or MS/MS. Note that if there is no data for an instrument, no spot overlays will be shown.

Next, select the type of score by selecting Joint Probability or MOWSE.

Enter the threshold value for the score in the Threshold field. Spots above this threshold can be considered confident hits; spots below this threshold are not.

Click on the Format buttons next to Above Threshold, Below Threshold, and No Hits to select the appearance of the different overlays.

No Hits are spots with no hits or with an unknown protein identification. This means that the mass spec tried and failed to recognize the sequence.

11.4 Protein Probe

If you have imported mass spec data for spots using Import MassLynx Results, you can launch the Protein Probe browser from within PDQUEST to review the protein information for those spots.

Note: Protein Probe must be installed on the same computer as PDQUEST to use this tool.

With your matchset open, select Protein Probe from the Identification menu and click on a spot for which you've imported data. The Protein Probe browser will open.

See your Protein Probe documentation for information on how to use the browser.

12. Graphs and Reports

The graphs and reports described in this section are all available from the Reports menu.

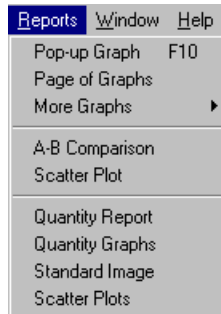


Fig. 12-1. Reports menu.

12.1 Histogram Graphs

Histogram graphs can be used to quickly compare a spot's quantity in each member of a matchset. A brief look at a histogram can give you a sense of the general trends in spot quantity.

Histograms are also useful as tools for detecting problems such as spots that are not matched and spots whose quantitation is suspiciously off the average. Reviewing the histograms of spots in an Analysis Set is a quick and easy way to confirm that the members of the Analysis Set meet the criteria that was specified when the set was created.

12.1.a Data Displayed in the Histogram

Each bar in a histogram for a spot represents the spot's quantity in a member of a matchset. If replicate group quantitation is selected, each bar represents the spot's quantity in a replicate group.

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The bars from left to right represent the gels in the order in which they were loaded. To check the order, use Assign (ALT+A). If you would like to change the order, use the function Reorder Members (Match > Create/Edit Matchset > Reorder Members).

As long as the spot is matched to the Standard, you can display a histogram for it.

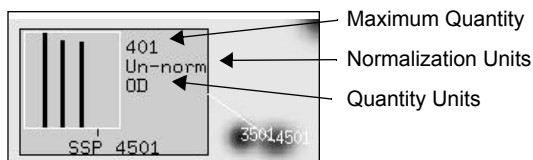


Fig. 12-2. Sample histogram with individual quantitation.

The Standard Spot (SSP) number is displayed beneath the histogram.

The number at the upper right of the histogram is the quantitation of the maximum bar in the graph. The other bars are drawn proportional to the highest bar.

Beneath the maximum quantitation, the normalization units are displayed. Normalization is a process by which quantitative data from different gels are adjusted so that you can compare different samples to one another (for more information on normalization, see section 7.6).

If a spot is saturated, the bar representing it on the histogram will be shaded red instead of black.

If replicate groups are defined for the matchset (see section 7.5), you can look at the quantitation of individual spots or the average data of duplicate gels. You can specify which way you want to review spot quantitation by selecting Quantitation Mode from the Analysis menu.

If you specify Replicate Group Quantitation, the bars on the histograms will display not only the maximum spot quantitation for each replicate group, but also the average spot quantitation and the standard deviation. Each bar represents one replicate group. The order of the bars corresponds to the order in which the replicate groups were made.

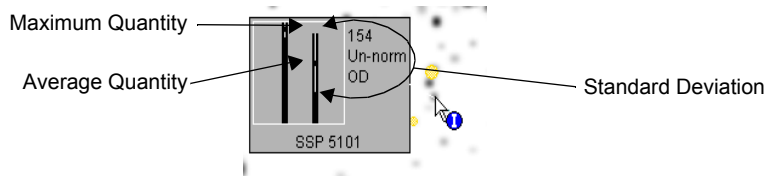


Fig. 12-3. Sample histogram with Replicate Group Quantitation.

12.1.b Types of Graphs

Spot Review Tool

Spot Review Tool on the Analysis menu (section 7.1) displays a page of graphs alongside the matchset. You can use this tool interactively with the matchset to review spots.

Pop-up Graph

Pop-up Graph (F10) shows a histogram next to a selected spot. Select Pop-up Graph from the menu or toolbar, then click on a spot.

With Pop-up Graph, only one histogram can be displayed in a window at a time. If you select another spot, the first histogram will disappear and a new one will be displayed.

Margin Graph

Margin Graph (SHIFT+F10) on the Reports menu and Graph toolbar will display histograms for several spots along the left and the right margins of the screen. Select Margin Graph, then click on several spots. Their histograms will be displayed in the left and right margins.

Depending on the size of your window, you will be able to display up to 10 margin graphs at once. If you keep clicking on spots after you've displayed 10 graphs, you will begin replacing graphs with your more recent selections.

Column Graphs

Column Graph on the menu and toolbar will fill the right and left margins with histograms of spots in the Standard. Select Column Graph; the spots whose graphs are displayed are all located in a narrow column of the image, centered around the location of the cursor.

To change the column of spots, use the Graph Advance/Graph Retreat tools (see below).

Graph Set

Graph Set (Analysis > Analysis Sets > Graph Set submenu) will graph the spots in an analysis set. With the analysis set selected, select the command. Use Graph Advance/Graph Retreat to graph different regions of spots in the set.

Graph Partial Matches

Graph Partial Matches (Match > Edit Matches submenu) displays histograms of each partially matched spot in the standard. Select the command from the submenu and click in the standard. Use Graph Advance/Graph Retreat to graph different regions of spots.

Graph Erratic Responses

Graph Erratic Responses (Match > Edit Matches submenu) displays histograms of each erratic spot in the standard. Select the command from the submenu, click in the standard, enter a fold change factor in the pop-up box, and click on Apply to display the graphs. Use Graph Advance/Graph Retreat to graph different regions of spots.

Graph Advance/Graph Retreat

If you are displaying Column Graphs or graphing spots using the Graph Partial Matches or Graph Analysis Set commands, you will need to shift the display to graph spots in different regions of the image. The commands for doing this are located on the Reports menu or the Graph toolbar.

Graph Advance (ALT+.) will shift the column of spots being graphed to the right. Graph Retreat (ALT+,) will shift the column of spots being graphed to the left.

12.2 A-B Comparison

The A-B Comparison function is used on matchset Standards to highlight and compare the spots that appear in two different matchset members.

Select A-B Comparison from the Reports menu and click on the Standard of the matchset in which you want to perform the comparison. A pop-box will prompt you to select the first member you want to compare (Member A). Click on the member name. A second pop-up box will prompt you to select the second member you want to compare (Member B).

When you select the second member, the box will disappear and the spots in the Standard will appear highlighted.

Spots that are marked in both members will be highlighted in green, spots that appear only in Member A will appear highlighted in red, and spots that appear only in Member B will appear highlighted in blue.

Spots that appear in neither member will not be highlighted.

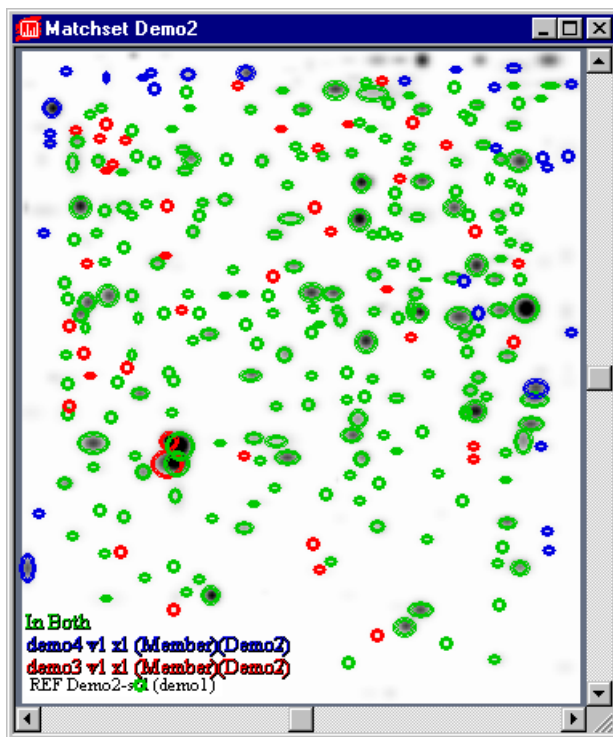


Fig. 12-4. Example of an A-B Comparison display.

A color key will appear at the bottom of the Standard.

12.3 Scatter Plots

The Scatter Plots command on the Reports menu allows you to print out scatter plots for all the members of a matchset. A scatter plot shows the relatedness of two different gels that have been matched in a matchset.

Note: To display a scatter plot for any two gels in a matchset, use the Scatter Plot Tool on the Analysis menu (section 7.3).

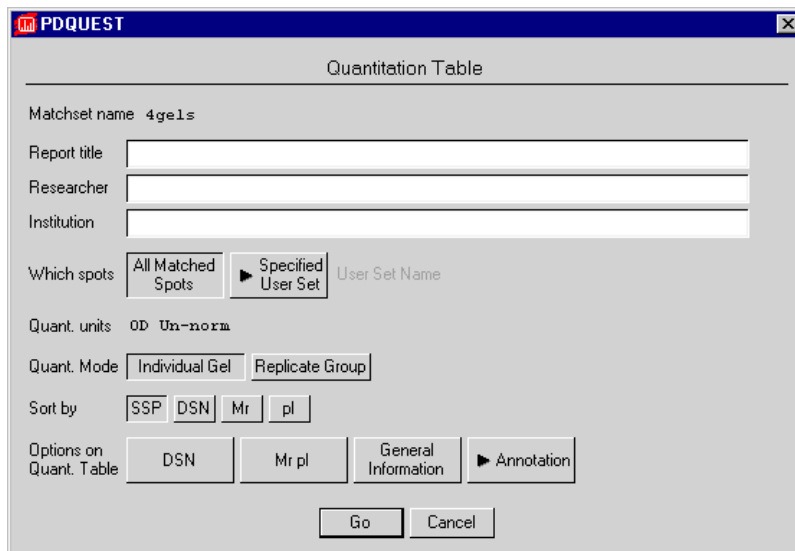
When you select Scatter Plots from the Reports menu, a dialog will open in which you can enter a report title, researcher name, and the name of the research institution.

Next, specify whether you want scatter plots of All Matched Spots or only the spots in a Specified Analysis Set. If you select the analysis set option, a pop-up box will list the available analysis sets in the matchset. Select a set and click on OK.

Click on the Go button to print. The plots will be printed in a report format.

12.4 Quantity Reports

If you select Quantity Report from the Reports menu, the general report information dialog box will open.



The image shows a software dialog box titled "PDQUEST" with a subtitle "Quantitation Table". The dialog box contains several input fields and buttons. The "Matchset name" field is pre-filled with "4gels". Below it are three empty text input fields for "Report title", "Researcher", and "Institution". The "Which spots" section has two buttons: "All Matched Spots" and "Specified User Set", with the latter being selected. To the right of the "Specified User Set" button is a text field labeled "User Set Name". The "Quant. units" field is set to "OD Un-norm". The "Quant. Mode" section has two buttons: "Individual Gel" and "Replicate Group", with the latter being selected. The "Sort by" section has four buttons: "SSP", "DSN", "Mr", and "pl", with "SSP" being selected. The "Options on Quant. Table" section has four buttons: "DSN", "Mr pl", "General Information", and "Annotation", with "Annotation" being selected. At the bottom are "Go" and "Cancel" buttons.

Fig. 12-5. Quantitation Table dialog box.

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You can type in a report title, the name of the researcher, and the name of the research institution in the appropriate fields.

You can specify the spots that will be included in your report: All Matched Spots or the spots in a Specified Analysis Set. If you click on the Specified Analysis Set button, a list of available Analysis Sets will be displayed. The name of the set you choose will appear next to the Analysis Set Name prompt.

If you want to see spot quantitation for each individual member of the matchset, select Individual Gel next to the Quant. Mode prompt. If you want to see the average quantitation for a Replicate Group, click on the Replicate Group button.

You can sort your spots by SSP number, DSN, molecular weight, or isoelectric point by clicking on the appropriate "Sort by" button.

Other Options

If DSN numbers are assigned to the protein spots and you would like to see them on the report, click on the DSN button.

Mr and pI data will be reported if you click on the Mr pI button.

Clicking on the General Information button will include a title page, a list of symbols and abbreviations, and information about the matchset, member gels, and their calibration data in the report.

You can print spot annotations by clicking on the Annotation button and selecting from the list of annotation categories. The report will include annotation information for the spots that have annotation data in the selected category.

Once you have selected your options, click on the Go button at the bottom of the form. This will open a smaller version of the standard print dialog, in which you can select your printer settings and proceed to print.

12.5 Quantity Graphs

If you select Quantity Graphs from the Reports menu, a dialog box will open in which you can specify the format of your quantitation histograms.

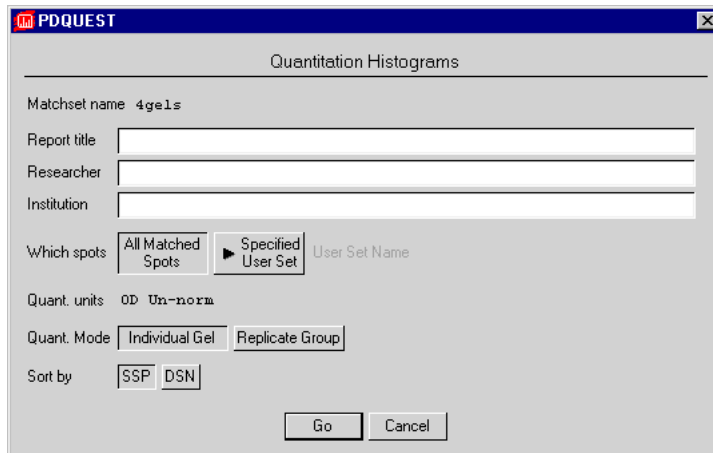


Fig. 12-6. Quantitation Histograms dialog box.

You can type in a report title, the name of the researcher, and the name of the research institution in the appropriate fields.

You can specify the spots that will be included in your report: All Matched Spots or the spots in a Specified Analysis Set. If you click on the Specified Analysis Set button, a list of available Analysis Sets will be displayed. The name of the set you choose will appear next to the Analysis Set Name prompt.

If you want to see spot quantitation for each individual member of the matchset, select Individual Gel next to the Quant. Mode prompt. If you want to see the average quantitation for a Replicate Group, click on the Replicate Group button.

You can sort your spots by SSP number or DSN by clicking on the appropriate "Sort by" button.

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Once you have selected your options, click on the Go button at the bottom of the form. This will open a smaller version of the standard print dialog, in which you can select your printer settings and proceed to print.

12.6 Standard Image Reports

Selecting Standard Image from the Reports menu allows you to print a map of up to 300 spots from the matchset Standard.

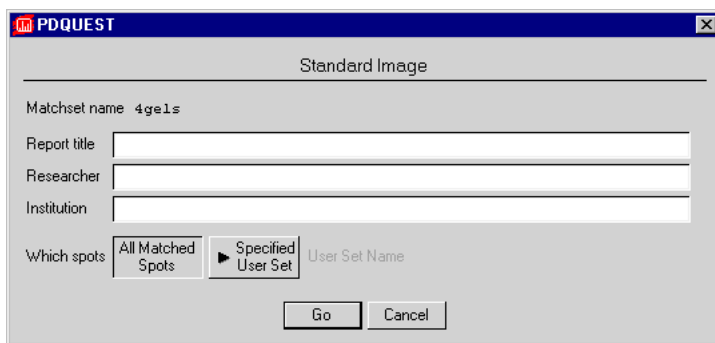
The image shows a software dialog box titled "PDQUEST" with a subtitle "Standard Image". It contains several input fields: "Matchset name" with the value "4ge1s", "Report title", "Researcher", and "Institution". Below these is a section labeled "Which spots" with two radio buttons: "All Matched Spots" (selected) and "Specified User Set". To the right of the "Specified User Set" button is a text field labeled "User Set Name". At the bottom of the dialog are "Go" and "Cancel" buttons.

Fig. 12-7. Standard Image dialog box.

You can type in a report title, the name of the researcher, and the name of the research institution in the appropriate fields.

You can specify the spots that will be included in your report: All Matched Spots or the spots in a Specified Analysis Set. If you click on the Specified Analysis Set button, a list of available Analysis Sets will be displayed. The name of the set you choose will appear next to the Analysis Set Name prompt.

Once you have selected your options, click on the Go button at the bottom of the form. This will open a smaller version of the standard print dialog, in which you can select your printer settings and proceed to print.

13. Printing and Exporting

This chapter describes the commands for printing and exporting images and data. These commands are located on the File menu.

13.1 Printing

The commands for printing images, actual-size images, image reports, and video printouts are located on the File > Print submenu.

1. Print Image prints a copy of the image and any overlays that are currently displayed.
2. Print Actual Size prints an actual-size copy of the image.
3. Image Report prints the image and information about its scan history, number of pixels, data range, etc.
4. Video Print prints images and reports to a video printer.*

*Video printing requires installation of the video board and cable that come with the Gel Doc 1000/2000 or Chemi Doc gel documentation system. The video board and cable can also be ordered separately.

13.1.a Printing Images

Print Image prints a copy of the active image window and any image overlays that are displayed.

File > Print > Print Image opens the Print Image dialog box.

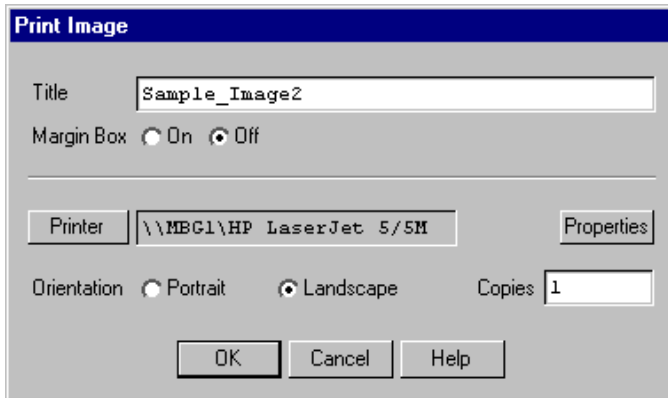


Fig. 13-1. Print Image dialog box.

At the top of the dialog box, enter the title that you want to appear above the image when you print.

Choose whether or not a margin border should appear around the edge of the image by clicking either the On or Off button next to the Margin Box prompt.

Click on Printer to display a list of your available printers. After you have selected a printer, you can click on Properties to select the standard Windows print settings for that printer.

Note: If you are using Windows 95 with certain printer drivers, you may experience difficulty printing some overlays and overlay symbols on images. While there is no single solution to this problem, you should try adjusting some settings in the Printer Properties dialog box, under the Graphics and Advanced tabs—specifically, the Dithering, DPI, Rasterizing, and RET settings.

Select the paper orientation by clicking on Portrait or Landscape next to the Layout prompt. (This can also be set in the Properties dialog box.)

Indicate how many copies should be printed in the text box next to the Copies prompt.

When you are satisfied with all the print parameters, click on the OK button to send the image to the printer.

13.1.b Print Actual Size

You can print your images at their actual size using the Print Actual Size command.

Note: If you are using the Gel Doc, Chemi Doc, Fluor-S, or Fluor-S MAX, you must specify the correct image area size when capturing your images to ensure accurate 1:1 printing. You can specify the image area size in the acquisition window for the instrument. See the chapter on each imaging device for more information.

Select Print Actual Size from the File > Print menu. The standard Print Image dialog box will open (see previous section).

13.1.c Image Report

Image Report allows you to print out a single-page report of an image and its associated information. The format of the report is designed to provide a concise yet thorough summary of the most relevant features of an image for documentation purposes.

The report includes the following information:

- The image.
- The report date.
- The image title and a brief description.
- The directory location.
- The date the image was scanned.
- The type of imaging device.
- The imaging area and number of pixels.
- Pixel size.
- The intensity range, image color, and memory size.
- Image background information.
- Relevant lane and band information.

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To print a image report of a particular gel, select Image Report from the File > Print submenu. This will open a smaller version of the standard print dialog box (see Print Image, above).

Note: TIFF images do not contain all the tagged information that would normally be included in an image file (for example, imaging device, scan date, image color, etc.). For this reason, the Image Report may list this information as “Unknown” for imported TIFF files.

13.1.d Video Printing

The Video Print command allows you to print images and reports on a video printer. To create a video printout of the active window, select Video Print from the File > Print submenu.

Note: Video printing requires installation of the video board and cable that come with the Gel Doc 1000/2000 and Chemi Doc gel documentation systems. The video board and cable can also be ordered separately.

Settings for the Mitsubishi P90W/P91W Video Printer

There are three settings for the Mitsubishi P90W/P91W video printer. Set Contrast to 0, Brightness to 0, and Gamma to 5.

The dip switches should stay in the orientation in which they are shipped: Pin 1 is up (on), and Pins 2–10 are down (off).

13.2 Exporting Matchset Data

Data from a matchset can be exported as an ASCII text file for further analysis using spreadsheet programs.

With the matchset loaded, select Export Matchset from the File > Export submenu.

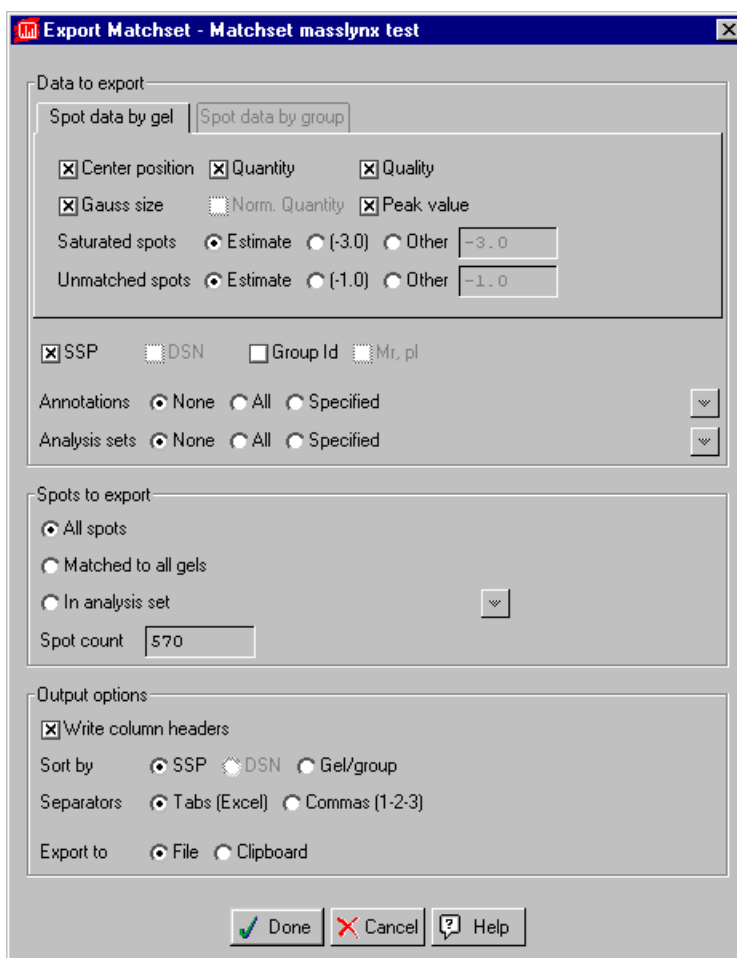


Fig. 13-2. Exporting matchset data.

In the dialog, you can export data for each gel in a matchset (Spot Data by Gel), or for each replicate group in the matchset (Spot Data by Group).

Under Spot Data by Gel, select the data to export for each spot (position, quantity, quality, Gaussian size, normalized quantity, and/or peak value).

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The quantities of saturated spots can be estimated, reported as the value -3.0, or reported in some other way that you specify. If you opt for the latter, click on the Other button and enter the value that should be reported.

If a spot is unmatched in one or more members of a matchset, it can be reported as the minimum quantity needed for a spot to be detected for the gel (Minimum Spot). Alternatively, it can be reported as the value -1.0 or as a value that you enter in the text box next to the Other button.

If you have created replicate groups for the matchset, you can export the averaged data for each group. Under Spot Data by Group, select the data to export (standard position, average quantity, coefficient of variation, standard Gaussian size, average normalized quantity, normalized coefficient of variation, and/or the counts in the member gels).

For each spot, you can include the SSP number, DSN (if designated), replicate group ID, and/or MrpI value (if calculated) by selecting the appropriate checkboxes.

You can export annotations in all categories or a specified category. If you select Specified, select the category from the pulldown list.

You can export all analysis sets or a specified analysis set. To specify an analysis set, select one from the pulldown list.

Under Spots to Export, specify the spots whose data should be exported. You can choose all the spots in the matchset, those spots that are matched in every member of the matchset, or the spots in a particular analysis set. The spot count will automatically adjust to reflect the total number of spots selected.

Under Output Options, you can include headers for each column in the export file. Select how you want to sort the spot data: by SSP number, DSN, or gel/replicate group.

Select the field separators you want to use to separate columns of data in your spreadsheets: commas or tabs.

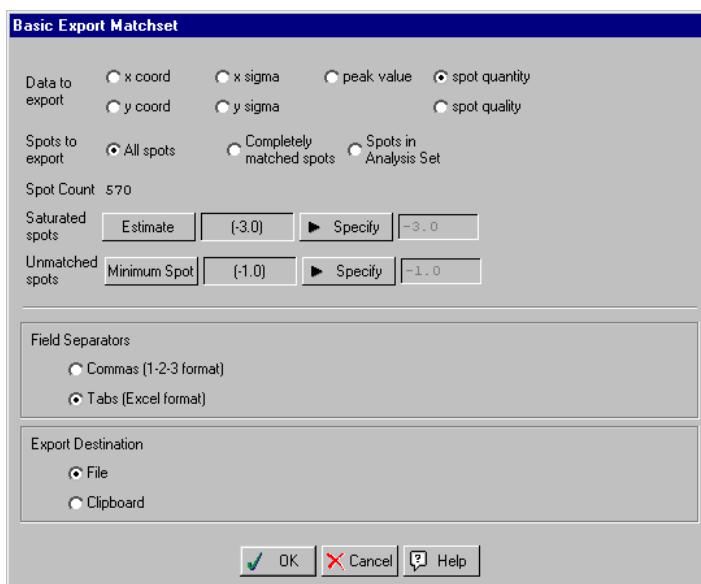
Your data can be exported as a file or to the clipboard. If you choose the file option button, clicking on Done will open a version of the Save As dialog box, in which you can enter a name and export location for the file.

If you choose the clipboard option, clicking on Done will save your data to the system clipboard.

Basic Export Matchset

The Basic Export Matchset dialog allows you to export the individual characteristics of spots in a matchset. This is a more basic version of the standard Export Matchset dialog. Data will be exported in ASCII text format.

With the matchset loaded, select Basic Export Matchset from the File > Export submenu.



The dialog box is titled "Basic Export Matchset". It contains several sections for configuring the export process:

- Data to export:** Radio buttons for "x coord", "x sigma", "peak value", "spot quantity", "y coord", "y sigma", and "spot quality". "spot quantity" is selected.
- Spots to export:** Radio buttons for "All spots", "Completely matched spots", and "Spots in Analysis Set". "All spots" is selected.
- Spot Count:** A text field containing "570".
- Saturated spots:** A section with "Estimate" and "Specify" buttons. The "Specify" button is active, showing a value of "-3.0" in a text field.
- Unmatched spots:** A section with "Minimum Spot" and "Specify" buttons. The "Specify" button is active, showing a value of "-1.0" in a text field.
- Field Separators:** Radio buttons for "Commas (1-2-3 format)" and "Tabs (Excel format)". "Tabs (Excel format)" is selected.
- Export Destination:** Radio buttons for "File" and "Clipboard". "File" is selected.
- Buttons:** "OK", "Cancel", and "Help" buttons at the bottom.

Fig. 13-3. Export Matchset Data dialog box.

Next to the "Data to export" prompt, select the characteristic of the spot that you want to export.

Specify the spots whose data should be exported. You can choose either all the spots in the matchset, those spots that are matched in every member of the matchset or the spots in a particular Analysis Set.

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The Spot Count will automatically adjust to reflect the total number of spots selected.

If you are exporting spot quantity data, select the way in which saturated spots and unmatched spots are reported.

The quantitation of saturated spots can be estimated, reported as (-3.0), or reported in some other way that you specify. If you opt for the latter, click on the Other button and enter the value that should be reported.

If a spot is unmatched in one or more members of a matchset, it can be reported as the minimum spot quantitation value (the minimum quantity needed for a spot to be detected for the gel). Alternatively, it can be reported as (-1.0) or as a value that you enter in the text box next to the Specify button.

Next, select the separators you want to use to separate columns of data in your spreadsheets: commas or tabs.

Your data can be exported as a file or to the clipboard.

If you choose the File option button, clicking on OK will open a version of the Save As dialog box, in which you can enter a name and export location for the file.

If you choose the Clipboard option button, clicking on OK will save your data to the clipboard.

13.2.a Exporting an Image

To export a 2-D scan, Filtered image, or Gaussian image as a TIFF image, select Export to TIFF Image from the File > Export submenu, then click on the image in a matchset or scanset.

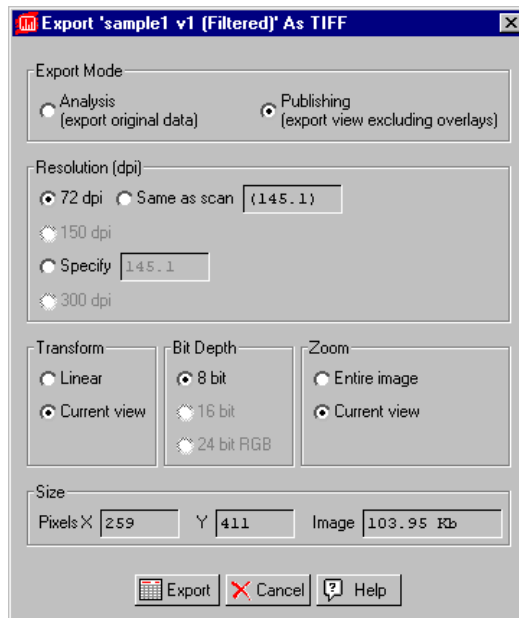


Fig. 13-4. Exporting TIFF Images.

In the export dialog, first select whether you want to export the original image data (Analysis option) or the currently displayed view of the image (Publishing option).

Note: Only the Publishing option is available if you are exporting from the Mutli-Channel Viewer.

Analysis Export Mode

The Analysis option exports the image data unmodified by any viewing adjustments you may have made (such as Transform or Zoom). If you select this option, the other controls in the dialog box will become inactive.

Publishing Export Mode

If you select the Publishing export option, the exported TIFF image will look like the image as it is currently displayed on the computer screen.

You can specify a resolution for the exported image by selecting 72 dpi (typical computer screen resolution), 150 dpi or 300 dpi (standard printing resolutions), Same as scan, or any resolution you Specify (up to the resolution of the scan).

If you have log transformed the image, you can specify a Linear transform, or preserve the Current view.

If your image has a pixel depth of 16 bits, you can compress it to 8 bit data for export to TIFF by selecting the 8 bit option.

Note: TIFF images are exported from the Multi-channel Viewer in 24 bit RGB mode to preserve the colors displayed in the viewer.

Finally, if you are only displaying part of the image due to magnification or repositioning, you can preserve the Current view or export the Entire image.

Exporting the Image

The size of the pixels in the image and the file size of the image are listed at the bottom of the dialog. When you are ready to export, click on the Export button.

A Save As dialog box will open. The default file name will have a .tif extension, and the file type will indicate that this is a TIFF image. You can change the file name or select a different directory to save in.

13.2.b Exporting Annotations

You can export matchset annotations by category to a text file for inclusion in a document or spreadsheet.

With the matchset open, select Export Annotations from the File > Export submenu. A box will open prompting you to select the annotation category or categories to export.

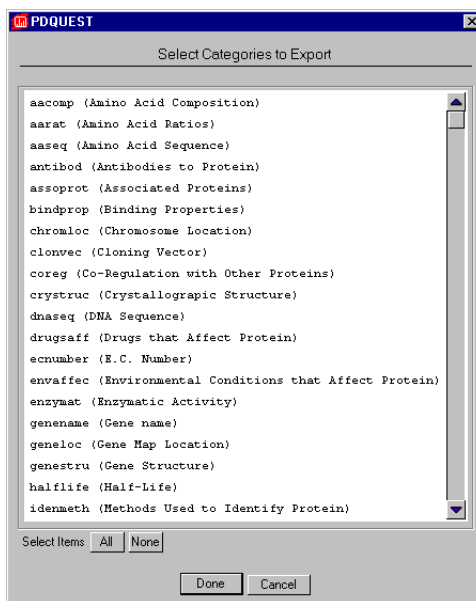


Fig. 13-5. Export Annotations pop-up box.

Click on an individual category name to select it, or shift-click or ctrl-click to select multiple categories.

When you have made your selection(s), click on Export. In the file name dialog, enter a name for the export file and select the directory to save to.

The data in the text file is tab-delimited. Each row includes the annotation category, SSP number for a spot in that category, and the associated annotation entry.

Text files can be opened in standard Windows and Macintosh documents and spreadsheets. If you selected multiple categories to export, the annotations will be sorted by category.

Appendix A

Gel Doc 1000/2000



Fig. A-1. Gel Doc.

Before you can begin acquiring images, the Gel Doc system must be properly installed and connected with the host computer. See the Gel Doc hardware manual for installation, startup, and operating instructions.

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To use the Gel Doc, you will need to have the Bio-Rad-supplied acquisition board installed in your PC or Macintosh. The drivers for this board will be installed when you install the main software application.

Make sure that your Gel Doc camera is turned on. If the camera is not turned on, the Gel Doc acquisition window will open but the screen will be black, and you will be unable to Video Print.

Simulation Mode

Any of the imaging device acquisition windows can be opened in “simulation mode.” In this mode, an acquisition window will open and the controls will appear active, but instead of capturing real images, the window will create “dummy” images of manufactured data.

You do not need to be connected to an imaging device to open a simulated acquisition window. This is useful for demonstration purposes or practice scans.

To enter simulation mode, hold down the CTRL key and select the name of the device from the File menu. The title of the acquisition window will indicate that it is simulated.

A.1 Gel Doc Acquisition Window

To acquire images using the Gel Doc, go to the File menu and select Gel Doc.... The acquisition window for the instrument will open, displaying a control panel and a video display window.

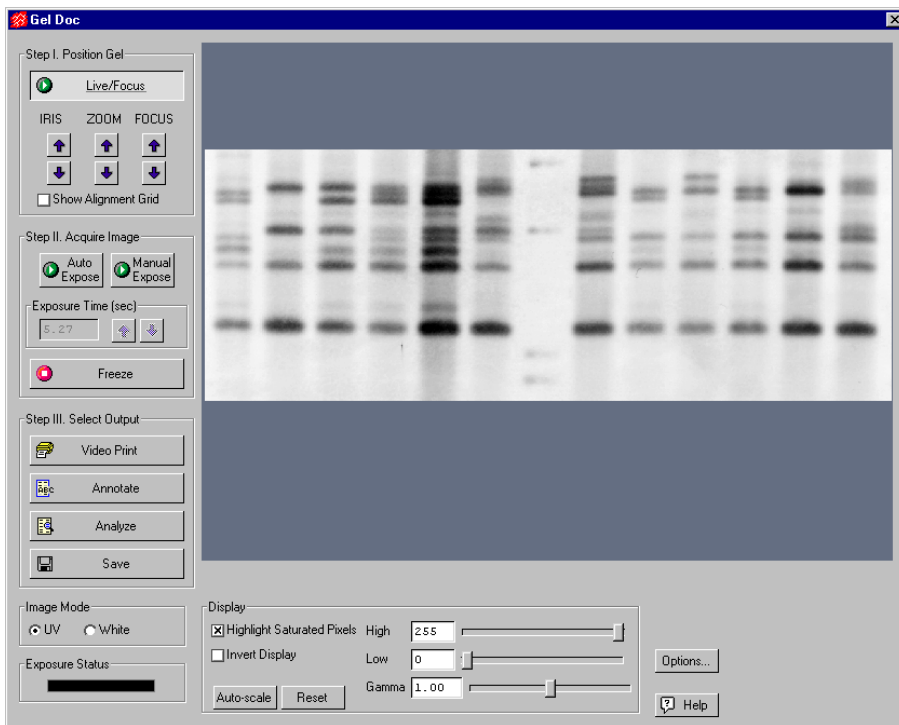


Fig. A-2. Gel Doc acquisition window

The Gel Doc video display window will open in “live” mode, giving you a live video display of your sample. If no image is visible, make sure the camera is on, check the cable connections, make sure the f-stop on the camera is not closed, and make sure that the protective cap is off the camera lens. Also check to see that the transilluminator is on and working.

The control panel has been arranged from top to bottom to guide you through the acquisition procedure. There are three basic steps to acquiring an image using the Gel Doc:

1. Position and focus the gel or other object to be imaged.
2. Acquire the image.

3. Select the output.

A.2 Step I. Position Gel

The Gel Doc window will open in “live” mode, giving you a live video display of your sample. In this mode, the Live/Focus button will appear selected, and frames will be captured and displayed at about 10 frames per second, depending on the speed of your computer.

You can use live mode to zoom, focus, and adjust the aperture on the camera, while positioning the sample within the area.

Note: Newer versions of the Gel Doc feature a motorized zoom lens that can be controlled directly from the acquisition window using the Iris, Zoom, and Focus arrow buttons. Click on the Up/Down buttons while viewing your sample in the window to adjust the lens. These buttons will not be visible if you are connected to older versions of the Gel Doc without the motorized zoom lens.

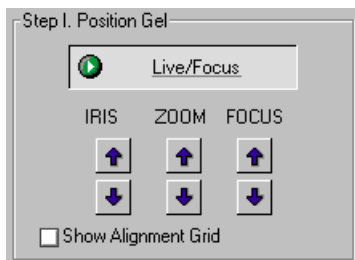


Fig. A-3. Newer Gel Docs feature camera control buttons in the acquisition window.

You can also select the Show Alignment Grid checkbox to facilitate positioning.

The image should stay in focus while zooming. If focus is not maintained, refer to your Gel Doc hardware manual.

Note: After positioning your sample, you should check the Imaging Area dimensions under Options (see below) to make sure that they conform to

the size of the area you are focusing on. To determine the size of the area you are focusing on, you can place a ruler in the Gel Doc box so that it is visible by the camera.

A.3 Step II. Acquire Image

For many white light applications, you can skip this step and save and print images directly from Live/Focus mode.

For UV light, chemiluminescent applications, or faint samples, you can take an automatic exposure based on the number of saturated pixels in the image or you can enter a specific exposure time.

Note: “Exposure” refers to the integration of the image on the camera CCD over a period of time. The effect is analogous to exposing photographic film to light over a period of time.

Auto Expose

Auto Expose will take an exposure whose time length is determined by the number of saturated pixels in the image. This is useful if you are uncertain of the optimal exposure length.

Note: If you know the approximate exposure time you want (± 3 seconds), you can skip this step and go directly to Manual Expose.

Click on the Auto Expose button to cancel Live/Focus mode and begin an automatic exposure. The Auto Expose button will appear selected throughout the exposure.

During the auto exposure, the image is continuously integrated on the camera CCD until it reaches a certain percentage of saturated pixels. This percentage is set in the Options dialog box. (Default = 0.75 percent. See Options below.)

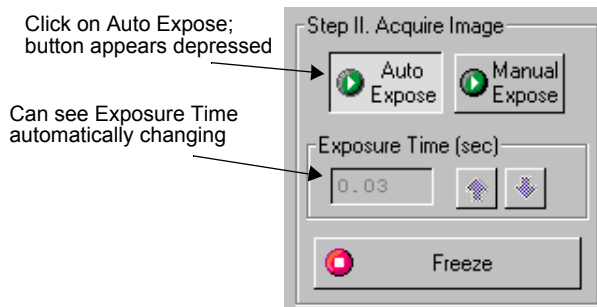


Fig. A-4. Auto Expose.

Once an image has reached the specified percent of saturated pixels, it is captured and displayed in the video display window, Auto Expose is automatically deactivated, the exposure time appears active in the Exposure Time field, and Manual Expose is activated.

Note: If you are having difficulty auto-exposing your sample, you can use Manual Expose to adjust your exposure time directly. Most non-chemical applications only require an exposure time of a few seconds, which can be quickly adjusted using Manual Expose.

Manual Expose

If you know the approximate exposure time you want, you can click on the Manual Expose button. Manual Expose is automatically activated after Auto Expose has deactivated.

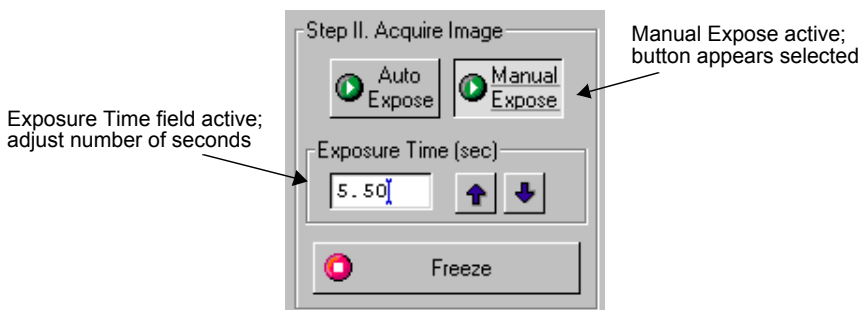


Fig. A-5. Manual Expose.

With Manual Expose activated, you can adjust the exposure time directly by changing the number of seconds in the Exposure Time field. Type in a number or use the arrow buttons next to the field.

When the specified exposure time is reached, the last captured image will be displayed in the Gel Doc image window. The camera continues to integrate the image on the CCD, updating the display whenever the specified number of seconds is reached.

Once you are satisfied with the quality of the displayed image, click on the Freeze button to stop the exposure process. The last full exposure will be displayed in the image window.

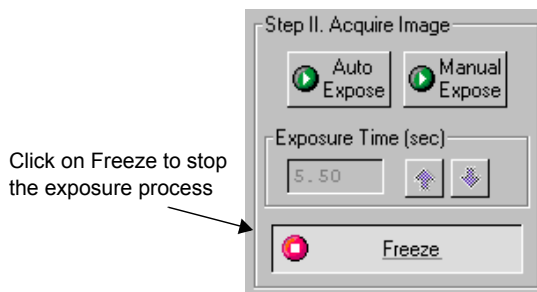


Fig. A-6. Freezing the exposure.

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Note: Freeze is automatically activated if you adjust any of the subsequent controls (e.g., Video Print, Image Mode, Display controls, etc.).

A.4 Step III. Select Output

The Gel Doc window has several output options.

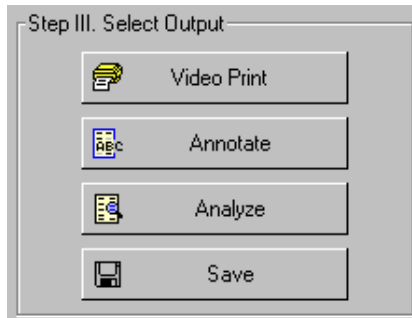


Fig. A-7. Output options.

Video Print

Clicking on Video Print will automatically send the currently displayed frame (either live or integrated) to a video printer. You can add information about your image to the bottom of the printout by selecting the appropriate checkboxes in the Options dialog box. (See Options, below.)

Annotate

Clicking on Annotate will open a separate image window displaying the captured image. The default name for the image will include the date, time, and user (if known).

The Text Overlay toolbar will also pop up to allow you to annotate your image.

The image will not be saved until you select Save or Save As from the File menu.

Analyze

Clicking on Analyze will open a separate image window displaying the captured image. The default name for the image will include the date, time, and user (if known).

You can then analyze the image using the menu and toolbar functions.

The image will not be saved until you select Save or Save As from the File menu.

Save

Clicking on Save will open a separate image window displaying the captured image. A Save As dialog box will automatically open displaying the default file name for the image, which will include the date, time, and user (if known). You can then change the file name and storage directory.

You can also save your image as a TIFF image for export to other applications.

A.5 Image Mode

The Image Mode option buttons allow you to set the type and scale of your data.

UV

Select this mode for fluorescent and chemiluminescent samples. With this mode selected, the data will be measured in linear intensity units.

White Light

Select this mode for reflective and transmissive samples. With this mode selected, the data will be measured in uncalibrated optical density (uOD) units.

A.6 Exposure Status

The Exposure Status bar shows the progress of your exposure. If your exposure time is greater than 1 second, the status bar display will give you a graphical representation of the remaining time before exposure is complete.

If the exposure time is less than 1 second, the status bar will not refresh itself for each exposure; it will remain at 100 percent.

A.7 Display

The Display controls are useful for quickly adjusting the appearance of your image for output to a video printer. Adjusting these controls will automatically freeze the video display and allow you to alter the image within the Gel Doc window.

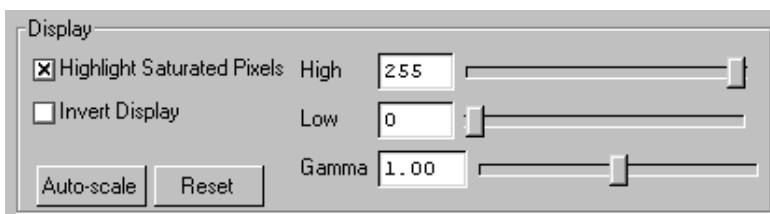


Fig. A-8. Display controls.

These controls are similar to those in the Transform dialog box.

Note: The Display controls will only change the appearance of the image. They will not change the underlying data.

Highlight Saturated Pixels

When this box is checked, any saturated pixels in the image will appear highlighted in red in the scan window and in the pop-up image window. To view/hide saturated pixels in the pop-up image window, use the Image > Transform command.

Invert Display

This checkbox will switch light spots on a dark background to dark spots on a light background, and visa versa. This will only affect how the image is displayed on the screen, not the actual image data.

Auto-scale

Clicking on Auto-scale will adjust your displayed image automatically. The lightest part of the image will be set to the minimum intensity (e.g., white), and the darkest will be set to the maximum intensity (e.g., black). You can then “fine tune” the display using the High, Low, and Gamma sliders described below.

High/Low Sliders

If Auto-scale doesn’t give you the appearance you want, you can use the High and Low sliders to redraw the image yourself. Dragging the High slider handle to the left will make weak signals appear darker. Dragging the Low slider handle to the right will reduce background noise.

You can also type specific High and Low values in the text boxes next to the sliders. Clicking anywhere on the slider bars will move the sliders incrementally.

Gamma Slider

Some images may be more effectively visualized if their data are mapped to the computer screen in a nonlinear fashion. Adjusting the Gamma slider handle changes the light and dark contrast nonlinearly.

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Reset

Reset will return the image to its original, unmodified appearance.

A.8 Options

Click on the Options button to open the Options dialog box. Here you can specify certain settings for your Gel Doc system.

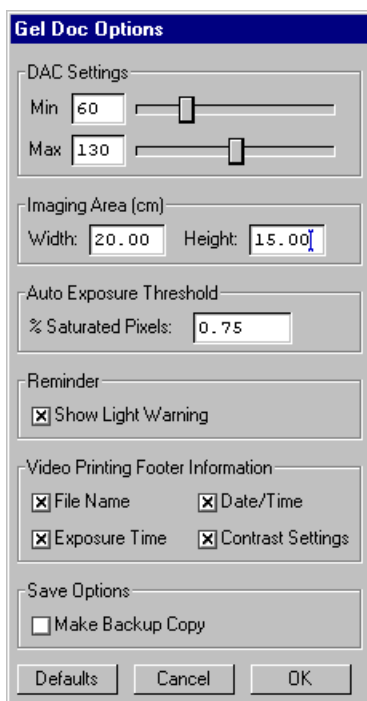


Fig. A-9. Available options in the Gel Doc acquisition window.

Click on OK to implement any changes you make in this box. Clicking on Defaults restores the settings to the factory defaults.

DAC Settings

Note: The default DAC settings are highly recommended and should be changed with caution.

These sliders may be used to adjust the minimum and maximum voltage settings of your video capture board. The minimum slider defines the pixel value that will appear as white in the image, while the maximum slider defines the pixel value that will appear as black. The slider scale is 0–255, with the defaults set to 60 minimum and 130 maximum.

Imaging Area

These fields are used to specify the size of your imaging area in centimeters, which in turns determines the size of the pixels in your image (i.e., resolution). When you adjust one imaging area dimension, the other dimension will change to maintain the aspect ratio of the camera lens.

Note: Your imaging area settings must be correct if you want to do 1:1 printing. These are also important if you are comparing the size of objects (e.g., using the Volume Tools) between images.

Auto Exposure Threshold

When you click on Auto Expose, the exposure time is determined by the percentage of saturated pixels you want in your image. This field allows you to specify that percentage.

Typically, you will want less than 1 percent of the pixels in your image saturated. Consequently, the default value for this field is 0.75 percent.

Reminder

When this checkbox is selected, the software will warn you to turn off your transilluminator light when you exit the Gel Doc acquisition window or when your system is “idle” for more than 5 minutes.

Note: If you are performing experiments that are longer than 5 minutes (e.g., chemiluminescence), this should be deselected.

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Video Printing Footer Information

The checkboxes in this group allow you to specify the information that will appear at the bottom of your video printer printouts.

Save Options

To automatically create a backup copy of any scan you create, select the Make Backup Copy checkbox.

With this checkbox selected, when you save a scan, a backup copy will be placed in the same directory as the scanned image. Windows backup files will have an “.sbk” extension. Macintosh backup files will have the word “backup” after the file name.

This backup copy will be read-only, which means that you cannot make changes to it. You can open it like a normal file, but you must save it under a different file name before editing the image or performing analysis.

Appendix B

Chemi Doc



Fig.B-1. Chemi Doc.

Before you can begin acquiring images, the Chemi Doc system must be properly installed and connected with the host computer. See the Chemi Doc hardware manual for installation, startup, and operating instructions.

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To use the Chemi Doc, you will need to have the Bio-Rad-supplied acquisition board installed in your PC or Macintosh. The drivers for this board will be installed when you install the main software application.

Make sure that your Chemi Doc camera is turned on. If the camera is not turned on, the Chemi Doc acquisition window will open but the screen will be black, and you will be unable to Video Print.

Simulation Mode

Any of the imaging device acquisition windows can be opened in "simulation mode." In this mode, an acquisition window will open and the controls will appear active, but instead of capturing real images, the window will create "dummy" images of manufactured data.

You do not need to be connected to an imaging device to open a simulated acquisition window. This is useful for demonstration purposes or practice scans.

To enter simulation mode, hold down the CTRL key and select the name of the device from the File menu. The title of the acquisition window will indicate that it is simulated.

B.1 Chemi Doc Acquisition Window

To acquire images using the Chemi Doc, go to the File menu and select Chemi Doc.... The acquisition window for the instrument will open, displaying a control panel and a video display window.

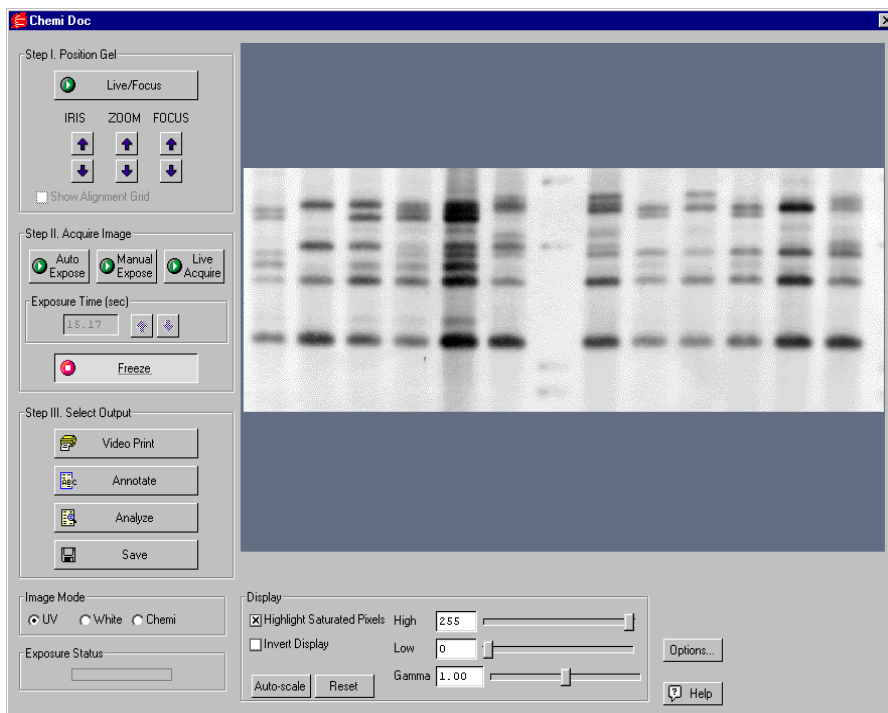


Fig.B-2. Chemi Doc acquisition window

The Chemi Doc video display window will open in “live” mode, giving you a live video display of your sample. If no image is visible, make sure the camera is on, check the cable connections, make sure the f-stop on the camera is not closed, and make sure that the protective cap is off the camera lens. Also check to see that the transilluminator is on and working.

The control panel has been arranged from top to bottom to guide you through the acquisition procedure. There are three basic steps to acquiring an image using the Chemi Doc:

1. Position and focus the gel or other object to be imaged.
2. Acquire the image.

3. Select the output.

B.2 Step I. Position Gel

The Chemi Doc window will open in “live” mode, giving you a live video display of your sample. In this mode, the Live/Focus button will appear selected, and frames will be captured and displayed at about 10 frames per second, depending on the speed of your computer.

You can use live mode to zoom, focus, and adjust the aperture on the camera, while positioning the sample within the area.

Note: Newer versions of the Chemi Doc feature a motorized zoom lens that can be controlled directly from the acquisition window using the Iris, Zoom, and Focus arrow buttons. Click on the Up/Down buttons while viewing your sample in the window to adjust the lens. These buttons will not be visible if you are connected to older versions of the Chemi Doc without the motorized zoom lens.

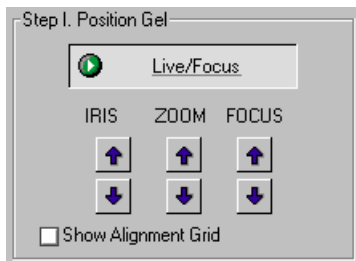


Fig.B-3. Live/Focus button and camera control buttons.

You can also select the Show Alignment Grid checkbox to facilitate positioning.

Note: After positioning your sample, you should check the Imaging Area dimensions under Options (see below) to make sure that they conform to the size of the area you are focusing on. To determine the size of the area you are focusing on, you can place a ruler in the Chemi Doc box so that it is visible by the camera.

B.3 Step II. Acquire Image

For many white light applications, you can skip this step and save and print images directly from Live/Focus mode.

For UV light, chemiluminescent applications, or faint samples, the Chemi Doc control panel has several features for creating image exposures. You can take an automatic exposure based on the number of saturated pixels in the image, you can enter a specific exposure time, or you can take a series of exposures and select the best one.

Note: “Exposure” refers to the integration of the image on the camera CCD over a period of time. The effect is analogous to exposing photographic film to light over a period of time.

Auto Expose

Use Auto Expose if you want to take a single exposure but are uncertain of the optimal exposure time.

Note: If you know the approximate exposure time you want (± 3 seconds), you can skip this step and go directly to Manual Expose.

Click on the Auto Expose button to cancel Live/Focus mode and begin an automatic exposure. The Auto Expose button will appear selected throughout the exposure.

During the auto exposure, the image is continuously integrated on the camera CCD until it reaches a certain percentage of saturated pixels. This percentage is set in the Options dialog box. (Default = 0.75 percent. See Options below.)

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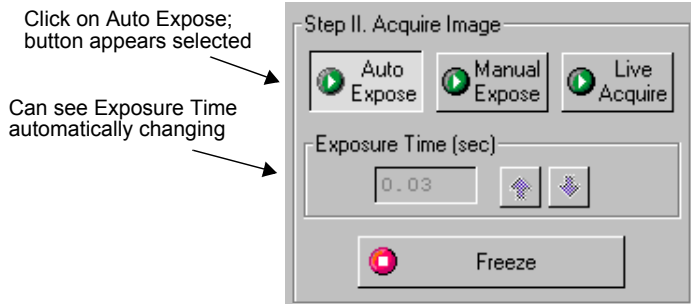


Fig.B-4. Selecting Auto Expose.

Once an image has reached the specified percentage of saturated pixels, it is captured and displayed in the video display window, Auto Expose is automatically deactivated, and the exposure time appears active in the Exposure Time field.

At this point, if you are in UV or White image mode, Manual Expose will be automatically activated. If you are in Chemi mode, the Freeze button will be automatically activated.

Note: If you are having difficulty auto-exposing your sample, you can use Manual Expose to adjust your exposure time directly. Most non-chemi applications only require an exposure time of a few seconds, which can be quickly adjusted using Manual Expose.

Manual Expose

If you know the approximate exposure time you want, you can click on the Manual Expose button. In UV or White image mode, Manual Expose is automatically activated after Auto Expose is complete.

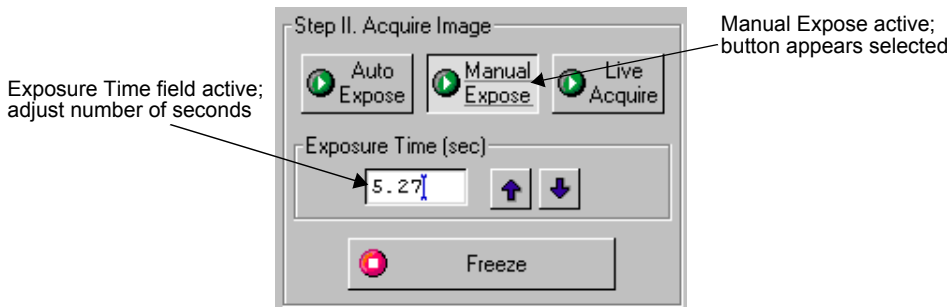


Fig.B-5. Setting a manual exposure.

With Manual Expose activated, you can adjust the exposure time directly by changing the number of seconds in the Exposure Time field. Type in a number or use the arrow buttons next to the field.

In UV or White image mode, when the specified exposure time is reached, the last captured image will be displayed in the Chemi Doc image window. The camera will continue to integrate the image on the CCD, updating the display whenever the specified number of seconds is reached.

Once you are satisfied with the quality of the displayed image, click on the Freeze button to stop the exposure process. The last full exposure will be displayed in the image window.

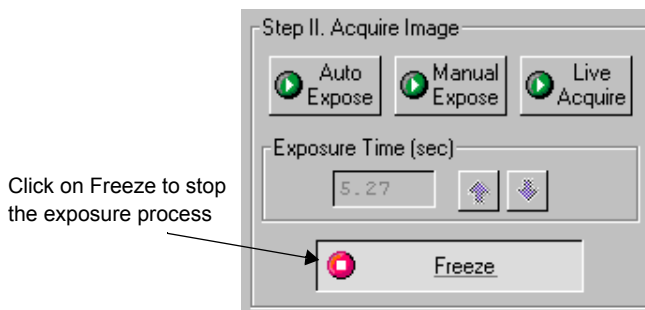


Fig.B-6. Freezing the manual exposure.

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In Chemi mode, Manual Expose will expose an image over the specified exposure time and then stop automatically.

Note: Freeze is automatically activated if you adjust any of the subsequent controls (e.g., Video Print, Image Mode, Display controls, etc.).

Live Acquire

Live Acquire mode allows you to specify an interval over which a series of progressively longer exposures are taken. All exposures are then displayed on the screen, and you can choose the one with the best image.

Click on the Live Acquire button. A settings dialog box will open in which you can specify the total exposure time, starting exposure time, and number of exposures.

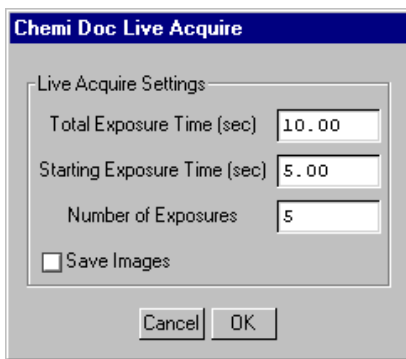


Fig.B-7. Live Acquire settings.

Note: You should specify no more than 10 exposures in the Live Acquire Settings dialog, to avoid excessive build-up of image background in later exposures. The fewer the exposures, the less background will be added to the image. See the Release Notes for additional instructions on reducing background in images captured using Live Acquire.

Select the Save Images checkbox if you want to automatically save each exposure as it is taken.

Click on OK in the settings dialog to begin taking exposures. If you selected Save Images, a Save dialog box will open in which you can specify the base file name and location of the exposure files. When you click on Save, the exposures will be taken.

The specified number of exposures will be taken at equal intervals between the starting exposure time and total exposure time. When each exposure is complete, an image window containing that exposure will open behind the Chemi Doc window. When the full exposure time has lapsed, all the image windows will move in front of the Chemi Doc window.

Note that the first exposure will have the base file name (the default base file name is the computer user name and a time stamp). Each subsequent exposure will have a version number (v2, v3, v4, etc.) appended to the base file name. The highest version number will be the final exposure. If you did not elect to auto-save the exposures as they were created, then each image will be unsaved.

To stop the Live Acquire, click on the Freeze button or adjust any of the subsequent controls (e.g., Video Print, Image Mode, Display controls, etc.).

Note: Exposures captured before freezing will be displayed in image windows.

Study the different images and select the best exposure(s) to keep. You can then proceed to the next step.

B.4 Step III. Select Output

The Chemi Doc window has several output options.

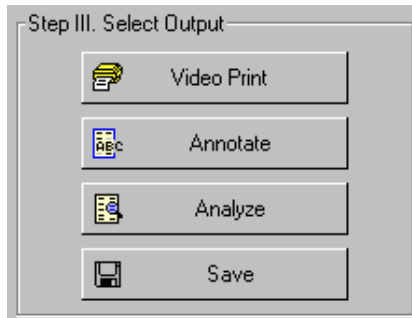


Fig.B-8. Output options.

Video Print

Clicking on Video Print will automatically send the currently displayed frame (either live or integrated) to a video printer. You can add information about your image to the bottom of the printout by selecting the appropriate checkboxes in the Options dialog box. (See Options, below.)

Annotate

Clicking on Annotate will open a separate image window displaying the captured image. The default name for the image will include the date, time, and user (if known).

The Text Overlay toolbar will also pop up to allow you to annotate your image.

The image will not be saved until you select Save or Save As from the File menu.

Analyze

Clicking on Analyze will open a separate image window displaying the captured image. The default name for the image will include the date, time, and user (if known).

You can then analyze the image using the other features in the main application.

The image will not be saved until you select Save or Save As from the File menu.

Save

Clicking on Save will open a separate image window displaying the captured image. A Save As dialog box will automatically open displaying the default file name for the image, which will include the date, time, and user (if known). You can then change the file name and storage directory.

You can also save your image as a TIFF image for export to other applications.

B.5 Image Mode

The Image Mode option buttons change the type and scale of the data, as well as the behavior of the Chemi Doc when acquiring an image.

UV

Select this mode for fluorescent and chemiluminescent samples. With this mode selected, the data will be measured in linear intensity units.

White Light

Select this mode for reflective and transmissive samples. With this mode selected, the data will be measured in uncalibrated optical density (uOD) units.

Chemi

This mode is designed for chemiluminescent samples. With this mode selected, the data is measured in linear intensity units; however, the data is inverted, so that samples will appear dark on a light background.

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Note: This setting will invert the data in your image, so that a pixel with an intensity of 255 will be changed to 0, and visa versa.

Also, Chemi mode changes the behavior of the Auto and Manual Expose functions, as described above.

B.6 Exposure Status

The Exposure Status bar shows the progress of your exposure. If your exposure time is greater than 1 second, the status bar display will give you a graphical representation of the remaining time before exposure is complete.

If the exposure time is less than 1 second, the status bar will not refresh itself for each exposure; it will remain at 100 percent.

B.7 Display

The Display controls are useful for quickly adjusting the appearance of your image for output to a video printer. Adjusting these controls will automatically freeze the video display and allow you to alter the image within the Chemi Doc window.

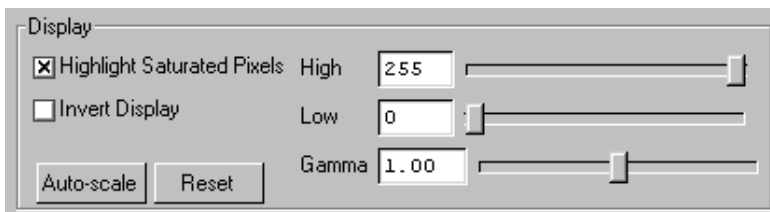


Fig.B-9. Display controls.

These controls are similar to those in the Transform dialog box.

Note: The Display controls will only change the appearance of the image. They will not change the underlying data.

Highlight Saturated Pixels

When this box is checked, any saturated pixels in the image will appear highlighted in red in the scan window and in the pop-up image window. To view/hide saturated pixels in the pop-up image window, use the Image > Transform command.

Invert Display

This checkbox will switch light spots on a dark background to dark spots on a light background, and visa versa.

Note: This will only affect how the image appears on your screen; it will not change the image data.

Auto-scale

Clicking on Auto-scale will adjust your displayed image automatically. The lightest part of the image will be set to the minimum intensity (e.g., white), and the darkest will be set to the maximum intensity (e.g., black). You can then “fine tune” the display using the High, Low, and Gamma sliders described below.

High/Low Sliders

If Auto-scale doesn't give you the appearance you want, you can use the High and Low sliders to redraw the image yourself. Dragging the High slider handle to the left will make weak signals appear darker. Dragging the Low slider handle to the right will reduce background noise.

You can also type specific High and Low values in the text boxes next to the sliders. Clicking anywhere on the slider bars will move the sliders incrementally.

Gamma Slider

Some images may be more effectively visualized if their data are mapped to the computer screen in a nonlinear fashion. Adjusting the Gamma slider handle changes the light and dark contrast nonlinearly.

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Reset

Reset will return the image to its original, unmodified appearance.

B.8 Options

Click on the Options button to open the Options dialog box. Here you can specify certain settings for your Chemi Doc system.

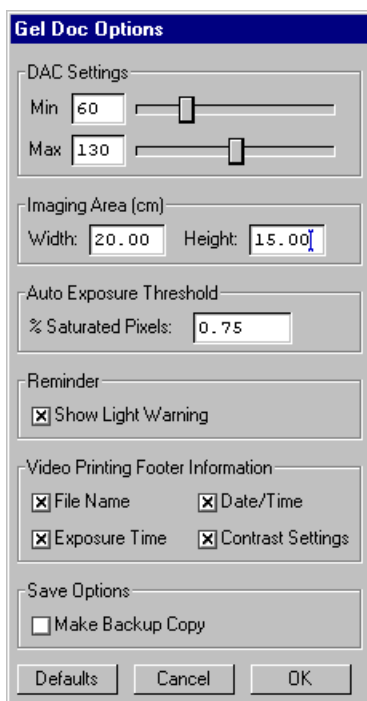


Fig.B-10. Available options in the Chemi Doc acquisition window.

Click on OK to implement any changes you make in this box. Clicking on Defaults restores the settings to the factory defaults.

DAC Settings

Note: The default DAC settings are highly recommended and should be changed with caution.

These sliders may be used to adjust the minimum and maximum voltage settings of your video capture board. The minimum slider defines the pixel value that will appear as white in the image, while the maximum slider defines the pixel value that will appear as black. The slider scale is 0–255, with the defaults set to 60 minimum and 130 maximum.

Imaging Area

These fields are used to specify the size of your imaging area in centimeters, which in turns determines the size of the pixels in your image (i.e., resolution). When you adjust one imaging area dimension, the other dimension will change to maintain the aspect ratio of the camera lens.

Note: Your imaging area settings must be correct if you want to do 1:1 printing. These are also important if you are comparing the size of objects (e.g., using the Volume Tools) between images.

Auto Exposure Threshold

When you click on Auto Expose, the exposure time is determined by the percentage of saturated pixels you want in your image. This field allows you to specify that percentage.

Typically, you will want less than 1 percent of the pixels in your image saturated. Consequently, the default value for this field is 0.75 percent.

Reminder

When this checkbox is selected, the software will warn you to turn off your transilluminator light when you exit the Chemi Doc acquisition window or when your system is “idle” for more than 5 minutes.

Note: If you are performing experiments that are longer than 5 minutes (e.g., chemiluminescence), this should be deselected.

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Video Printing Footer Information

The checkboxes in this group allow you to specify the information that will appear at the bottom of your video printer printouts.

Save Options

To automatically create a backup copy of any scan you create, select the Make Backup Copy checkbox.

With this checkbox selected, when you save a scan, a backup copy will be placed in the same directory as the scanned image. Windows backup files will have an “.sbk” extension. Macintosh backup files will have the word “backup” after the file name.

This backup copy will be read-only, which means that you cannot make changes to it. You can open it like a normal file, but you must save it under a different file name before editing the image or performing analysis.

Appendix C

GS-700 Imaging Densitometer



Fig. C-1. GS-700 Imaging Densitometer

Before you can begin scanning images with the GS-700 Imaging Densitometer[®], your instrument must be properly installed and connected with the host computer. See the hardware manual for installation, startup, and operating instructions.

PC Only: A Note About SCSI Cards

The GS-700 is connected to your computer by a Small Computer System Interface (SCSI) cable. To use the GS-700, you must have a SCSI card installed in your PC. If you have an older PC, you may also need to load the SCSI and WinASPI drivers that came with your card.

Simulation Mode

Any of the imaging device acquisition windows can be opened in “simulation mode.” In this mode, an acquisition window will open and the controls will

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appear active, but instead of capturing real images, the window will create “dummy” images of manufactured data.

You do not need to be connected to an imaging device to open a simulated acquisition window. This is useful for demonstration purposes or practice scans.

To enter simulation mode, hold down the CTRL key and select the name of the device from the File menu. The title of the acquisition window will indicate that it is simulated.

Note: There is no simulated calibration for the GS-700 and GS-710 Imaging Densitometers.

C.1 GS-700 Acquisition Window

To acquire images using the GS-700, go to the File menu and select GS-700.... The acquisition window for the densitometer will open, displaying a control panel and a scanning window.

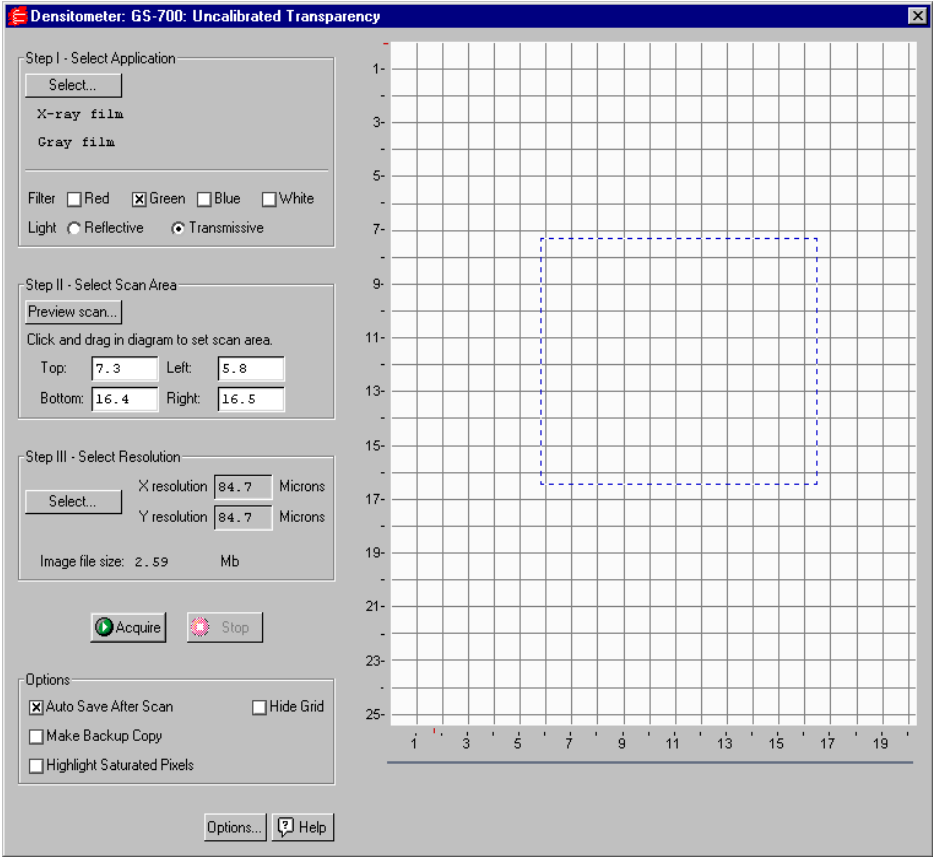


Fig. C-2. GS-700 acquisition window.

The scanning window is marked by grid lines that divide the area into square centimeters. These are numbered 1–35 top to bottom and 1–21 left to right if the light source is uncalibrated reflective, and 1–25 top to bottom and 1–20 left to right if the light source is uncalibrated transmissive (see below for details).

To hide the gridlines, click on the Hide Grid checkbox under Options.

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The control panel has been arranged from top to bottom to guide you through the acquisition procedure. There are four basic steps to scanning an image using the GS-700:

1. Select the application.
2. Select the scan area.
3. Select the resolution of the scan.
4. Acquire the image.

C.2 Step I. Select Application

To set the parameters for your particular scan, you can:

1. Select from a list of possible applications, or
2. Choose your own filter and light source settings.

Selecting from the List of Applications

When you select from the list of applications, the software automatically sets the appropriate filter(s) and other parameters for that particular application.

To select from the list of applications, click on the Select button under Select Application.

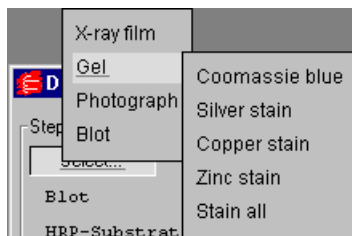


Fig. C-3. Example of the application tree in the GS-700 dialog box.

The applications are listed in a tree that expands from left to right. First you select the category of your application, then you select your particular application.

To exit the tree without selecting, press the ESC key.

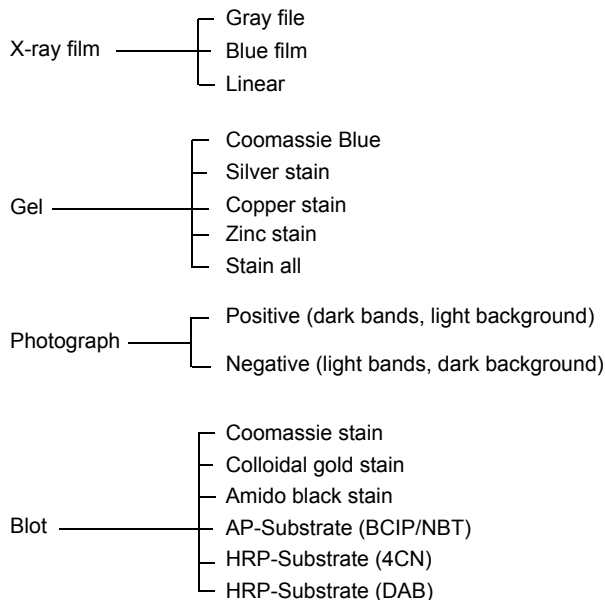


Fig. C-4. Applications available in the GS-700 acquisition window.

Choosing Your Own Settings

If you know the filter and light source settings you want, or want to experiment with different settings, you can choose them yourself.

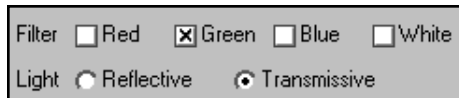


Fig. C-5. GS-700 Custom Application controls.

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Next to Filter, click on either the Red, Green, Blue, or White checkbox, or a combination of two of the first three (Red-Green, Green-Blue, Red-Blue).

Filter Color	Wavelength	Application Examples
Red	595–750 nm	Coomassie G-250, BCIP/NBT, Fast Green FCF, Methylene Blue
Green	520–570 nm	Coomassie R-250, Basic Fuchsin
Blue	400–530 nm	Crocein Scarlet
Gray Scale	400–750 nm	Silver Stains, Copper Stains, Film

Fig. C-6. Examples of filter colors and applications.

Next to Light, select the appropriate light source.

- Reflective mode is for scanning opaque mediums such as dried gels on filter paper, arrays, TLC plates, and photographs. The scanning dimensions in this mode are 21 cm x 35 cm (uncalibrated).
- Transmissive mode is for scanning transparent mediums such as films, gels, and slides. The scanning dimensions in this mode are 20 cm x 25 cm (uncalibrated).

C.3 Step II. Select Scan Area

Preview Scan

Before selecting the particular area to scan, you can preview the entire scanning area to determine the exact position of your sample.

Click on Preview Scan. A quick, low-resolution scan of the entire scanning area will appear in the scanning window. Your sample should be visible within a portion of this scan.

Selecting an Area

Using the preview scan as a guide, select your scan area by dragging your mouse within the scanning window. The border of the scan area you are selecting will be marked by a frame.

Note: The scan area you select must be at least 1 cm wide.

When you release the mouse button, the border changes to a dashed blue line, indicating a selected area.

- To *reposition* the scanning area box you have created, position your cursor inside the box and drag. The entire box will move.
- To *resize* the box, position your cursor on a box side and drag. The side you have selected will move.
- To *redo* the box entirely, position your cursor outside the box and drag. The old box will disappear and a new box will be created.

You can also select the scanning area by entering coordinates in the appropriate fields (Top, Bottom, Left, Right). As you enter a coordinate, the position of the scanning area box will change accordingly.

When selecting, be sure to include the entire area of interest, and be generous with borders. You can always crop the image later.

C.4 Step III. Select Resolution

To select from a list of possible scanner resolutions, click on the Select button under Select Resolution. This will open the Select Scan Resolution dialog box.

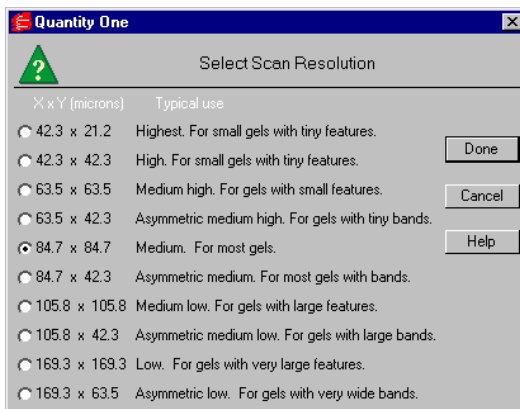


Fig. C-7. Select Scan Resolution dialog box.

Available resolutions are listed from highest to lowest in terms of the dimensions of the resulting pixels (in microns). Smaller pixels equal higher resolution. Each resolution is listed with its typical use.

In general, the size of your pixels should be one-tenth the height of your smallest object.

Some of the resolutions are asymmetrical, meaning that the resolution is higher in the vertical dimension (i.e., the pixels in the resulting image are larger in the horizontal dimension than in the vertical dimension). This is useful for gels with bands, where you are more interested in resolving in the y dimension to determine the size of bands and the spacing between them.

Specifying Your Own Resolution

If you select Oversample under More Options (see section C.7 for details), you can specify your own resolution within the range of 43–169 microns (micrometers). With Oversample selected, enter values directly in the fields next to X resolution and Y resolution in the main acquisition window.

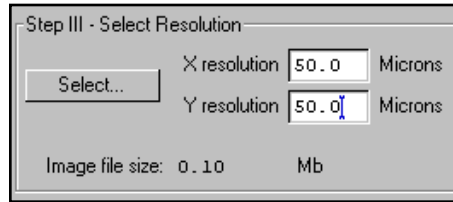


Fig. C-8. Entering a custom resolution (with Oversample selected).

Image File Size

Image File Size (under Select Resolution) shows the size of the scan file you are about to create. If you do not have enough computer memory for the specified file size, an error message will appear when you attempt to scan. If this happens, select a lower resolution or decrease the size of the area to be scanned. (Macintosh users can also increase the application memory partition. See your Macintosh computer documentation for guidance.)

C.5 Acquire the Image

If you want to calibrate your scans, read the following section on calibration before scanning.

To begin to scan, click on the Acquire button. The scanned image will begin to appear in the scanning window, line by line.

To interrupt a scan, click on the Stop button. A message will ask you to confirm the interrupt, and then you will be asked if you want to keep the partial scan. This feature is useful if you overestimated the size of the area you selected.

After the scan is complete, a window will open displaying the scan image, at which point you can analyze and save it.

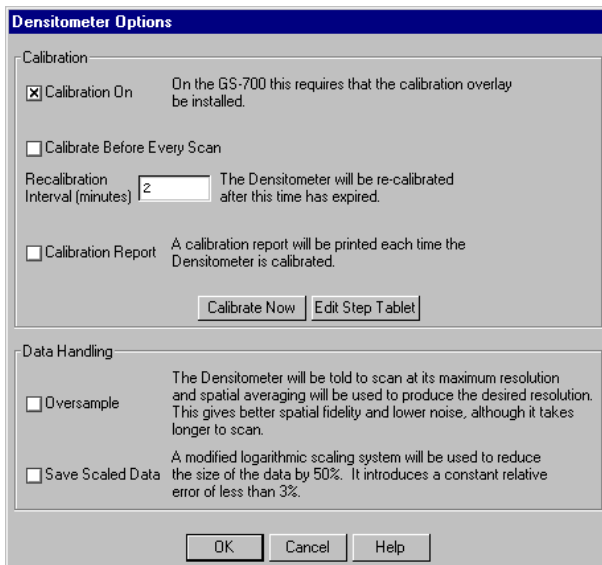
Note: The image will open with a default file name that includes the date, time, and (if applicable) user name. However, unless you have selected Auto

Save After Scan, the file will not be saved until you select Save or Save As from the File menu.

C.6 Calibration

If you have installed a calibration overlay, you can automatically calibrate your transmissive and reflective scans. (Calibration overlays for the GS-700 can be ordered from Bio-Rad.)

To set the automatic calibration settings, click on the More Options button in the GS-700 acquisition window. This will open the Densitometer Options dialog box.



The Densitometer Options dialog box is divided into two main sections: Calibration and Data Handling. The Calibration section includes a 'Calibration On' checkbox (checked), a 'Calibrate Before Every Scan' checkbox (unchecked), a 'Recalibration Interval (minutes)' field set to 2, and a 'Calibration Report' checkbox (unchecked). The Data Handling section includes an 'Oversample' checkbox (unchecked) and a 'Save Scaled Data' checkbox (unchecked). At the bottom are 'OK', 'Cancel', and 'Help' buttons.

Densitometer Options	
Calibration	
☒ Calibration On	On the GS-700 this requires that the calibration overlay be installed.
☐ Calibrate Before Every Scan	
Recalibration Interval (minutes) 2	The Densitometer will be re-calibrated after this time has expired.
☐ Calibration Report	A calibration report will be printed each time the Densitometer is calibrated.
Calibrate Now Edit Step Tablet	
Data Handling	
☐ Oversample	The Densitometer will be told to scan at its maximum resolution and spatial averaging will be used to produce the desired resolution. This gives better spatial fidelity and lower noise, although it takes longer to scan.
☐ Save Scaled Data	A modified logarithmic scaling system will be used to reduce the size of the data by 50%. It introduces a constant relative error of less than 3%.
OK Cancel Help	

Fig. C-9. More Options in the GS-700 acquisition window.

To enable automatic calibration, click on the Calibration On checkbox.

Note: Before you can calibrate, you must enter the correct step tablet values for your transmissive calibration strip into the Step Tablet Values form, as

described below. You do not need to change the values in the reflective step tablet form; you can use the default values.

With Calibration On selected, the other calibration settings become active.

Calibration Strip Window

When calibration is turned on, a calibration strip window will appear below the main scanning window and the length of the main scanning window will be reduced to 29 cm reflective and 23 cm transmissive.

Every time the densitometer calibrates, an image of the calibration strip will appear in the calibration strip window.

C.6.a Editing the Step Tablet

Before you can calibrate, you must enter the step tablet values for your transmissive calibration strip into the Step Tablet Values form.

In the package with your calibration strip, you will find a printout of the diffuse density values for each segment of your transmissive calibration strip. Those exact density values must be entered into the computer for the software to associate a correct density value with each step on the step tablet.

Note: Scanning in transmissive mode with incorrect step tablet values entered into the form can cause significant errors in the reported densities of your scans.

To enter the density values for your step tablet, click on the Edit Step Tablet button in the Densitometer Options box. The Step Tablet Values dialog box for the GS-700 will open.

Step	Diffuse	Specular	Step	Diffuse	Specular
1.	0.06	0.07	12.	1.72	2.43
2.	0.20	0.28	13.	1.88	2.66
3.	0.36	0.51	14.	2.03	2.87
4.	0.52	0.74	15.	2.18	3.08
5.	0.66	0.93	16.	2.34	3.31
6.	0.81	1.15	17.	2.49	3.52
7.	0.96	1.36	18.	2.64	3.73
8.	1.11	1.57	19.	2.80	3.96
9.	1.27	1.80	20.	2.94	4.16
10.	1.41	1.99	21.	3.08	4.36
11.	1.56	2.21			

Fig. C-10. Step Tablet dialog box.

The dialog box will indicate whether the displayed form is for the transmissive or reflective step tablet, depending on whether you are in transmissive or reflective mode.

Note: In transmissive scanning, the scanner is calibrated against a transmissive step tablet. In reflective scanning, the scanner is calibrated against a reflective step tablet, unless the reflective calibration strip is uncalibrated. If the reflective calibration strip does not have step tablet values associated with it, you can use the default values in the form.

In the Tablet Serial Number field, you can type in the serial number for the step tablet values you are entering. (The reflective step tablet will probably not have a serial number.)

Your calibration strip will have a clear plastic cover. If the quantity offset value for the clear cover is included in your calibration strip package, you can enter this number in the Quantity Offset field. Otherwise, use a value of 0.045.

Finally, enter the values in the appropriate fields under the Diffuse column. After the step tablet is scanned, the software will associate each density value with its corresponding segment on the step tablet.

The step tablet density values do not need to be entered each time you calibrate. Once they have been entered and saved, they will be automatically recalled when the calibration strip is scanned. You only need to enter new values if you use a new step tablet.

When you finished filling out the Step Tablet Values form, click on OK.

Diffuse Versus Specular O.D.

In the step tablet form, you enter O.D. as diffuse density, and then the software automatically calculates the specular density.

Specular density is a measure of the light that passes directly through a medium. Diffuse density includes light that is scattered as it passes through the medium. Step tablet values are given in diffuse density, but are measured by the scanner in specular density, and therefore must be converted according to the specular / diffuse O.D. ratio. This conversion does not affect quantitation.

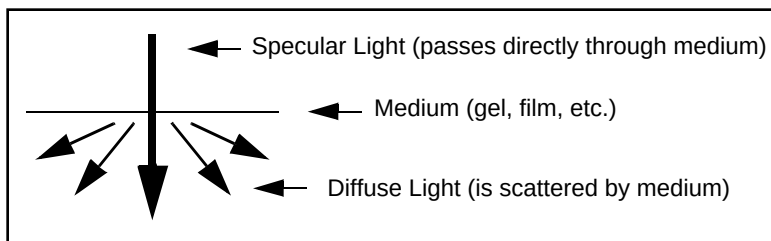


Fig. C-11. Specular and diffuse density

Diffuse density values are converted to specular optical density units according to the following formula:

$$\text{Specular OD} = 1.4 \text{ Diffuse OD}$$

C.6.b Calibration Settings

After you have entered the step tablet values, you can immediately calibrate by clicking on the Calibrate Now button (in the Densitometer Options dialog box).

You can also specify how often you want the densitometer to automatically recalibrate. Either click on the Calibrate Before Every Scan checkbox or enter a Recalibration Interval (in minutes) in the appropriate field.

Note: With calibration turned on, the scanner will automatically recalibrate each time you change your filter or your reflective/transmissive setting. (If you select a different application with the same filter and light settings, it will not auto recalibrate.)

Calibration Report

To print out a calibration report each time the densitometer calibrates, click on the Calibration Report checkbox.

C.7 Other Options

Oversample

This feature allows you to scan at the maximum resolution of the GS-700 (42.3 x 42.3 microns) and then use spatial averaging to create an image with lower resolution. This can result in better images at lower resolution—however, it takes longer to scan.

To turn on oversampling, click on the More Options button in the acquisition window and click on the Oversample checkbox.

With oversampling on, you can specify your own resolution within the range of 43–169 microns by entering values directly in the fields next to X resolution and Y resolution in the main acquisition window.

Auto Save After Scan

To automatically save any scan you create, click on the Auto Save After Scan checkbox.

With this checkbox selected, when you click on Acquire, a Save As dialog box will open asking you to specify a file name and location for the image you are about to create. The scan will begin when you click on the Save button.

Make Backup Copy

If you have checked Auto Save After Scan, you can also automatically create a backup copy of any scan you create.

Click on the Make Backup Copy checkbox. With this checkbox selected, when a scan is created and saved, a backup copy will be placed in the same directory as the scanned image. Windows backup files will have an “.sbk” extension. Macintosh backup files will have the word “backup” after the file name.

This backup copy will be read-only, which means that you cannot make changes to it. You can open it like a normal file, but you must save it under a different file name before editing the image or performing analysis.

Highlight Saturated Pixels

When this box is checked, any saturated pixels in the image will appear highlighted in red in the scan window and in the pop-up image window. To view/hide saturated pixels in the pop-up image window, use the Image > Transform command.

Hide Grid

To hide the gridlines in the scanning area window, click on the Hide Grid checkbox.

Appendix D

GS-710 Imaging Densitometer



Fig. D-1. GS-710 Imaging Densitometer

Before you can begin scanning images with the GS-710 Imaging Densitometer[®], your instrument must be properly installed and connected with the host computer. See the hardware manual for installation, startup, and operating instructions.

PC Only: A Note About SCSI Cards

The GS-710 is connected to your computer by a Small Computer System Interface (SCSI) cable. To use the GS-710, you must have a SCSI card installed in your PC. If you have an older PC, you may also need to load the SCSI and WinASPI drivers that came with your card.

Simulation Mode

Any of the imaging device acquisition windows can be opened in “simulation mode.” In this mode, an acquisition window will open and the controls will

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appear active, but instead of capturing real images, the window will create “dummy” images of manufactured data.

You do not need to be connected to an imaging device to open a simulated acquisition window. This is useful for demonstration purposes or practice scans.

To enter simulation mode, hold down the CTRL key and select the name of the device from the File menu. The title of the acquisition window will indicate that it is simulated.

Note: There is no simulated calibration for the GS-700 and GS-710 Imaging Densitometers.

D.1 GS-710 Acquisition Window

To acquire images using the GS-710, go to the File menu and select GS-710.... The acquisition window for the densitometer will open, displaying a control panel and a scanning window.

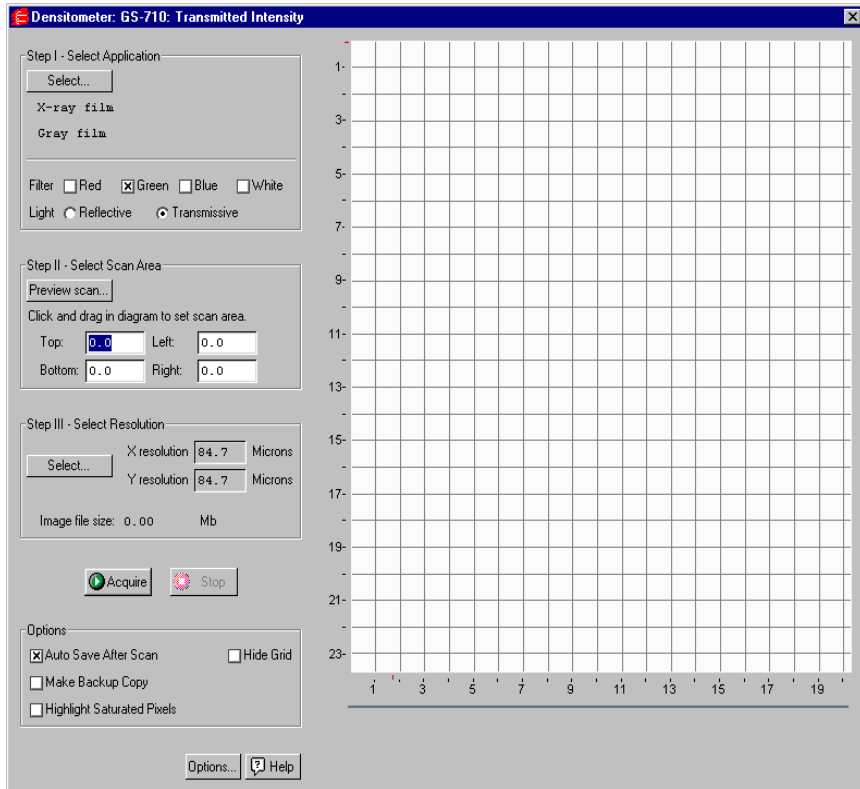


Fig. D-2. GS-710 acquisition window

The scanning window is marked by grid lines that divide the area into square centimeters. These are numbered 1–29 top to bottom and 1–21 left to right if the light source is reflective, and 1–23 top to bottom and 1–20 left to right if the light source is transmissive (see below).

To hide the gridlines, click on the Hide Grid checkbox under Options.

Below the main scanning window is the calibration strip window. Every time the densitometer calibrates, an image of the calibration strip will appear in this window.

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The control panel has been arranged from top to bottom to guide you through the acquisition procedure. There are five basic steps to scanning an image using the GS-710:

1. Select the application.
2. Select the scan area.
3. Select the resolution of the scan.
4. Calibrate the instrument. (This is automatic, after you enter the step tablet values before you first scan after installation.)
5. Acquire the image.

D.2 Step I. Select Application

To set the parameters for your particular scan, you can:

1. Select from a list of possible applications, or
2. Choose your own filter and light source settings.

Selecting from the List of Applications

When you select from the list of applications, the software automatically sets the appropriate filter(s) and other parameters for that particular application.

To select from the list of applications, click on the Select button under Select Application.

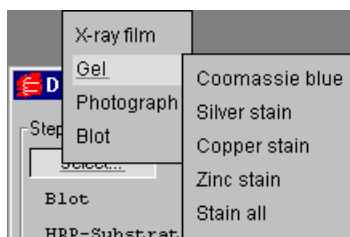


Fig. D-3. Example of the application tree in the GS-710 dialog box.

The applications are listed in a tree that expands from left to right. First you select the category of your application, then you select your particular application.

To exit the tree without selecting, press the ESC key.

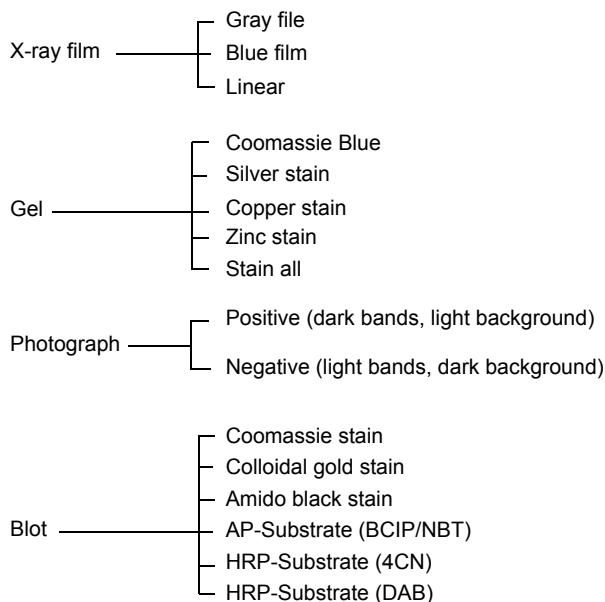


Fig. D-4. Applications available in the GS-710 acquisition window.

Choosing Your Own Settings

If you know the filter and light source settings you want, or want to experiment with different settings, you can choose them yourself.



Fig. D-5. GS-710 Custom Application controls.

Next to Filter, click on either the Red, Green, Blue, or White checkbox, or a combination of two of the first three (Red-Green, Green-Blue, Red-Blue).

Filter Color	Wavelength	Application Examples
Red	595–750 nm	Coomassie G-250, BCIP/NBT, Fast Green FCF, Methylene Blue
Green	520–570 nm	Coomassie R-250, Basic Fuchsin
Blue	400–530 nm	Crocein Scarlet
Gray Scale	400–750 nm	Silver Stains, Copper Stains, Film

Fig. D-6. Examples of filter colors and applications.

Next to Light, select the appropriate light source.

- Reflective mode is for scanning opaque mediums such as dried gels on filter paper, arrays, or photographs. The scanning dimensions in this mode are 21 cm x 29 cm.
- Transmissive mode is for scanning transparent mediums such as films, gels, or slides. The scanning dimensions in this mode are 20 cm x 23 cm.

D.3 Step II. Select Scan Area

Preview Scan

Before selecting the particular area to scan, you can preview the entire scanning area to determine the exact position of your sample.

Click on Preview Scan. A quick, low-resolution scan of the entire scanning area will appear in the scanning window. Your sample should be visible within a portion of this scan.

Selecting an Area

Using the preview scan as a guide, select your scan area by dragging your mouse within the scanning window. The border of the scan area you are selecting will be marked by a frame.

Note: The scan area you select must be at least 1 cm wide.

When you release the mouse button, the border changes to a dashed blue line, indicating a selected area.

- To *reposition* the scanning area box you have created, position your cursor inside the box and drag. The entire box will move.
- To *resize* the box, position your cursor on a box side and drag. The side you have selected will move.
- To *redo* the box entirely, position your cursor outside the box and drag. The old box will disappear and a new box will be created.

You can also select the scanning area by entering coordinates in the appropriate fields (Top, Bottom, Left, Right). As you enter a coordinate, the position of the scanning area box will change accordingly.

When selecting, be sure to include the entire area of interest, and be generous with borders. You can always crop the image later.

D.4 Step III. Select Resolution

To select from a list of possible scanner resolutions, click on the Select button under Select Resolution. This will open the Select Scan Resolution dialog box.

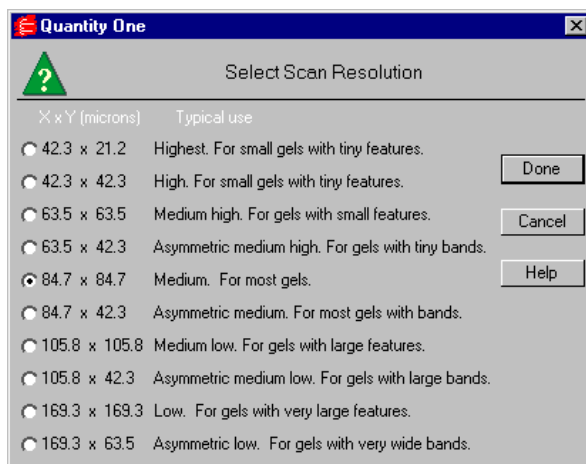


Fig. D-7. Select Scan Resolution dialog box.

Available resolutions are listed from highest to lowest in terms of the dimensions of the resulting pixels (in microns). Smaller pixels equal higher resolution. Each resolution is listed with its typical use.

In general, the size of your pixels should be one-tenth the height of your smallest object.

Some of the resolutions are asymmetrical, meaning that the resolution is higher in the vertical dimension (i.e., the pixels in the resulting image are larger in the horizontal dimension than in the vertical dimension). This is useful for gels with bands, where you are more interested in resolving in the y dimension to determine the size of bands and the spacing between them.

Specifying Your Own Resolution

If you select Oversample under More Options, you can specify your own resolution within the range of 43–169 microns (micrometers). With Oversample selected, enter values directly in the fields next to X resolution and Y resolution in the main acquisition window.

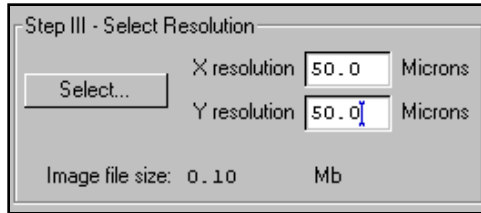


Fig. D-8. Entering a custom resolution (with Oversample selected).

Image File Size

Image File Size (under Select Resolution) shows the size of the scan file you are about to create. If you do not have enough computer memory for the specified file size, an error message will appear when you attempt to scan. If this happens, select a lower resolution or decrease the size of the area to be scanned. (Macintosh users can also increase the application memory partition. See your Macintosh computer documentation for guidance.)

D.5 Calibration

Images scanned using the GS-710 are automatically calibrated. The scanner plate has a built-in calibration overlay for both transmissive and reflective scanning.

- The transmissive calibration strip calibrates the densitometer from 0 to 3.0 O.D.
- The reflective calibration strip calibrates the densitometer to 2.0 O.D.

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The first time you use the GS-710, you must select some calibration settings to ensure that your calibration is accurate.

To set the automatic calibration settings, click on the More Options button in the GS-710 acquisition window. This will open the Densitometer Options dialog box.

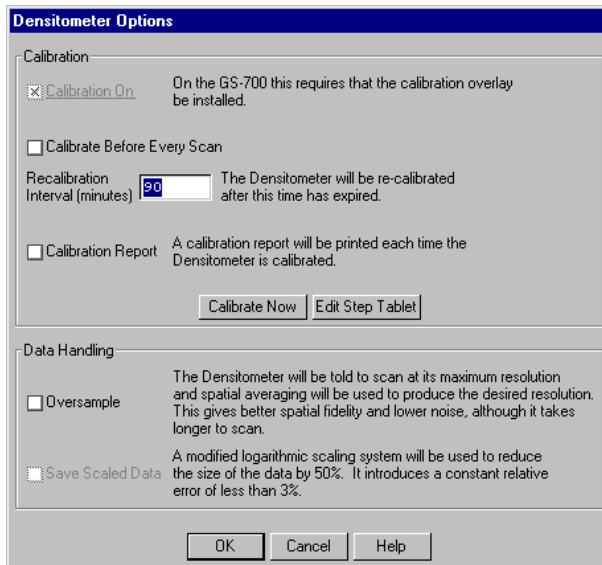


Fig. D-9. Densitometer Options dialog box.

D.5.a Editing the Step Tablet

Before you scan for the first time, you must enter the step tablet values for your transmissive calibration strip into the Step Tablet Values form.

Note: In reflective scanning, the scanner is calibrated against the reflective step tablet. However, you do not need to enter the values for this step tablet; you can use the default values in the Step Tablet Values form.

Attached to the outside of your GS-710, you will find a copy of the manufacturer's printout of the diffuse density values for each segment of your transmissive step tablet, as well as a serial number that identifies the

step tablet to which those density values belong. Those exact density values must be entered into the computer for the software to associate a correct density value with each step on the step tablet.

Note: Scanning in transmissive mode with incorrect step tablet values entered into the computer can cause significant errors in the reported densities of your scans.

To enter the density values for your step tablet, click on the Edit Step Tablet button in the Densitometer Options box. The Step Tablet Values dialog box for the GS-710 will open.

Step	Diffuse	Specular	Step	Diffuse	Specular
1.	0.05	0.07	12.	1.72	2.43
2.	0.20	0.28	13.	1.88	2.66
3.	0.36	0.51	14.	2.03	2.87
4.	0.52	0.74	15.	2.18	3.08
5.	0.66	0.93	16.	2.34	3.31
6.	0.81	1.15	17.	2.49	3.52
7.	0.96	1.36	18.	2.64	3.73
8.	1.11	1.57	19.	2.80	3.96
9.	1.27	1.80	20.	2.94	4.16
10.	1.41	1.99	21.	3.08	4.36
11.	1.56	2.21			

Fig. D-10. Step Tablet dialog box.

The dialog box will indicate whether the values are for the transmissive or reflective step tablet, depending on whether you are in transmissive or reflective mode.

If you are in reflective mode, you do not need to change the default step tablet values.

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If you are in transmissive mode, you do need to enter the correct step tablet values that were shipped with your GS-710. You can type the serial number for these values into the Tablet Serial Number field.

The Quantity Offset field does not apply in the GS-710. This value should be set to zero.

Next, enter the values for the transmissive step tablet in the appropriate fields under the Diffuse column. After the step tablet is scanned, the software will associate each density value with its corresponding segment on the step tablet.

The step tablet density values do not need to be entered each time you calibrate. Once they have been entered and saved, they will be automatically recalled when the calibration strip is scanned.

When you are finished filling out the Step Tablet Values form, click on OK.

Diffuse Versus Specular O.D.

In the step tablet form, you enter O.D. as diffuse density, and then the software automatically calculates the specular density.

Specular density is a measure of the light that passes directly through a medium. Diffuse density includes light that is scattered as it passes through the medium. Step tablet values are given in diffuse density, but are measured by the scanner in specular density, and therefore must be converted according to the specular / diffuse O.D. ratio. This conversion does not affect quantitation.

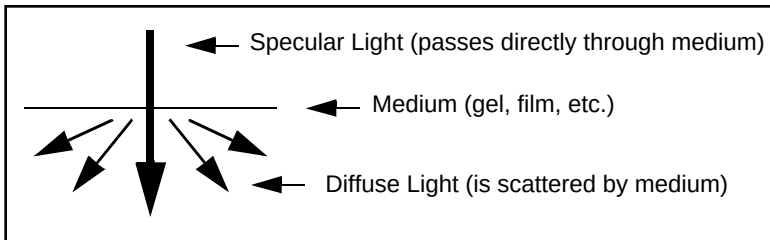


Fig. D-11. Specular and diffuse density

Diffuse density values are converted to specular optical density units according to the following formula:

$$\text{Specular OD} = 1.4 \text{ Diffuse OD}$$

D.5.b Calibration Settings

After you have entered the step tablet values, you can immediately calibrate by clicking on the Calibrate Now button (in the Densitometer Options dialog box).

You can also specify how often you want the GS-710 to automatically recalibrate. Either click on the Calibrate Before Every Scan checkbox or enter a Recalibration Interval (in minutes) in the appropriate field.

Note: The scanner will automatically recalibrate each time you change your filter or your reflective/transmissive setting. (If you select a different application with the same filter and light settings, it will not auto recalibrate.)

Calibration Report

To print out a calibration report each time the densitometer calibrates, click on the Calibration Report checkbox.

D.6 Acquire the Image

Note: Before scanning in transmissive mode, make sure the white balance region of the scanning area is not covered or obstructed in any way.

To begin to scan, click on the Acquire button. The scanned image will begin to appear in the scanning window, line by line.

To interrupt a scan, click on the Stop button. A message will ask you to confirm the interrupt, and then you will be asked if you want to keep the partial scan. This feature is useful if you overestimated the size of the area you selected.

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After the scan is complete, a window will open displaying the scan image, at which point you can analyze and save it.

Note: The image will open with a default file name that includes the date, time, and (if applicable) user name. However, unless you have selected Auto Save After Scan, the file will not be saved until you select Save or Save As from the File menu.

D.7 Other Options

Oversample

This feature allows you to scan at the maximum resolution of the GS-710 (42.3 x 42.3 microns) and then use spatial averaging to create an image with lower resolution. This can result in better images at lower resolution—however, it takes longer to scan.

To turn on oversampling, click on the More Options button in the acquisition window and click on the Oversample checkbox.

With oversampling on, you can specify your own resolution within the range of 43–169 microns by entering values directly in the fields next to X resolution and Y resolution in the main acquisition window.

Auto Save After Scan

To automatically save any scan you create, click on the Auto Save After Scan checkbox.

Note: In PDQUEST, this option is preselected and cannot be turned off. All images must be automatically saved when acquired.

With this checkbox selected, when you click on Acquire, a Save As dialog box will open asking you to specify a file name and location for the image you are about to create. The scan will begin when you click on the Save button.

Make Backup Copy

If you have checked Auto Save After Scan, you can also automatically create a backup copy of any scan you create.

Click on the Make Backup Copy checkbox. With this checkbox selected, when a scan is created and saved, a backup copy will be placed in the same directory as the scanned image. Windows backup files will have an “.sbk” extension. Macintosh backup files will have the word “backup” after the file name.

This backup copy will be read-only, which means that you cannot make changes to it. You can open it like a normal file, but you must save it under a different file name before editing the image or performing analysis.

Highlight Saturated Pixels

When this box is checked, any saturated pixels in the image will appear highlighted in red in the scan window and in the pop-up image window. To view/hide saturated pixels in the pop-up image window, use the Image > Transform command.

Hide Grid

To hide the gridlines in the scanning area window, click on the Hide Grid checkbox.

Appendix E

GS-800 Imaging Densitometer

Before you can begin scanning images with the GS-800 Imaging Densitometer®, your instrument must be properly installed and connected with the host computer. See the hardware manual for installation, startup, and operating instructions.

PC Only: A Note About SCSI Cards

The GS-800 is connected to your computer by a Small Computer System Interface (SCSI) cable. To use the GS-800, you must have a SCSI card installed in your PC. If you have an older PC, you may also need to load the SCSI and WinASPI drivers that came with your card.

Simulation Mode

Any of the imaging device acquisition windows can be opened in “simulation mode.” In this mode, an acquisition window will open and the controls will appear active, but instead of capturing real images, the window will create “dummy” images of manufactured data.

You do not need to be connected to an imaging device to open a simulated acquisition window. This is useful for demonstration purposes or practice scans.

To enter simulation mode, hold down the CTRL key and select the name of the device from the File menu. The title of the acquisition window will indicate that it is simulated.

Note: There is no simulated calibration for densitometers.

E.1 GS-800 Acquisition Window

To begin acquiring images, go to the File menu and select GS-800.... The acquisition window for the densitometer will open, displaying a control panel and a scanning window.

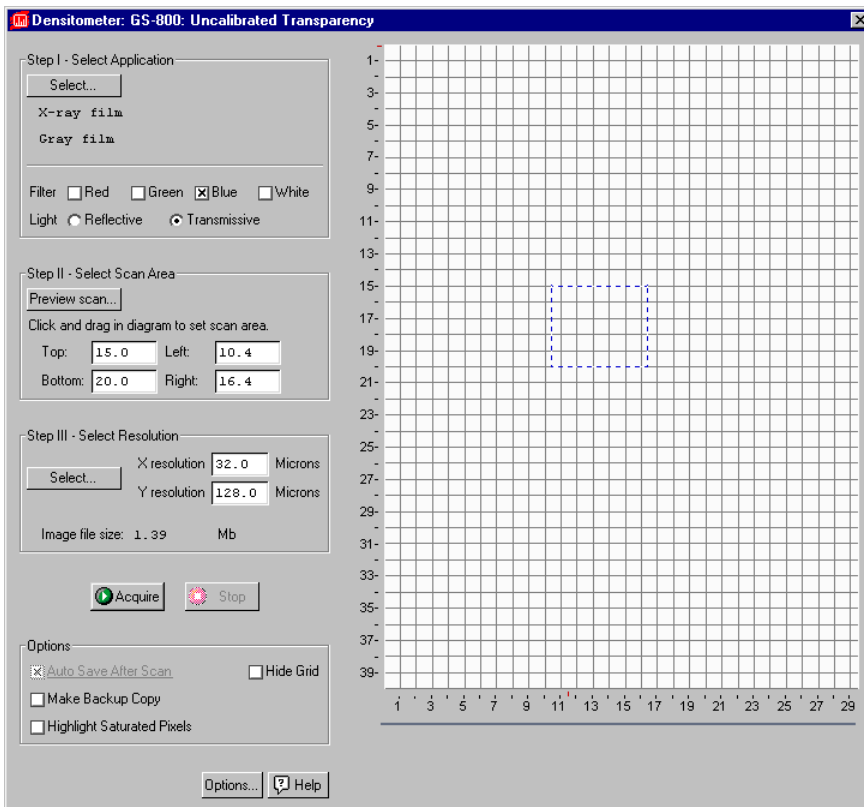


Fig. E-1. GS-800 acquisition window

The scanning window is marked by grid lines that divide the area into square centimeters. These are numbered 1-40 top to bottom and 1-30 left to right if the light source is reflective, and 1-40 top to bottom and 1-29 left to right if the light source is transmissive.

To hide the gridlines, click on the Hide Grid checkbox under Options.

Below the main scanning window is the calibration strip window. Every time the densitometer calibrates, an image of the calibration strip will appear in this window.

The control panel has been arranged from top to bottom to guide you through the acquisition procedure. There are five basic steps to scanning an image using the GS-800:

1. Select the application.
2. Select the scan area.
3. Select the resolution of the scan.
4. Calibrate the instrument. (This is automatic, after you enter the step tablet values before you first scan after installation.)
5. Acquire the image.

E.2 Step I. Select Application

To set the parameters for your particular scan, you can:

1. Select from a list of possible applications, or
2. Choose your own filter and light source settings.

Selecting from the List of Applications

When you select from the list of applications, the software automatically sets the appropriate filter(s) and other parameters for that particular application.

To select from the list of applications, click on the Select button under Select Application.

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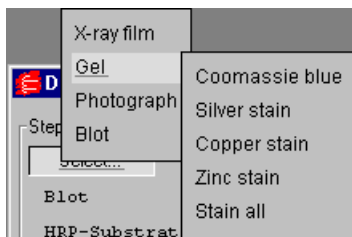


Fig. E-2. Example of the application tree in the GS-800 dialog box.

The applications are listed in a tree that expands from left to right. First you select the category of your application, then you select your particular application.

To exit the tree without selecting, press the ESC key.

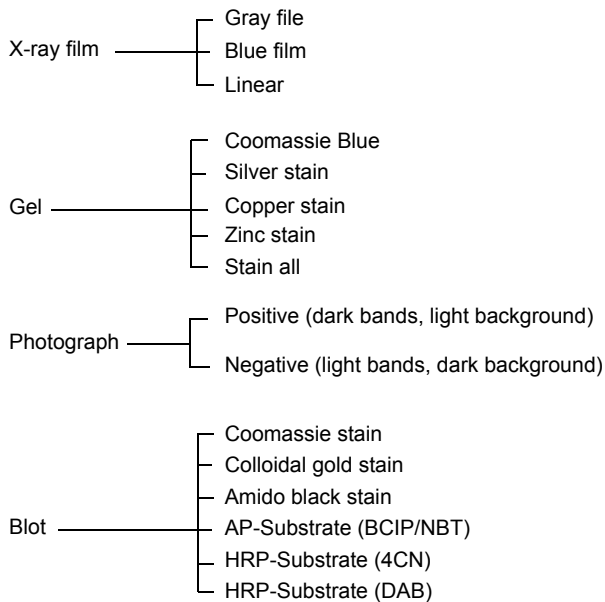


Fig. E-3. Applications available in the GS-800 acquisition window.

Choosing Your Own Settings

If you know the filter and light source settings you want, or want to experiment with different settings, you can choose them yourself.

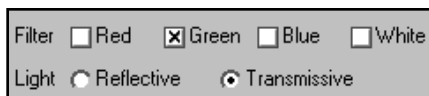


Fig. E-4. GS-800 Custom Application controls.

Next to Filter, click on either the Red, Green, Blue, or White checkbox, or a combination of two of the first three (Red-Green, Green-Blue, Red-Blue).

Filter Color	Wavelength	Application Examples
Red	595–750 nm	Coomassie G-250, BCIP/NBT, Fast Green FCF, Methylene Blue
Green	520–570 nm	Coomassie R-250, Basic Fuchsin
Blue	400–530 nm	Crocein Scarlet
Gray Scale	400–750 nm	Silver Stains, Copper Stains, Film

Fig. E-5. Examples of filter colors and applications.

Next to Light, select the appropriate light source.

- Reflective mode is for scanning opaque mediums such as dried gels on filter paper, arrays, or photographs. The scanning dimensions in this mode are 21 cm x 29 cm.
- Transmissive mode is for scanning transparent mediums such as films, gels, or slides. The scanning dimensions in this mode are 20 cm x 23 cm.

E.3 Step II. Select Scan Area

Preview Scan

Before selecting the particular area to scan, you can preview the entire scanning area to determine the exact position of your sample.

Click on Preview Scan. A quick, low-resolution scan of the entire scanning area will appear in the scanning window. Your sample should be visible within a portion of this scan.

Selecting an Area

Using the preview scan as a guide, select your scan area by dragging your mouse within the scanning window. The border of the scan area you are selecting will be marked by a frame.

Note: The scan area you select must be at least 1 cm wide.

When you release the mouse button, the border changes to a dashed blue line, indicating a selected area.

- To *reposition* the scanning area box you have created, position your cursor inside the box and drag. The entire box will move.
- To *resize* the box, position your cursor on a box side and drag. The side you have selected will move.
- To *redo* the box entirely, position your cursor outside the box and drag. The old box will disappear and a new box will be created.

You can also select the scanning area by entering coordinates in the appropriate fields (Top, Bottom, Left, Right). As you enter a coordinate, the position of the scanning area box will change accordingly.

When selecting, be sure to include the entire area of interest, and be generous with borders. You can always crop the image later.

E.4 Step III. Select Resolution

To select from a list of possible scanner resolutions, click on the Select button under Select Resolution. This will open the Select Scan Resolution dialog box.

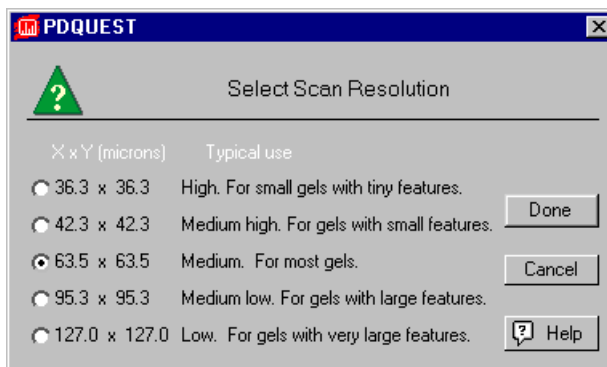


Fig. E-6. Select Scan Resolution dialog box.

Available resolutions are listed from highest to lowest in terms of the dimensions of the resulting pixels (in microns). Smaller pixels equal higher resolution. Each resolution is listed with its typical use.

In general, the size of your pixels should be one-tenth the height of your smallest object.

Specifying Your Own Resolution

If you select Oversample under More Options, you can specify your own resolution within the range of 43–169 microns (micrometers). With Oversample selected, enter values directly in the fields next to X resolution and Y resolution in the main acquisition window.

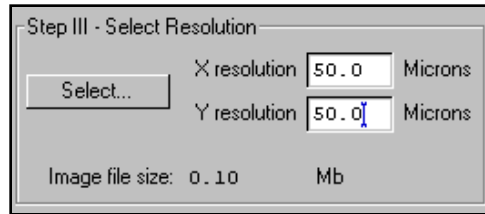


Fig. E-7. Entering a custom resolution (with Oversample selected).

Image File Size

Image File Size (under Select Resolution) shows the size of the scan file you are about to create. If you do not have enough computer memory for the specified file size, an error message will appear when you attempt to scan. If this happens, select a lower resolution or decrease the size of the area to be scanned. (Macintosh users can also increase the application memory partition. See your Macintosh computer documentation for guidance.)

E.5 Calibration

Images scanned by the GS-800 in transmissive mode are automatically calibrated. The scanner plate has a built-in calibration strip for transmissive scanning.

Note: Calibration is unavailable for the GS-800 in reflective mode.

The first time you use the GS-800, you must select some calibration settings to ensure that your calibration is accurate.

To set the automatic calibration settings, click on the More Options button in the GS-800 acquisition window. This will open the Densitometer Options dialog box.

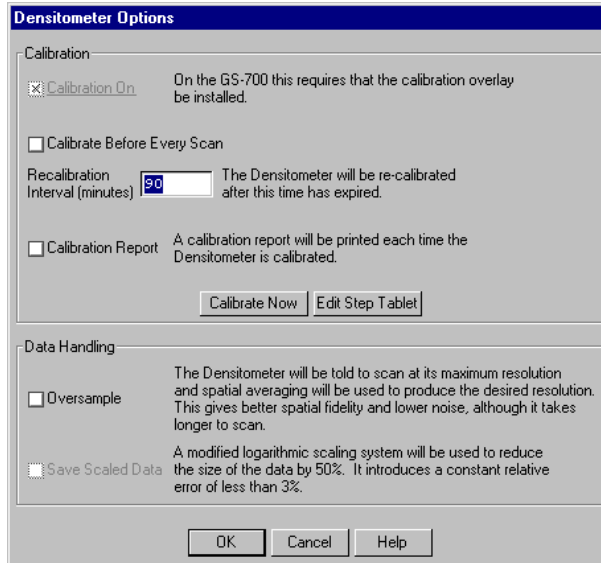


Fig. E-8. Densitometer Options dialog box.

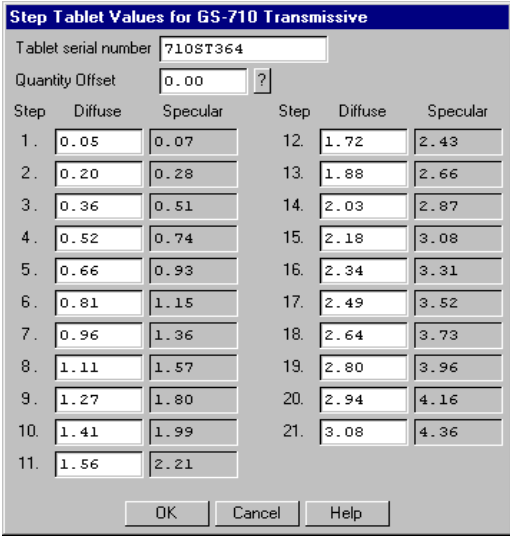
E.5.a Editing the Step Tablet

Before you scan for the first time, you must enter the step tablet values for your transmissive calibration strip into the Step Tablet Values form.

Attached to the outside of your GS-800, you will find a copy of the manufacturer's printout of the diffuse density values for each segment of your transmissive step tablet, as well as a serial number that identifies the step tablet to which those density values belong. Those exact density values must be entered into the computer for the software to associate a correct density value with each step on the step tablet.

Note: Scanning in transmissive mode with incorrect step tablet values can cause significant errors in the reported densities of your scans.

To enter the density values for your transmissive cal strip, select Transmissive in the main acquisition window and click on the Edit Step Tablet button in the Densitometer Options box.



The dialog box is titled "Step Tablet Values for GS-710 Transmissive". It contains a "Tablet serial number" field with the value "7108T364" and a "Quantity Offset" field with the value "0.00" and a question mark icon. Below these fields is a table with two columns of data. The first column has three sub-columns: "Step", "Diffuse", and "Specular". The second column also has three sub-columns: "Step", "Diffuse", and "Specular". The table contains 21 rows of data, numbered 1 through 21. At the bottom of the dialog box are three buttons: "OK", "Cancel", and "Help".

Step	Diffuse	Specular	Step	Diffuse	Specular
1.	0.05	0.07	12.	1.72	2.43
2.	0.20	0.28	13.	1.88	2.66
3.	0.36	0.51	14.	2.03	2.87
4.	0.52	0.74	15.	2.18	3.08
5.	0.66	0.93	16.	2.34	3.31
6.	0.81	1.15	17.	2.49	3.52
7.	0.96	1.36	18.	2.64	3.73
8.	1.11	1.57	19.	2.80	3.96
9.	1.27	1.80	20.	2.94	4.16
10.	1.41	1.99	21.	3.08	4.36
11.	1.56	2.21			

Fig. E-9. Step Tablet dialog box.

In the step tablet dialog box, type the serial number for the printout into the Tablet Serial Number field.

The Quantity Offset field does not apply in the GS-800. This value should be set to zero.

Next, enter the values for the transmissive step tablet in the appropriate fields under the Diffuse column. After the step tablet is scanned, the software will associate each density value with its corresponding segment on the step tablet.

The step tablet density values do not need to be entered each time you calibrate. Once they have been entered and saved, they will be automatically recalled when the calibration strip is scanned.

When you are finished filling out the Step Tablet Values form, click on OK.

Diffuse Versus Specular O.D.

In the step tablet form, you enter O.D. as diffuse density, and then the software automatically calculates the specular density.

Specular density is a measure of the light that passes directly through a medium. Diffuse density includes light that is scattered as it passes through the medium. Step tablet values are given in diffuse density, but are measured by the scanner in specular density, and therefore must be converted according to the specular / diffuse O.D. ratio. This conversion does not affect quantitation.

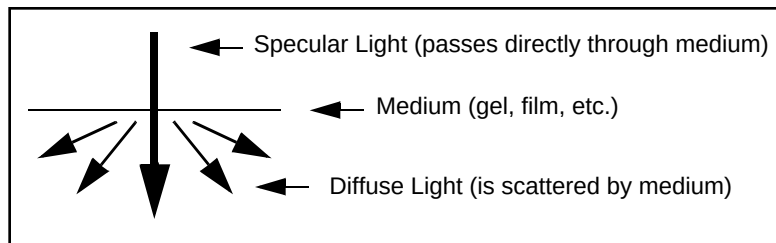


Fig. E-10. Specular and diffuse density

Diffuse density values are converted to specular optical density units according to the following formula:

$$\text{Specular OD} = 1.4 \text{ Diffuse OD}$$

E.5.b Calibration Settings

After you have entered the step tablet values, you can immediately calibrate by clicking on the Calibrate Now button (in the Densitometer Options dialog box).

You can also specify how often you want the GS-800 to automatically recalibrate. Either click on the Calibrate Before Every Scan checkbox or enter a Recalibration Interval (in minutes) in the appropriate field.

Note: The scanner will automatically recalibrate each time you change your filter or your reflective/transmissive setting. (If you select a different

application with the same filter and light settings, it will not auto recalibrate.)

Calibration Report

To print out a calibration report each time the densitometer calibrates, click on the Calibration Report checkbox.

E.6 Acquire the Image

Note: Before scanning in transmissive mode, make sure the white balance region of the scanning area is not covered or obstructed in any way.

To begin to scan, click on the Acquire button. The scanned image will begin to appear in the scanning window, line by line.

To interrupt a scan, click on the Stop button. A message will ask you to confirm the interrupt, and then you will be asked if you want to keep the partial scan. This feature is useful if you overestimated the size of the area you selected.

After the scan is complete, a window will open displaying the scan image, at which point you can analyze and save it.

Note: The image will open with a default file name that includes the date, time, and (if applicable) user name. However, unless you have selected Auto Save After Scan, the file will not be saved until you select Save or Save As from the File menu.

E.7 Other Options

Oversample

This feature allows you to scan at the maximum resolution of the scanner and then use spatial averaging to create an image with lower resolution. This can result in better images at lower resolution—however, it takes longer to scan.

To turn on oversampling, click on the More Options button in the acquisition window and click on the Oversample checkbox.

With oversampling on, you can specify your own resolution within the range of the densitometer by entering values directly in the fields next to X resolution and Y resolution in the main acquisition window.

Auto Save After Scan

To automatically save any scan you create, click on the Auto Save After Scan checkbox.

Note: In PDQUEST, this option is preselected and cannot be turned off. All images must be automatically saved when acquired.

With this checkbox selected, when you click on Acquire, a Save As dialog box will open asking you to specify a file name and location for the image you are about to create. The scan will begin when you click on the Save button.

Make Backup Copy

If you have checked Auto Save After Scan, you can also automatically create a backup copy of any scan you create.

Click on the Make Backup Copy checkbox. With this checkbox selected, when a scan is created and saved, a backup copy will be placed in the same directory as the scanned image. Windows backup files will have an “.sbk” extension. Macintosh backup files will have the word “backup” after the file name.

This backup copy will be read-only, which means that you cannot make changes to it. You can open it like a normal file, but you must save it under a different file name before editing the image or performing analysis.

Highlight Saturated Pixels

When this box is checked, any saturated pixels in the image will appear highlighted in red in the scan window and in the pop-up image window. To view/hide saturated pixels in the pop-up image window, use the Image > Transform command.

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Hide Grid

To hide the gridlines in the scanning area window, click on the Hide Grid checkbox.

Appendix F

Fluor-S Multimager



Fig.F-1. Fluor-S Multimager.

Before you can begin acquiring images using the Fluor-S[®] MultiImager, the imaging system must be properly installed and connected with the host computer. See the Fluor-S hardware manual for installation, startup, and operating instructions.

Note: Make sure that the temperature light on the Fluor-S is green before attempting to capture an image. If you are using a PC, the Fluor-S should be turned on and the initialization sequence completed *before* the host computer is turned on. See the hardware manual for more details.

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PC Only: A Note About SCSI Cards

The Fluor-S is connected to your computer by a Small Computer System Interface (SCSI) cable. To use the Fluor-S, you must have a SCSI card installed in your PC. If you have an older PC, you may also need to load the SCSI and WinASPI drivers that came with your card.

Simulation Mode

Any of the imaging device acquisition windows can be opened in “simulation mode.” In this mode, an acquisition window will open and the controls will appear active, but instead of capturing real images, the window will create “dummy” images of manufactured data.

You do not need to be connected to an imaging device to open a simulated acquisition window. This is useful for demonstration purposes or practice scans.

To enter simulation mode, hold down the CTRL key and select the name of the device from the File menu. The title of the acquisition window will indicate that it is simulated.

F.1 Fluor-S Acquisition Window

To acquire images using the Fluor-S, go to the File menu and select Fluor-S.... The acquisition window for the imager will open, displaying a control panel and an image display window.



Fig.F-2. Fluor-S acquisition window

When the Fluor-S window first opens, no image will be displayed.

The control panel has been arranged from top to bottom to guide you through the acquisition procedure. There are four basic steps to acquiring an image using the Fluor-S:

1. Select the application.
2. Position and focus the gel or other object to be imaged.
3. Set the exposure time.
4. Acquire the image.

F.2 Step I. Select Application

To set the appropriate filter and other parameters for the type of object you are imaging, click on the Select button under Select Application. The available applications and their associated settings are listed in a tree that expands from left to right.

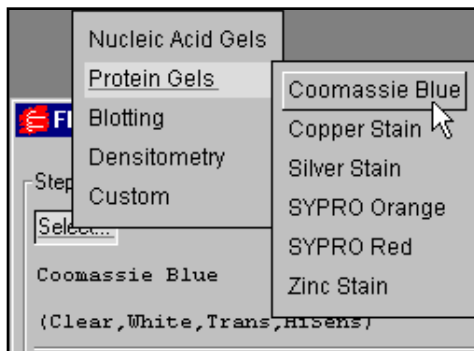


Fig.F-3. The application tree in the Fluor-S acquisition window.

First select your general application, then select the particular stain or medium you are using. If you select the chemiluminescent application under "Blotting," you also have the option of selecting High Resolution or High Sensitivity (see below).

When you select an application, the software automatically sets the appropriate standard filter (520LP, 530DF60, 610LP, clear, or none), light type (UV, white, or none), and light source (trans, epi, or neither) in the Fluor-S for that particular application.

For applications involving trans illumination, you must also specify a scan dimension (see below).

Your selection will be displayed below the Select button. To exit the tree without selecting, press the ESC key.

Custom Applications

If your application is not listed, if you want to use a user-installed filter, or if you want to access High Sensitivity mode (see below), you can create and save your own custom application.

From the application tree, select Custom, then Create. This will open a dialog box in which you can name your application and select your settings.

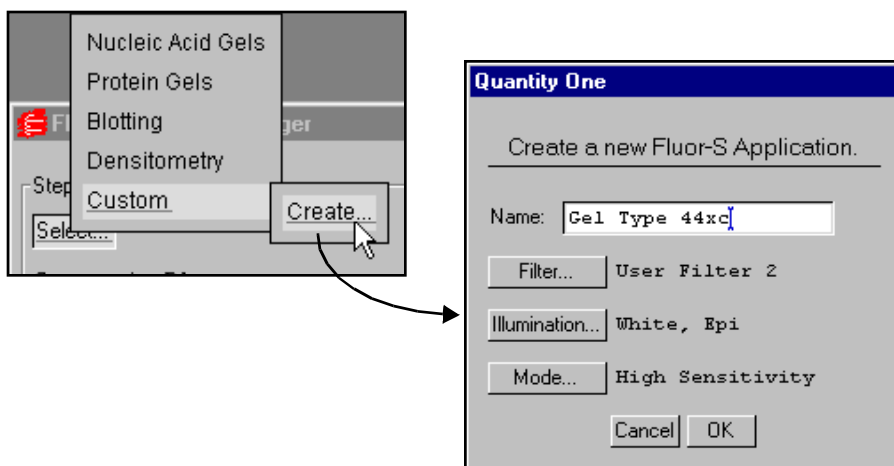


Fig.F-4. Creating a new custom application.

To select the filter (including user-defined), type of illumination, and camera mode, click on the appropriate buttons and select from the pop-up lists.

Note: Under Illumination, there is a listing for a spare UV light source. This selects the spare UV bulb in the Fluor-S. Select this light source if your main UV bulb fails.

Enter a name for your application in the Name field. Click on OK to implement your changes.

After you have created an application, you can select it from the application tree by selecting Custom and the name you created. You can delete the application by selecting Custom, Delete, and the name of the application.

Scan Dimension

If an application uses trans illumination, the Scan Dimension buttons become active. The scan dimension is the distance traveled by the transilluminating light source as it scans horizontally across the platen.

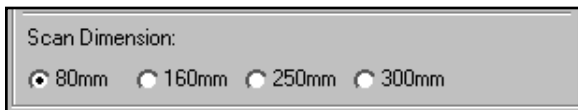


Fig.F-5. Scan dimension settings for trans illumination.

The full scanning range is 300 mm. Select a smaller range if your sample is small and you do not want to wait while the light source travels over the maximum scan width.

High Resolution Versus High Sensitivity

High Resolution and High Sensitivity are mutually exclusive options. High Resolution is the normal operating mode for the Fluor-S. High Sensitivity provides optimal sensitivity for low-light applications. It is the default selection for the chemiluminescence application, and may be selected for a custom application.

In High Resolution mode, images are captured at the maximum resolution of the camera.

In High Sensitivity mode, the pixels in the camera are “binned” (i.e., four pixels are combined into one) to increase the amount of signal per pixel. However, combining the pixels results in a reduction in the resolution of the image.

F.3 Step II. Position/Focus

Note: When you click on the Position or Focus button, the light inside the Fluor-S box automatically turns on. To turn the light off while positioning or focusing, hold down the SHIFT key when clicking on the button.

Position

After you have selected your application, you are ready to center your gel or other object within the camera frame. To do so, click on the Position button. The Fluor-S will begin capturing a “live” image and updating it every second.

With the Position button selected, look at the image in the acquisition window while you position your object in the center of the platen. If you have a zoom lens on the camera, you can adjust the magnification while you position.

You can select the Show Alignment Grid checkbox to facilitate positioning.

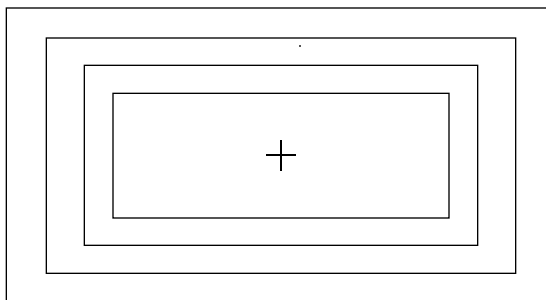


Fig.F-6. Fluor-S alignment grid.

When you are finished positioning, click on the Stop button.

Focus

Note: Before focusing, you should adjust the f-stop on the camera to the lowest setting (i.e., the maximum aperture). This reduces the depth of field, allowing you to more accurately focus the camera. Then, after focusing, increase the f-stop to the desired setting. (See the following table on recommended f-stops.)

After you have positioned your sample, click on the Focus button and look at the image in the acquisition window while aligning the two focusing arrows on the camera lens. While focusing, the camera will limit its focus to a small

portion of the sample (this will not affect any zoom lens adjustments you may have made.)

When you are finished focusing, click on the Stop button.

F.4 Step III. Set Exposure Time

When you are ready to capture an image, you will need to select an exposure time. “Exposure” refers to the integration of image captures on the CCD over a set period of time. The effect is analogous to exposing photographic film to light.

The exposure time you select should be based on your application and your “best guess” as to what exposure will give you the best image.

Note: The minimum exposure time in trans illuminated mode is 1 second. The minimum exposure time in epi illuminated mode is 0.1 second.

You can enter an exposure time (in seconds) directly in the field, or use the Arrow buttons to adjust the exposure time in 10 percent increments.



Fig.F-7. Selecting an exposure time.

The following table provides recommended exposure times for various applications.

Recommended Exposure Times and Lenses

Sample	Recommended Exposure ¹	Lens & Filter ²	Accessories Used
Fluorescent Stain Gel	3–40 sec.	Zoom/IR	None
Fluorescence End-Label Gel	30 sec.–3 min.	Zoom/IR	None
Fluorescent Blot	0.5–5 sec.	Zoom/IR	None
Chemifluorescent Blot	0.5–5 sec.	Zoom/IR	None
Colorimetric Gel	1–10 sec.	Zoom/IR	White Diffusion Plate
Colorimetric Blot	0.5–20 sec.	Zoom/IR	None
X-ray film	1–10 sec.	Zoom/IR	White Diffusion Plate
Weak Chemiluminescence ³	5–20 min.	50 mm	Chemi Tray (if sample is small)
Strong Chemiluminescence ³	30 sec.–2 min.	50 mm	Chemi Tray (if sample is small)
¹ Increase exposure time two fold for each step increase in f-stop. ² For sharper focusing, close the f-stop down 1–2 stops from full open while focusing. ³ For chemi applications, the 50mm lens is recommended. Always remove the 660 filter.			

For most applications, you can select an exposure time, capture an image, study it, then adjust the exposure time accordingly. Repeat this procedure as many times as necessary to obtain a good image.

For chemiluminescent samples, which degrade over time and emit low levels of light, you can select a high exposure time initially and use Live Acquire mode to save intermediate exposures (see following section).

Preview

For shorter exposures, you can use Preview to test different exposure times. Click on the Preview button create a preview exposure and display it in the acquisition window.

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A preview scan takes only half as long to create as a real scan, because the preview scan does not capture a “dark” image (see the following section on Options). The progress of the exposure will be displayed in the Exposure Status bar at the bottom of the dialog box.

You cannot save preview scans.

If you want to stop a preview scan that is in progress, click on the Stop button.

F.5 Acquire the Image

The Fluor-S gives you the option of simply acquiring and displaying a fully exposed image, or preserving intermediate exposures.

Acquire

To acquire and display a fully exposed image, click on the Acquire button. An exposure will be taken based on the time selected in Step III. This is appropriate for most short exposures.

The progress of the exposure will be displayed in the Exposure Status bar at the bottom of the acquisition window.

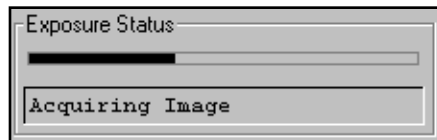


Fig.F-8. Exposure Status bar when acquiring an image.

Depending on which dark subtraction type you have selected, a dark count may be acquired immediately following image acquisition. See Dark Subtraction Type under Options, below.

If you want to stop a scan that is in progress, click on the Stop button. The acquisition will be terminated.

After an image has been acquired, a separate window will pop up containing the new image. The window will have a default file name that includes the date, time, and user (if known). To save the image, select Save or Save As from the File menu.

You can then analyze the image using the analysis functions.

Live Acquire

Live Acquire allows you to view and preserve intermediate exposures leading up to a full exposure. This is useful for longer exposure times, such as those required by chemiluminescent samples, where there is the potential for image saturation.

When you click on the Live Acquire button, the exposure time you selected is divided by the number of exposure counts set in the Fluor-S Options dialog box (see Options, below). For example, if you enter an exposure time of 10 minutes and an exposure count of 20, then 20 intermediate exposures will be produced at 30-second intervals. Each intermediate exposure will be displayed in the scan window. The final, full exposure will be displayed in a separate image window.

Note: The first intermediate exposure will take longer than 30 seconds to display if dark subtraction is performed.

You can automatically save your intermediate exposures as separate files for later review using the Auto Save After Scan option. See Options, below.

If you see an intermediate exposure that you like in the scan window, click on the Stop button. Live Acquire mode will end and the last intermediate exposure to be completed will open in a separate image window. You can then save it for analysis.

F.6 Options

Click on the Options button to open the Fluor-S Options dialog box.

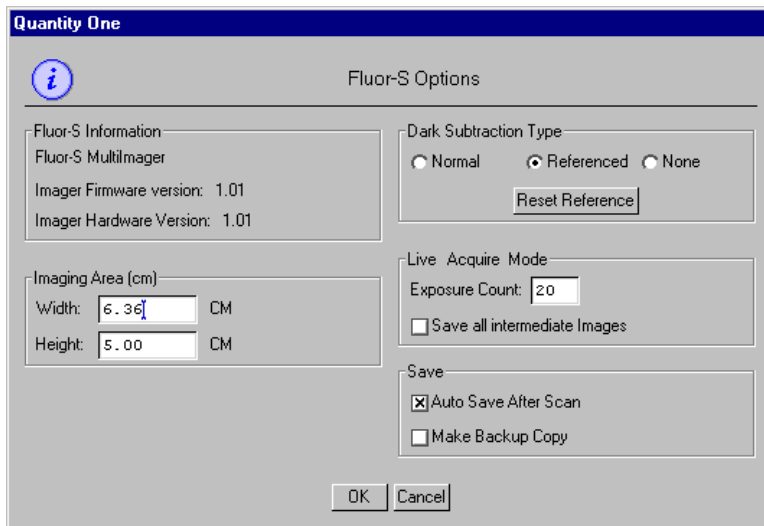


Fig.F-9. Options dialog box.

F.6.a Dark Subtraction Type

All CCD cameras accumulate electrons that produce a signal that is indistinguishable from light. This “dark count” adds to the noise in your images. In most cases, you will want to subtract this dark count from your images.

Normal

The Normal option button selects the default dark subtraction type. In this mode, after you acquire an image, a “dark” image of the same exposure length will be taken, and this will be subtracted from your image.

The progress of the dark exposure will be displayed in the Exposure Status bar following the regular image exposure.

Note: In Normal mode, a dark image is only acquired the first time you perform a scan with particular application and exposure settings. If you perform subsequent scans with the same settings, no dark exposure will be taken.

Referenced

If you do not want to perform a dark exposure with each acquisition, you can take a “reference” dark exposure that will be saved and subtracted from all subsequent acquisitions. Click on the Referenced button to activate this feature.

The first time you acquire an image after selecting this option, the Fluor-S will take a 60-second dark exposure that will be saved and used to subtract the dark count from all subsequent acquisitions.

For image exposures of greater or less than 60 seconds, the reference dark will be scaled accordingly and then subtracted. You can change the default reference dark exposure time using the Reset Reference button (see below).

If you deselect the Referenced button and then reselect it, the old reference dark exposure will still be available.

Note: Separate reference dark exposures will be taken for High Resolution mode and High Sensitivity mode. Once you have created a reference dark in each of these modes, each reference dark will be used according to the mode you are in.

Reset Reference

If you would like a reference dark with an exposure time that more closely matches that of your typical scans, click on the Reset Reference button.

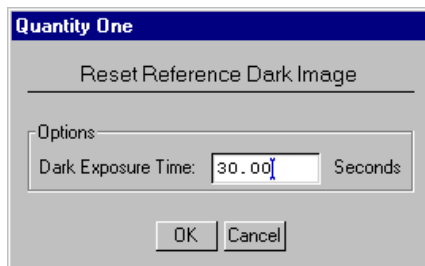


Fig.F-10. Reset Reference Dark pop-up box.

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A pop-up box will prompt you to enter a new reference dark exposure time in seconds. Click on OK to implement your change. The new reference dark will be created when you acquire your next image.

Note: Because of the high sensitivity of the CCD, fluctuations in background radiation and/or temperature in the room can affect the level of dark count. If you feel that radiation/temperature conditions have changed in the room since your last reference dark was created, use the Reset Reference button to delete your old reference and create a new one under current conditions.

None

If you do not want to perform dark subtraction, select None. No dark exposure will be acquired or subtracted.

F.6.b Live Exposure Mode

Exposure Count

If you are using the Live Acquire function (see previous section), you need to specify how many intermediate exposures you want to view/save during acquisition. Enter this number in the Exposure Count field.

The total exposure time will be divided by the number you enter in the Exposure Count field. If you enter an exposure time of 10 minutes and a count of 10, you will create 10 intermediate exposures at 1 minute intervals.

Note: Do not enter a count that will result in an intermediate exposure time that is less than the minimum exposure time for the mode you are in. The minimum exposure time in trans illuminated mode is 1 second, and the minimum exposure time in epi illuminated mode is 0.1 second. (Example: For a trans illuminated application, an exposure time of 20 seconds and an exposure count of 21 would result in an error.)

Save All Intermediate Images

If Auto Save After Scan is selected (see following section), the Save All Intermediate Images checkbox will become active. If you select this checkbox,

all your intermediate exposures will be saved as separate files. These files will have the same root name appended by a number indicating the exposure sequence. The final, full exposure will have the root name only, with no exposure number.

F.6.c Save

Auto Save After Scan

To automatically save any image you create, click on the Auto Save After Scan checkbox.

Note: In PDQUEST, this option is preselected and cannot be turned off. All images must be automatically saved when acquired.

With this checkbox selected, when you click on Acquire or Live Acquire, a Save As dialog box will open asking you to specify a file name and location for the image you are about to create. The scan will begin when you click on the Save button.

Note that in Live Acquire mode you can save your intermediate exposures by selecting Auto Save After Scan and then Save All Intermediate Images.

Make Backup Copy

You can automatically create a backup copy of any scan you create. To do so, first select Auto Save After Scan (see above), then select the Make Backup Copy checkbox.

With this checkbox selected, when you save a scan, a backup copy will be placed in the same directory as the scanned image. Windows backup files will have an “.sbk” extension. Macintosh backup files will have the word “backup” after the file name.

This backup copy will be read-only, which means that you cannot make changes to it. You can open it like a normal file, but you must save it under a different file name before editing the image or performing analysis.

F.6.d Imaging Area Size

The imaging area is the area of the sample that is captured by the camera and displayed in the scan window. To specify the size of this area, enter a dimension in the appropriate field under Imaging Area.

When you change one imaging area dimension, the other will change to maintain the aspect ratio of the camera lens.

The imaging area will change depending on your zoom factor. For example, if you have zoomed in on a area that is 4.5 x 3.5 cm, then you would enter 4.5 for the width (3.5 for the height would be calculated automatically).

Note: Your imaging area settings must be correct if you want to do 1:1 printing. They must also be correct if you want to compare the quantities of objects (e.g., using the Volume Tools) in different images.

The imaging area dimensions also determine the size of the pixels in your image (i.e., resolution). A smaller imaging area will result in a higher resolution.

F.7 Other Features

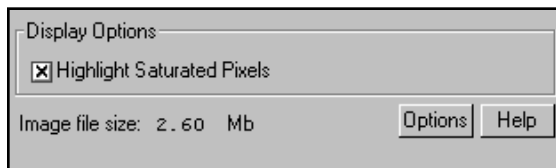


Fig.F-11. Other Fluor-S acquisition window features.

Highlight Saturated Pixels

When this box is checked, any saturated pixels in the image will appear highlighted in red in the scan window and in the pop-up image window. To view /hide saturated pixels in the pop-up image window, use the Image > Transform command.

File Size of Images

Image File Size shows the size of the image file you are about to create. This size is determined by whether the image was created in High Resolution or High Sensitivity mode.

If you do not have enough computer memory for the specified file size, an error message will appear when you attempt to acquire an image. (Macintosh users can increase the application memory partition. See your Macintosh computer documentation for guidance.)

Appendix G

Fluor-S MAX MultiImager



Fig.G-1. Fluor-S MAX MultiImager.

Before you can begin acquiring images using the Fluor-S[®] MAX MultiImager, the imaging system must be properly installed and connected with the host computer. See the Fluor-S MAX hardware manual for installation, startup, and operating instructions.

Note: Make sure that the temperature light on the Fluor-S MAX is green before attempting to capture an image. If you are using a PC, the Fluor-S MAX should be turned on and the initialization sequence completed *before* the host computer is turned on. See the hardware manual for more details.

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PC Only: A Note About SCSI Cards

The Fluor-S MAX is connected to your computer by a Small Computer System Interface (SCSI) cable. To use the Fluor-S MAX, you must have a SCSI card installed in your PC. If you have an older PC, you may also need to load the SCSI and WinASPI drivers that came with your card.

Simulation Mode

Any of the imaging device acquisition windows can be opened in “simulation mode.” In this mode, an acquisition window will open and the controls will appear active, but instead of capturing real images, the window will create “dummy” images of manufactured data.

You do not need to be connected to an imaging device to open a simulated acquisition window. This is useful for demonstration purposes or practice scans.

To enter simulation mode, hold down the CTRL key and select the name of the device from the File menu. The title of the acquisition window will indicate that it is simulated.

G.1 Fluor-S MAX Acquisition Window

To acquire images using the Fluor-S MAX, go to the File menu and select Fluor-S MAX.... The acquisition window for the imager will open, displaying a control panel and an image display window.

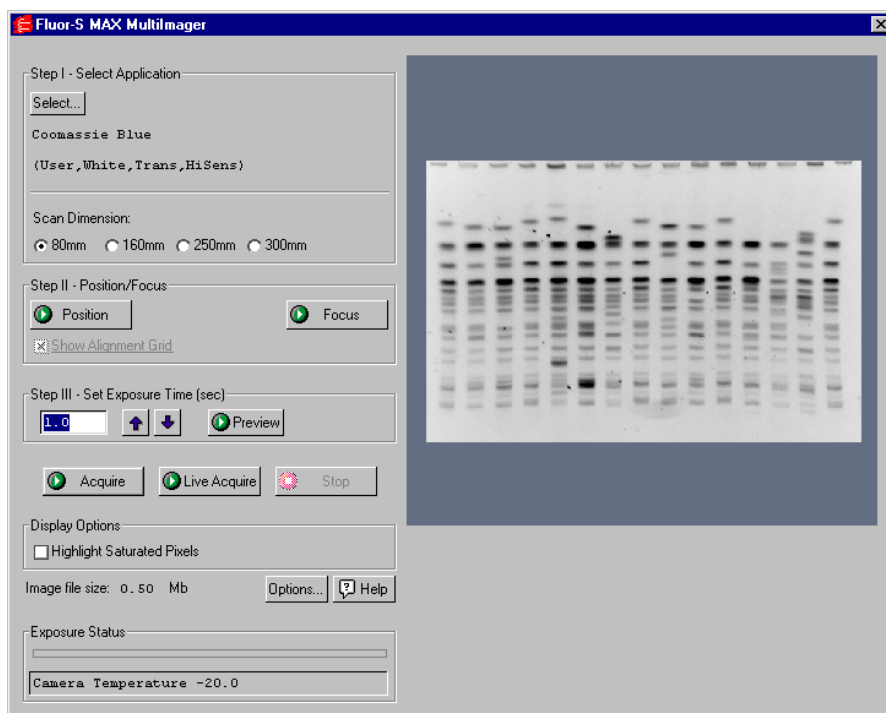


Fig.G-2. Fluor-S MAX acquisition window.

When the Fluor-S MAX window first opens, no image will be displayed.

The control panel has been arranged from top to bottom to guide you through the acquisition procedure. There are four basic steps to acquiring an image using the Fluor-S MAX:

1. Select the application.
2. Position and focus the gel or other object to be imaged.
3. Set the exposure time.
4. Acquire the image.

G.2 Step I. Select Application

To set the appropriate filter and other parameters for the type of object you are imaging, click on the Select button under Select Application. The available applications and their associated settings are listed in a tree that expands from left to right.

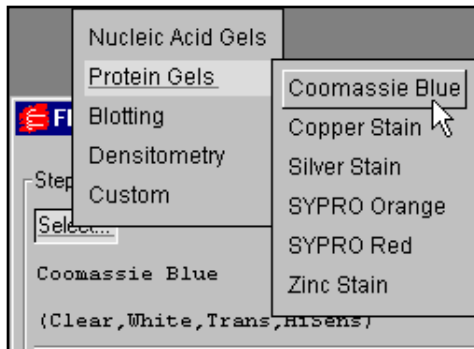


Fig.G-3. The application tree in the Fluor-S MAX acquisition window.

First select your general application, then select the particular stain or medium you are using. If you select the chemiluminescent application under "Blotting," you also have the option of selecting High Sensitivity or Ultra Sensitivity (see below).

When you select an application, the software automatically sets the appropriate standard filter (520LP, 530DF60, 610LP, clear, or none), light type (UV, white, or none), and light source (trans, epi, or neither) in the Fluor-S MAX for that particular application.

For applications involving trans illumination, you must also specify a scan dimension (see below).

Your selection will be displayed below the Select button. To exit the tree without selecting, press the ESC key.

Custom Applications

If your application is not listed, if you want to use a user-installed filter, or if you want to access Ultra Sensitivity mode (see below), you can create and save your own custom application.

From the application tree, select Custom, then Create. This will open a dialog box in which you can name your application and select your settings.

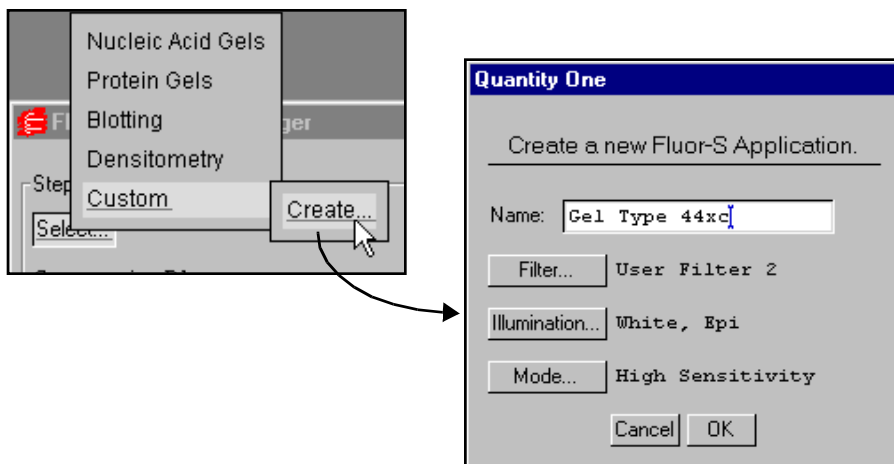


Fig.G-4. Creating a new custom application.

To select the filter (including user-defined), type of illumination, and camera mode, click on the appropriate buttons.

Note: Under Illumination, there is a listing for a spare UV light source. This selects the spare UV bulb in the Fluor-S MAX. Select this light source if your main UV bulb fails.

Enter a name for your application in the Name field. Click on OK to implement your changes.

After you have created an application, you can select it from the application tree by selecting Custom and the name you created. You can delete the application by selecting Custom, Delete, and the name of the application.

Scan Dimension

If an application uses trans illumination, the Scan Dimension buttons become active. The scan dimension is the distance traveled by the transilluminating light source as it scans horizontally across the platen.

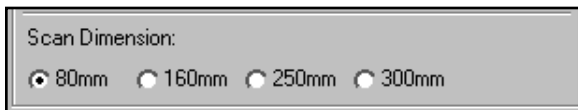


Fig.G-5. Scan dimension settings for trans illumination.

The full scanning range is 300 mm. Select a smaller range if your sample is small and you do not want to wait while the light source travels over the maximum scan width.

High Sensitivity and Ultra Sensitivity

High Sensitivity and Ultra Sensitivity are different camera modes. In Ultra Sensitivity mode, the Fluor-S MAX camera is cooled to -35.0 degrees C. In High Sensitivity mode, the camera is cooled to -20.0 degrees C. The current camera temperature is displayed at the bottom of the acquisition window.

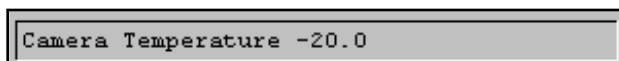


Fig.G-6. Camera Temperature display.

High Sensitivity is the normal operating mode of the Fluor-S MAX. Ultra Sensitivity provides optimal sensitivity for low-light applications. It is the default selection for the chemiluminescence application, and may be selected for a custom application.

Note: When you change from High to Ultra Sensitivity or visa versa, there will be a delay of several minutes while the Fluor-S MAX camera cools down or warms up. If you attempt to acquire an image during this period, you will be notified of the changing temperature. If you do not want to wait, you can cancel the mode change.

G.3 Step II. Position/Focus

Note: When you click on the Position or Focus button, the light inside the Fluor-S MAX box automatically turns off. This is because if you focus with the camera at maximum aperture (see note below), leaving the light on would make it difficult to view the image. To turn the light on while positioning or focusing, hold down the SHIFT key when clicking on the button.

Position

The next step in acquiring an image is centering your gel or other object within the camera frame. To do this, click on the Position button. The Fluor-S MAX will begin capturing a “live” image and updating it every second.

With the Position button selected, look at the image in the acquisition window while you position your object in the center of the platen. If you have a zoom lens on the camera, you can adjust the magnification while you position.

You can select the Show Alignment Grid checkbox to facilitate positioning.

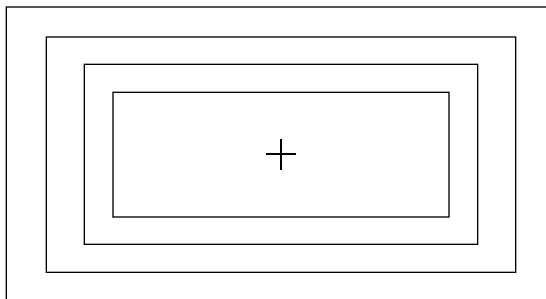


Fig.G-7. Fluor-S MAX alignment grid.

When you are finished positioning, click on the Stop button.

Focus

Note: Before focusing, you should adjust the f-stop on the camera to the lowest setting (i.e., the maximum aperture). This reduces the depth of field, allowing you to more accurately focus the camera. Then, after focusing, increase the f-stop to the desired setting.

After you have positioned your sample, click on the Focus button and look at the image in the acquisition window while aligning the two focusing arrows on the camera lens. While focusing, the camera will limit its focus to a small portion of the sample (this will not affect any zoom lens adjustments you may have made.)

When you are finished focusing, click on the Stop button.

G.4 Step III. Set Exposure Time

When you are ready to capture an image, you will need to select an exposure time. “Exposure” refers to the integration of image captures on the CCD over a set period of time. The effect is analogous to exposing photographic film to light.

The exposure time you select should be based on your application and your “best guess” as to what exposure will give you the best image.

Note: The minimum exposure time in trans illuminated mode is 1 second. The minimum exposure time in epi illuminated mode is 0.1 second.

You can enter an exposure time (in seconds) directly in the field, or use the Arrow buttons to adjust the exposure time in 10 percent increments.



Fig.G-8. Selecting an exposure time.

The following table provides recommended exposure times for various applications.

Recommended Exposure Times and Lenses

Sample	Recommended Exposure	Lens & Filter ¹	Accessories Used
Fluorescent Stain Gel	1–20 sec.	Zoom/IR	None
Fluorescence End-Label Gel	10 sec.–2 min.	Zoom/IR	None
Fluorescent Blot	0.1–3 sec.	Zoom/IR	None
Chemifluorescent Blot	0.1–3 sec.	Zoom/IR	None
Colorimetric Gel	1–5 sec.	Zoom/IR	White Diffusion Plate
Colorimetric Blot	0.2–10 sec.	Zoom/IR	None
X-ray film	0.1–5 sec.	Zoom/IR	White Diffusion Plate
Weak Chemiluminescence ²	2–10 min.	50 mm	Chemi Tray (if sample is small)
Strong Chemiluminescence ²	5 sec.–1 min.	50 mm	Chemi Tray (if sample is small)
¹ For sharper focusing, close the f-stop down 1–2 stops from full open while focusing. ² For chemi applications, the 50mm lens is recommended. Always remove the 660 filter.			

For most applications, you can select an exposure time, capture an image, study it, then adjust the exposure time accordingly. Repeat this procedure as many times as necessary to obtain a good image.

For chemiluminescent samples, which degrade over time and emit low levels of light, you can select a high exposure time initially and use Live Acquire mode to save intermediate exposures (see following section).

Preview

For shorter exposures, you can use Preview to test different exposure times. Click on the Preview button create a preview exposure and display it in the acquisition window.

A preview scan takes only half as long to create as a real scan, because the preview scan does not capture a “dark” image (see below). The progress of the exposure will be displayed in the Exposure Status bar at the bottom of the dialog box.

You cannot save preview scans.

If you want to stop a preview scan that is in progress, click on the Stop button.

G.5 Acquire the Image

The Fluor-S MAX gives you the option of simply acquiring and displaying a fully exposed image, or preserving intermediate exposures.

Acquire

To acquire and display a fully exposed image, click on the Acquire button. An exposure will be taken based on the time selected in Step III. This is appropriate for most short exposures.

The progress of each exposure will be displayed in the Exposure Status bar at the bottom of the acquisition window.

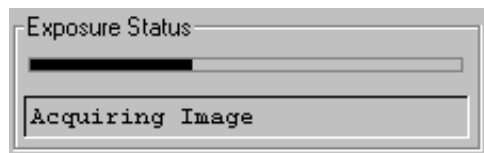


Fig.G-9. Exposure Status bar when acquiring an image.

Depending on which dark subtraction type you have selected, a dark count may be acquired immediately following image acquisition. See Dark Subtraction Type under Options, below.

If you want to stop a scan that is in progress, click on the Stop button. The acquisition will be terminated.

After an image has been acquired, a separate window will pop up containing the new image. The window will have a default file name that includes the date, time, and user (if known). To save the image, select Save or Save As from the File menu.

You can then analyze the image using the standard analysis functions.

Live Acquire

Live Acquire allows you to view and preserve intermediate exposures leading up to a full exposure. This is useful for longer exposure times, such as those required by chemiluminescent samples, where there is the potential for image saturation.

When you click on the Live Acquire button, the exposure time you selected is divided by the number of exposure counts set in the Options dialog box (see Options, below). For example, if you enter an Exposure Time of 10 minutes and an Exposure Count of 20, then 20 intermediate exposures will be produced at 30-second intervals. Each intermediate exposure will be displayed in the scan window. The final, full exposure will be displayed in a separate image window.

Note: The first intermediate exposure will take longer than 30 seconds to display if dark subtraction is performed.

You can automatically save your intermediate exposures as separate files for later review using the Auto Save After Scan option. See Options, below.

If you see an intermediate exposure that you like in the scan window, click on the Stop button. Live Acquire will end and the last intermediate exposure to be completed will open in a separate image window. You can then save it for analysis.

G.6 Options

Click on the Options button to open the Options dialog box.

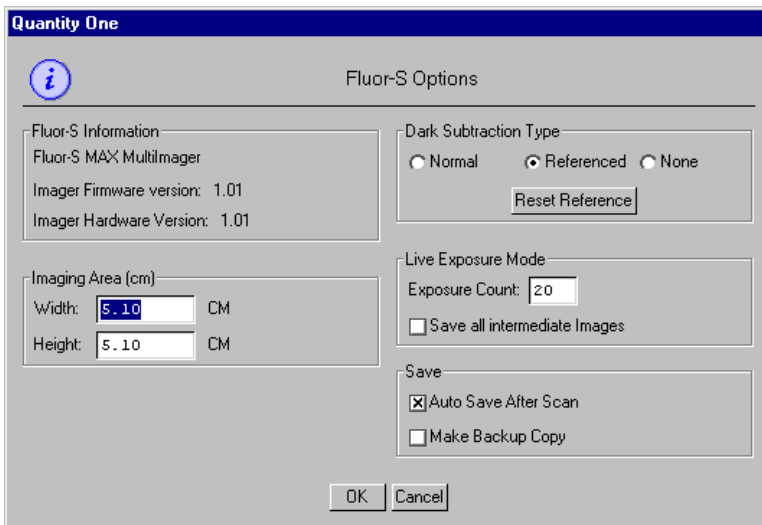


Fig.G-10. Options dialog box.

G.6.a Dark Subtraction Type

All CCD cameras accumulate electrons that produce a “signal” that is indistinguishable from light. This “dark count” adds to the noise in your images. In most cases, you will want to subtract this dark count from your images.

Normal

The Normal option button selects the default dark subtraction type. In this mode, after you acquire an image, a “dark” image of the same exposure length will be taken, and this will be subtracted from your image.

The progress of the dark exposure will be displayed in the Exposure Status bar following the regular image exposure.

Note: In Normal mode, a dark image is only acquired the first time you perform a scan with particular application and exposure settings. If you perform subsequent scans with the same settings, no dark exposure will be taken.

Referenced

If you do not want to perform a dark exposure with each acquisition, you can take a “reference” dark exposure that will be saved and subtracted from all subsequent acquisitions. Click on the Referenced button to activate this feature.

The first time you acquire an image after selecting this option, the Fluor-S MAX will take a 300-second dark exposure that will be saved and used to subtract the dark count from all subsequent acquisitions.

For image exposures of greater or less than 300 seconds, the reference dark will be scaled accordingly and then subtracted. You can change the default reference dark exposure time using the Reset Reference button (see below).

If you deselect the Referenced button and then reselect it, the old reference dark exposure will still be available.

Note: Separate reference dark exposures will be taken for High Sensitivity mode and Ultra Sensitivity mode. Once you have created a reference dark in each of these modes, each reference dark will be used according to the mode you are in.

Reset Reference Button

If you would like a reference dark with an exposure time that more closely matches that of your typical scans, click on the Reset Reference button.

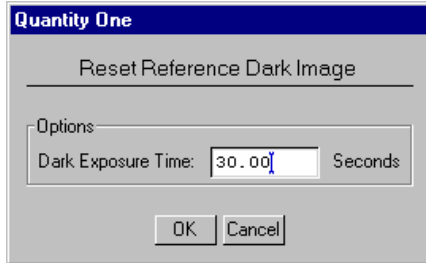


Fig.G-11. Reset Reference Dark pop-up box.

A pop-up box will prompt you to enter a new reference dark exposure time in seconds. Click on OK to implement your change. The new reference dark will be created when you acquire your next image.

Note: Because of the high sensitivity of the CCD, fluctuations in background radiation and/or temperature in the room can affect the level of dark count. If you feel that radiation/temperature conditions have changed in the room since your last reference dark was created, use the Reset Reference button to delete your old reference and create a new one under current conditions.

None

If you do not want to perform dark subtraction, select None. No dark exposure will be acquired or subtracted.

G.6.b Live Exposure Mode

Exposure Count

If you are using the Live Acquire function (see previous section), you need to specify how many intermediate exposures you want to view/save during acquisition. Enter this number in the Exposure Count field.

The total exposure time will be divided by the number you enter in the Exposure Count field. If you enter an exposure time of 10 minutes and a count of 10, you will create 10 intermediate exposures at 1 minute intervals.

Note: Do not enter a count that will result in an intermediate exposure time that is less than the minimum exposure time for the mode you are in. The minimum exposure time in trans illuminated mode is 1 second, and the minimum exposure time in epi illuminated mode is 0.1 second. (Example: For a trans illuminated application, an exposure time of 20 seconds and an exposure count of 21 would result in an error.)

Save All Intermediate Images

If Auto Save After Scan is selected (see following section), the Save All Intermediate Images checkbox will become active. If you select this checkbox, all your intermediate exposures will be saved as separate files. These files will have the same root name appended by a number indicating the exposure sequence. The final, full exposure will have the root name only, with no exposure number.

G.6.c Save

Auto Save After Scan

To automatically save any image you create, click on the Auto Save After Scan checkbox.

Note: In PDQUEST, this option is preselected and cannot be turned off. All images must be automatically saved when acquired.

With this checkbox selected, when you click on Acquire or Live Acquire, a Save As dialog box will open asking you to specify a file name and location for the image you are about to create. The scan will begin when you click on the Save button.

Note that in Live Acquire mode you can save your intermediate exposures by selecting Auto Save After Scan and then Save All Intermediate Images.

Make Backup Copy

You can automatically create a backup copy of any scan you create. To do so, first select Auto Save After Scan (see above), then select the Make Backup Copy checkbox.

With this checkbox selected, when you save a scan, a backup copy will be placed in the same directory as the scanned image. Windows backup files will have an “.sbk” extension. Macintosh backup files will have the word “backup” after the file name.

This backup copy will be read-only, which means that you cannot make changes to it. You can open it like a normal file, but you must save it under a different file name before editing the image or performing analysis.

G.6.d Imaging Area Size

The imaging area is the area of the sample that is captured by the camera and displayed in the scan window. To specify the size of your imaging area, enter a dimension in the appropriate field. When you change one imaging area dimension, the other will change to maintain the aspect ratio of the camera lens.

The imaging area will change depending on your zoom factor. For example, if you have zoomed in on a area that is 4.5 x 3.5 cm, then you would enter 4.5 for the width (3.5 for the height would be calculated automatically).

Note: Your imaging area settings must be correct if you want to do actual-size printing. They must also be correct if you want to compare the quantities of objects (e.g., using the Volume Tools) in different images.

The imaging area dimensions also determine the size of the pixels in your image (i.e., resolution). A smaller imaging area will result in a higher resolution.

G.7 Other Features

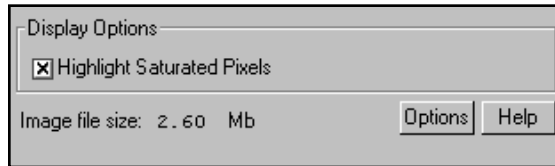


Fig.G-12. Other Fluor-S MAX acquisition window features.

Highlight Saturated Pixels

When this box is checked, any saturated pixels in the image will appear highlighted in red in the scan window and in the pop-up image window. To view/hide saturated pixels in the pop-up image window, use the Image > Transform command.

File Size of Images

Image File Size (below Options) shows the size of the image file you are about to create. This size is determined by whether the image was created in High Sensitivity or Ultra Sensitivity mode.

If you do not have enough computer memory for the specified file size, an error message will appear when you attempt to acquire an image. (Macintosh users can increase the application memory partition. See your Macintosh computer documentation for guidance.)

Appendix H

Personal Molecular Imager FX



Fig.H-1. Personal Molecular Imager FX

Before you can begin acquiring images using the Personal Molecular Imager[®] FX, the instrument must be properly installed and connected with the host computer. See the Personal FX hardware manual for installation, startup, and operating instructions.

Note: The Personal FX should be turned on and the initialization sequence completed *before* the host computer is turned on (except in the case of certain Power Macintosh configurations). See the hardware manual for more details.

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PC Only: A Note About SCSI Cards

The Personal FX is connected to your computer by a Small Computer System Interface (SCSI) cable. To use the Personal FX, you must have a SCSI card installed in your PC. If you have an older PC, you may also need to load the SCSI and WinASPI drivers that came with your card.

Simulation Mode

Any of the imaging device acquisition windows can be opened in “simulation mode.” In this mode, an acquisition window will open and the controls will appear active, but instead of capturing real images, the window will create “dummy” images of manufactured data.

You do not need to be connected to an imaging device to open a simulated acquisition window. This is useful for demonstration purposes or practice scans.

To enter simulation mode, hold down the CTRL key and select the name of the device from the File menu. The title of the acquisition window will indicate that it is simulated.

H.1 Personal FX Acquisition Window

To acquire images using the Personal FX, go to the File menu and select Personal FX.... The acquisition window for the imager will open, displaying a control panel and the scanning area window.

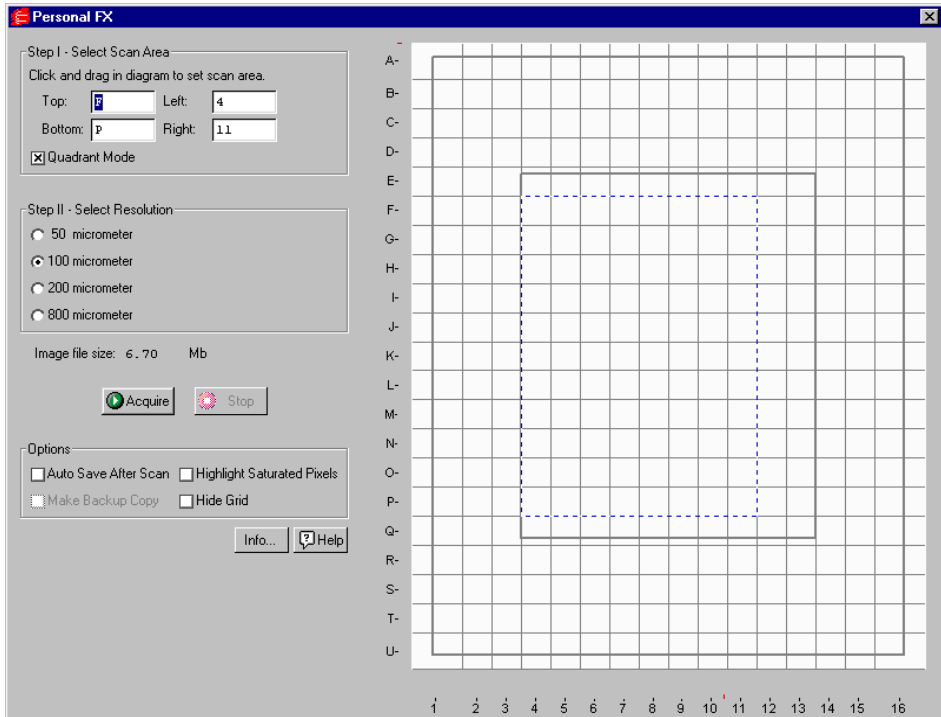


Fig.H-2. Personal FX acquisition window

The default scanning window is marked by grid lines that divide the area into quadrants. There is also an outer box and inner box marked by thicker lines. This conforms to the sample pad for the standard Bio-Rad Exposure Cassette that is supplied with the Personal FX. The quadrants are numbered 1–16 left to right and lettered A–U top to bottom.

If you prefer a scanning window measured in centimeters, deselect the Quadrant Mode checkbox in the control panel by clicking on it. To hide the gridlines, click on the Hide Grid checkbox under Options.

The control panel has been arranged from top to bottom to guide you through the acquisition procedure. There are three basic steps to scanning an image using the Personal FX:

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1. Select the scan area
2. Select the resolution
3. Acquire the image

H.2 Step I. Select Scan Area

To select a scan area, drag your mouse within the scanning window. (In the scanning window, your cursor appearance will change to a cross.) The border of the scan area you are selecting is marked by a frame.

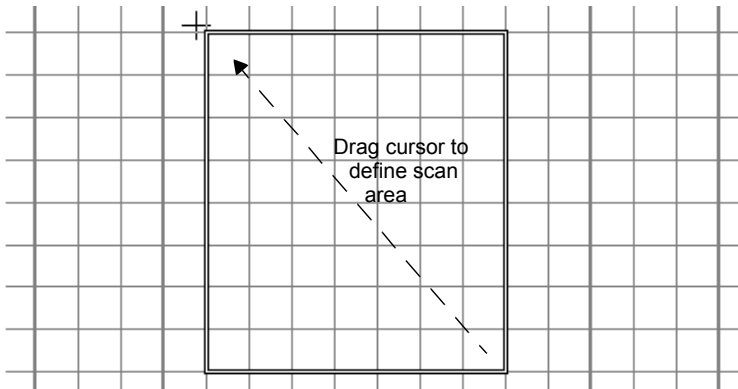


Fig.H-3. Selecting a scan area.

If you are in quadrant mode, note that the frame “locks” onto the next quadrant as you drag. When you release the mouse button, the border changes to a dashed blue line, indicating a selected area.

- To *reposition* the scanning box you have selected, position your cursor inside the box and drag. The entire box will move.
- To *resize* the box, position your cursor on a box side and drag. The side you have selected will move.
- To *redo* the box entirely, position your cursor outside the box and drag. The old box will disappear and a new box will be created.

You can also select the scanning area by entering coordinates in the appropriate fields (Top, Bottom, Left, Right). After you enter a coordinate, the position of the scanning area box will change accordingly.

When selecting, be sure to include the entire area of interest, and be generous with borders. You can always crop the image later.

H.3 Step II. Select Resolution

The Personal FX acquisition window allows you to scan at 50, 100, 200, or 800 micrometers. These resolutions are listed as option buttons in the control panel.

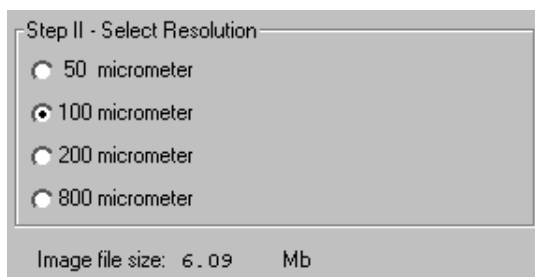


Fig.H-4. Resolution option buttons.

The resolution you select should be based on the size of the objects (e.g., bands, spots) you are interested in. For example:

- 50 micrometer resolution should be reserved for images requiring the highest level of detail, e.g., high density in situ samples, 1,536-well microplates, high density arrays, samples with very closely spaced bands. Files scanned at 50 micrometers can be very large.
- 100 micrometer resolution should be used for typical gels and arrays.
- 200 micrometer resolution is useful for gels with large bands and dot blots.
- 800 micrometer resolution should be reserved for very large objects, such as CAT assays.

File Size of Images

Image File Size (below Select Resolution) shows the size of the scan file you are about to create. If you do not have enough computer memory for the specified file size, an error message will appear when you attempt to scan. If this happens, select a lower resolution or decrease the size of the area to be scanned. (Macintosh users can also increase the application memory partition. See your Macintosh computer documentation for guidance.)

H.4 Acquire the Image

Once you have selected your scan area and resolution, you ready to acquire an image.

Click on the Acquire button. There may be a short delay while the image laser warms up; then the scanned image will begin to appear in the scanning window, line by line.

To interrupt a scan, click on the Stop button. A message will ask you to confirm the interrupt, and then you will be asked if you want to keep the partial scan. This feature is useful if you overestimated the size of the area you selected.

Note: If the image you are scanning has more than 10 saturated pixels, you will receive a warning message.

Saving the Image

After the scan is complete, a message will appear asking you if you want to keep the scan. If you select Yes, a separate window will pop up containing the new image.

You can then save and analyze the image using the standard menu and toolbar functions.

H.5 Options

Auto Save After Scan

To automatically save any scan you create, click on the Auto Save After Scan checkbox.

Note: In PDQUEST, this option is preselected and cannot be turned off. All images must be automatically saved when acquired.

With this checkbox selected, when you click on Acquire, a Save As dialog box will open asking you to specify a file name and location for the image you are about to create. The scan will begin when you click on the Save button.

Make Backup Copy

You can automatically create a backup copy of any scan you create. To do so, first select Auto Save After Scan (see above), then select the Make Backup Copy checkbox.

With this checkbox selected, when you save a scan, a backup copy will be placed in the same directory as the scanned image. Windows backup files will have an “.sbk” extension. Macintosh backup files will have the word “backup” after the file name.

This backup copy will be read-only, which means that you cannot make changes to it. You can open it like a normal file, but you must save it under a different file name before editing the image or performing analysis.

Highlight Saturated Pixels

When this box is checked, any saturated pixels in the image will appear highlighted in red in the scan window and in the pop-up image window. To view/hide saturated pixels in the pop-up image window, use the Image > Transform command.

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Hide Grid

To hide the gridlines in the scanning area window, click on the Hide Grid checkbox.

Appendix I

Molecular Imager FX



Fig.I-1. Molecular Imager FX

Before you can begin acquiring images using the Molecular Imager[®] FX, the instrument must be properly installed and connected with the host computer. See the FX hardware manual for installation, startup, and operating instructions.

Note: The FX should be turned on and the initialization sequence completed *before* the host computer is turned on (except in the case of certain Power Macintosh configurations). See the hardware manual for more details.

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PC Only: A Note About SCSI Cards

The FX is connected to your computer by a Small Computer System Interface (SCSI) cable. To use the FX, you must have a SCSI card installed in your PC. If you have an older PC, you may also need to load the SCSI and WinASPI drivers that came with your card.

Simulation Mode

Any of the imaging device acquisition windows can be opened in “simulation mode.” In this mode, an acquisition window will open and the controls will appear active, but instead of capturing real images, the window will create “dummy” images of manufactured data.

You do not need to be connected to an imaging device to open a simulated acquisition window. This is useful for demonstration purposes or practice scans.

To enter simulation mode, hold down the CTRL key and select the name of the device from the File menu. The title of the acquisition window will indicate that it is simulated.

I.1 FX Acquisition Window

To acquire images using the FX, go to the File menu and select FX.... The acquisition window for the imager will open, displaying the control panel for the imager and the scanning area window.

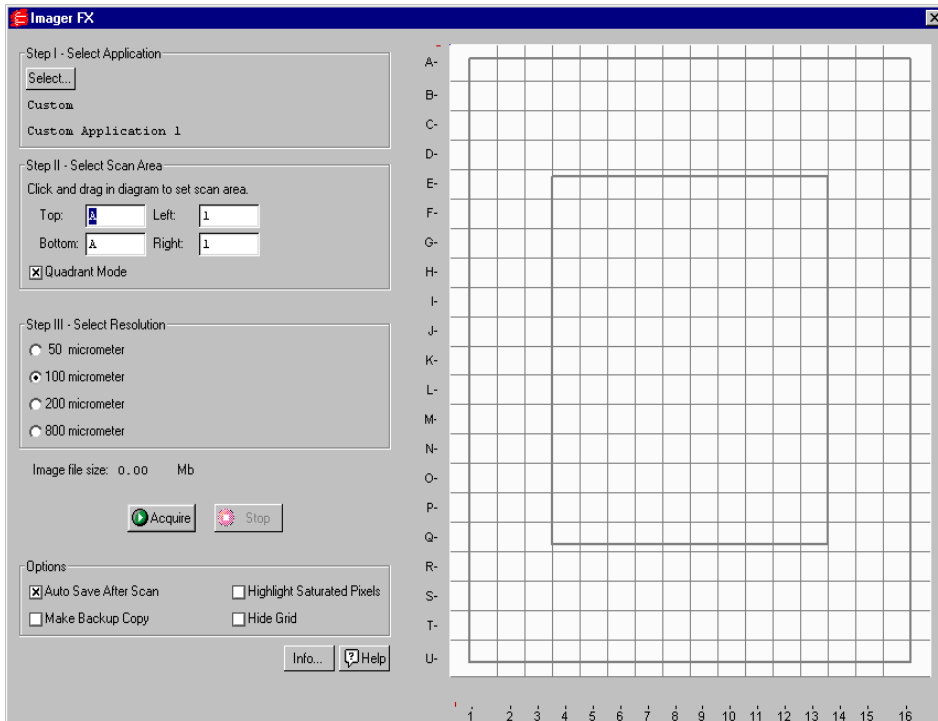


Fig.I-2. FX acquisition window

The default scanning window is marked by grid lines that divide the area into quadrants. There is also an outer box and inner box marked by thicker lines. This conforms to the sample pad for the standard Bio-Rad Exposure Cassette that is supplied with the FX. The quadrants are numbered 1–16 left to right and lettered A–U top to bottom.

If you prefer a scanning window measured in centimeters, deselect the Quadrant Mode checkbox in the control panel by clicking on it. To hide the gridlines, click on the Hide Grid checkbox under Options.

The control panel has been arranged from top to bottom to guide you through the acquisition procedure. There are four basic steps to scanning an image using the FX:

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1. Select the application.
2. Select the scan area.
3. Select the resolution.
4. Acquire the image.

I.2 Step I. Select Application

To select the appropriate filter and other parameters for the type of object you are imaging, click on the Select button under Select Application.

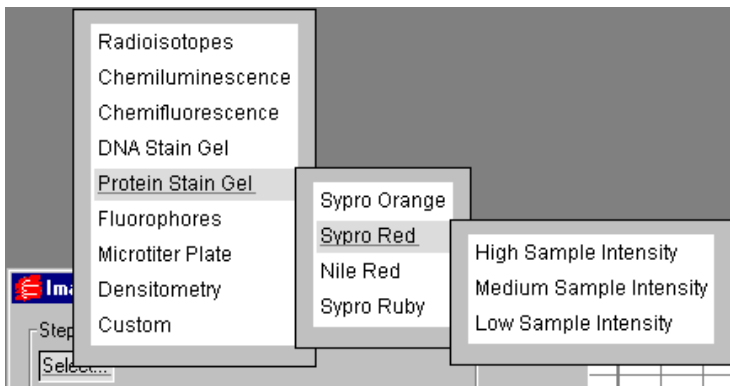


Fig.9-3. Example of an application tree: Ethidium Bromide gel.

Standard Applications

The standard applications and associated settings are listed in a tree that expands from left to right. When you select a standard application, the software automatically sets the appropriate filter(s) in the FX for that particular application.

Standard FX Applications

Category	Application
Radioisotopes	CS- or BI-Screen (Bio-Rad) K-Screen (Kodak) Fuji-Screen
Chemiluminescence	Chemi-Screen (Bio-Rad)
Chemifluorescence	ECL-Plus Attophos
DNA Stain Gel	Ethidium Bromide Sybr Green I & II Sybr Gold
Protein Stain Gel	Sypro Orange Sypro Red Nile Red Sypro Ruby
Fluorophores	Alexa 488 Alexa 532 Alexa 546 FITC FAM CY3 HEX R6G Texas Red
Microtiter Plate	DNA (Sybr Green I) Protein (Nano Orange) ssDNA (Oligreen) DNA (Picogreen) B-Gal (FDG) GUS (FDG)
Densitometry	Coomassie Blue Gel/Blot Copper Stain Gel/Blot Silver Stain Gel/Blot X-Ray Film (Grey Type)

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First select your general application, next select the particular stain or medium you are using, and finally (if appropriate) select the intensity of your samples.

Note: Some applications require an external laser. If you choose one of these but do not have an external laser attached, you will receive a warning.

To exit the tree without selecting, press the ESC key.

Your selection will be displayed below the Select button.

Sample Intensity

Most FX applications require that you specify a sample intensity. This is simply a rough estimate of how much sample is visible in your gel or other object.

If you are unsure of the level of intensity of your sample, you can always select a level, capture an image, then adjust the level and capture another image.

For example, if you select Low Sample Intensity and the resulting image has too many saturated pixels, you will receive a warning message. Simply change the setting to Medium Sample Intensity and rescan. If you select High Sample Intensity and the resulting image is too faint, select Medium or Low and rescan.

Custom Applications

If your application is not listed, if you want to use user-installed filters, or if you want to use an external laser, you can create and save your own custom application.

From the application tree, select Custom, then Create. This will open a dialog box in which you can name your application and select your settings.

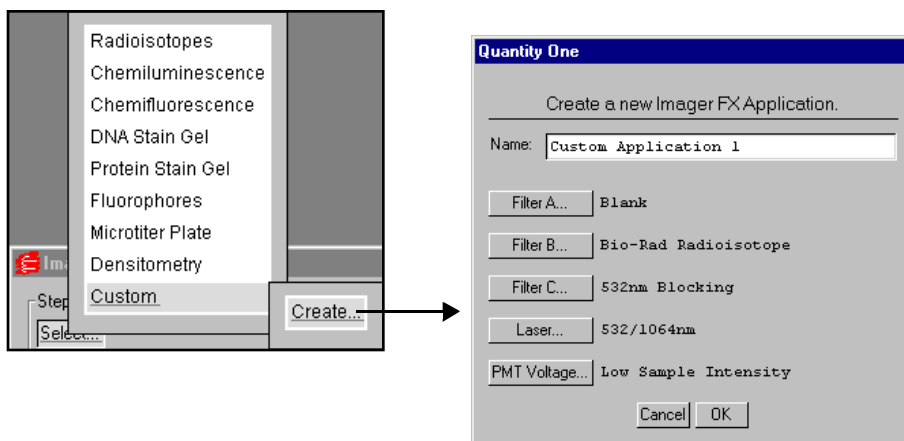


Fig.9-4. Creating a custom application.

To select a filter (including user-defined) or filter combination, click on the buttons for Filters A, B, and C, and make your choice from each pop-up list.

Note: The user-defined filters (User1, User2, etc.) cannot be renamed in the pop-up list, so be sure to remember which filter you insert into each position in the FX.

To use an external laser, click on the Laser button and select it from the pop-up list. Otherwise, use the default internal laser (532/1064nm).

Click on the PMT Voltage button to select a standard voltage for your custom application or create a custom PMT voltage.

To select a custom voltage, click on the Custom option. In the dialog box, adjust the slider to select a PMT voltage as a percentage of the maximum.

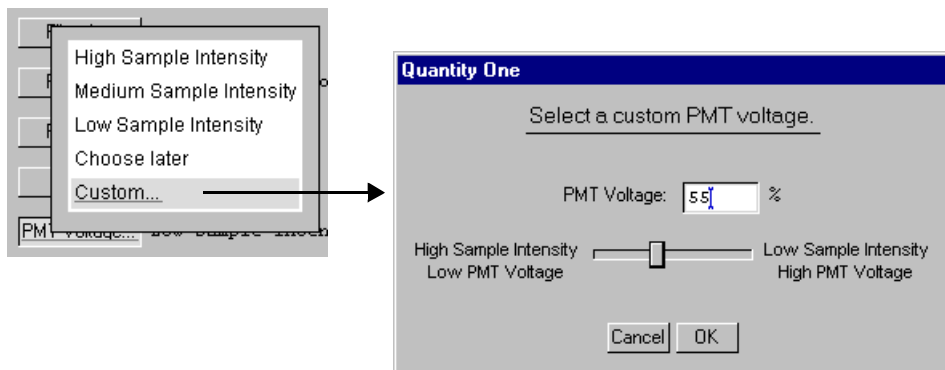


Fig.9-5. Selecting a custom PMT voltage.

Note: For voltages above 80% of maximum, you will receive a warning message that the high voltage could damage the PMT.

If you select Choose Later from the list of PMT voltages, the choices of sample intensity will be displayed when you select your custom application.

Finally, enter a name for your application in the Name field and click on OK to implement your changes.

After you have created an application, you can select it from the application tree by selecting Custom and the name you created. You can delete the application by selecting Custom, Delete, and the name of the application.

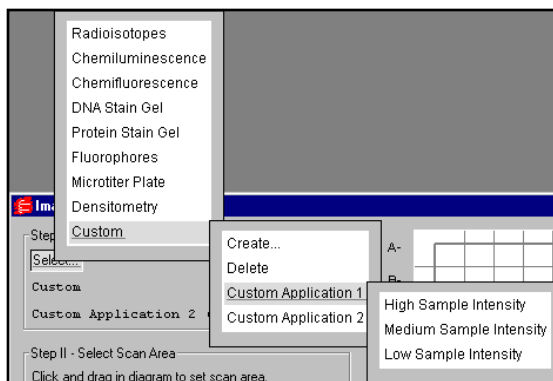


Fig.9-6. Selecting a custom application.

I.3 Step II. Select Scan Area

To select a scan area, drag your mouse within the scanning window. (In the scanning window, your cursor appearance will change to a cross.) The border of the scan area you are selecting is marked by a frame.

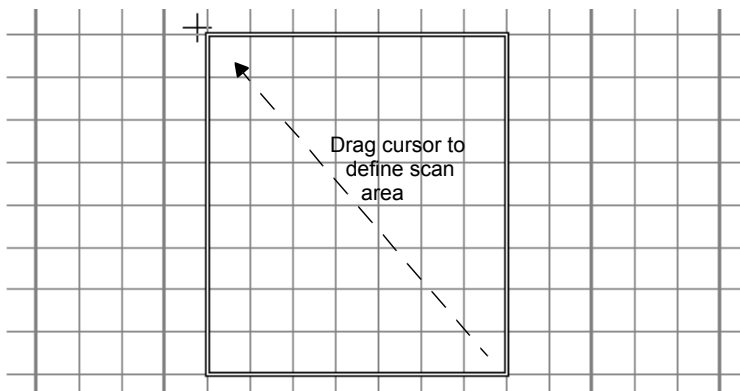


Fig.I-7. Selecting a scan area.

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If you are in quadrant mode, note that the frame “locks” onto the next quadrant as you drag. When you release the mouse button, the border changes to a dashed blue line, indicating a selected area.

- To *reposition* the scanning box you have selected, position your cursor inside the box and drag. The entire box will move.
- To *resize* the box, position your cursor on a box side and drag. The side you have selected will move.
- To *redo* the box entirely, position your cursor outside the box and drag. The old box will disappear and a new box will be created.

You can also select the scanning area by entering coordinates in the appropriate fields (Top, Bottom, Left, Right). After you enter a coordinate, the position of the scanning area box will change accordingly.

When selecting, be sure to include the entire area of interest, and be generous with borders. You can always crop the image later.

I.4 Step III. Select Resolution

The FX acquisition window allows you to scan at 50, 100, 200, or 800 micrometers. These resolutions are listed as option buttons in the control panel.

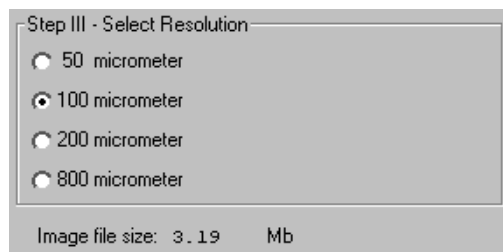


Fig.I-8. Resolution option buttons.

The resolution you select should be based on the size of the objects (e.g., bands, spots) you are interested in. For example:

- 50 micrometer resolution should be reserved for images requiring the highest level of detail, e.g., high density in situ samples, 1,536-well microplates, high density arrays, samples with very closely spaced bands. Files scanned at 50 micrometers can be very large.
- 100 micrometer resolution is useful for typical gels and arrays.
- 200 micrometer resolution is useful for gels with large bands and dot blots.
- 800 micrometer resolution should be reserved for very large objects, such as CAT assays.

File Size of Images

Image File Size (below Select Resolution) shows the size of the scan file you are about to create. If you do not have enough computer memory for the specified file size, an error message will appear when you attempt to scan. If this happens, select a lower resolution or decrease the size of the area to be scanned. (Macintosh users can also increase the application memory partition. See your Macintosh computer documentation for guidance.)

I.5 Acquire the Image

Once you have selected your application, scan area, and resolution, you are ready to acquire an image.

Click on the Acquire button. There may be a short delay while the image laser warms up; then the scanned image will begin to appear in the scanning window, line by line.

To interrupt a scan, click on the Stop button. A message will ask you to confirm the interrupt, and then you will be asked if you want to keep the partial scan. This feature is useful if you overestimated the size of the area you selected.

Note: If the image you are scanning has more than 10 saturated pixels, you will receive a warning message. If this happens, you can go back and select a higher sample intensity in the application tree.

Saving the Image

After the scan is complete, a message will appear asking you if you want to keep the scan. If you select Yes, a separate window will pop up containing the new image.

You can then save and analyze the image using the standard menu and toolbar functions.

I.6 Options

Auto Save After Scan

To automatically save any scan you create, click on the Auto Save After Scan checkbox.

With this checkbox selected, when you click on Acquire, a Save As dialog box will open asking you to specify a file name and location for the image you are about to create. The scan will begin when you click on the Save button.

Make Backup Copy

You can automatically create a backup copy of any scan you create. To do so, first select Auto Save After Scan (see above), then select the Make Backup Copy checkbox.

With this checkbox selected, when you save a scan, a backup copy will be placed in the same directory as the scanned image. Windows backup files will have an “.sbk” extension. Macintosh backup files will have the word “backup” after the file name.

This backup copy will be read-only, which means that you cannot make changes to it. You can open it like a normal file, but you must save it under a different file name before editing the image or performing analysis.

Highlight Saturated Pixels

When this box is checked, any saturated pixels in the image will appear highlighted in red in the scan window and in the pop-up image window. To view/hide saturated pixels in the pop-up image window, use the Image > Transform command.

Hide Grid

To hide the gridlines in the scanning area window, click on the Hide Grid checkbox.

Appendix J

Calibration and Merging

You can use calibration strips (calstrips) to calibrate your scans for more accurate quantitation. Calstrips are required for both gel calibration and merging multiple exposures.

J.1 Calibrating Gels

The commands for calibrating gels are located on the Edit > Calibrate Gels submenu.

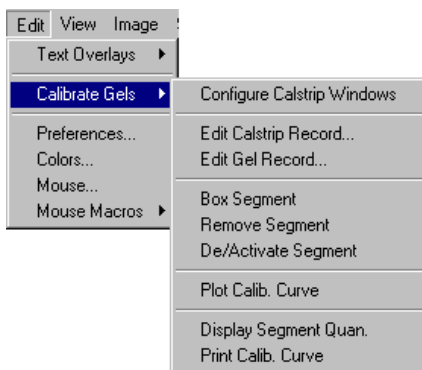


Fig. J-1. Edit > Calibrate Gels submenu.

Creating a Calstrip File

If calstrip images are currently part of your gel images, you will have to crop them and save them as separate files for use in PDQUEST. To do so, select Advanced Crop > Define Calstrip Area from the Image menu, draw a box around your calstrip, then crop it and save it as a calstrip file. See section 3.12.a for instructions.

Configuring Calstrip Subwindows

If you are displaying multiple calstrips linked to multiple exposures (see section J.2, Merging Multiple Exposures), Configure Calstrip Windows on the Edit > Calibrate Gels submenu will automatically arrange the calstrip images in subwindows.

J.1.a Calibration Strip Data

Values and units for each segment of your calstrip must be entered into a file called a Calstrip Record prior to calibration.

To do this, select Edit Calstrip Record from the Edit > Calibrate Gels submenu. The Calstrip Record dialog box will open.

Calstrip Record Editor

Name:

Description:

Detection: ☒ Silver Stain ☐ Fluorography ☐ Syber Fluorescent Stain
☐ Coomassie Stain ☐ Autoradiography
☐ Gelcode Stain

Isotope: ☐ 3H ☐ 14C ☐ 32P ☐ 33P ☐ 35S ☐ 125I

Counting date:

Units: ☐ DPM ☒ OD ☐ PSL ☐ Other
☐ CPM ☐ HLU ☐ RD ☐ INT
☐ ODu ☐ RDu ☐ CNT

Number of segments: 0

Segment values in quantity/cm²:

1:		7:		13:		19:	
2:		8:		14:		20:	
3:		9:		15:		21:	
4:		10:		16:		22:	
5:		11:		17:		23:	
6:		12:		18:		24:	

Load Reset Delete Save Cancel

Fig. J-2. Calstrip Record Editor dialog box.

Enter a title for the data record next to the Name prompt. You can enter descriptive information about the calstrip in the Description box.

Next, select the Detection method /stain that was used to make the calstrip.

For Fluorography and Autoradiography detection methods, select the Isotope used. Also enter the date the calstrip was made next to the Counting date prompt. The format in which the date should be entered is dd-mmm-yyyy (e.g., 01-Nov-1993 or 01-11-1993). The first three letters of the name of the month or the number of the month may be entered.

Specify the units for the quantitative values by clicking on one of the buttons next to the Units prompt or typing the units in the text box next to the Other button.

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Finally, enter the Segment values for each segment of the calstrip. These values should correspond to the calstrip segments as they appear from left to right (i.e., if the darkest segments are on the left, the values will decrease).

Range of Segment Values and Spot Quantities

The range of segment values should be approximately 2.0 to 2×10^4 . The minimum value the Calstrip Record will accept is 2.0. A typical maximum is 2×10^4 , but this value may vary. The range should span four orders of magnitude.

Typical spot quantities will range from 10^{-2} to 5×10^3 , depending on the size of the spot.

An example of segment values (DPM/cm²) for an ³⁵S calstrip are:

Segment #:	1.	278579.0	7.	14892.0	13.	827.0
	2.	159376.0	8.	9207.0	14.	516.0
	3.	100299.0	9.	5791.0	15.	315.0
	4.	67078.0	10.	3670.0	16.	134.0
	5.	40895.0	11.	2220.0	17.	93.0
	6.	24686.0	12.	1396.0	18.	44.0

Such a calstrip will produce spot quantitation values that typically range from 2.4 DPM to 502.6 DPM.

If you are working with Coomassie gels and you want to make a calstrip, you should begin with a stock solution and make approximately 10 serial dilutions, each of which will be used to make one segment of the calstrip.

For example, you might begin with a 50 µg/ml stock solution of BSA and make segments of the calstrip starting at 50 µg and progressing down to 5.0 ng in constant ratio steps.

In order to enter numerical values in the Calstrip Record that will produce interpretable spot quantitation values, convert all the numbers to ng (i.e., multiply all µg values by 10^3).

For the example mentioned above, the values would range from 50,000 ng to 5.0 ng.

Saving/loading Calstrip Records

When you have completed the information in the dialog box, click on the Save button at the bottom of the box.

To access the information in an existing Calstrip Record, click on the Load button. A list of saved Calstrip Records will be displayed. Select the one whose data you wish to see by clicking on it. The data associated with that record will be displayed on the form.

To delete a saved Calstrip Record, click on the Delete button.

Click on the Cancel button to close the Calstrip Record Editor form.

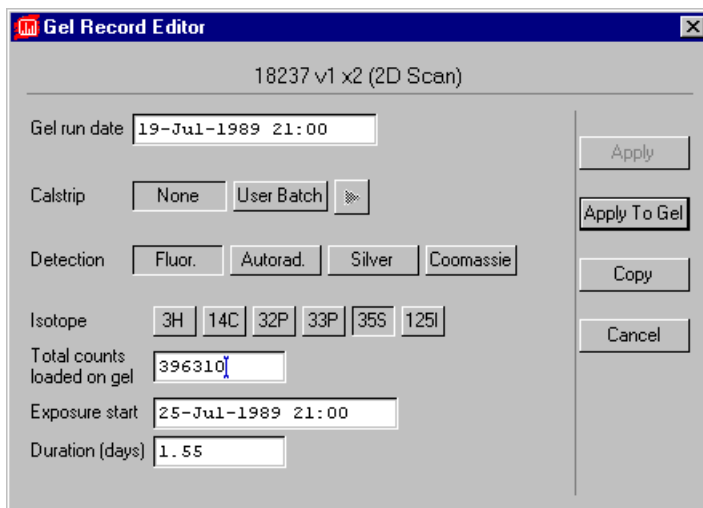
J.1.b Scan Information

After the calibration data has been entered, you will also need to enter some information about the gel that goes with it.

Start by loading the 2-D and calstrip scans. Display the calstrip scans by using the Configure Calstrip Windows function (Edit > Calibrate Gels > Configure Calstrip Windows).

Click on Gel Record in the Calibrate Gels submenu and click on the calstrip scan. The Gel Record dialog box will open, in which you can enter information about the calstrip.

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The screenshot shows the 'Gel Record Editor' window with the title bar '18237 v1 x2 (2D Scan)'. The window contains several input fields and buttons:

- Gel run date:** A text box containing '19-Jul-1989 21:00'.
- Calstrip:** A dropdown menu showing 'None', a 'User Batch' button, and a right-pointing arrow button.
- Detection:** A row of buttons: 'Fluor.', 'Autorad.', 'Silver', and 'Coomassie'.
- Isotope:** A row of buttons: '3H', '14C', '32P', '33P', '35S', and '125I'.
- Total counts loaded on gel:** A text box containing '396310'.
- Exposure start:** A text box containing '25-Jul-1989 21:00'.
- Duration (days):** A text box containing '1.55'.
- Buttons on the right:** 'Apply', 'Apply To Gel', 'Copy', and 'Cancel'.

Fig. J-3. Gel Record Editor.

Enter the date on which the gel was run in the text box next to the Gel run date prompt.

Next to the Calstrip prompt, click on the button that corresponds to the kind of calibration associated with this gel. In most cases, you or someone in your lab has made the calstrips, so the button labeled User Batch is the appropriate choice.

Next to the User Batch prompt, click on the arrow. From the list displayed, select the calibration record name for the calstrip. The Detection and Isotope fields are automatically obtained from the specified Calstrip Record.

Click on the appropriate button to indicate the type of detection used for your gel, if not already defined from the batch file.

Click on the Isotope button to indicate the isotope used in the gel. This is usually, though not necessarily, the same isotope as the one used in the calstrip.

Next, enter the number of counts loaded on the gel.

Enter the day on which you began exposing the gel to film in the text box next to the Exposure start prompt.

Note: Most calibration problems arise from incorrectly entering the dates used to calculate radioactive decay factors for the isotopes. Wrong values can make the calstrip appear to have decayed to nothing.

Next to Duration (days), enter the number of days that the film was exposed.

Once this data has been entered, you will want to apply it to the 2-D scan and the calstrip scan. To do so, click on the Apply button.

If you want to apply different calibration data to your 2-D scan than to your calstrip scan (if, for example, your 2-D scan and calstrip scans use different isotopes), use the Apply Here button to apply the gel record to the currently displayed image only (2-D or calstrip scan).

Copying Gel Record Information

If you have scanned several exposures of the same gel, most of the calibration data will be the same for all of them (except for the exposure start date and length). Instead of entering all the information for each exposure, you can enter it once, and then copy it between exposures using the Copy button.

Clicking on Copy will open a list of available gel records. Select the record that you would like to apply to the current image and click the mouse.

Click on the Cancel button to exit the operation.

J.1.c Sampling the Calibration Strip Segments

The next step in calibrating your gels involves sampling the densities in the segments of the calstrip to establish a correlation between the sampled intensities recorded by the scanner and the counts entered into the Calstrip Record.

If you have not already done so, display the calstrips. If you have gels with multiple exposures, use the Configure Calstrip Windows function to automatically divide the screen into enough windows to display all the calstrips for a given gel.

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Flip Calstrip (if necessary)

If the orientation of the scanned calstrip does not correspond to the order in which the segment activities are entered in the Calstrip Record, a message will be displayed, indicating that the calibration density decreases while the counts increase (or vice versa).

To change the orientation of the calstrip so it corresponds to the Calstrip Record segment activities, use the Horizontal Flip function (under the Image menu). If the dark segments were on the left, they will now be on the right, and vice versa. Again, we recommend that you enter your data from highest to lowest activity and display the calibration strip with the darkest segment on the left.

Box Segments in a Calstrip

Once the calstrip is in the correct orientation, go back to the Edit > Calibrate Gels submenu and select Box Segment.

Go to the first (leftmost) segment of your calstrip. Sampling boxes must be drawn beginning with the leftmost segment of the calstrip and moving right.

With Box Segment selected, drag a box on the calstrip. When you release the mouse button, the area will be sampled.

Note: When defining your box, be sure to avoid sampling regions that are very close to the borders of the segment or any irregular areas (i.e., areas that are overly dark or light).

The number of each segment will be displayed in the upper left corner of each sampling box.

Above the box will be a number between 0.00 and 2.00. This represents the O.D. value for the segment.

Beneath each box is the number of counts associated with that segment from the Calstrip Record.

If you make a mistake, simply redraw a new box over the old one. Repeat this procedure for all the segments of the calstrip.

Note: All segments entered in the calstrip record must be boxed starting from the left. The leftmost segment corresponds to Segment 1 on the Calstrip Record. Your calstrips and calstrip data should be designed so that either (1) the darkest segments are on the left and the count values decrease, or (2) the lightest values are on the left and the count values increase.

You can draw more boxes than the number of segments defined in the Calstrip Record. However, they are automatically and permanently deactivated.

J.1.d Plot the Calibration Curve

Once you have created the sampling boxes, select Plot Calib. Curve to display the calibration curve.

A graph will be displayed that correlates the optical density (usually between 0.00 and 2.00 O.D.) with DPMs or other specified units.

The Y axis represents the optical density (0.00–2.00) in each sampled segment, while the X axis represents the corresponding DPM or other unit per image unit area calculated from the calibration record data.

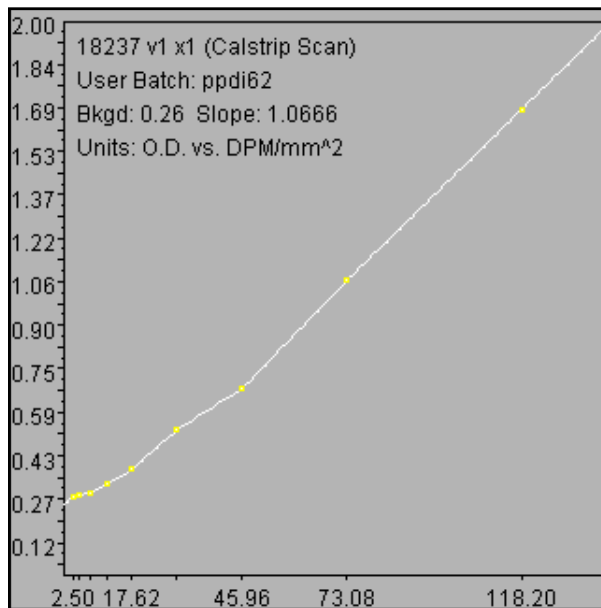


Fig. J-4. Example of a calibration curve.

Calibration Curve Problems

Find errant data points and their corresponding segments using the function Display Segment Quan.

Above each sampling box will be displayed the average O.D. value within the box, and just below the box will be displayed the corresponding DPM (or other specified unit). This should tell you which sampling boxes correspond to errant data points on the graph.

Usually, problems arise when a box straddles the boundary between two adjacent segments or the density is very uneven within the segment due to bleeding from an adjacent segment. These sorts of problems should be remedied by redoing the sampling boxes.

Use the function Remove Segment to remove an errant sampling box.

Draw a new sampling box on the segment using Box Segment, as described above.

Place the new box carefully to avoid repeating the problem. For example, keep the box away from segment boundaries. If the segment shows bleeding, place the smallest box possible in an area of the segment with representative O.D. Try not to sample blotchy areas.

Replot the calibration curve using the function Plot Calib. Curve.

If there is still a problem, you can opt not to include that particular segment in the calculation of the calibration curve. Select De/ Activate Segment and click on the problem segment. The segment will still exist, so its counts in the corresponding Calstrip Record do not need to be changed or deleted. However, the segment will not be included in the calculation of the calibration curve.

If you change your mind and want to reinclude the segment, select De/ Activate Segment again and click on the segment. The segment will be reactivated and included in calibration curve calculations.

J.1.e Repeat for Each Calstrip

If you have multiple film exposures of the same gel, repeat the steps outlined above for each exposure. The calibration strip batch number for each film exposure of a single gel should be the same since the same physical calibration strip was used to make each exposure.

J.2 Merging Multiple Exposures

If you are working with 8-bit images and are using calstrips on your scans, you can merge multiple exposures of the same gel into a single image with a greater dynamic range.

Note: You can only merge 8-bit images with calstrips. If you are not working with 8-bit images or are not using calstrips, you cannot merge exposures.

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To merge multiple exposures, make sure all your exposures and calstrips are loaded into the same image window, then simply auto-detect spots as you would normally (see the following chapter).

Creating Different Exposure Files

To identify different gel scans as different exposures of the same gel, open the scan files, select Change Version/Exposure from the File menu, and enter consecutive exposure numbers for the different scans. Make sure that the root names of the scan files are the same.

Note: You can give different exposures the same root name when you are scanning them, or you can change them later using the Save As command. After you change the exposure numbers, you will end up with files named RootName x1, RootName x2, RootName x3, etc.

Why Merge Exposures?

Because 8-bit scanners have a limited dynamic range, dark spots may saturate on long exposures, while faint spots may be missed on short exposures, and either extreme can be missed on medium duration exposures. Therefore, it is frequently necessary to combine the quantitative data from a number of 8-bit exposures of the same gel exposed for different lengths of time into a “merged” synthetic image. Merging creates such an image automatically as you spot detect.

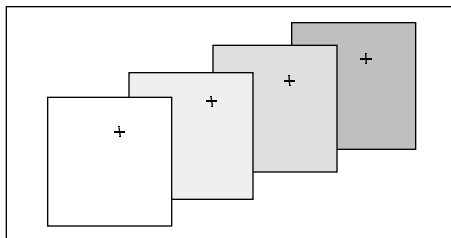


Fig. J-5. Diagrammatic representation of merging.

Cropping and Aligning Exposures of the Same Gel for Merging

Before you can merge multiple exposures, you must crop and align them so that they are exactly the same size and position. To do this, you will use the Advanced Crop commands on the Image menu. See section 3.12.a for instructions.

Appendix K

Cross-Platform File Exchange

It is possible to move image data between Discovery Series software applications on different platforms. There are different protocols depending on the platform (PC or Macintosh) you are transferring from and to.

K.1 Macintosh to PC

To transfer a file from a Discovery Series application running on a Macintosh to a Discovery Series application running on a PC, you need to tag the file name with the suffix appropriate to the image file type (e.g., 2-D scan).

For 2-D scans, use the suffix: .gsc

For Filtered images, use the suffix: .gim

For Gaussian images, use the suffix: .gsp

For calstrip scans, use the suffix: .csc

For matchsets, use the suffix: .ms

For high-level matchsets, use the suffix: .hms

K.2 PC to Macintosh

PC versions of Discovery Series files can be read directly by Macintosh applications, with no required modifications.

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